

**SOLVED
PAPER**

NEET (UG)
07th May 2023

**Code
F3**

Important Instructions

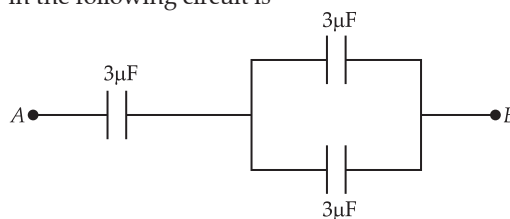
- The test is of **3 hours 20 minutes** duration and Test Booklet contains **200** multiple choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**. **50** Questions in each subject are divided into two **Section (A and B)** as per details given below:
(a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos- 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
(b) **Section B** shall consist of **15 (Fifteen)** Questions in each subject (Question Nos- 36 to 50, 86 to 100, 136 to 150 and 80 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.
Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, the first ten questions answered by the candidate shall be evaluated.
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, one mark will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
- Use of Electronic/Manual Calculator is prohibited.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

PHYSICS

Section A

- Q. 1. A football player is moving southward and suddenly turns eastward with the same speed to avoid an opponent. The force that acts on the player while turning is
(1) along north-ward (2) along north-east
(3) along south-west (4) along east-ward
- Q. 2. An ac source is connected to a capacitor C. Due to decrease in its operating frequency
(1) displacement current increases.
(2) displacement current decreases.
(3) capacitive reactance remains constant
(4) capacitive reactance decreases.
- Q. 3. The half life of a radioactive substance is 20 minutes. In how much time, the activity of substance drops to $\left(\frac{1}{16}\right)^{\text{th}}$ of its initial value?

- (1) 40 minutes (2) 60 minutes
(3) 80 minutes (4) 20 minutes
- Q. 4. The net magnetic flux through any closed surface is
(1) Positive (2) Infinity
(3) Negative (4) Zero
- Q. 5. The equivalent capacitance of the system shown in the following circuit is



- (1) $3\mu\text{F}$ (2) $6\mu\text{F}$
(3) $9\mu\text{F}$ (4) $2\mu\text{F}$

- Q. 6. The ratio of frequencies of fundamental harmonic produced by an open pipe to that of closed pipe having the same length is

(1) 2 : 1 (2) 1 : 3
(3) 3 : 1 (4) 1 : 2

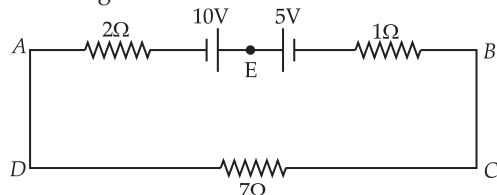
- Q. 7. In hydrogen spectrum, the shortest wavelength in the Balmer series is λ . The shortest wavelength in the Bracket series is

(1) 4λ (2) 9λ
(3) 16λ (4) 2λ

- Q. 8. A 12 V, 60 W lamp is connected to the secondary of a step down transformer, whose primary is connected to ac mains of 220 V. Assuming the transformer to be ideal, what is the current in the primary winding?

(1) 2.7 A (2) 3.7 A
(3) 0.37 A (4) 0.27 A

- Q. 9. The magnitude and direction of the current in the following circuit is



- (1) 0.5 A from A to B through E
(2) $\frac{5}{9}$ A from A to B through E
(3) 1.5 A from B to A through E
(4) 0.2 A from B to A through E
- Q. 10. The magnetic energy stored in an inductor of inductance $4\mu\text{H}$ carrying a current of 2 A is
- (1) 4 mJ (2) 8 mJ
(3) 8 μJ (4) 4 μJ
- Q. 11. The potential energy of a long spring when stretched by 2 cm is U. If the spring is stretched by 8 cm, potential energy stored in it will be
- (1) 4U (2) 8U
(3) 16U (4) 2U
- Q. 12. The errors in the measurement which arise due to unpredictable fluctuations in temperature and voltage supply are
- (1) Personal errors
(2) Least count errors
(3) Random errors
(4) Instrumental errors
- Q. 13. Resistance of a carbon resistor determined from colour codes is $(22000 \pm 5\%) \Omega$. The colour of third band must be
- (1) Green (2) Orange
(3) Yellow (4) Red
- Q. 14. A metal wire has mass (0.4 ± 0.002) g, radius (0.3 ± 0.001) mm and length (5 ± 0.02) cm. The maximum possible percentage error in the measurement of density will nearly be
- (1) 1.3% (2) 1.6%
(3) 1.4% (4) 1.2%

- Q. 15. Light travels a distance x in time t_1 in air and $10x$ in time t_2 in another denser medium. What is the critical angle for this medium?

(1) $\sin^{-1}\left(\frac{10t_2}{t_1}\right)$ (2) $\sin^{-1}\left(\frac{t_1}{10t_2}\right)$
(3) $\sin^{-1}\left(\frac{10t_1}{t_2}\right)$ (4) $\sin^{-1}\left(\frac{t_2}{t_1}\right)$

- Q. 16. In a plane electromagnetic wave travelling in free space, the electric field component oscillates sinusoidally at a frequency of 2.0×10^{10} Hz and amplitude 48 Vm^{-1} . Then the amplitude of oscillating magnetic field is (Speed of light in free space = $3 \times 10^8 \text{ m s}^{-1}$)

(1) $1.6 \times 10^{-8} \text{ T}$ (2) $1.6 \times 10^{-7} \text{ T}$
(3) $1.6 \times 10^{-6} \text{ T}$ (4) $1.6 \times 10^{-9} \text{ T}$

- Q. 17. The ratio of radius of gyration of a solid sphere of mass M and radius R about its own axis to the radius of gyration of the thin hollow sphere of same mass and radius about its axis is

(1) 5 : 3 (2) 2 : 5
(3) 5 : 2 (4) 3 : 5

- Q. 18. The amount of energy required to form a soap bubble of radius 2 cm from a soap solution is nearly (surface tension of soap solution = 0.03 N m^{-1})

(1) $5.06 \times 10^{-4} \text{ J}$ (2) $3.01 \times 10^{-4} \text{ J}$
(3) $50.1 \times 10^{-4} \text{ J}$ (4) $30.16 \times 10^{-4} \text{ J}$

- Q. 19. In a series LCR circuit, the inductance L is 10 mH, capacitance C is $1\mu\text{F}$ and resistance R is 100Ω . The frequency at which resonance occurs is

(1) 15.9 kHz (2) 1.59 rad/s
(3) 1.59 kHz (4) 15.9 rad/s

- Q. 20. If $\oint_s \vec{E} \cdot d\vec{S} = 0$ over a surface, then

(1) the magnitude of electric field on the surface is constant.
(2) all the charges must necessarily be inside the surface.
(3) the electric field inside the surface is necessarily uniform.
(4) the number of flux lines entering the surface must be equal to the number of flux lines leaving it.

- Q. 21. The venturi-meter works on

(1) Bernoulli's principle
(2) The principle of parallel axes
(3) The principle of perpendicular axes
(4) Huygen's principle

- Q. 22. Two bodies of mass m and $9m$ are placed at a distance R . The gravitational potential on the line joining the bodies where the gravitational field equals zero, will be (G = gravitational constant)

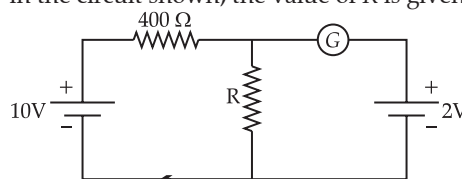
(1) $-\frac{12Gm}{R}$ (2) $-\frac{16Gm}{R}$
(3) $-\frac{20Gm}{R}$ (4) $-\frac{8Gm}{R}$

- Q. 23. The temperature of a gas is -50°C . To what temperature the gas should be heated so that the rms speed is increased by 3 times?
 (1) 3295°C (2) 3097K
 (3) 223K (4) 669°C
- Q. 24. The minimum wavelength of X-rays produced by an electron accelerated through a potential difference of V volts is proportional to
 (1) $\frac{1}{V}$ (2) $\frac{1}{\sqrt{V}}$
 (3) V^2 (4) $\sqrt{V}$
- Q. 25. A vehicle travels half the distance with speed v and the remaining distance with speed $2v$. Its average speed is
 (1) $\frac{2v}{3}$ (2) $\frac{4v}{3}$
 (3) $\frac{3v}{4}$ (4) $\frac{v}{3}$
- Q. 26. The angular acceleration of a body, moving along the circumference of a circle, is
 (1) along the radius towards the centre
 (2) along the tangent to its position
 (3) along the axis of rotation
 (4) along the radius, away from centre
- Q. 27. Let a wire be suspended from the ceiling (rigid support) and stretched by a weight W attached at its free end. The longitudinal stress at any point of cross-sectional area A of the wire is
 (1) $\frac{W}{A}$ (2) $\frac{W}{2A}$
 (3) Zero (4) $\frac{2W}{A}$
- Q. 28. A full wave rectifier circuit consists of two p-n junction diodes, a centre-tapped transformer, capacitor and a load resistance. Which of these components remove the ac ripple from the rectified output?
 (1) p-n junction diodes
 (2) Capacitor
 (3) Load resistance
 (4) A centre-tapped transformer
- Q. 29. An electric dipole is placed at an angle of 30° with an electric field of intensity $2 \times 10^5 \text{ NC}^{-1}$. It experiences a torque equal to 4 N m . Calculate the magnitude of charge on the dipole, if the dipole length is 2 cm .
 (1) 6 mC (2) 4 mC
 (3) 2 mC (4) 8 mC
- Q. 30. A Carnot engine has an efficiency of 50% when its source is at a temperature 327°C . The temperature of the sink is
 (1) 15°C (2) 100°C
 (3) 200°C (4) 27°C
- Q. 31. For Young's double slit experiment, two statements are given below:
Statement 1: If screen is moved away from the plane of slits, angular separation of the fringes remains constant.

Statement II: If the monochromatic source is replaced by another monochromatic source of larger wavelength, the angular separation of fringes decreases.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Both **Statement I** and **Statement II** are false.
 (2) **Statement I** is true but **Statement II** is false.
 (3) **Statement I** is false but **Statement II** is true.
 (4) Both **Statement I** and **Statement II** are true.
- Q. 32. The work functions of Caesium (Cs), Potassium (K) and Sodium (Na) are 2.14 eV , 2.30 eV and 2.75 eV respectively. If incident electromagnetic radiation has an incident energy of 2.20 eV , which of these photosensitive surfaces may emit photoelectrons?
 (1) Both Na and K (2) K only
 (3) Na only (4) Cs only
- Q. 33. Given below are two statements:
Statement 1: Photovoltaic devices can convert optical radiation into electricity.
Statement II: Zener diode is designed to operate under reverse bias in breakdown region.
 In the light of the above statements, choose the **most appropriate** answer from the options given below:
 (1) Both **Statement I** and **Statement II** are incorrect.
 (2) **Statement I** is correct but **Statement II** is incorrect.
 (3) **Statement I** is incorrect but **Statement II** is correct.
 (4) Both **Statement I** and **Statement II** are correct.
- Q. 34. If the galvanometer G does not show any deflection in the circuit shown, the value of R is given by

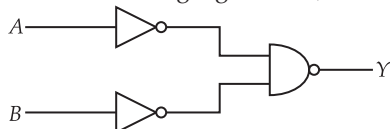


- (1) 50Ω (2) 100Ω
 (3) 400Ω (4) 200Ω
- Q. 35. A bullet is fired from a gun at the speed of 280 m s^{-1} in the direction 30° above the horizontal. The maximum height attained by the bullet is ($g=9.8 \text{ m s}^{-2}$, $\sin 30^{\circ} = 0.5$)
 (1) 2000 m (2) 1000 m
 (3) 3000 m (4) 2800 m

Section B

- Q. 36. A satellite is orbiting just above the surface of the earth with period T . If d is the density of the earth and G is the universal constant of gravitation, the quantity $\frac{3\pi}{Gd}$ represents
 (1) T^2 (2) T^3
 (3) $\sqrt{T}$ (4) T

Q. 37. For the following logic circuit, the truth table is



(1)	A	B	Y
	0	0	0
	0	1	1
	1	0	1
	1	1	1

(2)	A	B	Y
	0	0	1
	0	1	0
	1	0	1
	1	1	0

(3)	A	B	Y
	0	0	0
	0	1	0
	1	0	0
	1	1	1

(4)	A	B	Y
	0	0	1
	0	1	1
	1	0	1
	1	1	0

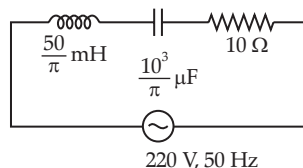
Q. 38. The radius of inner most orbit of hydrogen atom is 5.3×10^{-11} m. What is the radius of third allowed orbit of hydrogen atom?

- (1) $1.06 \text{ \AA}$ (2) $1.59 \text{ \AA}$
 (3) $4.77 \text{ \AA}$ (4) $0.53 \text{ \AA}$

Q. 39. A wire carrying a current I along the positive x -axis has length L . It is kept in a magnetic field $\vec{B} = (2\hat{i} + 3\hat{j} - 4\hat{k})$ T. The magnitude of the magnetic force acting on the wire is

- (1) $\sqrt{5} IL$ (2) $5 IL$
 (3) $\sqrt{3} IL$ (4) $3 IL$

Q. 40. The net impedance of circuit (as shown in figure) will be

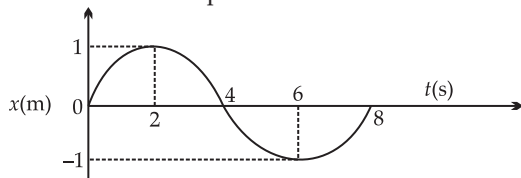


- (1) 15Ω (2) $5\sqrt{5} \Omega$
 (3) 25Ω (4) $10\sqrt{2} \Omega$

Q. 41. 10 resistors, each of resistance R are connected in series to a battery of emf E and negligible internal resistance. Then those are connected in parallel to the same battery, the current is increased n times. The value of n is

- (1) 100 (2) 1
 (3) 1000 (4) 10

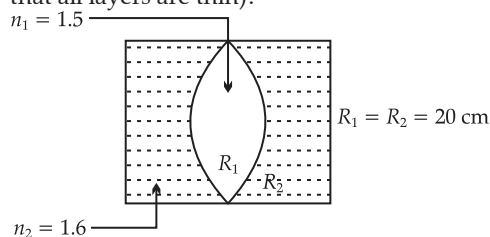
Q. 42. The x - t graph of a particle performing simple harmonic motion is shown in the figure. The acceleration of the particle at $t = 2$ s is



- (1) $-\frac{\pi^2}{8} \text{ m s}^{-2}$ (2) $\frac{\pi^2}{16} \text{ m s}^{-2}$

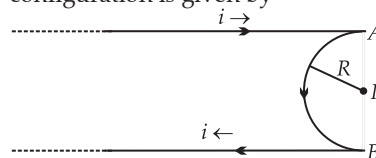
- (3) $-\frac{\pi^2}{16} \text{ m s}^{-2}$ (4) $\frac{\pi^2}{8} \text{ m s}^{-2}$

Q. 43. In the figure shown here, what is the equivalent focal length of the combination of lenses (Assume that all layers are thin)?



- (1) -40 cm (2) -100 cm
 (3) -50 cm (4) 40 cm

Q. 44. A very long conducting wire is bent in a semi-circular shape from A to B as shown in figure. The magnetic field at point P for steady current configuration is given by



- (1) $\frac{\mu_0 i}{4 R}$ pointed away from the page
 (2) $\frac{\mu_0 i}{4 R} \left[1 - \frac{2}{\pi} \right]$ pointed away from page
 (3) $\frac{\mu_0 i}{4 R} \left[1 - \frac{2}{\pi} \right]$ pointed into the page
 (4) $\frac{\mu_0 i}{4 R}$ pointed into the page

Q. 45. Calculate the maximum acceleration of a moving car so that a body lying on the floor of the car remains stationary. The coefficient of static friction between the body and the floor is 0.15 ($g = 10 \text{ ms}^{-2}$)

- (1) 150 ms^{-2} (2) 1.5 ms^{-2}
 (3) 50 ms^{-2} (4) 1.2 ms^{-2}

Q. 46. Two thin lenses are of same focal lengths (f), but one is convex and the other one is concave. When they are placed in contact with each other, the equivalent focal length of the combination will be

- (1) $\frac{f}{4}$ (2) $\frac{f}{2}$
 (3) Infinite (4) Zero

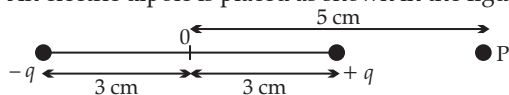
Q. 47. A horizontal bridge is built across a river. A student standing on the bridge throws a small ball vertically upwards with a velocity 4 m s^{-1} . The ball strikes the water surface after 4 s. The height of bridge above water surface is (Take $g = 10 \text{ ms}^{-2}$)

- (1) 60 m (2) 64 m
 (3) 68 m (4) 56 m

Q. 48. The resistance of platinum wire at 0°C is 2Ω and 6.8Ω at 80°C . The temperature coefficient of resistance of the wire is

- (1) $3 \times 10^{-3} \text{ }^\circ\text{C}^{-1}$ (2) $3 \times 10^{-2} \text{ }^\circ\text{C}^{-1}$
 (3) $3 \times 10^{-1} \text{ }^\circ\text{C}^{-1}$ (4) $3 \times 10^{-4} \text{ }^\circ\text{C}^{-1}$

Q. 49. An electric dipole is placed as shown in the figure.



The electric potential (in 10^2 V) at point P due to the dipole is (ϵ_0 = permittivity of free space and $\frac{1}{4\pi\epsilon_0} = K$)

- (1) $\left(\frac{5}{8}\right) qK$ (2) $\left(\frac{8}{5}\right) qK$

- (3) $\left(\frac{8}{3}\right) qK$ (4) $\left(\frac{3}{8}\right) qK$

Q. 50. A bullet from a gun is fired on a rectangular wooden block with velocity u . When bullet travels 24 cm through the block along its length horizontally, velocity of bullet becomes $\frac{u}{3}$. Then it further penetrates into the block in the same direction before coming to rest exactly at the other end of the block. The total length of the block is

- (1) 24 cm (2) 28 cm
 (3) 30 cm (4) 27 cm



CHEMISTRY

Section A

Q. 51. The element expected to form largest ion to achieve the nearest noble gas configuration is

- (1) F (2) N
 (3) Na (4) O

Q. 52. In Lassaigne's extract of an organic compound, both nitrogen and sulphur are present, which gives blood red colour with Fe^{3+} due to the formation of

- (1) NaSCN
 (2) $[\text{Fe}(\text{CN})_5 \text{NOS}]^{4-}$
 (3) $[\text{Fe}(\text{SCN})]^{2+}$
 (4) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3 \cdot x\text{H}_2\text{O}$

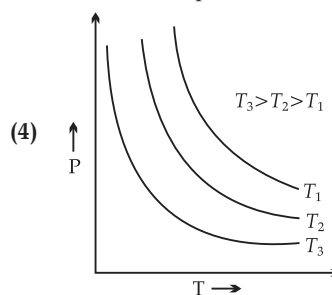
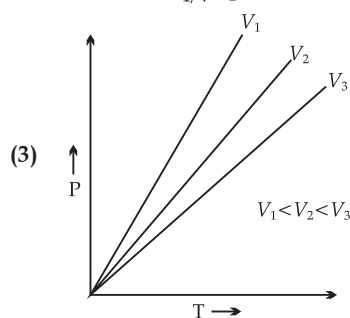
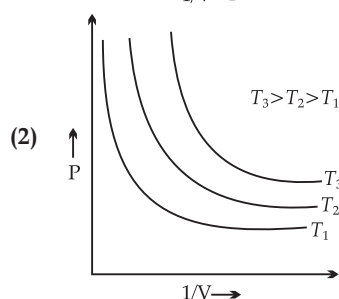
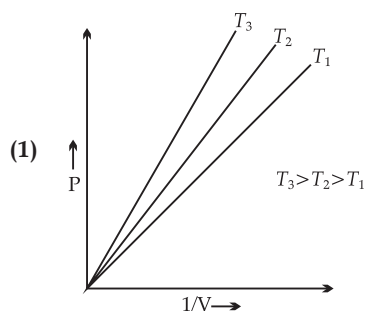
Q. 53. The relation between n_m , (n_m = the number of permissible values of magnetic quantum number (m)) for a given value of azimuthal quantum number (l), is

- (1) $l = 2n_m + 1$ (2) $n_m = 2l^2 + 1$
 (3) $n_m = l + 2$ (4) $l = \frac{n_m - 1}{2}$

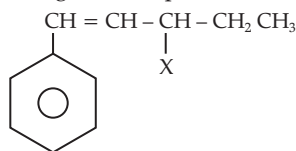
Q. 54. Which one is an example of heterogenous catalysis?

- (1) Hydrolysis of sugar catalysed by H^+ ions.
 (2) Decomposition of ozone in presence of nitrogen monoxide.
 (3) Combination between dinitrogen and dihydrogen to form ammonia in the presence of finely divided iron.
 (4) Oxidation of sulphur dioxide into sulphur trioxide in the presence of oxides of nitrogen.

Q. 55. Which amongst the following options is **correct** graphical representation of Boyle's Law?



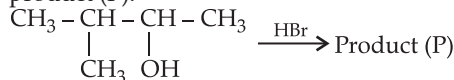
Q. 56. The given compound



is an example of _____.

- (1) aryl halide (2) allylic halide
(3) vinylic halide (4) benzylic halide

Q. 57. Consider the following reaction and identify the product (P).



3 - Methylbutan - 2 - ol

- (1) $\text{CH}_3\text{CH} = \text{CH} - \text{CH}_3$
(2) $\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CH} - \text{CH}_3 \\ | \quad | \\ \text{CH}_3 \quad \text{Br} \end{array}$

- (3) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{CH}_2 - \text{Br} \\ | \\ \text{CH}_3 \end{array}$

- (4) $\begin{array}{c} \text{Br} \\ | \\ \text{CH}_3 - \text{C} - \text{CH}_2 - \text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$

Q. 58. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**:
Assertion A: Helium is used to dilute oxygen in diving apparatus.

Reasons R: Helium has high solubility in O_2 .

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **A** and **R** are true and **R** is **NOT** the correct explanation of **A**.
(2) **A** is true but **R** is false.
(3) **A** is false but **R** is true.
(4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

Q. 59. The conductivity of centimolar solution of KCl at 25°C is $0.0210 \text{ ohm}^{-1} \text{ cm}^{-1}$ and the resistance of the cell containing the solution at 25°C is 60 ohm. The value of cell constant is

- (1) 3.28 cm^{-1} (2) 1.26 cm^{-1}
(3) 3.34 cm^{-1} (4) 1.34 cm^{-1}

Q. 60. The number of σ bonds, π bonds and lone pair of electrons in pyridine, respectively are

- (1) 12, 3, 0 (2) 11, 3, 1
(3) 12, 2, 1 (4) 11, 2, 0

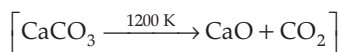
Q. 61. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**:
Assertion A: Metallic sodium dissolves in liquid ammonia giving a deep blue solution, which is paramagnetic.

Reasons R: The deep blue solution is due to the formation of amide.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**.
(2) **A** is true but **R** is false.
(3) **A** is false but **R** is true.
(4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

Q. 62. The **right** option for the mass of CO_2 produced by heating 20 g of 20% pure limestone is (Atomic mass of Ca = 40)



- (1) 1.76 g (2) 2.64 g
(3) 1.32 g (4) 1.12 g

Q. 63. Intermolecular forces are forces of attraction and repulsion between interacting particles that will include

- A. dipole-dipole forces.
B. dipole-induced dipole forces.
C. hydrogen bonding.
D. covalent bonding.
E. dispersion forces.

Choose the **most appropriate** answer from the options given below :

- (1) A, B, C, D are correct.
(2) A, B, C, E are correct.
(3) A, C, D, E are correct.
(4) B, C, D, E are correct.

Q. 64. For a certain reaction, the rate = $k [\text{A}]^2 [\text{B}]$, when the initial concentration of A is tripled keeping concentration of B constant, the initial rate would

- (1) increase by a factor of six.
(2) increase by a factor of nine.
(3) increase by a factor of three.
(4) decrease by a factor of nine.

Q. 65. Taking stability as the factor, which one of the following represents **correct** relationship?

- (1) $\text{InI}_3 > \text{InI}$
(2) $\text{AlCl} > \text{AlCl}_3$
(3) $\text{TlI} > \text{TlI}_3$
(4) $\text{TlCl}_3 > \text{TlCl}$

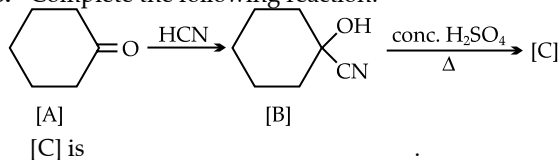
Q. 66. Which of the following statements are **NOT** correct?

- A. Hydrogen is used to reduce heavy metal oxides to metals.
B. Heavy water is used to study reaction mechanism.
C. Hydrogen is used to make saturated fats from oils.
D. The H-H bond dissociation enthalpy is lowest as compared to a single bond between two atoms of any element.
E. Hydrogen reduces oxides of metals that are more active than iron.

Choose the **most appropriate** answer from the options given below:

- (1) B, D only
(2) D, E only
(3) A, B, C only
(4) B, C, D, E only
- Q. 67. Weight (g) of two moles of the organic compound, which is obtained by heating sodium ethanoate with sodium hydroxide in presence of calcium oxide is
- (1) 32 (2) 30
(3) 18 (4) 16

Q. 68. Complete the following reaction:



- (1) (2)
(3) (4)
- Q. 69. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**:
Assertion A: In equation $\Delta_r G = -nFE_{\text{cell}}$, value of $\Delta_r G$ depends on n .
Reason R: E_{cell} is an intensive property and $\Delta_r G$ is an extensive property.
In the light of the above statements, choose the **correct** answer from the options given below:
- (1) Both **A** and **R** are true and **R** is **NOT** the correct explanation of **A**.
(2) **A** is true but **R** is false.
(3) **A** is false but **R** is true.
(4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

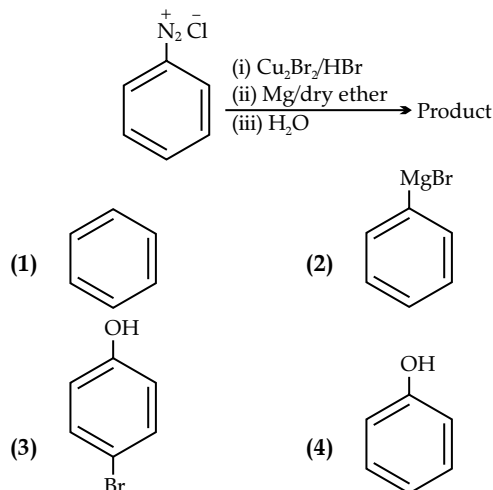
- Q. 70. Amongst the following, the total number of species NOT having eight electrons around central atom in its outer most shell, is
 NH_3 , AlCl_3 , BeCl_2 , CCl_4 , PCl_5 :
- (1) 2 (2) 4
(3) 1 (4) 3

- Q. 71. The **correct** order of energies of molecular orbitals of N_2 molecule, is:
- (1) $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$
(2) $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < \sigma^* 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y)$
(3) $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma 2p_z < \sigma^* 2p_z$
(4) $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$

- Q. 72. A compound is formed by two elements A and B. The element B forms cubic close packed structure and atoms of A occupy $1/3$ of tetrahedral voids.

If the formula of the compound is A_xB_y , then the value of $x + y$ is in option

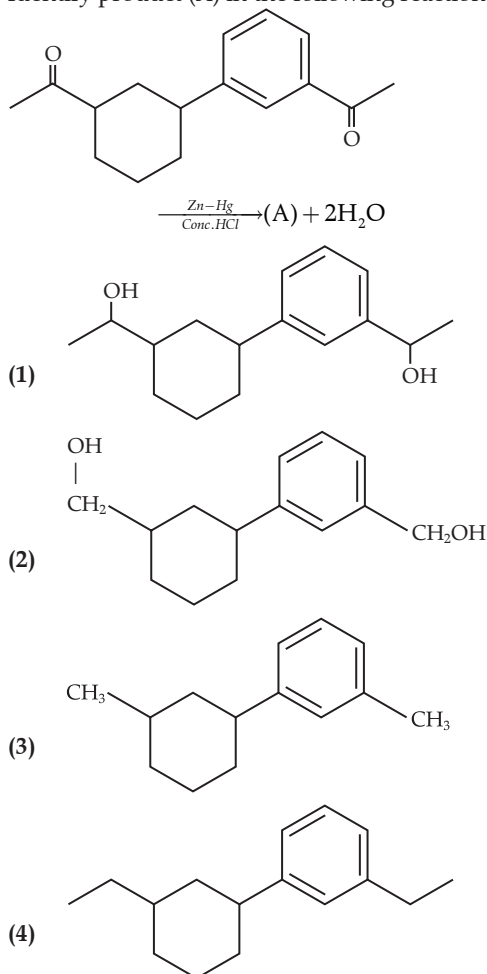
- (1) 4 (2) 3
(3) 2 (4) 5
- Q. 73. The stability of Cu^{2+} is more than Cu^+ salts in aqueous solution due to
- (1) enthalpy of atomization.
(2) hydration energy.
(3) second ionisation enthalpy.
(4) first ionisation enthalpy.
- Q. 74. Identify the product in the following reaction



- Q. 75. Given below are two statements:
Statement I: A unit formed by the attachment of a base to 1' position of sugar is known as nucleoside
Statement II: When nucleoside is linked to phosphorous acid at 5'-position of sugar moiety, we get nucleotide.
In the light of the above statements, choose the **correct** answer from the options given below:
- (1) Both **Statement I** and **Statement II** are false.
(2) **Statement I** is true but **Statement II** is false.
(3) **Statement I** is false but **Statement II** is true.
(4) Both **Statement I** and **Statement II** are true.
- Q. 76. Which one of the following statements is **correct**?
- (1) All enzymes that utilise ATP in phosphate transfer require Ca as the cofactor.
(2) The bone in human body is an inert and unchanging substance.
(3) Mg plays roles in neuromuscular function and interneuronal transmission.
(4) The daily requirement of Mg and Ca in the human body is estimated to be 0.2 - 0.3 g.
- Q. 77. Which of the following reactions will NOT give primary amine as the product?

- (1) $\text{CH}_3\text{CN} \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) LiAlH}_4} \text{Product}$
(2) $\text{CH}_3\text{NC} \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) LiAlH}_4} \text{Product}$
(3) $\text{CH}_3\text{CONH}_2 \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) LiAlH}_4} \text{Product}$
(4) $\text{CH}_3\text{CONH}_2 \xrightarrow{\text{Br}_2 / \text{KOH}} \text{Product}$

Q. 78. Identify product (A) in the following reaction:



Q. 79. Match List - I with List - II:

List - I	List-II
A. Coke	I. Carbon atoms are sp^3 hybridised.
B. Diamond	II. Used as a dry lubricant
C. Fullerene	III. Used as a reducing agent
D. Graphite	IV. Cage like molecules

Choose the **correct** answer from the options given below:

- (1) A-IV, B-I, C-II, D-III
 (2) A-III, B-I, C-IV, D-II
 (3) A-III, B-IV, C-I, D-II
 (4) A-II, B-IV, C-I, D-III

Q. 80. Amongst the given options which of the following molecules/ion acts as a Lewis acid?

- (1) H_2O (2) BF_3
 (3) OH^- (4) NH_3

Q. 81. Which amongst the following molecules on polymerization produces neoprene?

- (1) $H_2C = \overset{\text{Cl}}{\underset{|}{C}} - CH = CH_2$

- (2) $H_2C = CH - C \equiv CH$

- (3) $H_2C = \overset{\text{CH}_3}{\underset{|}{C}} - CH = CH_2$

- (4) $H_2C = CH - CH = CH_2$

Q. 82. Some tranquilizers are listed below. Which one from the following belongs to barbiturates?

- (1) Meprobamate
 (2) Valium
 (3) Veronal
 (4) Chlordiazepoxide

Q. 83. Homoleptic complex from the following complexes is

- (1) Diamminechloridonitrito - N- platinum (II)
 (2) Pentaamminecarbonatocobalt (III) chloride
 (3) Triamminetriaquachromium (III) chloride
 (4) Potassiumtrioxalatoaluminate (III)

Q. 84. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**:
Assertion A: A reaction can have zero activation energy.

Reasons R: The minimum extra amount of energy absorbed by reactant molecules so that their energy becomes equal to threshold value, is called activation energy.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **A** and **R** are true and **R** is **NOT** the correct explanation of **A**.
 (2) **A** is true but **R** is false.
 (3) **A** is false but **R** is true.
 (4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

Q. 85. Select the **correct** statements from the following:

- A. Atoms of all elements are composed of two fundamental particles.
 B. The mass of the electron is 9.10939×10^{-31} kg.
 C. All the isotopes of a given element show same chemical properties.
 D. Protons and electrons are collectively known as nucleons.
 E. Dalton's atomic theory, regarded the atom as an ultimate particle of matter.

Choose the **correct** answer from the options given below:

- (1) C, D and E only
 (2) A and E only
 (3) B, C and E only
 (4) A, B and C only

Section B

Q. 86. Match List-I with List - II:

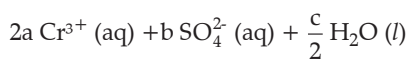
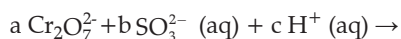
List-I (Oxoacids of Sulphur)	List-II (Bonds)
A. Peroxodisulphuric acid	I. Two S-OH, Four S=O, One S-O-S

B.	Sulphuric acid	II.	Two S-OH, One S=O
C.	Pyrosulphuric acid	III.	Two S-OH, Four S=O, One S-O-O-S
D.	Sulphurous acid	IV.	Two S-OH, Two S=O

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
 (2) A-I, B-III, C-IV, D-II
 (3) A-III, B-IV, C-II, D-I
 (4) A-1, B-III, C-II, D-IV

Q. 87. On balancing the given redox reaction,



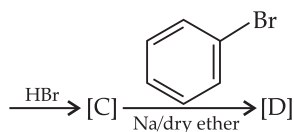
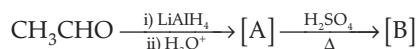
the coefficients a, b and c are found to be, respectively

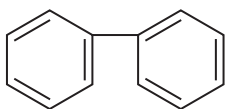
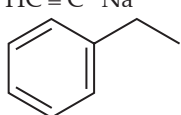
- (1) 3, 8, 1 (2) 1, 8, 3
 (3) 8, 1, 3 (4) 1, 3, 8

Q. 88. What fraction of one edge centred octahedral void lies in one unit cell of fcc?

- (1) $\frac{1}{3}$ (2) $\frac{1}{4}$
 (3) $\frac{1}{12}$ (4) $\frac{1}{2}$

Q. 89. Identify the final product [D] obtained in the following sequence of reactions.

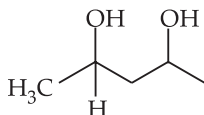


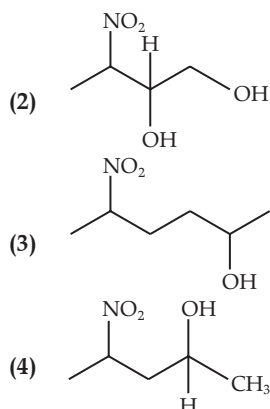
- (1) 
 (2) C_4H_{10}
 (3) $\text{HC} \equiv \text{C}^\ominus \text{Na}^+$
 (4) 

Q. 90. Which complex compound is most stable?

- (1) $[\text{Co}(\text{NH}_3)_3(\text{NO}_3)_3]$
 (2) $[\text{CoCl}_2(\text{en})_2] \text{NO}_3$
 (3) $[\text{Co}(\text{NH}_3)_6]_2(\text{SO}_4)_3$
 (4) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Br}](\text{NO}_3)_2$

Q. 91. Which amongst the following will be most readily dehydrated under acidic conditions?

- (1) 



Q. 92. Given below are two statements:

Statement I: The nutrient deficient water bodies lead to eutrophication.

Statement II: Eutrophication leads to decrease in the level of oxygen in the water bodies.

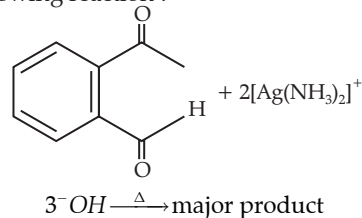
In the light of the above statements, choose the **correct** answer from the options given below:

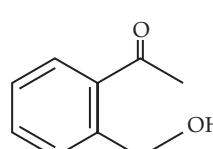
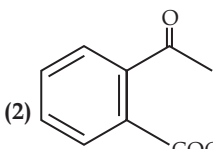
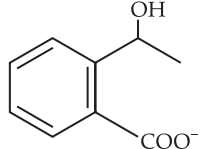
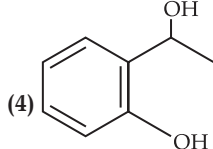
- (1) Both **Statement I** and **Statement II** are false.
 (2) **Statement I** is correct but **Statement II** is false.
 (3) **Statement I** is incorrect but **Statement II** is true.
 (4) Both **Statement I** and **Statement II** are true.

Q. 93. The reaction that does **NOT** take place in a blast furnace between 900 K to 1500 K temperature range during extraction of iron is

- (1) $\text{FeO} + \text{CO} \rightarrow \text{Fe} + \text{CO}_2$
 (2) $\text{C} + \text{CO}_2 \rightarrow 2\text{CO}$
 (3) $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$
 (4) $\text{Fe}_2\text{O}_3 + \text{CO} \rightarrow 2\text{FeO} + \text{CO}_2$

Q. 94. Identify the major product obtained in the following reaction :



- (1)  (2) 
 (3)  (4) 

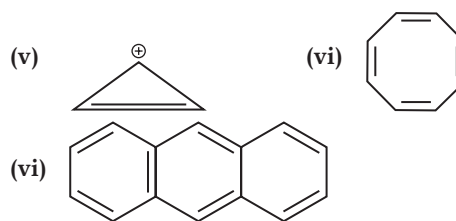
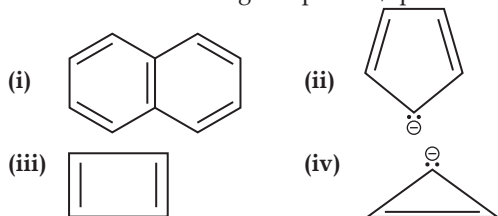
Q. 95. Which amongst the following options is the **correct** relation between change in enthalpy and change in internal energy?

- (1) $\Delta H = \Delta U + \Delta n_g RT$
 (2) $\Delta H - \Delta U = - \Delta n_g RT$

- (3) $\Delta H + \Delta U = \Delta nR$
 (4) $\Delta H = \Delta U - \Delta n_g RT$
- Q. 96. Which of the following statements are **INCORRECT**?
- A. All the transition metals except scandium form MO oxides which are ionic.
 B. The highest oxidation number corresponding to the group number in transition metal oxides is attained in Sc_2O_3 to Mn_2O_7 .
 C. Basic character increases from V_2O_3 to V_2O_4 to V_2O_5 .
 D. V_2O_4 dissolves in acids to give VO_4^{3-} salts.
 E. CrO is basic but Cr_2O_3 is amphoteric.
- Choose the **correct** answer from the options given below:
- (1) B and D only
 (2) C and D only
 (3) B and C only
 (4) A and E only

- Q. 97. The equilibrium concentrations of the species in the reaction $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$ are 2, 3, 10 and 6 mol L^{-1} , respectively at 300 K. ΔG° for the reaction is ($R = 2 \text{ cal/mol K}$)
- (1) -137.26 cal
 (2) -1381.80 cal
 (3) -13.73 cal
 (4) 1372.60 cal

- Q. 98. Consider the following compounds/species:

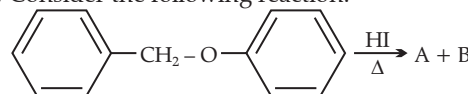


The number of compounds/species which obey Huckel's rule is _____.

- (1) 6 (2) 2
 (3) 5 (4) 4

- Q. 99. Pumice stone is an example of
- (1) gel (2) solid sol
 (3) foam (4) sol

- Q. 100. Consider the following reaction:



Identify products A and B.

- (1) $\text{A} = \text{C}_6\text{H}_5\text{CH}_2\text{OH}$ and $\text{B} = \text{C}_6\text{H}_5\text{I}$
 (2) $\text{A} = \text{C}_6\text{H}_5\text{CH}_2\text{I}$ and $\text{B} = \text{C}_6\text{H}_5\text{OH}$
 (3) $\text{A} = \text{C}_6\text{H}_5\text{CH}_3$ and $\text{B} = \text{C}_6\text{H}_5\text{I}$
 (4) $\text{A} = \text{C}_6\text{H}_5\text{CH}_3$ and $\text{B} = \text{C}_6\text{H}_5\text{OH}$

BOTANY

Section A

- Q. 101. The phenomenon of pleiotropism refers to
- (1) presence of two alleles, each of the two genes controlling a single trait.
 (2) a single gene affecting multiple phenotypic expression.
 (3) more than two genes affecting a single character.
 (4) presence of several alleles of a single gene controlling a single crossover.
- Q. 102. In tissue culture experiments, leaf mesophyll cells are put in a culture medium to form callus. This phenomenon may be called as :
- (1) Dedifferentiation
 (2) Development
 (3) Senescence
 (4) Differentiation
- Q. 103. Movement and accumulation of ions across a membrane against their concentration gradient

can be explained by

- (1) Facilitated Diffusion
 (2) Passive Transport
 (3) Active Transport
 (4) Osmosis.
- Q. 104. Among The Evil Quartet', which one is considered the most important cause driving extinction of species?
- (1) Over exploitation for economic gain
 (2) Alien species invasions
 (3) Co-extinctions
 (4) Habitat loss and fragmentation
- Q. 105. Upon exposure to UV radiation, DNA stained with ethidium bromide will show
- (1) Bright blue colour
 (2) Bright yellow colour
 (3) Bright orange colour
 (4) Bright red colour

- Q. 106.** Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**:
Assertion A: The first stage of gametophyte in the life cycle of moss is protonema stage.
Reason R: Protonema develops directly from spores produced in capsule.
 In the light of the above statements, choose the **most appropriate** answer from the options given below:
- (1) Both **A** and **R** are correct but **R** is NOT the correct explanation of **A**.
 - (2) **A** is correct but **R** is not correct.
 - (3) **A** is not correct but **R** is correct.
 - (4) Both **A** and **R** are correct and **R** is the correct explanation of **A**.
- Q. 107.** What is the role of RNA polymerase III in the process of transcription in eukaryotes?
- (1) Transcription of tRNA, 5 srRNA and snRNA
 - (2) Transcription of precursor of mRNA
 - (3) Transcription of only snRNAs
 - (4) Transcription of rRNAs (28S, 18S, and 5.8S)
- Q. 108.** The historic convention on Biological Diversity, 'The Earth Summit' was held in Rio de Janeiro in the year :
- | | |
|----------|----------|
| (1) 1992 | (2) 1986 |
| (3) 2002 | (4) 1985 |
- Q. 109.** Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R** :
Assertion A: ATP is used at two steps in glycolysis,
Reason R: First ATP is used in converting glucose into glucose-6-phosphate and second ATP is used in conversion of fructose-6- phosphate into fructose-1-6-diphosphate.
 In the light of the above statements, choose the correct answer from the options given below:
- (1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
 - (2) **A** is true but **R** is false.
 - (3) **A** is false but **R** is true.
 - (4) Both **A** and **R** are true and **R** is the correct explanation of **A**.
- Q. 110.** The thickness of ozone in a column of air in the atmosphere is measured in terms of:
- | | |
|--------------|------------------|
| (1) Decibels | (2) Decameter |
| (3) Kilobase | (4) Dobson units |
- Q. 111.** Given below are two statements:
Statement I: Endarch and exarch are the terms often used for describing the position of secondary xylem in the plant body.
Statement II: Exarch condition is the most common feature of the root system.
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) Both **Statement I** and **Statement II** are false.
 - (2) **Statement I** is correct but **Statement II** false.
 - (3) **Statement I** is incorrect but **Statement II** is true.
 - (4) Both **Statement I** and **Statement II** are true.
- Q. 112.** Identify the pair of heterosporous pteridophytes among the following:
- (1) *Selaginella* and *Salvinia*
 - (2) *Psilotum* and *Salvinia*
 - (3) *Equisetum* and *Salvinia*
 - (4) *Lycopodium* and *Selaginella*
- Q. 113.** Identify the **correct** statements :
- A.** Detrivores perform fragmentation.
 - B.** The humus is further degraded by some microbes during mineralization.
 - C.** Water soluble inorganic nutrients go down into the soil and get precipitated by a process called leaching.
 - D.** The detritus food chain begins with living organisms.
 - E.** Earthworms break down detritus into smaller particles by a process called catabolism.
- Choose the **correct** answer from the options given below:
- | | |
|------------------|------------------|
| (1) B, C, D only | (2) C, D, E only |
| (3) D, E, A only | (4) A, B, C only |
- Q. 114.** Axile placentation is observed in
- (1) China rose, Beans and Lupin
 - (2) Tomato, *Dianthus* and Pea
 - (3) China rose, *Petunia* and Lemon
 - (4) Mustard, Cucumber and Primrose
- Q. 115.** Family Fabaceae differs from Solanaceae and Liliaceae. With respect to the stamens, pick out the characteristics specific to family Fabaceae but not found in Solanaceae or Liliaceae.
- (1) Polyadelphous and epipetalous stamen
 - (2) Monadelphous and Monotheou anthers
 - (3) Epiphylous and Dithecous anthers
 - (4) Diadelphous and Dithecous anthers
- Q. 116.** The reaction centre in PS II has an absorption maxima at
- | | |
|------------|------------|
| (1) 700 nm | (2) 660 nm |
| (3) 780 nm | (4) 680 nm |
- Q. 117.** During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out
- | | |
|---------------------|--------------|
| (1) DNA | (2) Histones |
| (3) Polysaccharides | (4) RNA |
- Q. 118.** Among eukaryotes, replication of DNA takes place in
- | | |
|--------------------------|--------------------------|
| (1) S phase | (2) G ₁ phase |
| (3) G ₁ phase | (4) M phase |
- Q. 119.** The process of appearance of recombination nodules occurs at which sub stage of prophase I in meiosis?
- | | |
|----------------|---------------|
| (1) Pachytene | (2) Diplotene |
| (3) Diakinesis | (4) Zygotene |

Q. 120. How many ATP and NADPH_2 are required for the synthesis of one molecule of glucose during calvin cycle?

- (1) 18 ATP and 12 NADPH_2
- (2) 12 ATP and 16 NADPH_2
- (3) 18 ATP and 16 NADPH_2
- (4) 12 ATP and 12 NADPH_2

Q. 121. In the equation

$$\text{GPP} - \text{R} = \text{NPP}$$

GPP is Gross Primary Productivity

NPP is Net Primary Productivity

R here is _____.

- (1) Respiratory quotient
- (2) Respiratory loss
- (3) Reproductive allocation
- (4) Photosynthetically active radiation

Q. 122. Large, colourful, fragrant flowers with nectar are seen in:

- (1) bird pollinated plants
- (2) bat pollinated plants
- (3) wind pollinated plants
- (4) insect pollinated plants

Q. 123. Spraying of which of the following phytohormone on juvenile conifers helps in hastening the maturity period, that leads to early seed production?

- (1) Gibberellic Acid
- (2) Zeatin
- (3) Absciscic Acid
- (4) Indole-3-butyric Acid

Q. 124. Which micronutrient is required for splitting of water molecule during photosynthesis?

- (1) molybdenum
- (2) magnesium
- (3) copper
- (4) manganese

Q. 125. Expressed Sequence Tags (ESTs) refers to

- (1) All genes that are expressed as proteins.
- (2) All genes whether expressed or unexpressed.
- (3) Certain important expressed genes.
- (4) All genes that are expressed as RNA.

Q. 126. Frequency of recombination between gene pairs on same chromosome as a measure of the distance between genes to map their position on chromosome, was used for the first time by

- (1) Sutton and Boveri
- (2) Alfred Sturtevant
- (3) Henking
- (4) Thomas Hunt Morgan

Q. 127. Given below are two statements:

Statement I: The forces generated by transpiration can lift a xylem-sized column of water over 130 meters height.

Statement II: Transpiration cools leaf surfaces sometimes 10 to 15 degrees, by evaporative cooling.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are incorrect.
- (2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

Q. 128. In gene gun method used to introduce alien DNA into host cells, microparticles of _____ metal are used.

- (1) Zinc
- (2) Tungsten or gold
- (3) Silver
- (4) Copper

Q. 129. Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**:
Assertion A: Late wood has fewer xylary elements with narrow vessels.

Reason R: Cambium is less active in winters.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
- (2) **A** is true but **R** is false.
- (3) **A** is false but **R** is true.
- (4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

Q. 130. What is the function of tassels in the corn cob?

- (1) To trap pollen grains
- (2) To disperse pollen grains
- (3) To protect seeds
- (4) To attract insects

Q. 131. In angiosperm, the haploid, diploid and triploid structures of a fertilized embryo sac sequentially are:

- (1) Antipodals, synergids, and primary endosperm nucleus
- (2) Synergids, zygote and primary endosperm nucleus
- (3) Synergids, antipodals and polar nuclei
- (4) Synergids, primary endosperm nucleus and zygote

Q. 132. Which hormone promotes internode/ptiole elongation in deep water rice?

- (1) Kinetin
- (2) Ethylene
- (3) 2, 4-D
- (4) GA_3

Q. 133. Which of the following stages of meiosis involves division of centromere?

- (1) Metaphase II
- (2) Anaphase II
- (3) Telophase
- (4) Metaphase I

Q. 134. Unequivocal proof that DNA is the genetic material was first proposed by

- (1) Alfred Hershey and Martha Chase
- (2) Avery, Macleoid and McCarthy
- (3) Wilkins and Franklin
- (4) Frederick Griffith

Q. 135. Cellulose does not form blue colour with Iodine because

- (1) It is a helical molecule.
- (2) It does not contain complex helices and hence cannot hold iodine molecules.
- (3) It breaks down when iodine reacts with it.
- (4) It is a disaccharide.

Section B

Q. 136. Match List I with List II:

	List I		List II
A.	Oxidative decarboxylation	I.	Citrate synthase
B.	Glycolysis	II.	Pyruvate dehydrogenase
C.	Oxidative phosphorylation	III.	Electron transport system
D.	Tricarboxylic acid cycle	IV.	EMP pathway

Choose the correct answer from the option given below:

- (1) A-II, B-IV, C-I, D-III
- (2) A-III, B-I, C-II, D-IV
- (3) A-II, B-IV, C-III, D-I
- (4) A-III, B-IV, C-II, D-I

Q. 137. Which of the following combinations required for chemiosmosis?

- (1) Membrane, proton pump, proton gradient, NADP synthase
- (2) Proton pump, electron gradient, ATP synthase
- (3) Proton pump, electron gradient, NADP synthase
- (4) Membrane, proton pump, proton gradient, ATP synthase

Q. 138. Melonate inhibits the growth of pathogenic bacteria by inhibiting the activity of

- (1) Amylase
- (2) Lipase
- (3) Dinitrogenase
- (4) Succinic dehydrogenase

Q. 139. Given below are two statements : One labelled as **Assertion A** and the other labelled as **Reason R**:
Assertion A: In gymnosperms the pollen grains are released from the microsporangium and carried by air currents.

Reason R: Air currents carry the pollen grains to the mouth of the archegonia where the male gametes are discharged and pollen tube is not formed.

In the light of the above statements, choose the correct answer from the options give below:

- (1) Both A and R are true but R is NOT the correct explanation of A.
- (2) A is true but R is false.
- (3) A is false but R is true.
- (4) Both A and R are true and R is the correct explanation of A.

Q. 140. Main steps in the formation of Recombinant DNA are given below. Arrange these steps in a correct sequence.

- A. Insertion of recombinant DNA into the host cell.
- B. Cutting of DNA at specific location by restriction enzyme.
- C. Isolation of desired DNA fragment. D.

Amplification of gene of interest using PCR. Choose the correct answer from the options given below:

- (1) C, A, B, D
- (2) C, B, D, A
- (3) B, D, A, C
- (4) B, C, D, A

Q. 141. Match List I with List II :

	List I		List II
A.	Cohesion	I.	More attraction in liquid phase
B.	Adhesion	II.	Mutual attraction among water molecules
C.	Surface tension	III.	Water loss in liquid phase
D.	Guttation	IV.	Attraction towards polar surfaces

Choose the correct answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-III, B-I, C-IV, D-II
- (3) A-II, B-I, C-IV, D-III
- (4) A-II, B-IV, C-I, D-III

Q. 142. Which one of the following statements is NOT correct?

- (1) Algal blooms caused by excess of organic matter in water improve water quality and promote fisheries.
- (2) Water hyacinth grows abundantly in eutrophic water bodies and leads to an imbalance in the ecosystem dynamics of the water body.
- (3) The amount of some toxic substances of industrial waste water increases in the organisms at successive trophic levels.
- (4) The micro-organisms involved in biodegradation of organic matter in a sewage polluted water body consume a lot of oxygen causing the death of aquatic organisms.

Q. 143. How many different proteins does the ribosome consist of?

- (1) 60
- (2) 40
- (3) 20
- (4) 80

Q. 144. Match List I with List II:

	List I (Interaction)		List II (Species A and B)
A.	Mutualism	I.	+ (A), O(B)
B.	Commensalism	II.	- (A), O(B)
C.	Amensalism	III.	+ (A), -(B)
D.	Parasitism	IV.	+ (A), +(B)

Choose the correct answer from the options given below:

- (1) A-IV, B-I, C-II, D-III
- (2) A-IV, B-III, C-I, D-II
- (3) A-III, B-I, C-IV, D-II
- (4) A-IV, B-II, C-I, D-III

Q. 145. Given below are two statements : One is labelled as **Assertion A** and the other is labeled as **Reason R**:

Assertion A: A flower is defined as modified shoot wherein the shoot apical meristem changes to floral meristem.

Reason R: Internode of the shoot gets condensed to produce different floral appendages laterally at successive nodes instead of leaves.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
- (2) **A** is true but **R** is false.
- (3) **A** is false but **R** is true.
- (4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

Q. 146. Identify the **correct** statements:

- A. Lenticels are the lens-shaped openings permitting the exchange of gases.
- B. Bark formed early in the season is called hard bark.
- C. Bark is a technical term that refers to all tissues exterior to vascular cambium.
- D. Bark refers to periderm and secondary phloem.
- E. Phellogen is single-layered in thickness.

Choose the **correct** answer from the options given below:

- (1) A and D only
- (2) A, B and D only
- (3) B and C only
- (4) B, C and E only

Q. 147. Given below are two statements :

Statement I: Gause's 'Competitive Exclusion Principle states that two closely related species competing for the same resources cannot co-exist indefinitely and competitively inferior one will be eliminated eventually.

Statement II: In general, carnivores are more adversely affected by competition than herbivores. In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are false.
- (2) **Statement I** is correct but **Statement II** is false.
- (3) **Statement I** is incorrect but **Statement II** is true.
- (4) Both **Statement I** and **Statement II** are true.

Q. 148. Which of the following statements are correct

about Klinefelter's Syndrome?

- A. This disorder was first described by Langdon Down (1866).
- B. Such an individual has overall masculine development. However, the feminine development is also expressed.
- C. The affected individual is short statured.
- D. Physical, psychomotor and mental development is retarded.
- E. Such individuals are sterile.

Choose the **correct** answer from the options given below:

- (1) C and D only
- (2) B and E only
- (3) A and E only
- (4) A and B only

Q. 149. Match **List I** with **List II**:

	List I		List II
A.	M Phase	I.	Proteins are synthesized
B.	G ₂ Phase	II.	Inactive phase
C.	Quiescent stage	III.	Interval between mitosis and initiation of DNA replication
D.	G ₁ Phase	IV.	Equational division

Choose the correct answer from the options given below:

- (1) A-IV, B-II, C-I, D-III
- (2) A-IV, B-I, C-II, D-III
- (3) A-II, B-IV, C-I, D-III
- (4) A-III, B-II, C-IV, D-I

Q. 150. Match **List I** with **List II**:

	List I		List II
A.	Iron	I.	Synthesis of auxin
B.	Zinc	II.	Component of nitrate reductase
C.	Boron	III.	Activator of catalase
D.	Molybdenum	IV.	Cell elongation and differentiation

Choose the correct answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-III, B-I, C-IV, D-II
- (3) A-II, B-IV, C-I, D-III
- (4) A-III, B-II, C-I, D-IV

ZOOLOGY

Section A

Q. 151. Given below are two statements:

Statement I: In prokaryotes, the positively charged DNA is held with some negatively charged proteins in a region called nucleoid.

Statement II: In eukaryotes, the negatively charged DNA is wrapped around the positively charged histone octamer to form nucleosome.

In the light of the above statements, the **correct** answer from the options given below:

- (1) Both the **statements I** and **Statement II** are false.
- (2) **Statement I** is correct but **Statement II** is false.
- (3) **Statement I** incorrect but **Statement II** is true.
- (4) Both **Statement I** and **Statement II** are true.

Q. 152. In which blood corpuscles, the HIV undergoes replication and produces progeny viruses?

- (1) B-lymphocytes (2) Basophils
(3) Eosinophils (4) T_H cells

Q. 153. Which of the following statements is correct?

- (1) Biomagnification refers to increase in concentration of the toxicant at successive trophic levels.
(2) Presence of large amount of nutrients in water restricts 'Algal Bloom'.
(3) Algal Bloom decreases fish mortality.
(4) Eutrophication refers to increase in domestic sewage and waste water in lakes.

Q. 154. Match **List I** with **List II**.

	List I		List II
A.	P-wave	I.	Beginning of systole
B.	Q-wave	II.	Repolarisation of ventricles
X.	QRS complex	III.	Depolarisation of atria
Δ .	T-wave	iv.	Depolarisation of ventricles

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
(2) A-II, B-IV, C-I, D-III
(3) A-I, B-II, C-III, D-IV
(4) A-III, B-I, C-IV, D-II

Q. 155. Broad palm with single palm crease is visible in a person suffering from-

- (1) Turner's syndrome
(2) Klinefelter's syndrome
(3) Thalassemia
(4) Down's syndrome

Q. 156. Given below are two statements:

Statement I: RNA mutates at a faster rate.

Statement II: Viruses having RNA genome and shorter life span mutate and evolve faster.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are false.
(2) **Statement I** is true but **Statement II** is false.
(3) **Statement I** false but **Statement II** is true.
(4) Both **Statement I** and **Statement II** are true.

Q. 157. Given below are statements:

Statement I: Ligaments are dense irregular tissue.

Statement II: Cartilage is dense regular tissue.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are false.
(2) **Statement I** is true but **Statement II** is false.
(3) **Statement I** false but **Statement II** is true.
(4) Both **Statement I** and **Statement II** are true.

Q. 158. Given below are statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion A: Nephrons are of two types: Cortical & Juxta medullary, based on their relative position in cortex and medulla.

Reason R: Juxta medullary nephrons have short loop of Henle whereas, cortical nephrons have longer loop of Henle.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
(2) **A** is true but **R** is false.
(3) **A** is false but **R** is true.
(4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

Q. 159. Which of the following is not a cloning vector?

- (1) YAC (2) pBR322
(3) Probe (4) BAC

Q. 160. Given below are two statements:

Statement I: A protein is imagined as a line, the left end represented by first amino acid (C-terminal) and the right end represented by last amino acid (N-terminal).

Statement II: Adult human haemoglobin, consist of 4 subunits (two subunits of α type and two subunits of β type).

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are false.
(2) **Statement I** is true but **Statement II** is false.
(3) **Statement I** is false but **Statement II** is true.
(4) Both **Statement I** and **Statement II** are true.

Q. 161. Once the undigested and unabsorb substances enter the caecum, their back is prevented by-

- (1) Ileo-caecal valve
(2) Gastro-oesophageal sphincter
(3) Pyloric sphincter
(4) Sphincter of Oddi

Q. 162. Match **List I** with **List II**.

	List I		List II
A.	Vasectomy	I.	Oral method
B.	Coitus	II.	Barrier method
X.	Cervical caps	III.	Surgical method
Δ .	Saheli	IV.	Natural method

Choose the **correct** answer from the option given below:

- (1) A-III, B-IV, C-II, D-I
(2) A-II, B-III, C-I, D-IV
(3) A-IV, B-II, C-I, D-III
(4) A-III, B-I, C-IV, D-II

Q. 163. Which of the following functions is carried out by cytoskeleton in a cell?

- (1) Protein synthesis

- (2) Motility
(3) Transportation
(4) Nuclear division

Q. 164. Which one of the following common sexually transmitted diseases is completely curable when detected early and treated properly?

- (1) Gonorrhoea (2) Hepatitis-B
(3) HIV Infection (4) Genital herpes

Q. 165. Vital capacity of lung is _____.

- (1) IRV + ERV + TV + RV
(2) IRV + ERV + TV - RV
(3) IRV + ERV + TV
(4) IRV + ERV

Q. 166. Which of the following are NOT considered as the part of endomembrane system?

- A. Mitochondria
B. Endoplasmic Reticulum
C. Chloroplasts
D. Golgi complex

Choose the **most appropriate** answer from the options below:

- (1) A, C and E only
(2) A and D only
(3) A, D and E only
(4) B and D only

Q. 167. Match List I with List II.

List I		List II	
A.	CCK	I.	Kidney
B.	GIP	II.	Heart
C.	ANF	III.	Gastric gland
D.	ADH	IV.	Pancreas

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-IV, D-I
(2) A-II, B-IV, C-I, D-III
(3) A-IV, B-II, C-III, D-I
(4) A-IV, B-III, C-II, D-I

Q. 168. Match List I with List II.

List I		List II	
A.	Gene 'a'	I.	β -galactosidase
B.	Gene 'y'	II.	Transacetylase
C.	Gene 'i'	III.	Permease
D.	Gene 'z'	IV.	Repressor protein

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
(2) A-III, B-IV, C-I, D-II
(3) A-III, B-I, C-IV, D-II
(4) A-II, B-I, C-IV, D-III

Q. 169. Match List I with List II with respect to human eye.

List I		List II	
A.	Fovea	I.	Visible coloured portion of eye that regulates diameter of pupil.

B.	Iris	II.	External layer of eye formed of dense connective tissue.
C.	Blind spot	III.	Point of greatest visual acuity or resolution.
D.	Sclera	IV.	Point where optic nerve leaves the eyeball and photoreceptor cells are absent.

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
(2) A-I, B-IV, C-III, D-II
(3) A-II, B-I, C-III, D-IV
(4) A-III, B-I, C-IV, D-II

Q. 170. Select the correct group/set of Australian Marsupials exhibiting adaptive radiation.

- (1) Numbat, Spotted cuscus, Flying phalanger
(2) Mole, Flying squirrel, Tasmanian tiger cat
(3) Lemur, Anteater, Wolf
(4) Tasmanian wolf, Bobcat, Marsupial mole

Q. 171. Given below are two statements:

Statements I: Vas deferens receives a duct from seminal vesicle and opens into urethra as the ejaculatory duct.

Statements II: The cavity of the cervix is called cervical canal which along with vagina forms birth canal.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **Statements I** and **Statements II** are false.
(2) **Statements I** is correct but **Statements II** is false.
(3) **Statements I** incorrect but **Statement II** is true.
(4) Both **Statement I** and **Statement II** are true.

Q. 172. Match List I with List II.

List I		List II	
A.	Heroin	I.	Effect on cardiovascular system
B.	Morphine	II.	Slow down body function
C.	Cocaine	II.	Painkiller
D.	Morphine	III.	Interfere with transport of dopamine

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-I, C-IV, D-III

Q. 173. Given below are two statements:

	List I		List II
A.	Ringworm	I.	<i>Haemophilus influenzae</i>
B.	Filariasis	II.	<i>Trichophyton</i>
X.	Malaria	III.	<i>Wuchereria bancrofti</i>
Δ.	Pneumonia	IV.	<i>Plasmodium vivax</i>

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-III, B-II, C-I, D-IV
- (3) A-III, B-II, C-IV, D-I
- (4) A-II, B-II, C-IV, D-I

Q. 174. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**. **Assertion A:** Amniocentesis for sex determination is one of the strategies of Reproductive and Child Health Care Programme.

Reason R: Ban on amniocentesis checks increasing menace of female foeticide.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **A** and **R** are true and **R** is NOT the correct explanation of **A**.
- (2) **A** is true but **R** is false.
- (3) **A** is false but **R** is true.
- (4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

Q. 175. Radial symmetry is not found in adults of phylum_____.

- (1) Hemichordate
- (2) Coelenterata
- (3) Echinodermata
- (3) Ctenophora

Q. 176. Given below are two statements:

Statement I: Low temperature preserves the enzyme in a temporarily inactive state activity because proteins are denatured by whereas high temperature destroys enzymatic activity because proteins are denatured by heat.

Statement II: When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of the enzyme, it is known as competitive inhibitor.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **Statements I** and **Statements II** are false.
- (2) **Statements I** is true but **Statements II** is false.
- (3) **Statements I** is false but **statements II** is true.
- (4) Both **Statements I** and **statements II** are true.

Q. 177. Match **List I** and **List II**.

	List I		List II
A.	Taenia	I.	Nephridia
B.	<i>Paramoecium</i>	II.	Contractile vacuole
C.	<i>Periplaneta</i>	III.	Flame cells
D.	<i>Pheretima</i>	IV.	Urecose gland

Choose the **correct** answer from the options give below:

- (1) A-I, B-II, C-IV, D-III
- (2) A-III, B-II, C-IV, D-I
- (3) A-II, B-I, C-IV, D-III
- (4) A-I, B-II, C-III, D-IV

Q. 178. Which of the following statements are correct regarding female reproductive cycle?

- A. In non-primate mammals cyclical changes during reproduction are called oestrus cycle.
- B. First menstrual cycle begins at puberty and is called menopause.
- C. Lack of menstruation may be indicative of pregnancy.
- D. Cyclic menstruation extends between menarche and menopause.

Choose the **most appropriate** answer from the options given below:

- (1) A and B only
- (2) A, B and C only
- (3) A, C and D only
- (4) A and D only

Q. 179. Given below are two statements:

Statements I: Electrostatic precipitator is most widely used in thermal power plant.

Statement II: Electrostatic precipitator in thermal power plant removes ionising radiations

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are incorrect.
- (2) **Statement I** is correct but **Statement II** is incorrect.
- (3) **Statement I** incorrect but **Statement II** is correct.
- (4) Both **Statement I** and **Statement II** are correct.

Q. 180. Match **List I** and **List II**.

	List I (Cells)		List II (Secretion)
A.	Peptic cells	I.	Mucus
B.	Goblet cells	II.	Bile juice
C.	Oxyntic cells	III.	Proenzyme pepsinogen
D.	Hepatic cells	IV.	HCl and intrinsic factor for absorption of vitamin B ₁₂

Choose the **correct** answer from the options give below:

- (1) A-II, B-I, C-III, D-IV
- (2) A-III, B-I, C-IV, D-II
- (3) A-II, B-IV, C-I, D-III
- (4) A-IV, B-III, C-II, D-I

Q. 181. Given below are two statements: one labelled as **Assertion A** and the other is labelled as **Reason R**. **Assertion A:** Endometrium is necessary for implantation of blastocyst.

Reason R: In the absence of fertilization, the corpus luteum degenerates that causes disintegration of endometrium.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
- (2) **A** is true but **R** is false.
- (3) **A** is false but **R** is true.
- (4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

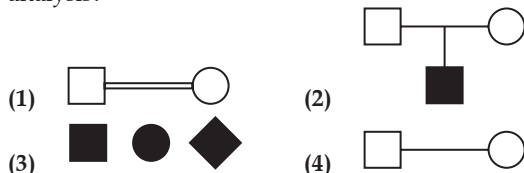
Q. 182. Match List I with List II.

List I (Type of Joint)		List II (Found between)	
A.	Cartilaginous	I.	Between flat skull bones
B.	Ball and Socket Joint	II.	Between adjacent vertebrae in vertebral column
X.	Fibrous Joint	III.	Between carpal and metacarpal of thumb
Δ.	Saddle Joint	IV.	Between humerus and pectoral girdle

Choose the **correct** answer from the options given below:

- (1) A-II, B-IV, C-I, D-III
- (2) A-I, B-IV, C-III, D-II
- (3) A-II, B-IV, C-III, D-I
- (4) A-III, B-I, C-II, D-IV

Q. 183. Which one of the following symbols represents mating between relatives in human pedigree analysis?



Q. 184. Which one of the following techniques does not serve the purpose of early diagnosis of a disease for its early treatment?

- (1) Serum and Urine analysis
- (2) Polymerase Chain Reaction (PCR) technique
- (3) Enzyme Linked Immuno-Sorbent Assay (ELISA) technique
- (4) Recombinant DNA Technology

Q. 185. Match List I with List II.

List I (Interacting species)		List II (Name of interaction)	
A.	A leopard and a lion in a forest/grassland	I.	Competition
B.	A cuckoo laying egg in a crow's nest	II.	Brood parasitism

C.	Fungi and root of a higher plant in mycorrhizae	III.	Mutualism
D.	A cattle egret and a cattle in a field	IV.	Commensalism

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-IV, D-III
- (2) A-III, B-IV, C-I, D-II
- (3) A-II, B-III, C-I, D-IV
- (4) A-I, B-II, C-III, D-IV

Section B

Q. 186. Given below are two statements:

Statement I: During G_0 phase of cell cycle, the cell is metabolically inactive.

Statement II: The centrosome undergoes duplication during S phase of interphase.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are incorrect.
- (2) **Statement I** is correct but
- (3) **Statement I** is incorrect but **Statement II** is correct.
- (4) Both **Statement I** and **Statement II** are correct.

Q. 187. Match List I with List II.

List I		List II	
A.	Logistic growth	I.	Unlimited resource availability condition
B.	Exponential	II.	Limited resource availability condition
C.	Expanding age pyramid	III.	The percent individuals of pre reproductive age is largest followed by reproductive and post reproductive age groups
D.	Stable age pyramid	IV.	The percent individuals of pre-reproductives and reproductive age group are same

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-II, B-IV, C-I, D-III
- (3) A-II, B-IV, C-III, D-I
- (4) A-II, B-I, C-III, D-IV

Q. 188. In cockroach, excretion is brought about by-

- A. Phallic gland
- B. Ureose gland
- C. Nephrocytes
- D. Fat body
- E. Collateral glands

Choose the **correct** answer from the options given below:

- (1) A, B and E only (2) B, C and D only
(3) B and D only (4) A and E only

Q. 189. Select the correct statements.

- A. Tetrad formation is seen during leptotene.
B. During anaphase, the centromeres split and chromatids separate.
C. Terminalization takes place during pachytene.
D. Nucleolus, golgi complex and ER are reformed during telophase.
E. Crossing over takes place between sister chromatids of homologous chromosome.

Choose the **correct** answer from the options given below:

- (1) B and D only
(2) A, C and E only
(3) B and E only
(4) A and C only

Q. 190. Which of the following are NOT under the control of thyroid hormone?

- A. Maintenance of water and electrolyte balance.
B. Regulation of basal metabolic rate.
C. Normal rhythm of sleep-wake cycle.
D. Development of immune system.
E. Support the process of R.B.Cs formation.

Choose the **correct** answer from the options given below:

- (1) B and C only
(2) C and D only
(3) D and E only
(4) A and D only

Q. 191. which of the following is characteristic feature of cockroach regarding sexual dimorphism?

- (1) Presence of anal styles
(2) Presence of sclerites
(3) Presence of anal cerci
(4) Dark brown body colour and anal cerci

Q. 192. Which of the following statements are correct?

- A. Basophils are most abundant cells of the total WBCs.
B. Basophils secrete histamine, serotonin and heparin.
C. Basophils are involved in inflammatory response.
D. Basophils have kidney shaped nucleus.
E. Basophils are agranulocytes.

Choose the **correct** answer from the options given below:

- (1) C and E only (2) B and C only
(3) A and B only (4) D and E only

Q. 193. Which of the following statements are correct?

- A. An excessive loss of body fluid from the body switches off osmoreceptors.
B. ADH facilitates water reabsorption to

prevent diuresis.

- C. ANF causes vasodilation.
D. ADH causes increase in blood pressure.
E. ADH is responsible for decrease in GFR.

Choose the **correct** answer from the options given below:

- (1) B, C and D only
(2) A, B and E only
(3) C, D and E only X
(4) A and B only

Q. 194. Match List I with List II.

List I		List II	
A.	Mast cells	I.	Ciliated epithelium
B.	Inner surface of bronchiole	II.	Areolar
C.	Blood.	III.	Cuboidal epithelium
D.	Tubular parts	IV.	specialised connective tissue

Choose the **correct** answer from the options give below:

- (1) A-II, B-III, C-I, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-III, B-IV, C-II, D-I
(4) A-I, B-II, C-IV, D-III

Q. 195. Select the correct statements with reference to chordates.

- A. Presence of a mid-dorsal, solid and double nerve cord.
B. Presence of closed circulatory system.
C. Presence of paired pharyngeal gillslits.
D. Presence of dorsal heart
E. Triploblastic pseudocoelomate animals.

Choose the **correct** answer from the options given below:

- (1) B and C only
(2) B, D and E only
(3) C, D and E only
(4) A, C and D only

Q. 196. The unique mammalian characteristics are:

- (1) hairs, pinna and mammary glands
(2) hairs, pinna and indirect development
(3) pinna, monocondylic skull and mammary glands
(4) hairs, tympanic membrane and mammary glands

Q. 197. The parts of human brain that helps in regulation of sexual behaviour, expression of excitement, pleasure, rage, fear etc. are :

- (1) Corpora quadrigemina & hippocampus
(2) Brain stem & epithalamus
(3) Corpus callosum and thalamus
(4) Limbic system & hypothalamus

Q. 198. Which one of the following is the sequence on corresponding coding strand, if the sequence on mRNA formed follows

5'AUCGAUCGAUCGAUCGAUGG AUCG AUCG 3'?

- (1) 3' UAGCUAGCUAGCUAGCUA GCUAGCUAGC 5'
- (2) 5' ATCGATCGATCGATCGATCG ATCGATCG 3'
- (3) 3' ATCGATCGATCGATCGATG ATCGATCG 5'
- (4) 5' UAGCUAGCUAGCUAGCUA GCUAGC UAGC 3'

Q. 199. Which of the following statements are correct regarding skeletal muscle?

- A. Muscle bundles are held together by collagenous connective tissue layer called fascicle.
- B. Sarcoplasmic reticulum of muscle fiber is a store house of calcium ions.
- C. Striated appearance of skeletal muscle fibre is due to distribution pattern actin and myosin proteins.

D. M line is considered as functional of contraction called sarcomere.

Choose the **most appropriate** answer from the options given below:

- (1) B and C only
- (2) A, C and D only
- (3) C and D only
- (4) A, B and C only

Q. 200. Which one of the following is NOT an advantage of inbreeding?

- (1) It exposes harmful recessive genes that are eliminated by selection.
- (2) Elimination of less desirable genes and accumulation of superior genes takes place due to it.
- (3) It decreases the productivity of inbred population, after continuous inbreeding.
- (4) It decreases homozygosity.



Booklet	Batch	RollNumber	Name	Test Date	Invigilator's Signature	Test Center Code
(A) 10+1 (S)		0 0 0 0 0 0 0 0 0 0				0 0
(B) 10-2 (V)		1 1 1 1 1 1 1 1 1 1				1 1
(C) 10-3 (A)		2 2 2 2 2 2 2 2 2 2				2 2
(D) Crash (C)		3 3 3 3 3 3 3 3 3 3				3 3
(E) Paper		4 4 4 4 4 4 4 4 4 4				4 4
(F) Paper		5 5 5 5 5 5 5 5 5 5				5 5
(G) Paper		6 6 6 6 6 6 6 6 6 6				6 6
(H) Paper 1 (1)		7 7 7 7 7 7 7 7 7 7				7 7
(I) Paper 2 (2)		8 8 8 8 8 8 8 8 8 8				8 8
(J) Paper		9 9 9 9 9 9 9 9 9 9				9 9

Student's Signature

Invigilator's Signature

Test Date

Student's Signature

The OMR Sheet will be computer checked. Fill the circles completely and dark enough for proper detection. Use ballpen (black or blue) for marking.

Avoid Improper Marking

Partially Filled

Lightly Filled

Certified that all the entries in this section have been properly filled by the student

1 (a) (b) (c) (d)	6 (a) (b) (c) (d)	11 (a) (b) (c) (d)	16 (a) (b) (c) (d)	21 (a) (b) (c) (d)	26 (a) (b) (c) (d)
2 (a) (b) (c) (d)	7 (a) (b) (c) (d)	12 (a) (b) (c) (d)	17 (a) (b) (c) (d)	22 (a) (b) (c) (d)	27 (a) (b) (c) (d)
3 (a) (b) (c) (d)	8 (a) (b) (c) (d)	13 (a) (b) (c) (d)	18 (a) (b) (c) (d)	23 (a) (b) (c) (d)	28 (a) (b) (c) (d)
4 (a) (b) (c) (d)	9 (a) (b) (c) (d)	14 (a) (b) (c) (d)	19 (a) (b) (c) (d)	24 (a) (b) (c) (d)	29 (a) (b) (c) (d)
5 (a) (b) (c) (d)	10 (a) (b) (c) (d)	15 (a) (b) (c) (d)	20 (a) (b) (c) (d)	25 (a) (b) (c) (d)	30 (a) (b) (c) (d)
31 (a) (b) (c) (d)	36 (a) (b) (c) (d)	41 (a) (b) (c) (d)	46 (a) (b) (c) (d)	51 (a) (b) (c) (d)	56 (a) (b) (c) (d)
32 (a) (b) (c) (d)	37 (a) (b) (c) (d)	42 (a) (b) (c) (d)	47 (a) (b) (c) (d)	52 (a) (b) (c) (d)	57 (a) (b) (c) (d)
33 (a) (b) (c) (d)	38 (a) (b) (c) (d)	43 (a) (b) (c) (d)	48 (a) (b) (c) (d)	53 (a) (b) (c) (d)	58 (a) (b) (c) (d)
34 (a) (b) (c) (d)	39 (a) (b) (c) (d)	44 (a) (b) (c) (d)	49 (a) (b) (c) (d)	54 (a) (b) (c) (d)	59 (a) (b) (c) (d)
35 (a) (b) (c) (d)	40 (a) (b) (c) (d)	45 (a) (b) (c) (d)	50 (a) (b) (c) (d)	55 (a) (b) (c) (d)	60 (a) (b) (c) (d)
61 (a) (b) (c) (d)	66 (a) (b) (c) (d)	71 (a) (b) (c) (d)	76 (a) (b) (c) (d)	81 (a) (b) (c) (d)	86 (a) (b) (c) (d)
62 (a) (b) (c) (d)	67 (a) (b) (c) (d)	72 (a) (b) (c) (d)	77 (a) (b) (c) (d)	82 (a) (b) (c) (d)	87 (a) (b) (c) (d)
63 (a) (b) (c) (d)	68 (a) (b) (c) (d)	73 (a) (b) (c) (d)	78 (a) (b) (c) (d)	83 (a) (b) (c) (d)	88 (a) (b) (c) (d)
64 (a) (b) (c) (d)	69 (a) (b) (c) (d)	74 (a) (b) (c) (d)	79 (a) (b) (c) (d)	84 (a) (b) (c) (d)	89 (a) (b) (c) (d)
65 (a) (b) (c) (d)	70 (a) (b) (c) (d)	75 (a) (b) (c) (d)	80 (a) (b) (c) (d)	85 (a) (b) (c) (d)	90 (a) (b) (c) (d)
91 (a) (b) (c) (d)	96 (a) (b) (c) (d)	101 (a) (b) (c) (d)	106 (a) (b) (c) (d)	111 (a) (b) (c) (d)	116 (a) (b) (c) (d)
92 (a) (b) (c) (d)	97 (a) (b) (c) (d)	102 (a) (b) (c) (d)	107 (a) (b) (c) (d)	112 (a) (b) (c) (d)	117 (a) (b) (c) (d)
93 (a) (b) (c) (d)	98 (a) (b) (c) (d)	103 (a) (b) (c) (d)	108 (a) (b) (c) (d)	113 (a) (b) (c) (d)	118 (a) (b) (c) (d)
94 (a) (b) (c) (d)	99 (a) (b) (c) (d)	104 (a) (b) (c) (d)	109 (a) (b) (c) (d)	114 (a) (b) (c) (d)	119 (a) (b) (c) (d)
95 (a) (b) (c) (d)	100 (a) (b) (c) (d)	105 (a) (b) (c) (d)	110 (a) (b) (c) (d)	115 (a) (b) (c) (d)	120 (a) (b) (c) (d)
121 (a) (b) (c) (d)	126 (a) (b) (c) (d)	131 (a) (b) (c) (d)	136 (a) (b) (c) (d)	141 (a) (b) (c) (d)	146 (a) (b) (c) (d)
122 (a) (b) (c) (d)	127 (a) (b) (c) (d)	132 (a) (b) (c) (d)	137 (a) (b) (c) (d)	142 (a) (b) (c) (d)	147 (a) (b) (c) (d)
123 (a) (b) (c) (d)	128 (a) (b) (c) (d)	133 (a) (b) (c) (d)	138 (a) (b) (c) (d)	143 (a) (b) (c) (d)	148 (a) (b) (c) (d)
124 (a) (b) (c) (d)	129 (a) (b) (c) (d)	134 (a) (b) (c) (d)	139 (a) (b) (c) (d)	144 (a) (b) (c) (d)	149 (a) (b) (c) (d)
125 (a) (b) (c) (d)	130 (a) (b) (c) (d)	135 (a) (b) (c) (d)	140 (a) (b) (c) (d)	145 (a) (b) (c) (d)	150 (a) (b) (c) (d)
151 (a) (b) (c) (d)	156 (a) (b) (c) (d)	161 (a) (b) (c) (d)	166 (a) (b) (c) (d)	171 (a) (b) (c) (d)	176 (a) (b) (c) (d)
152 (a) (b) (c) (d)	157 (a) (b) (c) (d)	162 (a) (b) (c) (d)	167 (a) (b) (c) (d)	172 (a) (b) (c) (d)	177 (a) (b) (c) (d)
153 (a) (b) (c) (d)	158 (a) (b) (c) (d)	163 (a) (b) (c) (d)	168 (a) (b) (c) (d)	173 (a) (b) (c) (d)	178 (a) (b) (c) (d)
154 (a) (b) (c) (d)	159 (a) (b) (c) (d)	164 (a) (b) (c) (d)	169 (a) (b) (c) (d)	174 (a) (b) (c) (d)	179 (a) (b) (c) (d)
155 (a) (b) (c) (d)	160 (a) (b) (c) (d)	165 (a) (b) (c) (d)	170 (a) (b) (c) (d)	175 (a) (b) (c) (d)	180 (a) (b) (c) (d)
181 (a) (b) (c) (d)	186 (a) (b) (c) (d)	191 (a) (b) (c) (d)	196 (a) (b) (c) (d)		
182 (a) (b) (c) (d)	187 (a) (b) (c) (d)	192 (a) (b) (c) (d)	197 (a) (b) (c) (d)		
183 (a) (b) (c) (d)	188 (a) (b) (c) (d)	193 (a) (b) (c) (d)	198 (a) (b) (c) (d)		
184 (a) (b) (c) (d)	189 (a) (b) (c) (d)	194 (a) (b) (c) (d)	199 (a) (b) (c) (d)		
185 (a) (b) (c) (d)	190 (a) (b) (c) (d)	195 (a) (b) (c) (d)	200 (a) (b) (c) (d)		

Q. No.	Answer Key	Topic's Name	Chapter Name
PHYSICS			
1	2	Acceleration	Kinematics
2	2	Displacement Current	Alternating Current
3	3	Radioactivity	Nucleus
4	4	Gauss Law In Magnetism	Magnetism
5	4	Combination of Capacitors	Capacitors
6	1	Fundamental Mode Of Vibration In Organ Pipe	Waves
7	1	Hydrogen Spectrum	Atoms
8	4	Transformer	Electro Magnetic Induction
9	1	Electrical Circuits	Current Electricity
10	3	Magnetic Energy	Electromagnetic Induction
11	3	Spring Potential Energy	Work Power And Energy
12	3	Errors	Units And Measurement
13	2	Colour Code of Resistance	Current Electricity
14	2	Errors	Units And Measurement
15	3	Total Internal Reflection	Geometrical Optics
16	2	Electromagnetic Wave	Electromagnetic Wave
17	4	Moment of Inertia	Rigid Body Mechanics
18	2	Surface Tension	Fluid Mechanics
19	3	LCR Series Circuit	Alternating Current
20	4	Gauss Law I Electrostatics	Electrostatics
21	1	Venturimeter	Fluid Mechanics
22	2	Gravitational Potential	Gravitation
23	1	Molecular Speeds Of Ideal Gases	Kinetic Theory of Gases
24	1	X-Rays	Dual Nature of Matter And Radiation
25	2	Average Velocity	One Dimensional Motion
26	3	Circular Motion	Circular Motion
27	1	Stress	Mechanics Of Solid
28	2	Rectifier	Semiconductors
29	3	Torque Experienced by an Electric Dipole Placed in External Field	Electrostatics
30	4	Carnot Engine	Thermodynamics
31	2	Youngs Double Slit Experiment	Wave Optics
32	4	Photoelectric Effect	Dual Nature of Matter And Radiation
33	4	Optoelectronic Devices	Semiconductors
34	2	Current Circuits	Current Electricity
35	2	Projectile Motion	Motion in a Plane
36	1	Gravitation	Gravitation

Q. No.	Answer Key	Topic's Name	Chapter Name
37	1	Logic Gate	Semiconductors
38	3	Bohrs Model	Atoms
39	2	Force on a Current Carrying Conductor in A magnetic Field	Magnetism
40	2	LCR Series Circuit	Alternating Current
41	1	Current Circuits	Current Electricity
42	3	SHM	SHM
43	2	Combination of Lens	Geometrical Optics
44	3	Magnetic Field	Magnetic Effects of Current
45	2	Friction	Laws of Motion
46	3	Combination of Lens	Geometrical Optics
47	2	Equations of Kinematics	Kinematics
48	2	Resistance Dependent on Temperature	Current Electricity
49	4	Electrostatic Potential	Electrostatic
50	4	Equation of Kinematics	Kinematics
CHEMISTRY			
51	2	Atomic Size	Classification of Elements and Periodicity in Properties
52	3	Qualitative analysis	Organic Chemistrysome Basic Principles and Techniques
53	4	Quantum number	Structure of Atom
54	3	Catalysis	Surface chemistry
55	1	Gas laws	States of matter
56	2	Aryl halide naming	Haloalkanes and Haloarenes
57	4	Addition of HBr	Alcohol phenol ether
58	2	Henry's law	Solution
59	2	Conductance	Electrochemistry
60	2	Sigma pi bond	Chemical Bonding and Molecular structure
61	2	Physical properties of alkali metal	s block
62	1	Mole concept	Some Basic Concepts of Chemistry
63	2	Intermolecular forces	States of Matter : Gases and Liquids
64	2	Rate of reaction	Chemical kinetics
65	3	Inert pair effect	p block
66	2	Properties of hydrogen	Hydrogen
67	1	Decarboxylation	Hydrocarbons
68	3	Nucleophilic addition reaction	Aldehydes, Ketones and Carboxylic Acids
69	1	Free energy	Thermodynamics
70	4	Octet rule	Chemical Bonding and Molecular structure
71	4	Molecular Orbital Theory	Chemical Bonding and Molecular structure

Q. No.	Answer Key	Topic's Name	Chapter Name
72	4	Octahedral Void	Solid State
73	2	Hydration Energy	Chemical Bonding and Molecular structure
74	1	Diazonium Salt	Organic Compounds Containing Nitrogen
75	2	Nucleoside and nucleotide	Biomolecules
76	4	Role of mg and ca	s block element
77	2	Amines	Organic Compounds Containing Nitrogen
78	4	Wittmann reduction	Aldehyde, ketone and carboxylic acid
79	2	Allotrope of carbon	p block
80	2	Acids and bases	Equilibrium
81	1	Monomer	Polymer
82	3	Tranquillizer	Chemistry in Everyday Life
83	4	Homoleptic complexes	Coordination Compounds
84	1	Activation energy	Chemical kinetics
85	3	Fundamental particle	Atomic structure
86	1	Oxo acids of sulphur	p block
87	4	Balancing of equation	Redox reaction
88	2	Void	Solid state
89	4	Reduction of aldehyde	Aldehyde, ketone and carboxylic acid, Haloalkanes
90	2	Stability of complexes	Coordination compound
91	1	Dehydration of alcohol	Alcohol phenol ether
92	3	Eutrophication	Environmental Chemistry
93	4	Blast furnace	General Principles and Processes of Isolation of Elements
94	2	Tollens reagent	Aldehyde, ketone and carboxylic acid
95	1	Internal energy	Thermodynamics
96	2	Properties of d block element	d block
97	2	Equilibrium constant	Equilibrium
98	4	Huckel rule	Organic Chemistry: Some Basic Principles and Techniques
99	2	Types of soil	Surface chemistry
100	2	Cleavage of ether	Alcohol phenol ether
BOTANY			
101	2	Pleiotropism	Principles of Inheritance and Variation
102	1	Tissue culture	Strategies for Enhancement in Food Production
103	3	Movement of ions	Transport in Plants
104	4	Loss of biodiversity	Biodiversity and Conservation
105	3	Recombinant DNA Technology	Biotechnology : Principles and Processes
106	4	Bryophytes	Plant Kingdom
107	1	Transcription	Molecular basis of inheritance
108	1	Conservation	Biodiversity and Conservation

Q. No.	Answer Key	Topic's Name	Chapter Name
109	4	Glycolysis	Respiration in Plant
110	4	Ozone	Environmental Issues
111	3	Vascular tissue	Anatomy of Flowering plants
112	1	Pteridophytes	Plant Kingdom
113	4	Food Chain	Organisms and Populations
114	3	Placentation	Morphology of Flowering Plant
115	4	Families	Morphology of Flowering Plant
116	4	Photosystems	Photosynthesis in Higher Plants
117	1	DNA Purification	Biotechnology and its Applications
118	1	Replication	Molecular basis of inheritance
119	1	Cell cycle	Cell Cycle and Cell Division
120	1	Glycolysis	Respiration in Plant
121	2	Productivity	Organisms and Populations
122	1	Pollination	Morphology of Flowering plants
123	1	Hormones	Plant growth and development
124	4	Photosynthesis	Photosynthesis in Higher Plants
125	4	Gene expression	Molecular basis of inheritance
126	2	Recombination	Principles of inheritance and variation
127	4	Transpiration	Transport in plants
128	2	Transformation technique	Biotechnology: Principles and processes
129	4	Tissue System	Anatomy of flowering plants
130	1	Pollination	Morphology of flowering plants
131	2	Female Reproductive System	Respiration in plants
132	2	Hormones	Plants growth and development
133	2	Cell Cycle	Cell cycle and cell division
134	1	Genetic Material	Principles of inheritance and variation
135	2	Plant Tissues	Anatomy of flowering plants
136	3	Respiration	Respiration in plants
137	4	Atp Synthesis	Respiration in plants
138	4	Microbes	Microbes in human welfare
139	2	Gymnosperms	Photosynthesis in higher plants
140	2	Recombinant DNA technology	Biotechnology and its Applications
141	4	Transport	Transport in plants
142	1	Algal blooms	Environmental issues
143	4	Ribosomes	Cell: the unit of life
144	1	Biotic interactions	Ecosystem
145	4	Structure of flower	Sexual Reproduction in Flowering Plants
146	1	Secondary growth	Plant growth and development

Q. No.	Answer Key	Topic's Name	Chapter Name
147	2	Biotic interactions	Ecosystem
148	2	Genetic Variation	Principles of inheritance and variation
149	2	Cell cycle	Cell cycle and cell division
150	2	Nutrition	Plant growth and development
ZOOLOGY			
151	3	Nucleus	Cell: The Unit of Life
152	4	HIV	Reproductive Health
153	1	Eutrophication	Environmental Issues
154	4	Heart	Body fluids and Circulation
155	4	Down's Syndrome	Principles of Inheritance and Variation
156	4	Mutation	Molecular Basis of Inheritance
157	1	Bones	Locomotion and Movement
158	2	Nephrons	Excretory Products and their Elimination
159	3	Vectors	Biotechnology : Principles and Processes
160	3	Proteins	Biomolecules
161	1	Digestive sysytem	Digestion and Absorption
162	1	STDs	Reproductive Health
163	2	Cytoskeleton	Locomotion and Movement
164	1	STDs	Reproductive Health
165	3	Lungs	Breathing and Exchange of Gases
166	1	Endomembrane system	Cell: The Unit of Life
167	4	Hormones	Chemical Coordination and Integration
168	1	Lac Operon	Molecular Basis of Inheritance
169	4	Eye Structure	Neural Control and Coordination
170	1	Adaptive Radiation	Evolution
171	4	Female reproductive system	Human Reproduction
172	4	Chemicals	Human Health and Disease
173	4	Diseases	Human Health and Disease
174	3	Sex determination	Reproductive Health
175	1	Phylums	Animal Kingdom (Part- I) Invertebrata
176	4	Enzymes	Biomolecules
177	2	Excretion	Animal Kingdom (Part- I) Invertebrata
178	3	Mensuration	Human Reproduction
179	2	Pollution	Environmental Issues
180	2	Digestive enzymes	Digestion and Absorption
181	1	Mensuration	Human Reproduction
182	1	Joints	Locomotion and Movement
183	1	Pedigree Analysis	Principles of Inheritance and Variation

Q. No.	Answer Key	Topic's Name	Chapter Name
184	1	Diagnostic techniques	Biotechnology and its Applications
185	4	Biotic interactions	Ecosystem
186	3	Cell Cycle	Cell Cycle and Cell Division
187	4	Population growth	Organisms and Populations
188	2	Cockroach	Animal Kingdom (Part- I) Invertebrata
189	1	Cell Cycle	Cell Cycle and Cell Division
190	2	Hormones	Chemical Coordination and Integration
191	1	Cockroach	Animal Kingdom (Part- I) Invertebrata
192	2	Blood cells	Body Fluids and Circulation
193	1	Hormones	Chemical Coordination and Integration
194	2	Nephrons	Excretory Products and their Elimination
195	1	Chordates	Animal Kingdom (Part -II) Vertebrata
196	1	Ear	Neural Control and Coordination
197	4	Brain	Neural Control and Coordination
198	2	Transcription	Molecular Basis of Inheritance
199	1	Muscles	Locomotion and Movement
200	3	Inbreeding	Strategies for Enhancement in Food Production

NEET (UG)

7th May 2023 Paper

ANSWERS WITH EXPLANATION

PHYSICS

Section—A

1. **Option (2) is correct.**

Explanation: Let's his speed is V

Force will act along the direction of acceleration.

And acceleration = $\frac{\text{Change in velocity}}{\text{time interval}}$

Change in velocity,

$$\Delta \vec{v} = \vec{v} - \vec{u}$$

$$\vec{u} = v(-\hat{j})$$

$$\vec{v} = v\hat{i}$$

$$\text{So, } \Delta \vec{v} = v\hat{i} - v(-\hat{j})$$

$$= v\hat{i} + v\hat{j}$$

So, $\Delta \vec{v}$ is along N-E direction.

2. **Option (2) is correct.**

Explanation: Displacement current, $(i_d) =$
Conduction current (i_c)

$$i_d = i_c = \frac{V}{X_c}, \text{ where } X_c = \text{Capacitive reactance.}$$

$$X_c = \frac{1}{\omega c}, \text{ where, } \omega = \text{angular frequency and}$$

$C = \text{capacitance.}$

$$\omega = 2\pi f, \text{ where } f = \text{frequency,}$$

When f decreases, ω decreases, then X_c increases
so i_c or i_d decreases.

3. **Option (3) is correct.**

Explanation:

$$\frac{A}{A_0} = \frac{1}{2^n}$$

$n = \text{no. of half - lives.}$

$$\frac{A}{A_0} = \frac{1}{16} = \frac{1}{2^4}$$

$n = 4 \text{ half - lives}$

$$\begin{aligned} \text{So total time, } t &= n \times \frac{T_1}{2} \\ &= 4 \times 20 \\ &= 80 \text{ minute.} \end{aligned}$$

4. **Option (4) is correct.**

Explanation:

Magnetic monopole doesn't exist.

$$\text{So, } \oint \vec{B} \cdot d\vec{s} = 0$$

5. **Option (4) is correct.**

Explanation: The equivalent capacitance in parallel is $3\mu F + 3\mu F = 6\mu F$
& $6\mu F$ is in series with $3\mu F$.

$$\text{So, } C_{\text{equivalent}} = \frac{3 \times 6}{3 + 6} = 2\mu F.$$

6. **Option (1) is correct.**

Explanation: Fundamental mode of vibration for open pipe is $\frac{V}{2l}$ and the that of closed pipe is $\frac{V}{4l}$.

So, their ratio $\frac{\frac{V}{2l}}{\frac{V}{4l}} = 2:1$. Where V is the speed of sound & l in length.

7. **Option (1) is correct.**

Explanation:

$$\frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_f^2} - \frac{1}{n_i^2} \right]$$

For minimum Wavelength, Energy is maximum,

So, $n_i = \infty$

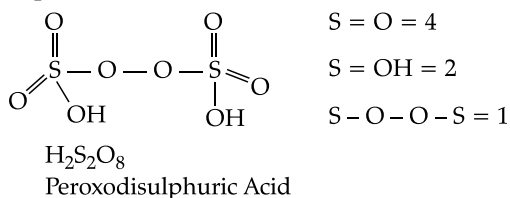
For Balmer series, $n_f = 2$, and for Brackett series, $n_f = 4$.

$$\frac{1}{\lambda} = \frac{R}{4}$$

$$\frac{1}{\lambda_{\text{brackett}}} = \frac{R}{16}$$

$$\frac{\lambda_{\text{brackett}}}{\lambda} = \frac{R}{4} \times \frac{16}{R}$$

$$\lambda_{\text{brackett}} = 4\lambda$$

8. Option (4) is correct.**Explanation:**

As transformer is ideal.

$$\text{So, } P_{\text{output}} = P_{\text{input}} = 60$$

$$\Rightarrow I_p V_p = 60$$

$$\begin{aligned}
 \Rightarrow I_p &= \frac{60}{V_p} \\
 &= \frac{60}{220} = \frac{3}{11} \\
 &= 0.27 \text{ A}
 \end{aligned}$$

9. Option (1) is correct.**Explanation:** The two batteries are opposing, So

$$\epsilon_{\text{net}} = 10\text{V} - 5\text{V} = 5\text{V} \text{ and } R_{\text{net}} = 2 + 1 + 7 = 10\Omega$$

$$i = \frac{\epsilon_{\text{net}}}{R_{\text{net}}} = \frac{5}{10} = 0.5\text{A, from A to B though E}$$

10. Option (3) is correct.**Explanation:** Magnetic energy stored in an inductor is

$$\frac{1}{2} Li^2 = \frac{1}{2} \times (4\mu\text{H}) \times 2^2 = 8\mu\text{J}$$

11. Option (3) is correct.**Explanation:** PE stored in a stretched spring in

$$\frac{1}{2} kx^2, \text{ where } k = \text{spring const. and } x \text{ is extension.}$$

$$U = \frac{1}{2} k(2)^2$$

$$\text{And } U' = \frac{1}{2} k(8)^2$$

$$\Rightarrow \frac{U}{U'} = \frac{1}{16}$$

$$\Rightarrow U' = 16U$$

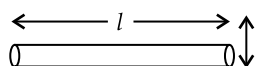
12. Option (3) is correct.**Explanation:** As the cause of errors in unpredictable, this kind of error is random error.**13. Option (2) is correct.****Explanation:**

Colour	Number	Multiplies	Tolerance
Black	0	10^0	
Brown	1	10^1	

Red	2	10^2	
Orange	3	10^3	
Yellow	4	10^4	
Green	5	10^5	
Blue	6	10^6	
Violet	7	10^7	
Gray	8	10^8	
White	9	10^9	
Gold		10^{-1}	5%
Silver		10^{-2}	10%

So, the colour of third band is only.
which is of orange.**14. Option (2) is correct.****Explanation:**

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

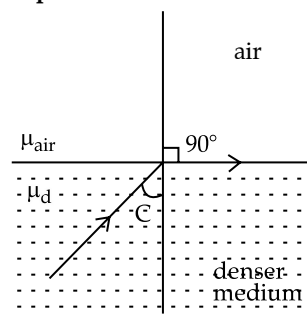


$$d = \frac{m}{\pi r^2 l}$$

$$\left(\pm \frac{\Delta d}{d} \right) \% = \pm \left(\frac{\Delta m}{m} + 2 \frac{\Delta r}{r} + \frac{\Delta l}{l} \right) \times 100$$

$$= \pm \left(\frac{0.002}{0.4} + 2 \frac{0.001}{0.3} + \frac{0.02}{5} \right) \times 100$$

$$= 1.6 \%$$

15. Option (3) is correct.**Explanation:**

$$\mu_d \sin C = \mu_{\text{air}} \sin 90^\circ$$

$$\frac{e}{V_d} \sin C = \frac{C}{V_{\text{air}}} \quad \left[\because \mu_{\text{medium}} = \frac{C}{V_{\text{medium}}} \right]$$

$$\sin C = \frac{V_d}{V_{\text{air}}} = \frac{\frac{10x}{t_2}}{\frac{x}{t_1}} = \frac{10t_1}{t_2}$$

$$C = \sin^{-1} \left(\frac{10t_1}{t_2} \right)$$

- 16.
- Option (2) is correct.**

Explanation:

$$C = \frac{E_0}{B_0}$$

Where symbols have their usual meanings.

$$\text{So, } B_0 = \frac{E_0}{C}$$

$$= \frac{48}{3 \times 10^8}$$

$$= 1.6 \times 10^{-7} \text{ T}$$

- 17.
- Option (4) is correct.**

Explanation:

$$I_{\text{solid sphere}} = \frac{2}{5} MR^2 = MK_{\text{solid}}^2$$

$$\Rightarrow K_{\text{solid}} = \sqrt{\frac{2}{5}} R$$

Similarly, that of hollow sphere

$$I_{\text{hollow}} = \frac{2}{3} MR^2 = MK_{\text{hollow}}^2$$

$$\Rightarrow K_{\text{hollow}} = \sqrt{\frac{2}{3}}$$

$$\text{So, } \frac{K_{\text{solid}}}{K_{\text{hollow}}} = \frac{\sqrt{\frac{2}{5}}}{\sqrt{\frac{2}{3}}} = \sqrt{\frac{3}{5}} = \sqrt{3} : \sqrt{5}$$

- 18.
- Option (2) is correct.**

Explanation: Surface tension = $\frac{\text{Work done}}{\Delta \text{Area}}$

A soap bubble has 2 free surface.

$$\text{So, area, } A = 2 \times 4\pi r^2 = 8\pi r^2$$

$$\begin{aligned} \Rightarrow \text{Work done} &= \text{surface tension} \times 8\pi r^2 \\ &= (0.03) \times 8 \times \pi \times (2 \times 10^{-2})^2 \\ &= 3.01 \times 10^{-4} \text{ J.} \end{aligned}$$

- 19.
- Option (3) is correct.**

Explanation: In Resonance condition of LCR circuit.

$$\begin{aligned} X_L &= X_C \\ \Rightarrow \omega_r L &= \frac{1}{\omega_r C} \end{aligned}$$

$$\Rightarrow \omega_r = \frac{1}{\sqrt{LC}}$$

$$\Rightarrow f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$= \frac{1}{2\sqrt{10 \times 10^{-3} \times 10^{-6}}}$$

$$= 1.59 \text{ kHz.}$$

(Where symbols have their usual meanings.)

- 20.
- Option (4) is correct.**

Explanation: According to gauss law of electrostatics.

$$\text{The net flux } \phi_E = \oint \vec{E} \cdot d\vec{S} = \frac{q_{\text{encl}}}{\epsilon_0}$$

So, here net flux is zero, that means net charge enclosed by the surface is Zero.

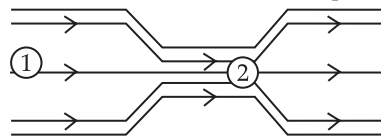
So the flux entering the surface is equal to flux leaving the surface in same.

- 21.
- Option (1) is correct.**

Explanation: Bernoulli's principle is based on conservation of energy, that is the total work done by pressure, gravity and change in KE is constant for an ideal fluid flowing in a closed tube.

$$P + \frac{1}{2} \rho v^2 + \rho gh = \text{const}$$

Venturie-meter works on this principle.

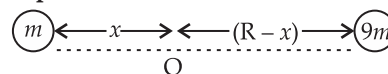


$$P_1 + \frac{1}{2} \rho v_1^2 + \rho gh = P_2 + \frac{1}{2} \rho v_2^2 + \rho gh$$

Perpendicular and parallel axes theorem helps to find moment inertia at different axes of rotation.

Huygens theory explain the wave nature of light.

- 22.
- Option (2) is correct.**

Explanation:At 'O' field due to mass $m \times 9m$ is zero. So

$$\frac{Gm}{x^2} = \frac{G(9m)}{(R-x)^2}$$

$$\Rightarrow \frac{1}{x} = \frac{3}{R-x}$$

$$\Rightarrow x = \frac{R}{4} \times (R-x) = \frac{3R}{4}$$

So, potential at that point will be

$$\frac{-Gm}{\left(\frac{R}{4}\right)} + \frac{-G9m}{\left(\frac{3R}{4}\right)} = \frac{-16Gm}{R}$$

- 23.
- Option (1) is correct.**

Explanation: The rms speed is

$$V = \sqrt{\frac{3RT}{M}}$$

When it is increased by 3 times, i.e.,

$$4V = \sqrt{\frac{3RT'}{M}}$$

(where symbols have their usual meanings)

$$\begin{aligned}\Rightarrow T' &= 16 \text{ T} \\ &= 16 (273-50) \\ &= 3568 \text{ K} \\ &= 3295^\circ \text{ C.}\end{aligned}$$

24. Option (1) is correct.

Explanation: Minimum wave length of X-Rays is

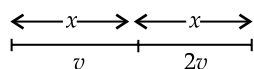
$$\lambda_{\min} = \frac{hc}{eV},$$

$$\text{Hence } \lambda_{\min} \propto \frac{1}{V}$$

25. Option (2) is correct.

Explanation: Average speed,

$$v_{\text{avg}} = \frac{\text{Total distance}}{\text{Total time}}$$



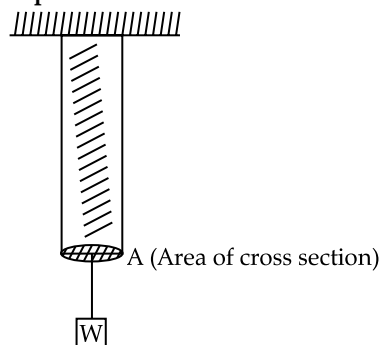
$$\text{So, } v_{\text{avg}} = \frac{2x}{\frac{x}{v} + \frac{x}{2v}} = \frac{2(v)(2v)}{v + 2v} = \frac{4v}{3}$$

26. Option (3) is correct.

Explanation: Angular displacement, angular velocity and angular acceleration are axial vectors, So, their direction is always along the axis of rotation.

27. Option (1) is correct.

Explanation:

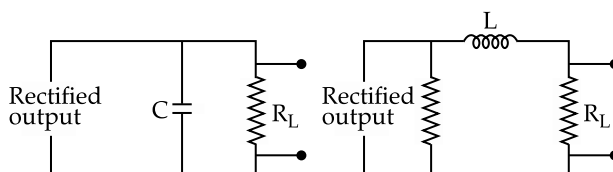


Stress is, restoring force per unit area. As there is no mass of Rod. Restoring force is equal to the applied force in magnitude.

$$\text{So, stress} = \frac{W}{A}$$

28. Option (2) is correct.

Explanation: To remove the ac ripples from the rectified output, a filter circuit is connected, which contains capacitor or inductor, as capacitor allows the ac while inductor allows the dc.



29. Option (3) is correct.

Explanation: $\tau = p E \sin \theta$ & $p = q \times l$

Where symbols have their usual meanings.

$$\begin{aligned}\Rightarrow 4 &= (q \times 2 \times 10^{-2}) (2 \times 10^5) \times \sin 30^\circ \\ \Rightarrow q &= 2 \times 10^{-3} \text{ C} \\ &= 2 \text{ m C}\end{aligned}$$

30. Option (4) is correct.

Explanation: Efficiency of carnot engine,

$$\begin{aligned}\eta &= \left(1 - \frac{T_L}{T_H}\right) = \frac{50}{100} \\ \Rightarrow \frac{1}{2} &= 1 - \frac{T_L}{T_H} \\ \Rightarrow \frac{T_L}{T_H} &= \frac{1}{2} \\ \Rightarrow \frac{T_L}{273 + 327} &= \frac{1}{2}\end{aligned}$$

$$\begin{aligned}\Rightarrow T_2 &= 300 \text{ K} \\ &= 300 - 273 \\ &= 27^\circ \text{ C}\end{aligned}$$

Where T_L = sink temp (in K)

& T_H = source temp (in K)

31. Option (2) is correct.

Explanation: Angular fringe width, $\theta = \frac{\lambda}{d}$, where

λ = wavelength of light & d is the slit separation. So θ is independent of separation between screen and slit (D).

If λ increases, then θ must increase.

32. Option (4) is correct.

Explanation: According to Einstein's Photo electric equation.

$$KE_{\max} = h\nu - \phi_0$$

So, photoelectric effect is only possible, when the energy of incident radiation is larger than work function (ϕ_0).

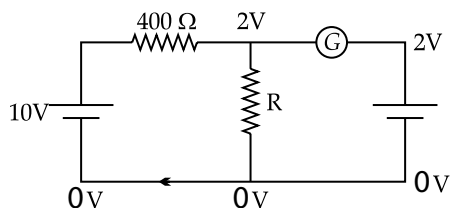
Hence, in the given situation, Photoelectric effect is possible in case of Cs only, as 2.20 eV is larger than 2.14 eV, and lower from 2.30 eV and 2.75 eV as well.

33. Option (4) is correct.

Explanation: Photo voltaic cells (or solar cells). Convert radiation energy into electrical energy. Zener diodes are heavily doped to operate at breakdown voltage in reverse bias.

34. Option (2) is correct.

Explanation:



As Galvanometer has no current, voltage across R will be 2V as shown.

So voltage applied is 10V and voltage across 400 Ω is $10 - 2 = 8\text{V}$.

So current in the circuit is $i = \frac{8}{400}\text{amp}$.

That current will flow through R,

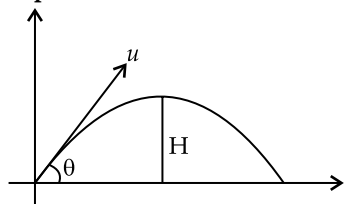
So, $2 = iR$

$$= \frac{8}{400} R$$

$$\Rightarrow R = 100 \Omega$$

35. Option (2) is correct.

Explanation:



$$H = \frac{u^2 \sin^2 \theta}{2g}$$

$$\Rightarrow H = \frac{(280)^2 (\sin 30^\circ)^2}{2 \times 9.8}$$

$$= 1000 \text{ m.}$$

Section—B

36. Option (1) is correct.

Explanation: It can be solved by dimensional analysis.

$$\frac{3\pi}{Gd} = \frac{1}{[M^{-1}L^3T^{-2}][ML^{-3}]} = [T^2]$$

OR

It can be derived from the formula.

$$\begin{aligned} T &= 2\pi \sqrt{\frac{R}{g}} \\ &= 2\pi \sqrt{\frac{R^3}{GM}} \\ &= 2\pi \sqrt{\frac{R^3}{G \times \frac{4}{3}\pi R^3 \times d}} \end{aligned}$$

$$\begin{aligned} \Rightarrow T^2 &= \frac{4\pi^2 \times 3}{4\pi Gd} \\ &= \frac{3\pi}{Gd} \end{aligned}$$

37. Option (1) is correct.

Explanation:

$$y = \overline{A \cdot B} = \overline{A} + \overline{B}$$

$$y = A + B$$

Which is 'OR' gate.

38. Option (3) is correct.

Explanation: In bohr's model, the radius of n^{th} or bit is

$$r_{n^{\text{th}}} = r_0 \frac{A}{z}$$

(Symbols have their usual meanings)

For H – atom, and for $n = 3$

$$\begin{aligned} r_{n^{\text{th}}} &= r_0 (3)^2 \\ &= 0.53 \times 9 \\ &= 4.77 \text{ Å.} \end{aligned}$$

39. Option (2) is correct.

Explanation: The magnetic force (F), on a current (I) carrying conductor of length (L) in a magnetic field (B) is

$$\vec{F} = I\vec{L} \times \vec{B}$$

$$= IL\hat{i} \times (2\hat{i} + 3\hat{j} - 4\hat{k})$$

$$= 3IL\hat{k} + 4IL\hat{j}$$

$$|\vec{F}| = \sqrt{(3IL)^2 + (4IL)^2}$$

$$= 5IL$$

40. Option (2) is correct.

Explanation: $Z = \sqrt{R^2 + (X_C - X_L)^2}$

$$R = 10 \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1 \times 10^6}{2\pi(50) \times \frac{10^3}{\pi}}$$

$$= 10 \Omega$$

$$X_L = \omega L$$

$$= 2\pi(50) \times \frac{50}{\pi} \times 10^{-3}$$

$$= 5\Omega$$

$$\text{So } Z = \sqrt{10^2 + (10 - 5)^2}$$

$$= \sqrt{125}$$

$$= 5\sqrt{5} \Omega$$

Symbols have their usual meanings.

41. Option (1) is correct.

Explanation: Current, $i_s = \frac{E_{net}}{R_{net}}$

$$\text{In series, } i_s = \frac{E}{10R}$$

$$\text{In parallel, } i_p = \frac{E}{\frac{R}{10}}$$

$$\text{So, } \frac{i}{i_p} = \frac{E}{10R} \cdot \frac{R}{E} \cdot \frac{10}{1} = \frac{1}{100}$$

$$i_p = 100 i_s, n = \frac{i_p}{i_s}$$

$$\text{So, } n = 100.$$

42. Option (3) is correct.

Explanation: In SHM, acceleration,
 $\vec{a} = -\omega^2 \vec{x}$

$$T = 8 \text{ sec,}$$

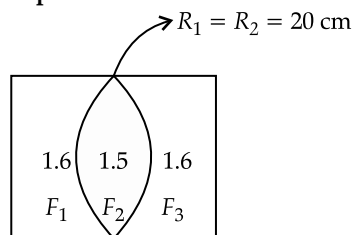
$$\omega = \frac{2\pi}{8}$$

$$\text{at } t = 2s, x = +1$$

$$\text{So } a = -\left(\frac{2\pi}{8}\right)^2 (+1) = \frac{-\pi^2}{16} \text{ m/s}^2$$

43. Option (2) is correct.

Explanation:



$$\frac{1}{f_1} = (1.6 - 1) \left(\frac{1}{\infty} - \frac{1}{20} \right) = \frac{1}{f_3}$$

$$\Rightarrow \frac{1}{f_1} = \frac{1}{f_3} = \frac{-0.3}{10}$$

$$\begin{aligned} \frac{1}{f_2} &= (1.5 - 1) \left[\frac{1}{20} + \frac{1}{20} \right] \\ &= \frac{0.5}{10} \end{aligned}$$

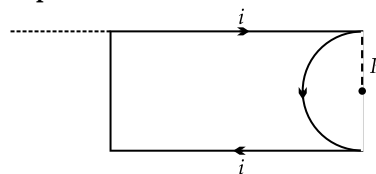
For combination of lens.

$$\begin{aligned} \frac{1}{F} &= \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} \\ &= \frac{-0.3}{10} + \frac{-0.3}{10} + \frac{0.5}{10} \\ &= \frac{-1}{100} \end{aligned}$$

$$\Rightarrow F = -100 \text{ cm.}$$

44. Option (3) is correct.

Explanation:



At the end of an current carrying conductor, magnetic field.

$$B_1 = \frac{\mu_0 i}{4\pi R} \otimes \text{ (into the page) and for semicircles}$$

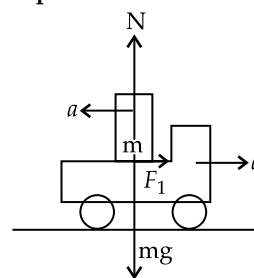
$$\text{part, magnetic field, } B_2 = \frac{\mu_0 i}{4R} \odot \text{ (Out of the page)}$$

There are two straight conductors (having same direction of $\vec{B}$) and a circular wire.

$$\text{So } B_{net} = \frac{\mu_0 i}{4R} \left(1 - \frac{2}{\pi} \right) \odot \text{ out of the page.}$$

45. Option (2) is correct.

Explanation:



The block to move with the truck without sliding.

$$ma \leq F_1$$

$$ma_{max} = \mu N$$

$$\Rightarrow \cancel{m} a_{max} = \mu \cancel{m} g$$

$$= 0.15 \times 10$$

$$= 1.5 \text{ ms}^{-2}$$

46. Option (3) is correct.

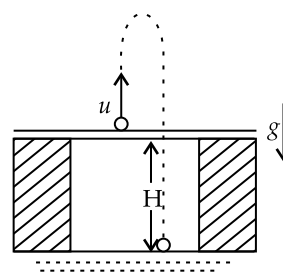
$$\begin{aligned} \text{Explanation: } \frac{1}{F} &= \frac{1}{f_1} + \frac{1}{f_2} \\ &= \frac{1}{f} - \frac{1}{f} = 0 \end{aligned}$$

$$\Rightarrow F = \infty$$

47. Option (2) is correct.

Explanation: $u = 4 \text{ m/s}$

$$t = 4s$$



Applying equation of motion

$$\vec{s} = \vec{u}t + \frac{1}{2}\vec{a}t^2$$

$$-H = 4 \times 4 + \frac{1}{2}(-10) \times 4^2$$

$$-H = -64 \text{ m, } H = 64 \text{ m}$$

So height of the bridge is 64m.

48. Option (2) is correct.

Explanation: $R_{80^\circ} = 6.8$

$$R_0 = 2$$

$$R_t = R_0(1 + \alpha\Delta t)$$

$$\Rightarrow 6.8 = 2[1 + \alpha(80 - 0)]$$

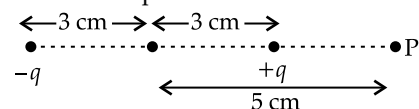
$$\Rightarrow \alpha = \frac{2.4}{80} = 3 \times 10^{-2} \text{ } ^\circ\text{C}^{-1}$$

(Symbols have their usual meanings)

49. Option (4) is correct.

Explanation:

Electrostatic potential



$$V_P = \frac{k(-q)}{8 \text{ cm}} + \frac{kq}{2 \text{ cm}}$$

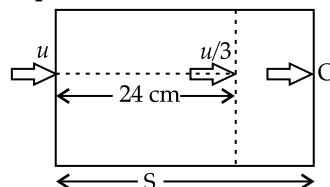
$$= Kq \left(\frac{1}{2 \text{ cm}} - \frac{1}{8 \text{ cm}} \right)$$

$$= \frac{3Kq}{8 \times 10^{-2}} V$$

$$= \frac{3}{8} Kq \times 10^2 V$$

50. Option (4) is correct.

Explanation:



$$\text{By } v^2 - u^2 = 2as$$

$$\left(\frac{u}{3} \right)^2 - u = 2(-a) \times 24$$

$$\frac{u^2}{9} - u^2 = -48a$$

$$\frac{-8u^2}{9} = -48a$$

$$\frac{8u^2}{9} = 48a$$

Similarly

$$0 = u^2 - 2as$$

$$u^2 = 2as$$

$$\frac{8u^2}{9u^2} = \frac{48a}{2as}$$

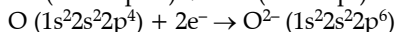
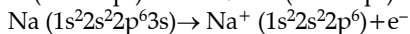
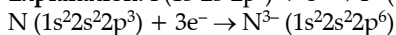
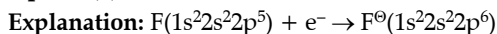
$$s = 27 \text{ m}$$

□□□

CHEMISTRY

Section A

51. Option (2) is correct.

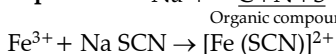


$F^\ominus, N^{3-}, Na^+, O^{2-}$ have same number of electrons.

So they are known as iso electronic species. As the charge on anion increases, ionic size also increases.

$\therefore N^{3-}$ is the anion with largest ionic size.

52. Option (3) is correct.



Blood Red Colour

In case of organic compound containing both Nitrogen and sulphur, they react with sodium and lead to the formation of sodium thiocyanate which gives blood red colour on reaction with Fe^{3+} .

53. Option (4) is correct.

Explanation: For given value of l , m_l can take values.

$$m_l = -l, \dots, 0, \dots, +l$$

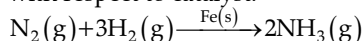
$$\text{if } l = 1, m_l = -1, 0, +1$$

$$\therefore n_m = 2l + 1$$

$$l = \frac{n_m - 1}{2}$$

54. Option (3) is correct.

Explanation: Heterogeneous catalyst in that catalyst in which Reactant is present in different state with respect to catalyst.

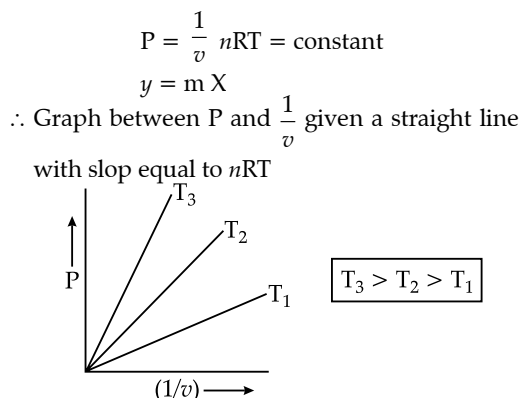


55. Option (1) is correct.

Explanation: According to Boyle's law at constant temp, pressure is inversely proportional to volume.

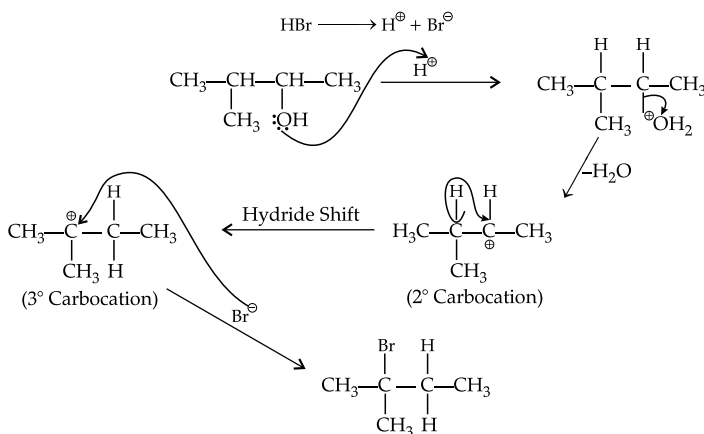
$$PV = nRT$$

$$P = \frac{nRT}{V}$$



57. **Option (4) is correct.**

Explanation:



Reaction proceed through Carbocation intermediate, less stable Carbocation rearrange itself to more stable Carbocation.

58. **Option (2) is correct.**

Explanation: Helium is used to dilute oxygen in diving apparatus because it has low solubility in blood.

$\therefore$ Assertion is true but reason is false.

59. **Option (2) is correct.**

Explanation: $(k) = g \times \frac{l}{a}$... (i)

Conductivity = conductance $\times$ cell constant

Since, we know that $g = \frac{1}{R}$

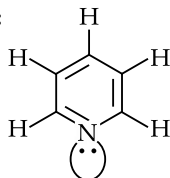
$$k = \frac{1}{R} \times \frac{l}{a} \Rightarrow \frac{l}{a} = R \times k$$

$$R = 60 \, \Omega, k = 0.0210$$

$$\therefore \frac{l}{a} = 0.0210 \times 60 = 1.26 \, \text{cm}^{-1}$$

60. **Option (2) is correct.**

Explanation:



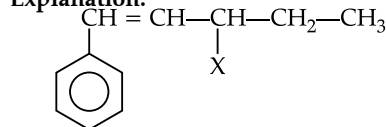
No. of π bond = 3

No. of lone pair = 1

No. of σ bond = 11

56. **Option (2) is correct.**

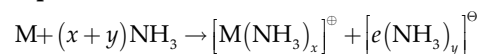
Explanation:



α' Carbon Atom attached to multiple bond is called Allylic carbon Atom.

61. **Option (2) is correct.**

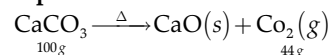
Explanation:



When metal is dissolved in liquid ammonia, a blue colour paramagnetic solution is formed. This blue colour paramagnetic solution is due to the solvated electron.

62. **Option (1) is correct.**

Explanation:



$$\text{Amount of CaCO}_3 = 20 \times \frac{20}{100} = 4\text{g}$$

($\therefore$ CaCO_3 from 20% pure)

$$\therefore 100\text{g of CaCO}_3 \text{ produce } \text{CO}_2 = 44\text{g}$$

$$1\text{g of CaCO}_3 \text{ produce } \text{CO}_2 = \frac{44}{100}$$

$$4\text{g of CaCO}_3 \text{ produce } \text{CO}_2 = \frac{44}{100} \times 4 = 1.76\text{g}$$

63. **Option (2) is correct.**

Explanation: Since covalent bonds hold atoms within the molecule together, however intermolecular

forces of attraction and repulsion between the molecule.

∴ Intermolecular forces are dipole-dipole forces, dipole-induced dipole forces, Hydrogen Bonding dispersion forces.

64. **Option (2) is correct.**

Explanation: Since Rate = $K[A]^2[B]$

New concentration of A

$$[A] = 3 \times [A]$$

$$\therefore \text{Rate} = K[3A]^2[B]$$

$$\text{Rate} = 9K[A]^2[B]$$

∴ Rate increases by a factor of 9.

65. **Option (3) is correct.**

Explanation: B Al Ga In Tl

As we move down the group stability of +1 oxidation state increases due to inert pair effect.

∴ Tl in +1 oxidation state have more stability

∴ $TlI > TlI_3$

66. **Option (2) is correct.**

Explanation: Hydrogen is used to reduce heavy metal oxides to metal as it is a good reduction agent.

D_2O (Heavy water) is used in organic reaction mechanism to study reaction mechanism.

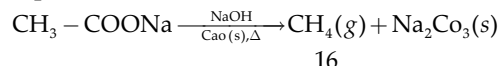
Hydrogen can also be used to make saturated fats from oils.

H-H bond dissociation energy is maximum.

Hydrogen reduces only those oxides of metal which are more reactive than iron.

67. **Option (1) is correct.**

Explanation:

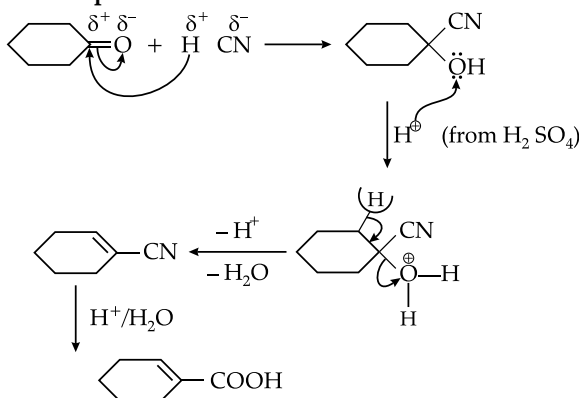


Weight of 1 mole of $CH_4 = 16g$

2 mole of $CH_4 = 2 \times 16 = 32g$

68. **Option (3) is correct.**

Explanation:



69. **Option (1) is correct.**

Explanation: Relationship between ΔG and E_{cell}° is given by

$$\Delta G = -nFE_{\text{cell}}$$

n = no. of electron

F = Faraday

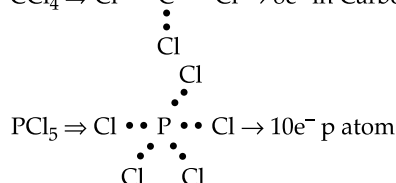
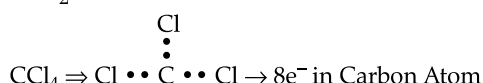
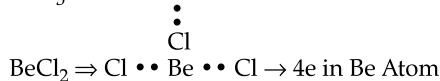
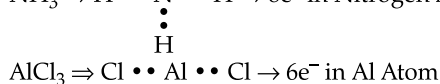
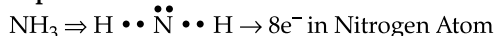
Also E_{cell}° is an intensive property

ΔG is an extensive property

∴ Both Assertion and Reason are correct but reason is not correct explanation of Assertion.

70. **Option (4) is correct.**

Explanation:



71. **Option (4) is correct.**

Explanation: For molecule up to N_2 increasing order of energies in various molecular orbital is $\sigma 1s, \sigma^* 1s, \sigma 2s, \sigma^* 2s, \pi 2p_x = \pi 2p_y, \sigma 2p_z, \pi^* 2p_x = \pi^* 2p_y, \sigma^* 2p_z$

72. **Option (4) is correct.**

Explanation: Let us suppose that number of B atom in CCP = n

Tetrahedral void = $2n$

Octahedral void = n

$$\therefore \text{No. of A atom} = \frac{1}{3} \times 2n = \frac{2n}{3}$$

$$\therefore A : B = \frac{2n}{3} : n$$

$$A : B = 2 : 3$$

Formula = A_2B_3

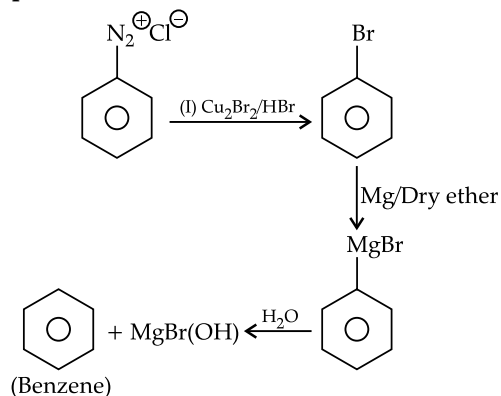
$$x=2, y=3 \Rightarrow x+y=2+3=5$$

73. **Option (2) is correct.**

Explanation: Stability of Cu^{2+} is more than Cu^+ salts in aqueous solution. This is due to high hydration energy of Cu^{2+} ion than Cu^+ .

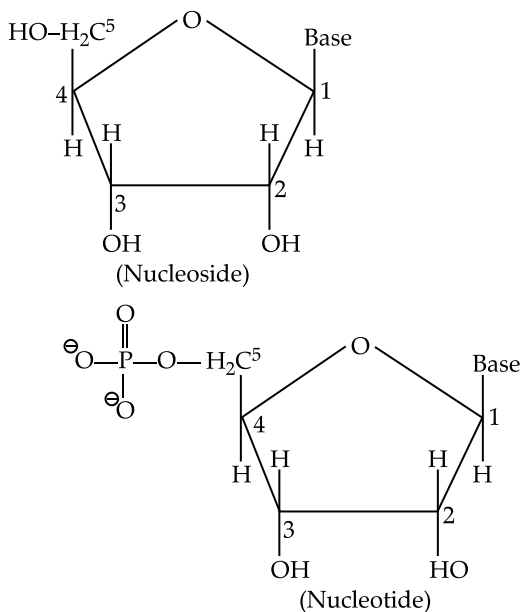
74. **Option (1) is correct.**

Explanation:



75. **Option (1) is correct.**

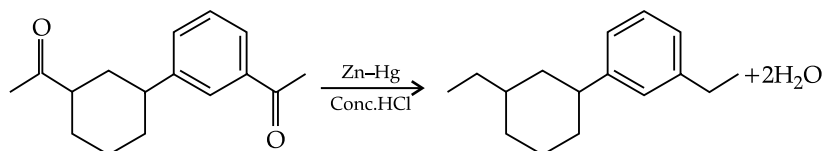
Explanation:



Nucleoside is formed by the attachment of a base to 1' position of sugar. Nucleotide informed

78. **Option (4) is correct.**

Explanation:



Zn.Hg/Conc. HCl reduces $\text{—}\overset{\text{O}}{\parallel}\text{C—}$ to —CH_2

79. **Option (2) is correct.**

Explanation:

- Coke is used as reducing agent in metallurgical process.
- Diamond each Carbon Atom is sp^3 hybridized.
- Fullerene consist of a Cage like structure.
- Graphite is used as lubricant.

80. **Option (2) is correct.**

Explanation: Lewis Acids are those substance which can accept a pair of electron, Lewis base are those substance which can donate a pair of electron.

$\text{H}_2\ddot{\text{O}}:$ $\rightarrow$ Lewis base
 $\text{BF}_3 \rightarrow 6e^-$, so it can accept two electrons.



$\text{:}\ddot{\text{O}}\text{H}$ $\rightarrow$ Lewis base

$\ddot{\text{N}}\text{H}_3 \rightarrow$ Lewis base

when nucleoside is attached to position 5' to the phosphoric acid.

76. **Option (4) is correct.**

Explanation: All enzyme that utilize ATP in phosphate transfer required Mg as the cofactor. Bone in human body is not an inert and unchanging substance. Ca play roles in neuromas cular function.

The daily requirement of Mg and Ca in human body in 200 – 300 mg.

77. **Option (2) is correct.**

Explanation: $\text{CH}_3 - \text{C} \equiv \text{N} \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) LiAlH}_4}$

$\text{CH}_3 - \text{CH}_2 - \text{NH}_2$ (1° Amine)

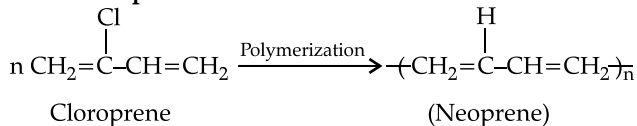
$\text{CH}_3 - \text{N} \equiv \text{C} \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) LiAlH}_4} \text{CH}_3 - \overset{\text{H}}{\underset{|}{\text{N}}} - \text{CH}_3$ (2° Amine)

$\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{NH}_2 \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) LiAlH}_4} \text{CH}_3 - \text{CH}_2 - \text{NH}_2$ (1° Amine)

$\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{NH}_2 + \text{Br}_2 + \text{KOH} \rightarrow \text{CH}_3 - \text{NH}_2$ (1° Amine)

81. **Option (1) is correct.**

Explanation:



Chloroprene produces neoprene with the help of free radical polymerization.

82. **Option (3) is correct.**

Explanation: Meprobamate, Valium and Chlordiazepoxide are tranquilizers.

Veronal is the derivatives of barbituric acid and considered as barbiturate.

83. **Option (4) is correct.**

Explanation: Homoleptic complexes are those complexes which contain only one types of ligand. $\text{K}_3 [\text{Al} (\text{OX})_3]$ – Potassium trioxalatoaluminate (III)

84. **Option (1) is correct.**

Explanation: Activation energy of certain reaction may be zero.e.g., diradical reaction.

Reason: Activation energy is defined as the minimum energy which must be supplied to the reactant so that there energy become equal to threshold energy.

85. **Option (3) is correct.**

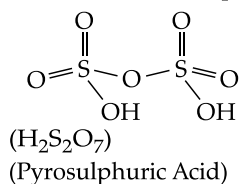
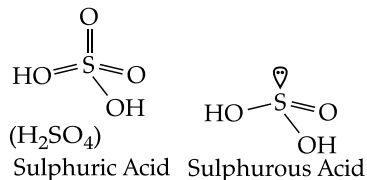
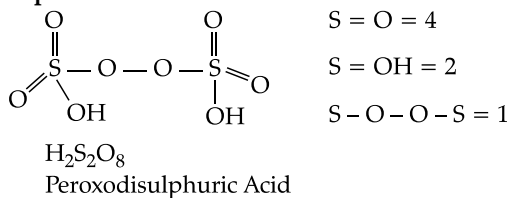
Explanation: Atom consist of three fundamental particle proton, electron and neutron.

- Mass of electron (m_e) = 9.10939×10^{-31} kg
- Isotope have same chemical properties.
- Nucleon = neutron + proton
- According to Dalton's Atomic theory atom can't be further subdivided.

Section B

86. **Option (1) is correct.**

Explanation:

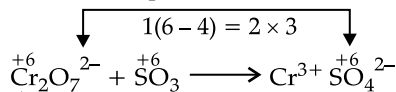


87. **Option (4) is correct.**

Explanation:

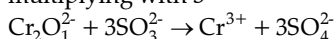


Step (i) to write balanced equation first of all write Skelton eqn.

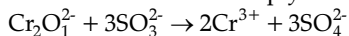


Change in O.N = $2(3-6) = -6 \times 1$

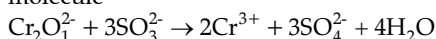
Step (ii) balance increase and decrease by multiplying with 3



Step (iii) balanced all other atoms except O. So in order to balance cr multiply it with 2

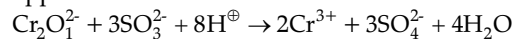


Step (iv) balance O atom by adding H_2O molecule



O = 16

Step (v) balance H atom by adding H^+ ions opposite site



$\therefore a = 1, b = 3, c = 8$

88. **Option (2) is correct.**

Explanation: No. of atom in fcc = 4

$\therefore$ Octahedral Void in fcc = 4

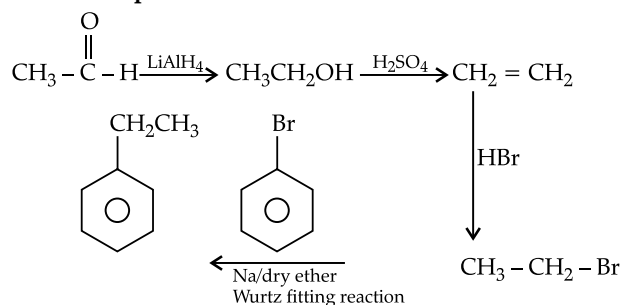
Octahedral void in fcc = Body centre + edge centre

Contribution at edge centre = $\frac{1}{4}$

$\therefore$ Fraction of one edge centered octahedral void in one unit of FCC = $\frac{1}{4}$

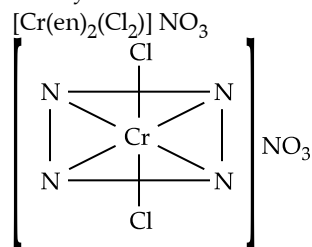
89. **Option (4) is correct.**

Explanation:



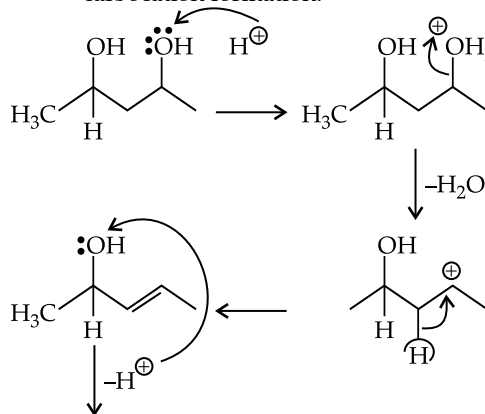
90. **Option (2) is correct.**

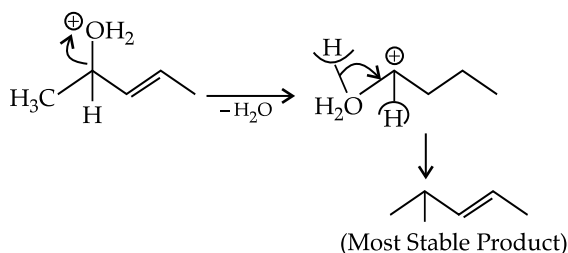
Explanation: Chelating complexes are more stable as they lead to the formation of cyclic structure.



91. **Option (1) is correct.**

Explanation: Acidic dehydration process through carbocation formation.



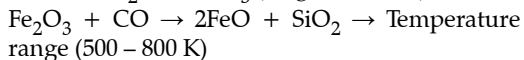
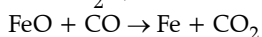
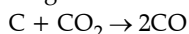


92. **Option (3) is correct.**

Explanation: Eutrophication is the excessive growth of plant in water bodies due to presence of nutrient. Due to Eutrophication level of dissolved oxygen decreases in water bodies.

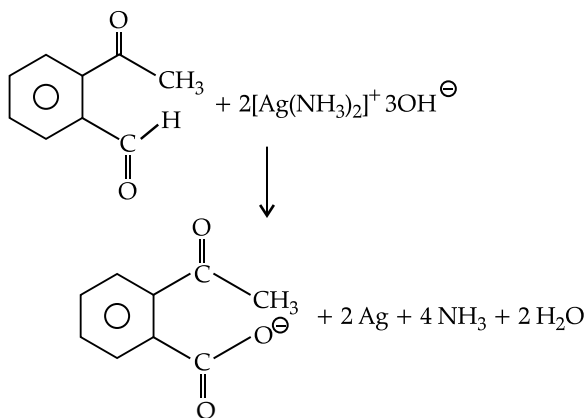
93. **Option (4) is correct.**

Explanation: Reaction which take place in a blast furnace between 900 K to 1500 K Temperature range are



94. **Option (2) is correct.**

Explanation: Tollen's Reagent used to differentiate Aldehyde and Ketone and it produces silver mirror.



95. **Option (1) is correct.**

Explanation: The correct relationship between ΔH and ΔU is given by eqn.

$$\Delta H = \Delta U + \Delta n_g RT$$

96. **Option (2) is correct.**

Explanation: All the transition metal except scandium form Mo which are ionic.

Highest Oxidation state corresponding to group number in transition metal oxide is attained in Sc_2O_3 to Mn_2O_7 .

Acidic character increases from V_2O_3 to V_2O_5

V_2O_4 gives VO^{2+} when dissolved in acid.

CrO is basic but Cr_2O_3 is Amphoteric

97. **Option (2) is correct.**

Explanation: $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$

$\frac{2}{\text{A}} \frac{3}{\text{B}} \frac{10}{\text{C}} \frac{6}{\text{D}} \rightarrow \text{at equilibrium}$

$$K_{\text{eq}} = \frac{[\text{C}][\text{D}]}{[\text{A}][\text{B}]} = \frac{10}{2} \frac{6}{3} = 10$$

$$\Delta G = -RT \ln K_{\text{eq}} = -2.303 RT \log_{10} K$$

$$= -2.303 \times 2 \times 300 \times \log(10)$$

$$= -1381.8 \text{ cal}$$

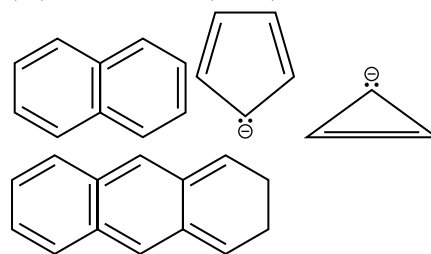
98. **Option (4) is correct.**

Explanation: Aromatic compound are those compound which satisfy Huckel's Rule

(i) Compound must be planar

(ii) Complete delocalisation.

(iii) Must Contain $(4n+2)$ π electron.



99. **Option (2) is correct.**

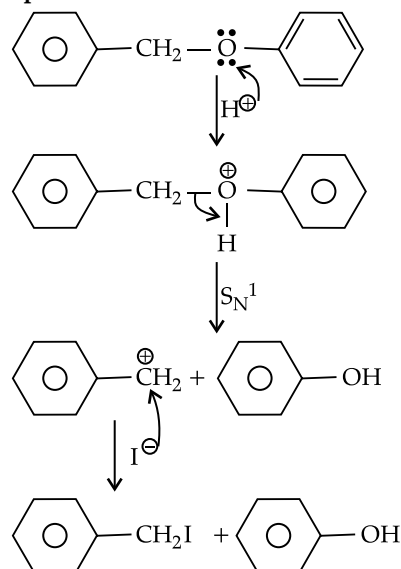
Explanation: Dispersed phase = Gas

Dispersion medium = Solid

$\therefore$ Pumic Stone in Solid Sol.

100. **Option (2) is correct.**

Explanation:



BOTANY

Section A

101. **Option (2) is correct.**

Explanation: In pleiotropism, a single gene affects multiple phenotypic expressions and the gene is called a pleiotropic gene.

102. **Option (1) is correct.**

Explanation: In tissue culture experiments, when leaf mesophyll cells are put in a culture medium, it will result in the formation of an undifferentiated mass of cells called callus. This phenomenon in which living plant cells that are differentiated by now and have lost the capacity to divide, regain the property of division is known as dedifferentiation.

103. **Option (3) is correct.**

Explanation: Active transport involves the movement and accumulation of ions across a membrane against the concentration gradient. It requires energy and special carrier proteins to transport molecules from their lower concentration to their higher concentration.

104. **Option (4) is correct.**

Explanation: The evil quartet is the term used for four major reasons for biodiversity loss. They include: a) habitat loss and fragmentation, b) alien species invasion, c) co-extinction, d) over-exploitation. Among these habitat loss and fragmentation is the most important cause driving plants and animals to extinction.

105. **Option (3) is correct.**

Explanation: Ethidium Bromide (EtBr) is added to the running buffer during the separation of DNA fragments by agarose gel electrophoresis. It is used because upon binding with the DNA molecule and subsequent illumination with a UV light source, the DNA will appear in bright orange in colour.

As a result, DNA banding pattern can be visualised.

106. **Option (4) is correct.**

Explanation: The main stage of the life cycle of a moss is the gametophyte which consists of two stages. The first stage is the protonema stage, which develops directly from a spore. Capsule of the sporophyte contains spore which gives rise to protonema. The second stage is the leafy stage, which develops from the protonema.

107. **Option (1) is correct.**

Explanation: In eukaryotes, there are three major types of RNA polymerases.

RNA polymerase I help in the transcription of 5.8S, 18S, and 28S rRNAs.

RNA polymerase II helps in the transcription of hnRNAs (precursor of mRNA).

RNA polymerase III helps in the transcription of tRNAs, ScRNA, 5S rRNA and snRNA.

108. **Option (1) is correct.**

Explanation: The historic convention on Biological Diversity, also called "The Earth Summit" was held in Rio de Janeiro in the year 1992. It called upon all nations to take the required measures for the conservation of biodiversity.

109. **Option (4) is correct.**

Explanation: ATP in glycolysis is used in two steps in the first or preparatory phase.

(i) Glucose → Glucose-6-phosphate

(ii) Fructose-6-phosphate → Fructose-1,6-bisphosphate

110. **Option (4) is correct.**

Explanation: The thickness of the ozone in a column of air is measured in terms of Dobson units (DU). The average amount of ozone in the atmosphere is roughly 300 Dobson Units, equivalent to a layer 3 millimeters (0.12 inches) thick.

111. **Option (3) is correct.**

Explanation: Endarch and exarch are the terms often used for describing the position of the primary xylem and not the secondary xylem in the plant body.

The primary xylem is of two types. protoxylem and metaxylem. The position of the protoxylem and metaxylem defines endarch or exarch conditions. In endarch type of primary xylem, the protoxylem lies towards the centre and metaxylem lies towards the periphery while in exarch type of primary xylem, the protoxylem lies towards the periphery and metaxylem lies towards the centre. Exarch type is the most common feature seen in the root system.

112. **Option (1) is correct.**

Explanation: Heterosporous pteridophyte produces two different kinds of spores. Examples include *Selaginella* and *Salvinia*. While homosporous pteridophytes produce a single type of spores. Examples include *Psilotum*, *Lycopodium* and *Equisetum*.

113. **Option (4) is correct.**

Explanation: Earthworms and other detritivores contribute to decomposition by fragmentation of detritus in which a part of detritus is eaten by them and comes out in highly pulverised state in their faeces. Mineralization is the process in

which humus is degraded to release inorganic nutrients. Leaching is the process in which water-soluble inorganic substances such as fertilizers or pesticides are washed out from the soil, go down and get precipitated. The detritus food chain starts with dead organic matter. The saprotrophic bacteria and fungi break down detritus into simpler inorganic substances by a process called catabolism. Earthworms do fragmentation not catabolism.

114. Option (3) is correct.

Explanation: China rose, Petunia and Lemon show axile placentation. In axile placentation, the ovary is partitioned into two or more chambers by septa. The placenta is formed in the central axial column where all the septa meet.

Dianthus and *Primrose* show free central placentation.

Peas, Lupin and Beans show marginal placentation. Cucumbers and mustard show parietal placentation.

115. Option (4) is correct.

Explanation: Fabaceae → Diadelphous and ditheous anther.

Solanaceae → Polyandrous, epipetalous and ditheous anther.

Liliaceae → Polyandrous, epiphyllous and ditheous anther.

116. Option (4) is correct.

Explanation: In Photosystem-I, the reaction centre chlorophyll a has an absorption peak at 700 nm, while in Photosystem-II, the reaction centre has an absorption maxima at 680 nm.

117. Option (1) is correct.

Explanation: On addition of chilled ethanol, purified DNA precipitates out. It can be seen as a collection of fine threads in the suspension.

118. Option (1) is correct.

Explanation: Replication of DNA takes place in the S-phase of the cell cycle in eukaryotes while other cell organelles duplicate in the G₁ phase of the cell cycle.

119. Option (1) is correct.

Explanation: The process of recombination occurs at the pachytene stage of prophase I. It is characterised by the appearance of recombination nodules.

120. Option (1) is correct.

Explanation: For one turn in the Calvin cycle:

ATP used = 3 molecules

NADPH used = 2 molecules

Thus for six cycles:

ATP used = $3 \times 6 = 18$ molecules

NADPH used = $2 \times 6 = 12$ molecules

Therefore, number of total ATP and NADPH required to produce one molecule of glucose during the Calvin cycle is 18 and 12 respectively.

121. Option (2) is correct.

Explanation: A considerable amount of GPP is utilised by plants in respiration. Gross primary productivity minus respiratory losses (R), is the net primary productivity.

So, R = Respiratory loss

122. Option (1) is correct.

Explanation: Colourful fragrant flowers with nectar help to attract biotic pollinators (insects, birds, bats, etc.). Bird pollinated plants are larger, and have more nectar. Insect pollinated plants can be small in size. Bat pollinated flowers are pale and nocturnal.

123. Option (1) is correct.

Explanation: Spraying of gibberellins (GAs) on juvenile conifers hastens the maturity period, thus leading to early seed production.

124. Option (4) is correct.

Explanation: Manganese plays a major role in the splitting of water to release oxygen during photosynthesis.

Copper is required for the overall metabolism in plants.

Molybdenum helps in nitrogen metabolism.

Magnesium activates enzymes involved in photosynthesis and respiration.

125. Option (4) is correct.

Explanation: All the genes that are expressed as RNA are known as Expressed Sequence Tags (ESTs).

126. Option (2) is correct.

Explanation: Alfred Sturtevant for the first time used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes and for mapping of their position on the chromosome.

Sutton and Boveri gave the chromosomal theory of inheritance.

Henking discovered X-chromosome.

Thomas Hunt Morgan proved the chromosomal theory of inheritance.

127. Option (4) is correct.

Explanation: Various measurements reveal that the force that is produced by transpiration can generate pressures sufficient to lift a xylem-

sized column of water up to 130 meters high. Transpiration is also known as a process that cools leaf surfaces, sometimes 10 to 15 degrees, by evaporative cooling.

128. Option (2) is correct.

Explanation: A gene gun is a method of recombinant DNA technology where the DNA, RNA, genes, or proteins are transferred to the target cells without the help of any vectors. In this method, microparticles of gold or tungsten are used. Gold and tungsten are inert so they do not react with the chemicals of cells.

129. Option (4) is correct.

Explanation: In winter, the activity of cambium is less and forms a lesser number of xylary elements that have narrow vessels, and the resulting wood is called autumn wood or latewood.

130. Option (1) is correct.

Explanation: Tassels in the corn cob are elongated exerted stigmas and styles which help to trap pollen grains.

131. Option (2) is correct.

Explanation: Synergids are the cells of gametophytes and so these are haploid. Zygote is formed by the fusion of two gametes and thus it is diploid. On the other hand primary endosperm nucleus is formed by the fusion of a diploid secondary nucleus with a male gamete. Therefore, it is triploid.

132. Option (2) is correct.

Explanation: Ethylene helps in rapid internode/ petiole elongation in deep-water rice plants.

133. Option (2) is correct.

Explanation: Anaphase II of meiosis involves the splitting of centromere. During Metaphase I and II, chromosomes align at the equator. During telophase, chromosomes reach the respective poles.

134. Option (1) is correct.

Explanation: The proof that DNA is the genetic material came first time from the experiment of Alfred Hershey and Martha Chase. Avery, Macleod and McCarty gave the biochemical characterisation of the Transforming Principle. The transformation experiments using *Pneumococcus* were conducted by Frederick Griffith. Wilkins and Franklin produced X-ray diffraction data of DNA.

135. Option (2) is correct.

Explanation: Cellulose does not form a blue colour with iodine because cellulose does not have a helical structure so it does not bind to iodine to form a coloured product.

Section B

136. Option (3) is correct.

Explanation: Pyruvate is formed by catabolism of glucose in the cytosol. After it enters the mitochondria, it undergoes oxidative decarboxylation with the help of complex set of reactions that are catalysed by pyruvate dehydrogenase. The scheme of glycolysis is often referred to as the EMP pathway. In an electron transport system, the energy produced is utilised for the production of the proton gradient required for phosphorylation. So, this process is also called oxidative phosphorylation. The TCA (tricarboxylic acid cycle) starts with the condensation of the acetyl group with oxaloacetic acid (OAA) and water to produce citric acid. The reaction uses the enzyme citrate synthase.

137. Option (4) is correct.

Explanation: The process of ATP synthesis using 'free energy' obtained when electrons are passed to several carriers (ETC) is known as chemiosmosis. Chemiosmosis requires a membrane, a proton pump, a proton gradient and ATP synthase.

138. Option (4) is correct.

Explanation: Melonate acts as an antibiotic which inhibits the growth of pathogenic bacteria by inhibiting the activity of Succinic dehydrogenase. Thus, it affects TCA cycle and hampers cellular respiration.

139. Option (2) is correct.

Explanation: In gymnosperms, the pollen grains are released from the microsporangium and they are carried in air currents. They come in touch with the opening of the ovules borne on megasporophylls. The *pollen tube* having the male gametes grows towards archegonia in the ovules and empties its contents near the mouth of the archegonia.

140. Option (2) is correct.

Explanation: Recombinant DNA technology involves several steps in specific sequences like isolation of DNA then fragmentation of DNA by restriction endonucleases followed by isolation of desired DNA fragment and then ligation of the DNA fragment into a vector following transferring the recombinant DNA into the host and finally culturing the host cells in a medium at large scale and extraction of the desired product.

141. Option (4) is correct.

Explanation: Cohesion involves mutual attraction between water molecules. Adhesion represents

the attraction of water molecules to polar surfaces. Surface tension involves the attraction of water molecules towards each other in the liquid phase more than to water in the gas phase. Guttation represents the loss of water in the liquid phase.

142. Option (1) is correct.

Explanation: Algal blooms are caused by the increase of inorganic pollutants in water. It gives a distinct colour to the water bodies. It decreases water quality and promotes fish mortality.

143. Option (4) is correct.

Explanation: The ribosome consists of about 80 different proteins.

144. Option (1) is correct.

Explanation: (+, +) Mutualism: In this interaction, both the interacting species are benefitted.

(+, 0) Commensalism: Only one species is benefitted and the other species remains unaffected.

(-, 0) Amensalism: One species is harmed and other is unaffected.

(+, -) Parasitism: One species is benefitted and the other is negatively affected.

145. Option (4) is correct.

Explanation: A flower is a modified shoot wherein the shoot apical meristem changes to a floral meristem. Internodes do not get elongated and as a result the axis gets condensed. The apex produces many kinds of floral appendages laterally at the successive nodes instead than of leaves.

146. Option (1) is correct.

Explanation: Lenticels are lens-shaped openings that help in the exchange of gases between the outer atmosphere and the internal tissue of the stem. A bark formed early in the season is called early or soft bark. Later on at the end of the season, late or hard bark is formed. The bark is a

non technical term for all tissues exterior to the vascular cambium. Bark includes periderm and secondary phloem. Phellogen is a double layered ring-shaped structure made up of narrow, thin-walled and nearly rectangular cells.

147. Option (2) is correct.

Explanation: Gause's 'Competitive Exclusion Principle' states that two species that are closely related and competing for the same resources cannot co-exist indefinitely and the species which is competitively inferior will get eliminated. Herbivores and plants appear to be more affected by competition than carnivores.

148. Option (2) is correct.

Explanation: Klinefelter's syndrome is the result of presence of an additional copy of the X-chromosome resulting in a karyotype of 47, XXY. Such an individual has overall masculine development, however, the feminine development is also expressed in these individuals. Such individuals are sterile.

149. Option (2) is correct.

Explanation: M phase or mitosis is the phase where the actual cell division occurs. Mitosis is also called equational division. During the G₂ phase DNA synthesis stops but the cell synthesises of RNA, proteins, etc. for the next phase. The quiescent stage is the inactive phase in which non-dividing cells enter. The G₁ phase represents the stage between mitosis and initiation of DNA replication.

150. Option (2) is correct.

Explanation: Iron activates the catalase enzyme. Zinc is required in the synthesis of auxin.

Boron is required for cell elongation and cell differentiation.

Molybdenum is a component of nitrogenase and nitrate reductase enzyme.

ZOOLOGY

Section A

151. Option (3) is correct.

Explanation: In prokaryotes, the negatively charged DNA is held by some positively charged non-histone proteins in a region known as the nucleoid. In eukaryotes, nucleosome represents the structure that is formed by the negatively charged DNA wrapped around the positively charged histone octamer.

152. Option (4) is correct.

Explanation: HIV enters into helper T-lymphocytes (T_H), undergoes replication and produces progeny viruses.

153. Option (1) is correct.

Explanation: An increase in the concentration of the toxicant at successive trophic levels is called biomagnification. Large amounts of inorganic nutrients in water lead to the growth of algae thus algal blooms occur. It increases fish deaths in the

water body. Eutrophication is a process in which water bodies become nutrient rich. Releasing sewage into rivers introduces a large amount of nutrients into the water also causes eutrophication is a process in which water bodies become nutrient rich.

154. Option (4) is correct.

Explanation: In an ECG:

P-wave represents the depolarisation of the atria. Q-wave represents the beginning of systole. QRS complex represents the depolarisation of ventricles.

T-wave represents the repolarisation of the ventricles.

155. Option (4) is correct.

Explanation: Down's syndrome is the result of an additional copy of chromosome number 21. Its symptoms are:

- Broad palm with characteristic palm crease
- Short statured with small round head
- Furrowed tongue and partially open mouth, etc.

156. Option (4) is correct.

Explanation: RNA being unstable and single-stranded, mutates at a faster rate. So viruses having RNA genome and shorter life span mutate and evolve very fast.

157. Option (1) is correct.

Explanation: A ligament is an example of dense regular connective tissue. Cartilage is a specialised connective tissue.

158. Option (2) is correct.

Explanation: Assertion is true as there are two types of nephrons, i.e., cortical nephrons and juxtamedullary nephrons based on their relative position in the cortex and medulla. The reason is incorrect as the loop of Henle in juxtamedullary nephrons is long.

159. Option (3) is correct.

Explanation: A probe is a radioactive molecule that helps in the identification of mutated genes. YAC, BAC, and pBR322 are vectors.

160. Option (3) is correct.

Explanation: A protein is imagined as a line, the left end represented by the first amino acid also called as N-terminal amino acid and the right end is represented by the last amino acid also called the C-terminal amino acid. Adult haemoglobin consists of 4 subunits.

161. Option (1) is correct.

Explanation: When the undigested food (faeces) enters into the caecum of the large Intestine, the

ileo-caecal valve prevents the backflow of the faecal matter.

162. Option (1) is correct.

Explanation:

- Vasectomy is a surgical technique of contraception.
- Coitus interruptus is a natural technique of contraception.
- Cervical cap is a barrier technique of contraception.
- Saheli pill is an oral method of contraception.

163. Option (2) is correct.

Explanation: Cytoskeleton which consists of microtubules, microfilaments and intermediate filaments is involved in many functions such as motility, mechanical support, and maintenance of the shape of the cell.

164. Option (1) is correct.

Explanation: Gonorrhoea is a bacterial STD that can be treated and cured completely. All others are viral diseases which can be managed but have no cure.

165. Option (3) is correct.

Explanation: Vital capacity is the maximum volume of air a person can inhale after a forced expiration. It is calculated as the sum of inspiratory reserve volume (IRV), tidal volume (TV), and expiratory reserve volume (ERV), i.e., $VC = IRV + TV + ERV$.

166. Option (1) is correct.

Explanation: Endoplasmic reticulum (ER), Golgi complex, lysosomes and vacuoles together are known as the endomembrane system. Mitochondria, Chloroplasts and peroxisomes are not included in it.

167. Option (4) is correct.

Explanation: Cholecystokinin (CCK) acts on both the pancreas and gall bladder and helps in the secretion of pancreatic and bile enzymes respectively. GIP inhibits gastric secretion and motility. Atrial Natriuretic Factor (ANF) is released from the atrial wall of the heart. Anti-diuretic hormone (ADH) acts mainly on the kidney.

168. Option (1) is correct.

Explanation: In a *lac* operon,

Gene *a* codes for enzyme transacetylase.

Gene *y* codes for enzyme permease.

Gene *i* codes for a repressor protein

Gene *z* codes for enzyme beta-galactosidase.

169. **Option (4) is correct.**

Explanation:

- (i) **Fovea:** Point of greatest visual acuity or resolution.
- (ii) **Iris:** Visible coloured portion of the eye that regulates the diameter of the pupil.
- (iii) **Blind spot :** Optic nerve leaves the eye-ball and photoreceptor cells are absent.
- (iv) **Sclera:** External layer of the eye formed of dense connective tissue.

170. **Option (1) is correct.**

Explanation: Numbat, spotted cuscus and flying phalanger are Australian marsupials exhibiting adaptive radiation.

171. **Option (4) is correct.**

Explanation: The Vas deferens receives a duct from the seminal vesicle and opens into the urethra as the ejaculatory duct. The cavity of the cervix is called the cervical canal which along with the vagina forms the birth canal.

172. **Option (4) is correct.**

Explanation: Heroin is an opioid and a depressant that slows down body functions. Marijuana produces effects on the cardiovascular system of the body. Cocaine interferes with the transport of the neurotransmitter dopamine. Morphine is used as a sedative and painkiller.

173. **Option (4) is correct.**

Explanation: (i) Ringworm is caused by *Trichophyton*.

(ii) Filariasis is caused by *Wuchereria bancrofti*.

(iii) Malaria is caused by *Plasmodium* species.

(iv) Pneumonia is caused by *Haemophilus influenzae*.

174. **Option (3) is correct.**

Explanation: Amniocentesis is used to test for the presence of certain genetic disorders such as Down's syndrome, and haemophilia, or to determine the survivability of the foetus but it is often misused as a tool for the detection of the sex of the child which results into increase in cases of female foeticide. Ban on this technique helps in preventing female foeticide.

175. **Option (1) is correct.**

Explanation: Radial symmetry is not found in adults of hemichordates as these are bilaterally symmetrical animals.

176. **Option (4) is correct.**

Explanation: Both statements are correct as low temperature preserves the enzyme in a temporarily inactive state whereas high temperature destroys

enzymatic activity because proteins are denatured by heat. A competitive inhibitor is structurally similar to the substrate and it competes with the same for the binding site of the enzyme.

177. **Option (2) is correct.**

Explanation: Flame cells are the excretory structures in Platyhelminthes like *Taenia*. *Paramoecium* uses contractile vacuoles for excretion. Nephridia are the excretory structures of annelids like *Pheretima*. Ureose glands are present in the cockroach.

178. **Option (3) is correct.**

Explanation: First menstrual cycle that begins at puberty is called menarche. Menopause is the stoppage of the menstrual cycle.

179. **Option (2) is correct.**

Explanation: Electrostatic precipitator is most widely used in thermal power plants. It can remove over 99 per cent of particulate matter present in the exhaust from a thermal power plant.

180. **Option (2) is correct.**

Explanation: Mucus neck cells secrete mucus. Peptic or chief cells secrete the proenzyme pepsinogen. Parietal or oxyntic cells secrete HCl and intrinsic factors for absorption of vitamin B₁₂. Hepatic cells secrete bile juice.

181. **Option (1) is correct.**

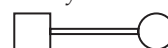
Explanation: Cells present on the outer layer of the blastocyst get attached to the outer layer of the endometrium for pregnancy to occur. They release a protein called L-selectin. The Corpus luteum secretes a large amount of progesterone which is required for the maintenance of the endometrium of the uterus. In the absence of fertilisation, the corpus luteum degenerates resulting in a decrease in the level of progesterone hormone that will cause the disintegration of endometrium followed by menstruation.

182. **Option (1) is correct.**

Explanation: A cartilaginous joint is present in between the adjacent vertebrae in the vertebral column. The ball and socket joint is present between the humerus and pectoral girdle. A fibrous joint is present between flat skull bones. The saddle joint is present between the carpal and metacarpal of the thumb.

183. **Option (1) is correct.**

Explanation: The symbol representing mating between relatives (consanguineous mating) in the human pedigree analysis is:



184. **Option (1) is correct.**

Explanation: Conventional methods of diagnosis like serum and urine analysis, etc, do not help in early diagnosis. While Recombinant DNA technology, Polymerase Chain Reaction [PCR] and Enzyme-Linked Immuno-Sorbent Assay (ELISA) are some of the techniques that help in early diagnosis.

185. **Option (4) is correct.**

Explanation: A leopard and a lion in a forest/grassland are an example of competition where both species are competing for the same resources. A cuckoo laying an egg in a crow's nest is an example of brood parasitism where the cuckoo is the bird that lays its egg in the nest of a crow.

Fungi and root of a higher plant in mycorrhizae is an example of mutualism where both species are benefitted. A cattle egret and cattle in a field is an example of commensalism where one species benefits and the other remains unaffected.

Section B

186. **Option (3) is correct.**

Explanation: Cells in the G_0 phase of the cell cycle remain metabolically active but no longer proliferate. In animal cells, DNA replication begins in the nucleus during the S phase and the centriole duplicates in the cytoplasm.

187. **Option (4) is correct.**

Explanation: Logistic growth occurs in case of limited resource availability conditions. Exponential growth occurs in case of unlimited resources. Expanding age pyramid shows a growing population where the per cent of individuals of pre-reproductive age is the largest followed by reproductive and post-reproductive age groups. A stable age pyramid shows a stable population where number of pre reproductive and reproductive age individuals is almost same.

188. **Option (2) is correct.**

Explanation: In cockroaches, excretion is brought about by Malpighian tubules, fat body, nephrocytes and urecose glands.

189. **Option (1) is correct.**

Explanation: Tetrad formation can be seen during the zygotene stage. Centromeres split and chromatids separate during Anaphase. Terminalisation of chiasmata takes place during diakinesis. Nucleolus, Golgi complex and ER are reformed during telophase. Crossing over takes place between non-sister chromatids.

190. **Option (2) is correct.**

Explanation: Thyroid hormones play an important role in the regulation of basal metabolic rate, maintenance of water and electrolyte balance and support the process of RBCs formation, whereas this hormone is not involved in regulating the normal rhythm of the sleep-wake cycle and development of the immune system.

191. **Option (1) is correct.**

Explanation: Cockroaches show sexual dimorphism because males and females have slightly different morphology. One of the main features is the presence of anal styles only in males. Females lack anal styles.

192. **Option (2) is correct.**

Explanation: Basophils secrete histamine, serotonin, heparin etc. and are involved in inflammatory responses. Basophils are granulocytes. Neutrophils are the most abundant cells (60–65 %) of the total WBCs. Basophils are the least (0.5–1%) abundant of all WBCs. Monocytes have a kidney-shaped nucleus.

193. **Option (1) is correct.**

Explanation: ADH helps in water reabsorption from the DCT of the nephron to prevent diuresis, which increases blood pressure. It causes constriction of blood vessels thus increases the GFR. ANF is secreted by the heart and it is a vasodilator. Excessive loss of body fluid from the body switches on the osmoreceptors.

194. **Option (2) is correct.**

Explanation: Areolar connective tissue contains mast cells, fibroblasts and macrophages. The inner surface of bronchioles is marked by ciliated epithelium. Blood is a specialised connective tissue. Tubular parts of the nephron are covered by cuboidal epithelium.

195. **Option (1) is correct.**

Explanation: Chordates have closed circulatory systems and pharyngeal gill slits. The nerve cord is dorsal, hollow and single. The heart is ventral. They are triploblastic and coelomate.

196. **Option (1) is correct.**

Explanation: The presence of hairs, pinna and mammary glands are unique features of mammals. Monocondylic skull is present in Aves and reptiles. Mammals have a dicondylic skull. The tympanic membrane is present in amphibians also. Indirect development is not seen in mammals.

197. **Option (4) is correct.**

Explanation: The limbic system along with

the hypothalamus regulates sexual behaviour, expression of excitement, pleasure, rage, fear, etc.

198. Option (2) is correct.

Explanation: The sequence of the coding strand is the same as transcribed mRNA except for thymine at the place of uracil.

Template strand → 3'-TAGCTAGCTAGCTAGC-TAGCTAGCTAGC-5'

Coding strand → 5'-ATCGATCGATCG-ATCGATCGATCGATCG-3'

mRNA → 5' AUCGAUCGAUCGAUCGAUC-GAUCG AUCG 3'

199. Option (1) is correct.

Explanation: Muscle bundles are joined together by a collagenous connective tissue layer called fascia. Muscle bundles are known as fascicles. The portion of the myofibril present between two successive 'Z' lines is the main functional unit of contraction called a sarcomere.

200. Option (3) is correct.

Explanation: A decrease in the productivity of the inbred population is not an advantage of inbreeding.



SOLVED PAPER	NEET (UG) (Re-Examination) 4th September 2022	Code U6
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Important Instructions:

- The test is of **3 hours 20 minutes** duration and Test Booklet contains **200** multiple choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**. 50 in each subject are divided into two **Section (A and B)** as per details given below:
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos- 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** Questions in each subject (Question Nos- 36 to 50, 86 to 100, 136 to 150 and 80 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, the first ten questions answered by the candidate shall be evaluated.
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, one mark will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Use of Electronic/Manual Calculator is prohibited.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

PHYSICS

Section-A

- Q. 1.** Identify the function which represents a non-periodic motion.
- $e^{-\omega t}$
 - $\sin \omega t$
 - $\sin \omega t + \cos \omega t$
 - $\sin (\omega t + \pi/4)$
- Q. 2.** The magnetic field of a plane electromagnetic wave is given by $\vec{B} = 3 \times 10^{-8} \cos (1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{j}$, then the associated electric field will be:
- $3 \times 10^{-8} \cos (1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{z} \text{ V/m}$
 - $3 \times 10^{-8} \sin (1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{z} \text{ V/m}$
 - $9 \sin (1.6 \times 10^3 x - 48 \times 10^{10} t) \hat{z} \text{ V/m}$
 - $9 \cos (1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{z} \text{ V/m}$
- Q. 3.** The incorrect statement about the property of a Zener diode is:

- Zener voltage remains constant at breakdown
 - It is designed to operate under reverse bias
 - Depletion region formed is very wide
 - p and n regions of zener diode are heavily doped
- Q. 4.** A cell of emf 4 V and internal resistance 0.5Ω is connected to a 7.5Ω external resistance. The terminal potential difference of the cell is:
- 3.75 V
 - 4.25 V
 - 4 V
 - 0.375 V
- Q. 5.** Given below are two statements:
- Statement-I:**
In an a.c. circuit, the current through a capacitor leads the voltage across it.
- Statement-II:**
In a.c. circuits containing pure capacitance only, the phase difference between the current and the voltage is π :

Code Q3

- (1)** v

(3) $2\sqrt{2} v$

(2) $2\sqrt{2} v$

(4) $3\sqrt{2} v$

Q. 4. As the temperature increases, the electrical resistance :

- (1) increases for both conductors and semiconductors
- (2) decreases for both conductors and semiconductors
- (3) increases for conductors but decreases for semiconductors
- (4) decreases for conductors but increases for semiconductors

Q. 5. A long solenoid of radius 1 mm has 100 turns per mm. If 1 A current flows in the solenoid, the magnetic field strength at the centre of the solenoid is :

- (1) $6.28 \times 10^{-2} \text{ T}$ (2) $12.56 \times 10^{-2} \text{ T}$
- (3) $12.56 \times 10^{-4} \text{ T}$ (4) $6.28 \times 10^{-4} \text{ T}$

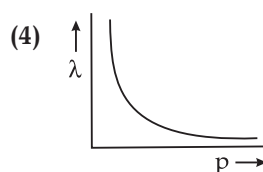
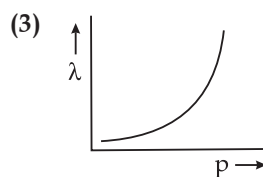
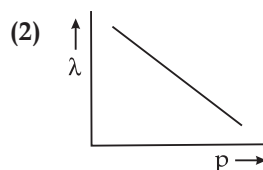
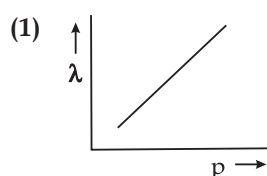
Q. 6. If the initial tension on a stretched string is doubled, then the ratio of the initial and final speeds of a transverse wave along the string is :

- (1) 1 : 1 (2) $\sqrt{2} : 1$
- (3) $1 : \sqrt{2}$ (4) 1 : 2

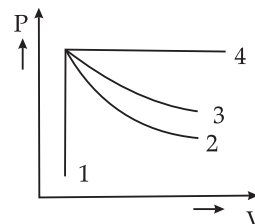
Q. 7. Two objects of mass 10 kg and 20 kg respectively are connected to the two ends of a rigid rod of length 10 m with negligible mass. The distance of the center of mass of the system from the 10 kg mass is :

- (1) $\frac{10}{3} \text{ m}$ (2) $\frac{20}{3} \text{ m}$
- (3) 10 m (4) 5 m

Q. 8. The graph which shows the variation of the de Broglie wavelength (λ) of a particle and its associated momentum



Q. 9. An ideal gas undergoes four different processes from the same initial state as shown in the figure below. Those processes are adiabatic, isothermal, isobaric and isochoric. The curve which represents the adiabatic process among 1, 2, 3 and 4 is :



- (1) 1 (2) 2
- (3) 3 (4) 4

Q. 10. Plane angle and solid angle have :

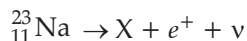
- (1) Units but no dimensions
- (2) Dimensions but no units
- (3) No units and no dimensions
- (4) Both units and dimensions

Q. 11. The angle between the electric lines of force and the equipotential surface is :

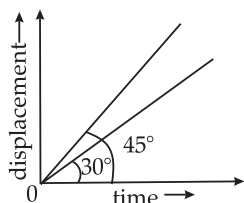
- (1) 0° (2) 45°
- (3) 90° (4) 180°

Q. 12. A copper wire of length 10 m and radius $(10^{-2} / \sqrt{\pi}) \text{ m}$ has electrical resistance of 10Ω . The current density in the wire for an electric field strength of 10 (V/m) is :

- (1) 10^4 A/m^2 (2) 10^6 A/m^2
 (3) 10^{-5} A/m^2 (4) 10^5 A/m^2
- Q. 13.** In the given nuclear reaction, the element X is :

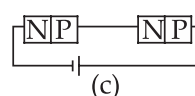
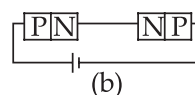
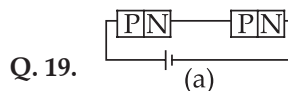


- (1) ${}_{11}^{23}\text{Na}$ (2) ${}_{10}^{23}\text{Ne}$
 (3) ${}_{10}^{22}\text{Ne}$ (4) ${}_{12}^{22}\text{Mg}$
- Q. 14.** The displacement-time graphs of two moving particles make angles of 30° and 45° with the x -axis as shown in the figure. The ratio of their respective velocity is :



- (1) $\sqrt{3} : 1$ (2) $1 : 1$
 (3) $1 : 2$ (4) $1 : \sqrt{3}$
- Q. 15.** In half wave rectification, if the input frequency is 60 Hz, then the output frequency would be :
- (1) zero (2) 30 Hz
 (3) 60 Hz (4) 120 Hz
- Q. 16.** The ratio of the radius of gyration of a thin uniform disc about an axis passing through its centre and normal to its plane to the radius of gyration of the disc about its diameter is :
- (1) $2 : 1$ (2) $\sqrt{2} : 1$
 (3) $4 : 1$ (4) $1 : \sqrt{2}$
- Q. 17.** When two monochromatic lights of frequency, ν and $\frac{\nu}{2}$ are incident on a photoelectric metal, their stopping potential becomes $\frac{V_s}{2}$ and V_s respectively. The threshold frequency for this metal is :
- (1) 2ν (2) 3ν
 (3) $\frac{2}{3}\nu$ (4) $\frac{3}{2}\nu$

- Q. 18.** The dimensions $[\text{MLT}^{-2} \text{A}^{-2}]$ belong to the :
- (1) magnetic flux
 (2) self inductance
 (3) magnetic permeability
 (4) electric permittivity



In the given circuits (a), (b) and (c), the potential drop across the two p - n junctions are equal in :

- (1) Circuit (a) only
 (2) Circuit (b) only
 (3) Circuit (c) only
 (4) Both circuits (a) and (c)
- Q. 20.** The energy that will be ideally radiated by a 100 kW transmitter in 1 hour is :
- (1) $36 \times 10^7 \text{ J}$ (2) $36 \times 10^4 \text{ J}$
 (3) $36 \times 10^5 \text{ J}$ (4) $1 \times 10^5 \text{ J}$
- Q. 21.** In a Young's double slit experiment, a student observes 8 fringes in a certain segment of screen when a monochromatic light of 600 nm wavelength is used. If the wavelength of light is changed to 400 nm, then the number of fringes he would observe in the same-region of the screen is :
- (1) 6 (2) 8
 (3) 9 (4) 12
- Q. 22.** The peak voltage of the ac source is equal to:
- (1) the value of voltage supplied to the circuit
 (2) the rms value of the ac source
 (3) $\sqrt{2}$ times the rms value of the ac source
 (4) $1/\sqrt{2}$ times the rms value of the ac source
- Q. 23.** If a soap bubble expands, the pressure inside the bubble :
- (1) decreases
 (2) increases
 (3) remains the same
 (4) is equal to the atmospheric pressure

Q. 24. A biconvex lens has radii of curvature, 20 cm each. If the refractive index of the material of the lens is 1.5, the power of the lens is :

- (1) + 2 D (2) + 20 D
(3) + 5 D (4) infinity

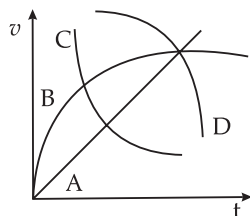
Q. 25. The ratio of the distances travelled by a freely falling body in the 1st, 2nd, 3rd and 4th second:

- (1) 1 : 2 : 3 : 4
(2) 1 : 4 : 9 : 16
(3) 1 : 3 : 5 : 7
(4) 1 : 1 : 1 : 1

Q. 26. The angular speed of a fly wheel moving with uniform angular acceleration changes from 1200 rpm to 3120 rpm in 16 seconds. The angular acceleration in rad/s^2 is :

- (1) 2π (2) 4π
(3) 12π (4) 104π

Q. 27. A spherical ball is dropped in a long column of a highly viscous liquid. The curve in the graph shown, which represents the speed of the ball (v) as a function of time (t) is :



- (1) A (2) B
(3) C (4) D

Q. 28. Let T_1 and T_2 be the energy of an electron in the first and second excited states of hydrogen atom, respectively. According to the Bohr's model of an atom, the ratio $T_1 : T_2$ is :

- (1) 1 : 4 (2) 4 : 1
(3) 4 : 9 (4) 9 : 4

Q. 29. A square loop of side 1 m and resistance 1 Ω is placed in a magnetic field of 0.5 T. If the plane of loop is perpendicular to the direction of magnetic field, the magnetic flux through the loop is :

- (1) 2 weber (2) 0.5 weber
(3) 1 weber (4) zero weber

Q. 30. Two resistors of resistance, 100 Ω and 200 Ω are connected in parallel in an electrical circuit. The ratio of the thermal energy developed in 100 Ω to that in 200 Ω in a given time is :

- (1) 1 : 2 (2) 2 : 1
(3) 1 : 4 (4) 4 : 1

Q. 31. Given below are two statements :

Statement I :

Biot-Savart's law gives us the expression for the magnetic field strength of an infinitesimal current element (Idl) of a current carrying conductor only.

Statement II :

Biot-Savart's law is analogous to Coulomb's inverse square law of charge q , with the former being related to the field produced by a scalar source. Idl while the latter being produced by a vector source, q .

In light of above statements choose the most appropriate answer from the options given below :

- (1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Statement I is correct and Statement II is incorrect
(4) Statement I is incorrect and Statement II is correct

Q. 32. Match List-I with List-II :

List-I (Electromagnetic Waves)	List-II (Wavelength)
(1) AM radio waves	(i) 10^{-10} m
(2) Microwaves	(ii) 10^2 m
(3) Infrared radiations	(iv) 10^{-2} m
(4) X-rays	(iv) 10^{-4} m
(1) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)	
(2) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)	
(3) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)	
(4) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)	

Q. 33. A light ray falls on a glass surface of refractive index $\sqrt{3}$, at an angle 60° . The angle between the refracted and reflected rays would be :

- (1) 30° (2) 60°
(3) 90° (4) 120°

Q. 34. When light propagates through a material medium of relative permittivity ϵ_r and relative permeability μ_r , the velocity of light, v is given : (c -velocity of light in vacuum) :

- (1) $v = c$ (2) $v = \sqrt{\frac{\mu_r}{\epsilon_r}}$
(3) $v = \sqrt{\frac{\epsilon_r}{\mu_r}}$ (4) $v = \frac{c}{\sqrt{\epsilon_r \mu_r}}$

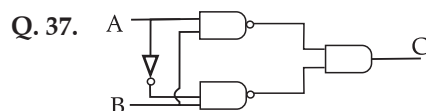
Q. 35. A body of mass 60 g experiences a gravitational force of 3.0 N, when placed at a particular point. The magnitude of the gravitational field intensity at that point is :

- (1) 0.05 N/kg (2) 50 N/kg
(3) 20 N/kg (4) 180 N/kg

Section—B

Q. 36. Two pendulums of length 121 cm and 100 cm start vibrating in phase. At some instant, the two are at their mean position in the same phase. The minimum number of vibrations of the shorter pendulum after which the two are again in phase at the mean position is :

- (1) 11 (2) 9
(3) 10 (4) 8



The truth table for the given logic circuit is :

- | (1) | A | B | C |
|-----|---|---|---|
| | 0 | 0 | 0 |
| | 0 | 1 | 1 |
| | 1 | 0 | 1 |
| | 1 | 1 | 0 |
- | (2) | A | B | C |
|-----|---|---|---|
| | 0 | 0 | 1 |
| | 0 | 1 | 0 |
| | 1 | 0 | 0 |
| | 1 | 1 | 1 |

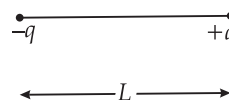
- | (3) | A | B | C |
|-----|---|---|---|
| | 0 | 0 | 1 |
| | 0 | 1 | 0 |
| | 1 | 0 | 1 |
| | 1 | 1 | 0 |

- | (4) | A | B | C |
|-----|---|---|---|
| | 0 | 0 | 0 |
| | 0 | 1 | 1 |
| | 1 | 0 | 0 |
| | 1 | 1 | 1 |

Q. 38. A series LCR circuit with inductance 10 H, capacitance $10 \mu\text{F}$, resistance 50Ω is connected to an ac source of voltage, $V = 200 \sin(100t)$ volt. If the resonant frequency of the LCR circuit is ν_o and the frequency of the ac source is ν , then:

- (1) $\nu_o = \nu = 50 \text{ Hz}$
(2) $\nu_o = \nu = \frac{50}{\pi} \text{ Hz}$
(3) $\nu_o = \frac{50}{\pi} \text{ Hz}, \nu = 50 \text{ Hz}$
(4) $\nu = 100 \text{ Hz}, \nu_o = \frac{100}{\pi} \text{ Hz}$

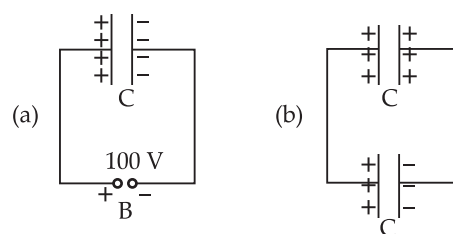
Q. 39. Two point charges $-q$ and $+q$ are placed at a distance of L , as shown in the figure :



The magnitude of electric field intensity at a distance R ($R \gg L$) varies as :

- (1) $\frac{1}{R^2}$ (2) $\frac{1}{R^3}$
(3) $\frac{1}{R^4}$ (4) $\frac{1}{R^6}$

Q. 40. A capacitor of capacitance $C = 900 \text{ pF}$ is charge fully by 100 V battery B as shown in figure (a). Then it is disconnected from the battery and connected to another uncharged capacitor of capacitance $C = 900 \text{ pF}$ as shown in figure (b). The electrostatic energy stored by the system (b) is:



- (1) $4.5 \times 10^{-6} \text{ J}$ (2) $3.25 \times 10^{-6} \text{ J}$
 (3) $2.25 \times 10^{-6} \text{ J}$ (4) $1.5 \times 10^{-6} \text{ J}$
- Q. 41.** Given below are two statements : One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The stretching of a spring is determined by the shear modulus of the material of the spring.

Reason (R) : A coil spring of copper has more tensile strength than a steel spring of same dimensions.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 (3) (A) is true but (R) is false
 (4) (A) is false but (R) is true
- Q. 42.** The transparent media A and B are separated by a plane boundary. The speed of light in those media are $1.5 \times 10^8 \text{ m/s}$ and $2.0 \times 10^8 \text{ m/s}$, respectively. The critical angle for a ray of light for these two media is :

- (1) $\sin^{-1}(0.500)$ (2) $\sin^{-1}(0.750)$
 (3) $\tan^{-1}(0.500)$ (4) $\tan^{-1}(0.750)$

- Q. 43.** The volume occupied by the molecules contained in 4.5 kg water at STP, if the intermolecular forces vanish away is :

- (1) $5.6 \times 10^6 \text{ m}^3$ (2) $5.6 \times 10^3 \text{ m}^3$
 (3) $5.6 \times 10^{-3} \text{ m}^3$ (4) 5.6 m^3

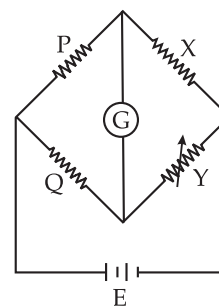
- Q. 44.** A nucleus of mass number of 189 splits into two nuclei having mass number 125 and 64. The ratio of radius of two daughter nuclei respectively is :

- (1) 1 : 1 (2) 4 : 5
 (3) 5 : 4 (4) 25 : 16

- Q. 45.** The area of a rectangular field (in m^2) of length 55.3 m and breadth 25 m after rounding off the value for correct significant digits is :

- (1) 138×10^1 (2) 1382
 (3) 1382.5 (4) 14×10^2

- Q. 46.** A wheatstone bridge is used to determine the value of unknown resistance X by adjusting the variable resistance Y as shown in the figure. For the most precise measurement of X, the resistances P and Q :



- (1) should be approximately equal to $2X$
 (2) should be approximately equal and are small
 (3) should be very large and unequal
 (4) do not play any significant role
- Q. 47.** A ball is projected with a velocity, 10 ms^{-1} , at an angle of 60° with the vertical direction. Its speed at the highest point of its trajectory will be :

- (1) Zero (2) $5\sqrt{3} \text{ ms}^{-1}$
 (3) 5 ms^{-1} (4) 10 ms^{-1}

- Q. 48.** Match List-I with List-II :

List-I	List-II
(1) Gravitation constant (G)	(i) $[L^2T^{-2}]$
(2) Gravitational potential energy	(ii) $[M^{-1}L^3T^{-2}]$
(3) Gravitational potential	(iii) $[LT^{-2}]$
(4) Gravitational intensity	(iv) $[ML^2T^{-2}]$

Choose the correct answer from the options given below :

- (1) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
- (2) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
- (3) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
- (4) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

Q. 49. From Ampere's circuital law for a long straight wire of circular cross-section carrying a steady current, the variation of magnetic field in the inside and outside region of the wire is :

- (1) uniform and remains constant for both the regions
- (2) a linearly increasing function of distance upto the boundary of the wire and then linearly decreasing for the outside region

- (3) a linearly increasing function of distance r upto the boundary of the wire and then decreasing one with $1/r$ dependence for the outside region.
- (4) a linearly decreasing function of distance upto the boundary of the wire and then a linearly increasing one for the outside region

Q. 50. A big circular coil of 1000 turns and average radius 10 m is rotating about its horizontal diameter at 2 rad s^{-1} . If the vertical component of earth's magnetic field at that place is $2 \times 10^{-5} \text{ T}$ and electrical resistance of the coil is 12.56Ω , then the maximum induced current in the coil will be :

- | | |
|------------|-----------|
| (1) 0.25 A | (2) 1.5 A |
| (3) 1 A | (4) 2 A |

□□□

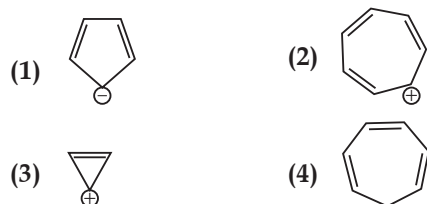
CHEMISTRY

Section A

Q. 51. Choose the correct statement :

- (1) Diamond and graphite have two dimensional network.
- (2) Diamond is covalent and graphite is ionic.
- (3) Diamond is sp^3 hybridized and graphite is sp^2 hybridized.
- (4) Both diamond and graphite are used as dry lubricants.

Q. 52. Which compound amongst the following is not an aromatic compound?



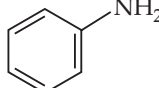
Q. 53. In one molal solution that contains 0.5 mole of a solute, there is

- (1) 500 mL of solvent
- (2) 500 g of solvent
- (3) 100 mL of solvent
- (4) 1000 g of solvent

Q. 54. The IUPAC name of an element with atomic number 119 is :

- (1) ununennium
- (2) unnilennium
- (3) unununnium
- (4) ununoctium

Q. 55. Which of the following is suitable to synthesize chlorobenzene ?

- (1) Benzene, Cl_2 , anhydrous FeCl_3
- (2) Phenol, NaNO_2 , HCl , CuCl
- (3) , HCl
- (4) , HCl , Heating

Q. 56. Given below are two statements :

Statements I :

Primary aliphatic amines react with HNO_2 to give unstable diazonium salts.

Statements II :

Primary aromatic amines react with HNO_2 to form diazonium salts which are stable even above 300 K.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Both **Statement I** and **Statement II** are correct
 - (2) Both **Statement I** and **Statement II** are incorrect
 - (3) **Statement I** is correct but **Statement II** is incorrect
 - (4) **Statement I** is incorrect but **Statement II** is correct
- Q. 57.** Gadolinium has a low value of third ionization enthalpy because of
- (1) small size
 - (2) high exchange enthalpy
 - (3) high electronegativity
 - (4) high basic character.
- Q. 58.** Identify the **incorrect** statement from the following :
- (1) All the five $5d$ orbitals are different in size when compared to the respective $4d$ orbitals.
 - (2) All the five $4d$ orbitals have shapes similar to the respective $3d$ orbitals.
 - (3) In an atom, all the five $3d$ orbitals are equal in energy in free state.
 - (4) The shapes of d_{yx} , d_{yz} , and d_{zx} orbitals are similar to each other; and $d_{x^2-y^2}$ and d_{z^2} are similar to each other.
- Q. 59.** Given below are two statements :
- Statements I :**
The acidic strength of monosubstituted nitro-phenol is higher than phenol because of electron withdrawing nitro group.
- Statement II :**
 o -nitrophenol, m -nitrophenol and p -nitrophenol will have same acidic strength as they have one nitro group attached to the phenolic ring.
- In the light of the above statements, choose the **most appropriate** answer from the options given below :
- (1) Both **Statement I** and **Statement II** are correct.
 - (2) Both **Statement I** and **Statement II** are incorrect.
 - (3) **Statement I** is correct but **Statement II** is incorrect.
 - (4) **Statement I** is incorrect but **Statement II** is correct.
- Q. 60.** At 298 K, the standard electrode potentials of Cu^{2+}/Cu , Zn^{2+}/Zn , Fe^{2+}/Fe and Ag^+/Ag are 0.34 V, -0.76 V, -0.44 V and

0.80 V, respectively.

On the basis of standard electrode potential, predict which of the following reaction can not occur ?

- (1) $\text{CuSO}_4(\text{aq}) + \text{Zn}(\text{s}) \rightarrow \text{ZnSO}_4(\text{aq}) + \text{Cu}(\text{s})$
 - (2) $\text{CuSO}_4(\text{aq}) + \text{Fe}(\text{s}) \rightarrow \text{FeSO}_4(\text{aq}) + \text{Cu}(\text{s})$
 - (3) $\text{FeSO}_4(\text{aq}) + \text{Zn}(\text{s}) \rightarrow \text{ZnSO}_4(\text{aq}) + \text{Fe}(\text{s})$
 - (4) $2\text{CuSO}_4(\text{aq}) + 2\text{Ag}(\text{s}) \rightarrow 2\text{Cu}(\text{s}) + \text{Ag}_2\text{SO}_4(\text{aq})$
- Q. 61.** Given below are two statements :
- Statement I :**
In the coagulation of a negative sol, the flocculating power of the three given ions is in the order :
 $\text{Al}^{3+} > \text{Ba}^{2+} > \text{Na}^+$
- Statement II :**
In the coagulation of a positive sol, the flocculating power of the three given salts is in the order :
 $\text{NaCl} > \text{Na}_2\text{SO}_4 > \text{Na}_3\text{PO}_4$
- In the light of the above statements, choose the **most appropriate** answer from the options given below :
- (1) Both **Statement I** and **Statement II** are correct
 - (2) Both **Statement I** and **Statement II** are incorrect
 - (3) **Statement I** is correct but **Statement II** is incorrect
 - (4) **Statement I** is incorrect but **Statement II** is correct.
- Q. 62.** Which statement regarding polymers is not correct ?
- (1) Elastomers have polymer chains held together by weak intermolecular forces.
 - (2) Fibers possess high tensile strength.
 - (3) Thermoplastic polymers are capable of repeatedly softening and hardening on heating and cooling respectively.
 - (4) Thermosetting polymers are reusable.

- Q. 63.** Match List-I with List-II :

List-I (Products formed)	List-II (Reaction of carbonyl compound with)
(a) Cyanohydrin	(i) NH_2OH
(b) Acetal	(ii) RNH_2
(c) Schiff's base	(iii) alcohol
(d) Oxime	(iv) HCN

Choose the correct answer from the options given below :

- (1) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
- (2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
- (3) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
- (4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

Q. 64. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) :

In a particular point defect, an ionic solid is electrically neutral, even if few of its cations are missing from its units cells.

Reason (R) :

In an ionic solid, Frenkel defect arises due to dislocation of cation from its lattice site to interstitial site, maintaining overall electrical neutrality.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct
- (4) (A) is not correct but (R) is correct

Q. 65. Which one is **not** correct mathematical equation for Dalton's Law of partial pressure? Here p = total pressure of gaseous mixture:

- (1) $p = p_1 + p_2 + p_3$
- (2) $p = n_1 \frac{RT}{V} + n_2 \frac{RT}{V} + n_3 \frac{RT}{V}$
- (3) $p_i = x_i p_i$ where p_i = partial pressure of i^{th} gas
 x_i = mole fraction of i^{th} gas in gaseous mixture
- (4) $p_i = x_i p_i^0$, where x_i = mole fraction of i^{th} gas in gaseous mixture
 p_i^0 = pressure of i^{th} gas in pure state

Q. 66. Given below are two statements :

Statements I :

The boiling points of aldehydes and ketones are higher than hydrocarbons of comparable molecular masses because of weak molecular association in aldehydes and ketones due to

dipole-dipole interactions.

Statements II :

The boiling points of aldehydes and ketones are lower than the alcohols of similar molecular masses due to the absence of H-bonding.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Both **Statement I** and **Statement II** are correct
- (2) Both **Statement I** and **Statement II** are incorrect
- (3) **Statement I** is correct but **Statement II** is incorrect
- (4) **Statement I** is incorrect but **Statement II** is correct

Q. 67. $\text{RMgX} + \text{CO}_2 \xrightarrow[\text{ether}]{\text{dry}} \text{Y} \xrightarrow{\text{H}_3\text{O}^+} \text{RCOOH}$

What is Y in the above reaction ?

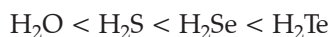
- (1) $\text{RCOO}^-\text{Mg}^+ \text{X}$
- (2) $\text{R}_3\text{CO}^-\text{Mg}^+ \text{X}$
- (3) RCOO^-X^+
- (4) $(\text{RCOO})_2\text{Mg}$

Q. 68. The incorrect statement regarding enzymes is :

- (1) Enzymes are bio-catalysts.
- (2) Like chemical catalysts enzymes reduce the activation energy of bio-processes.
- (3) Enzymes are polysaccharides.
- (4) Enzymes are very specific for a particular reaction and substrate.

Q. 69. Given below are two statements :

Statements I : The boiling points of the following hydrides of group 16 elements increases in the order:



Statements II : The boiling points of these hydrides increase with increase in molar mass.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Both **Statement I** and **Statement II** are incorrect
- (2) Both **Statement I** and **Statement II** are incorrect
- (3) **Statement I** is correct but **Statement II** is incorrect
- (4) **Statement I** is incorrect but **Statement II** is correct

Q. 70. Match **List-I** with **List-II**

List-I (Hydrides)	List-II (Nature)
(a) MgH_2	(i) Electron precise
(b) GeH_4	(ii) Electron deficient
(c) B_2H_6	(iii) Electron rich
(d) HF	(iv) Ionic

Choose the **correct answer** from the options below :

- (1) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
 (2) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
 (3) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)
 (4) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

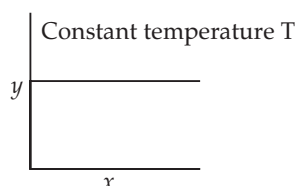
Q. 71. Match List-I with List-II :

List-I (Drug class)	List-II (Drug molecule)
(a) Antacids	(i) Salvarsan
(b) Antihistamines	(ii) Morphine
(c) Analgesics	(iii) Cimetidine
(d) Antimicrobials	(iv) Seldane

Choose the **correct answer** from the options given below :

- (1) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)
 (2) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
 (3) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)
 (4) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

Q. 72. The given graph is representation of kinetics of a reaction :



The y and x axes for zero and first order reactions, respectively are

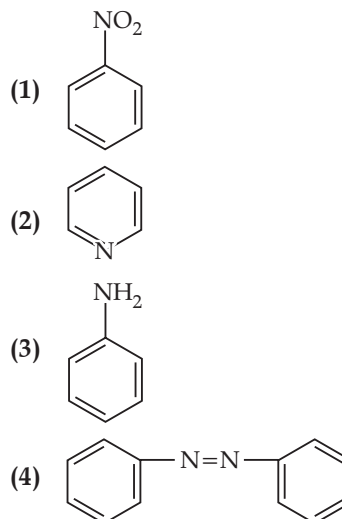
- (1) zero order (y = concentration and x = time), first order ($y = t_{1/2}$ and x = concentration)
 (2) zero order (y = concentration and x = time), first order (y = rate constant and x = concentration)
 (3) zero order (y = rate and x = concentration), first order ($y = t_{1/2}$ and x = concentration)
 (4) zero order (y = rate and concentration), first order (y = rate and $x = t_{1/2}$)

Q. 73. The pH of the solution containing 50 mL each of 0.10 M sodium acetate and 0.01 M acetic acid is

[Given pK_a of $\text{CH}_3\text{COOH} = 4.57$]

- (1) 5.57 (2) 3.57
 (3) 4.57 (4) 2.57

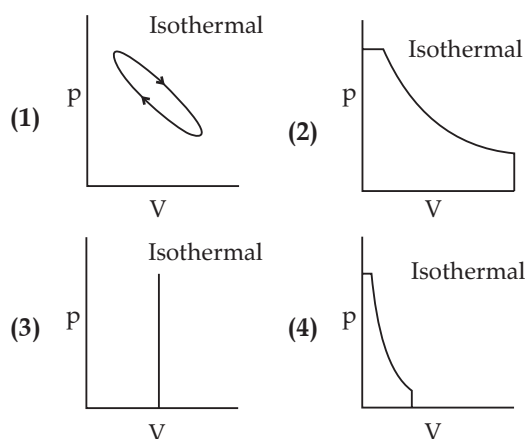
Q. 74. The Kjeldahl's method for the estimation of nitrogen can be used to estimate the amount of nitrogen in which one of the following compounds ?



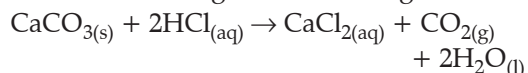
Q. 75. Amongst the following which one will have maximum 'lone pair-lone pair' electron repulsions ?

- (1) ClF_3 (2) IF_5
 (3) SF_4 (4) XeF_2

Q. 76. Which of the following p-V curve represents maximum work done ?



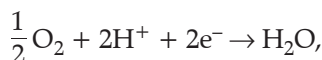
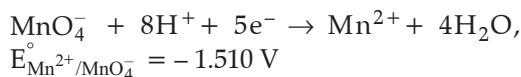
Q. 77. What mass of 95% pure CaCO_3 will be required to neutralise 50 mL of 0.5 M HCl solution according to the following reaction?



[Calculate upto second place of decimal point]

- (1) 1.25 g (2) 1.32 g
 (3) 3.65 g (4) 9.50 g

Q. 78. Given below are half cell reactions :



$$E_{\text{O}_2/\text{H}_2\text{O}}^\circ = +1.223 \text{ V}$$

Will the permanganate ion, MnO_4^- liberate O_2 from water in the presence of an acid ?

- (1) Yes, because $E_{\text{cell}}^\circ = +0.287 \text{ V}$
- (2) No, because $E_{\text{cell}}^\circ = -0.287 \text{ V}$
- (3) Yes, because $E_{\text{cell}}^\circ = +2.733 \text{ V}$
- (4) No, because $E_{\text{cell}}^\circ = -2.733 \text{ V}$

Q. 79. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) I-Cl is more reactive than I_2 .

Reason (R) : I-Cl bond is weaker than I-I bond.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both (A) and (R) are correct and (R) is not the correct explanation of (A).
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
- (3) (A) is correct but (R) is not correct.
- (4) (A) is not correct but (R) is correct.

Q. 80. Which amongst the following is incorrect statement ?

- (1) The bond orders of O_2^+ , O_2 , O_2^- and O_2^{2-} are 2.5, 2, 1.5 and 1, respectively.
- (2) C_2 molecules has four electrons in its two degenerate π molecular orbitals.
- (3) H_2^+ ion has one electron.
- (4) O_2^+ ion is diamagnetic.

Q. 81. The **incorrect** statement regarding chirality is :

- (1) $\text{S}_{\text{N}}1$ reaction yields 1 : 1 mixture of both enantiomers.
- (2) The product obtained by $\text{S}_{\text{N}}2$ reaction of haloalkane having chirality at the reactive site shows inversion of configuration.
- (3) Enantiomers are superimposable mirror images on each other.
- (4) A racemic mixture shows zero optical rotation.

Q. 82. Identify the **incorrect** statement from the following :

- (1) Alkali metals react with water to form their hydroxides.

- (2) The oxidation number of K in KO_2 is +4.
- (3) Ionisation enthalpy of alkali metals decreases from top to bottom in the group.

- (4) Lithium is the strongest reducing agent among the alkali metals.

Q. 83. Which of the following statement is not correct about diborane ?

- (1) There are two 3-centre-2-electron bonds.
- (2) The four terminal B-H bonds are two centre two electron bonds.
- (3) The four terminal Hydrogen atoms and the two Boron atoms lie in one plane.
- (4) Both the Boron atoms are sp^2 hybridised.

Q. 84. Match **List-I** with **List II**

List-I

List-II

- | | |
|---------|--|
| (a) Li | (i) absorbent for carbon dioxide |
| (b) Na | (ii) electrochemical cells |
| (c) KOH | (iii) coolant in fast breeder reactors |
| (d) Cs | (iv) photoelectric cell |

Choose the **correct answer** from the options given below :

- (1) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)
- (2) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
- (3) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)
- (4) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

Q. 85. The IUPAC name of the complex :



- (1) dicyanidosilver (II), diaquaargentate (II)
- (2) diaquasilver(II) dicyanidoargentate(II)
- (3) dicyanidosilver(I) diaquaargentate (I)
- (4) diaquasilver(I) discyanidoargentate (I)

Section B

Q. 86. $3\text{O}_2(\text{g}) \rightleftharpoons 2\text{O}_3(\text{g})$

for the above reaction at 298 K, K_c is found to be 3.0×10^{-59} . If the concentration of O_2 at equilibrium is 0.040 M then concentration of O_2 at equilibrium is 0.040 M then concentration of O_3 in M is :

- (1) 4.38×10^{-32}
- (2) 1.9×10^{-63}
- (3) 2.4×10^{31}
- (4) 1.2×10^{21}

Q. 87. Match **List-I** with **List-II**

List-I

List-II

(Ores)

(Composition)

- | | |
|---------------|--|
| (a) Haematite | (i) Fe_3O_4 |
| (b) Magnetite | (ii) ZnCO_3 |
| (c) Calamine | (iii) Fe_2O_3 |
| (d) Kaolinite | (iv) $[\text{Al}_2(\text{OH})_4\text{Si}_2\text{O}_5]$ |

Choose the **correct answer** from the option given below :

- (1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
 (2) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
 (3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
 (4) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

Q. 88. Given below are two statement :

Statement I :

In Lucas test, primary, secondary and tertiary alcohols are distinguished on the basis of their reactivity with conc. HCl + ZnCl₂, known as Lucas Reagent.

Statement II :

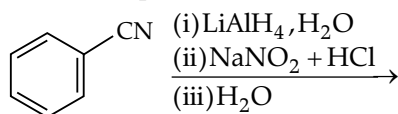
Primary alcohols are most reactive and immediately produce turbidity at room temperature on reaction with Lucas Reagent. In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Both **Statement I** and **Statement II** are correct.
 (2) Both **Statement I** and **Statement II** are incorrect
 (3) **Statement I** is correct but **Statement II** is incorrect
 (4) **Statement I** is incorrect but **Statement II** is correct.

Q. 89. If radius of second Bohr orbit of the He⁺ ion is 105.8 pm, what is the radius of third Bohr orbit of Li²⁺ ion ?

- (1) 158.7 pm (2) 15.87 pm
 (3) 1.587 pm (4) 158.7 Å

Q. 90. The product formed from the following reaction sequence is



- (1)
- (2)
- (3)
- (4)

Q. 91. Which one of the following is **not** formed when acetone with 2-pentanone in the presence of dilute NaOH followed by heating?

- (1)
- (2)
- (3)
- (4)

Q. 92. Compound X on reaction with O₃ followed by Zn/H₂O formaldehyde and 2-methyl propanal as products. The compound X is :

- (1) 3-Methylbut-1-ene
 (2) 2-Methylbut-1-ene
 (3) 3-Methylbut-2-ene
 (4) Pent-2-ene

Q. 93. In the neutral of faintly alkaline medium, KMnO₄ oxidizes iodide into iodate. The change in oxidation state of manganese in this reaction is from

- (1) + 7 to + 4
 (2) + 6 to + 4
 (3) + 7 to + 3
 (4) + 6 to + 5

Q. 94. A 10.0 L flask contains 64 g of oxygen at 27°C. (Assume O₂ gas is behaving ideally). The pressure inside the flask in bar is

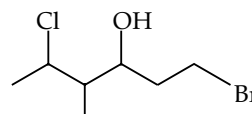
(Given R = 0.0831 L bar K⁻¹ mol⁻¹)

- (1) 2.5 (2) 498.6
 (3) 49.8 (4) 4.9

Q. 95. For a first order reaction A → Products, initial concentration of A is 0.1 M, which becomes 0.001 M after 5 minutes. Rate constant for the reaction in min⁻¹ is :

- (1) 1.3818 (2) 0.9212
 (3) 0.4606 (4) 0.2303

Q. 96. The correct IUPAC name of the following compound is :



- (1) 1-bromo-5-chloro-4-methylhexan-3-ol
 (2) 6-bromo-2-chloro-4-methylhexan-4-ol
 (3) 1-bromo-4-methyl-5-chlorohexan-3-ol
 (4) 6-bromo-4-methyl-2-chlorohexan-4-ol
- Q. 97.** Find the emf of the cell in which the following reacting takes place at 298 K
- $$\text{Ni(s)} + 2\text{Ag}^+ (0.001 \text{ M}) \rightarrow \text{Ni}^{2+} (0.001) + 2 \text{Ag(s)}$$
- (Given that $E^\circ_{\text{cell}} = 10.5 \text{ V}$, $\frac{2.303RT}{F} = 0.059$ at 298 K)
- (1) 1.0385 V
 (2) 1.385 V
 (3) 0.9615 V
 (4) 1.05 V
- Q. 98.** The order of energy absorbed which is responsible for the color of complexes
- (A) $[\text{Ni}(\text{H}_2\text{O})_2(\text{en})_2]^{2+}$
 (B) $[\text{Ni}(\text{H}_2\text{O})_4(\text{en})]^{2+}$ and
- (C) $[\text{Ni}(\text{en})_2]^{2+}$
 (1) (A) > (B) > (C)
 (2) (C) > (B) > (A)
 (3) (C) > (A) > (B)
 (4) (B) > (A) > (C)
- Q. 99.** Copper crystallizes in *fcc* unit cell with cell edge length of $3.608 \times 10^{-8} \text{ cm}$. The density of copper is 8.92 g cm^{-3} . Calculate the atomic mass of copper :
- (1) 63.1 u (2) 31.55 u
 (3) 60 u (4) 65 u
- Q. 100.** The pollution due to oxides of sulphur gets enhanced due to the presence of :
- (a) particulate matter
 (b) ozone
 (c) hydrocarbons
 (d) hydrogen peroxide
- (1) (a), (d) only (2) (a), (b), (d) only
 (3) (b), (c), (d) only (4) (a), (c), (d) only

□□□

BOTANY

Section A

- Q. 101.** Which one of the following statements cannot be connected to Predation ?
- (1) It helps in maintaining species diversity in a community
 (2) It might lead to extinction of a species.
 (3) Both the interacting species are negatively impacted.
 (4) It is necessitated by nature to maintain the ecological balance.
- Q. 102.** Given below are two statements :
- Statement I :** Decomposition is a process in which the detritus is degraded into simpler substances by microbes.
Statement II : Decomposition is faster if the detritus is rich in lignin and chitin
- In the light of the above statements, choose the correct answer from the options given below :
- (1) Both **Statement I** and **Statement II** are correct.
 (2) Both **Statement I** and **Statement II** are incorrect.
 (3) **Statement I** is correct but **Statement II** is incorrect.
 (4) **Statement I** is incorrect but **Statement II** is correct.
- Q. 103.** Read the following statements about the vascular bundles :
- (a) In rots, xylem and phloem in a vascular bundle are arranged in an alternate manner along the different radii.
 (b) Conjoint closed vascular bundles do not possess cambium.
 (c) In open vascular bundles, cambium is present in between xylem and phloem.
 (d) The vascular bundles of dicotyledonous stem possess endarch protoxylem.
 (e) In monocotyledonous root, usually there are more than six xylem bundles present.
- Choose the **correct answer** from the options given below :
- (1) (a), (b) and (d) Only
 (2) (b), (c), (d) and (e) Only
 (3) (a), (b), (c) and (d) Only
 (4) (a), (c), (d) and (e) Only
- Q. 104.** Which of the following is not observed during apoplastic pathway
- (1) Movement of water occurs through

- (4) Natality
- Q. 115. Which of the following is not true regarding the release of energy during ATP synthesis through chemiosmosis ? It involves :
- (1) Breakdown of proton gradient
 - (2) Breakdown of electron gradient
 - (3) Movement of protons across the membrane to the stroma
 - (4) Reduction of NADP to NADPH₂ on the stroma side of the membrane
- Q. 116. Which one of the following never occurs during mitotic cell division ?
- (1) Spindle fibres attach to kinetochores of chromosomes
 - (2) Movement of centrioles towards opposite poles
 - (3) Pairing of homologous chromosomes
 - (4) Coiling and condensation of the chromatids
- Q. 117. The appearance of recombination nodules on homologous chromosomes during meiosis characterizes :
- (1) Synaptonemal complex
 - (2) Bivalent
 - (3) Sites at which crossing over occurs
 - (4) Terminalization
- Q. 118. XO type of sex determination can be found in :
- (1) *Drosophila*
 - (2) Birds
 - (3) Grasshoppers
 - (4) Monkeys
- Q. 119. Production of Cucumber has increased manifold in recent years. Application of which of the following phytohormones has resulted in this increased yield as the hormone is known to produce female flowers in the plants :
- (1) ABA
 - (2) Gibberellin
 - (3) Ethylene
 - (4) Cytokinin
- Q. 120. Given below are two statements :
- Statement I :** Mendel studied seven pairs of contrasting traits in pea plants and proposed the Laws of Inheritance
- Statement II :** Seven characters examined by Mendel in his experiment on pea plants were seed shape and colour, flower colour, pod shape and colour, flower position and stem height
- In the light of the above statements, choose the **correct answer** from the options given below :
- (1) Both **Statement I** and **Statement II** are correct
 - (2) Both **Statement I** and **Statement II** are incorrect
 - (3) **Statement I** is correct but **Statement II** is incorrect
 - (4) **Statement I** is incorrect but **Statement II** is correct
- Q. 121. What amount of energy is released from glucose during lactic acid fermentation ?
- (1) Approximately 15%
 - (2) More than 18%
 - (3) About 10%
 - (4) Less than 7%
- Q. 122. Identify the **correct set** of statements :
- (a) The leaflets are modified into pointed hard thorns in *Citrus* and *Bougainvillea*
 - (b) Axillary buds form slender and spirally coiled tendrils in cucumber and pumpkin
 - (c) Stem is flattened and fleshy in *Opuntia* and modified to perform the function of leaves
 - (d) *Rhizophora* shows vertically upward growing roots that help to get oxygen for respiration
 - (e) Subaerially growing stems in grasses and strawberry help in vegetative propagation
- Choose the **correct answer** from the options given below :
- (1) (b) and (c) Only
 - (2) (a) and (d) Only
 - (3) (b), (c) (d) and (e) Only
 - (4) (a), (b), (d) and (e) Only
- Q. 123. Hydrocolloid carrageen is obtained from :
- (1) Chlorophyceae and Phaeophyceae
 - (2) Phaeophyceae and Rhodophyceae
 - (3) Rhodophyceae only
 - (4) Phaeophyceae only
- Q. 124. The device which can remove particulate matter present in the exhaust from a thermal power plant is :
- (1) STP
 - (2) Incinerator
 - (3) Electrostatic Precipitator
 - (4) Catalytic Converter
- Q. 125. Given below are two statements :
- Statement I :** Cleistogamous flowers are invariably autogamous
- Statement II :** Cleistogamy is disadvantageous as there is

- no chance for cross pollination
In the light of the above statements, choose the **correct** answer from the options given below:
- (1) Both **Statement I** and **Statement II** are correct
 - (2) Both **Statement I** and **Statement II** are incorrect
 - (3) **Statement I** is correct but **Statement II** is incorrect
 - (4) **Statement I** is incorrect but **Statement II** is correct
- Q. 126.** "Girdling Experiment" was performed by Plant Physiologists to identify the plant tissue through which :
- (1) water is transported
 - (2) food is transported
 - (3) for both water and food transportation
 - (4) osmosis is observed
- Q. 127.** Which one of the following plants show vexillary aestivation and diadelphous stamens ?
- (1) *Colchicum autumnale*
 - (2) *Pisum sativum*
 - (3) *Allium cepa*
 - (4) *Solanum nigrum*
- Q. 128.** Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Polymerase chain reaction is used in DNA amplification
Reason (R) : The ampicillin resistant gene is used as a selectable marker to check transformation.
In the light of the above statements, choose the **correct answer** from the options given below :
- (1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
 - (2) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
 - (3) **(A)** is correct but **(R)** is not correct
 - (4) **(A)** is not correct but **(R)** is correct
- Q. 129.** The gaseous plant growth regulator is used in plants to :
- (1) speed up the malting process
 - (2) promote root growth and root hair formation to increase the absorption surface
 - (3) help overcome apical dominance
 - (4) kill dicotyledonous weeds in the fields
- Q. 130.** DNA polymorphism forms the basis of :
- (1) Genetic mapping
 - (2) DNA fingerprinting
 - (3) Both genetic mapping and DNA fingerprinting
 - (4) Translation
- Q. 131.** Given below are two statements :
- Statement I :** The primary CO_2 acceptor in C_4 plants is phosphoenolpyruvate and is found in the mesophyll cells
- Statement II :** Mesophyll cells of C_4 plants lack RuBisCo enzyme. In the light of the above statements, choose the **correct** answer from the options given below :
- (1) Both **Statement I** and **Statement II** are correct
 - (2) Both **Statement I** and **Statement II** are incorrect
 - (3) **Statement I** is correct but **Statement II** is incorrect
 - (4) **Statement I** is incorrect but **Statement II** is correct
- Q. 132.** Which one of the following produces nitrogen fixing nodules on the roots of *Alnus* ?
- (1) *Rhizobium*
 - (2) *Frankia*
 - (3) *Rhodospirillum*
 - (4) *Beijernickia*
- Q. 133.** Which of the following is not a method of *ex-situ* conservation ?
- (1) *In vitro* fertilization
 - (2) National Parks
 - (3) Micropropagation
 - (4) Cryopreservation
- Q. 134.** The process of translation of mRNA to proteins begins as soon as :
- (1) The small subunit of ribosome encounters mRNA
 - (2) The larger subunit of ribosome encounters mRNA
 - (3) Both the subunits join together to bind with mRNA
 - (4) The tRNA is activated and the larger subunit of ribosome encounters mRNA
- Q. 135.** Match **List-I** with **List-II**.
- | | List-I | | List-II |
|-----|---------------|-------|----------------------------------|
| (a) | Manganese | (i) | Activates the enzyme catalase |
| (b) | Magnesium | (ii) | Required for pollen germination |
| (c) | Boron | (iii) | Activates enzymes of respiration |

(d)	Iron	(iv)	Functions in splitting of water during photosynthesis
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Choose the **correct answer** from the options given below.

- (1) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
 (2) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
 (3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
 (4) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

Section B

Q. 136. The entire fleet of buses in Delhi were converted to CNG from diesel. In reference to this, which one of the following statements is **false** ?

- (1) CNG burns more efficiently than diesel
 (2) The same diesel engine is used in CNG buses making the cost of conversion low
 (3) It is cheaper than diesel
 (4) It can not be adulterated like diesel

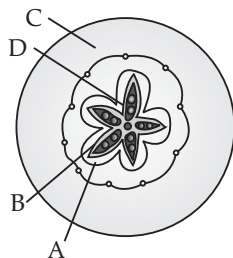
Q. 137. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Mendel's law of Independent assortment does not hold good for the genes that are located closely on the same chromosome.

Reason (R) : Closely located genes assort independently. In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 (3) (A) is correct but (R) is not correct
 (4) (A) is not correct but (R) is correct

Q. 138. Which part of the fruit, labelled in the given figure makes it a false fruit ?



- (1) A :: Mesocarp
 (2) B :: Endocarp

- (3) C :: Thalamus
 (4) D :: Seed

Q. 139. If a geneticist uses the blind approach for sequencing the whole genome of an organism, followed by assignment of function to different segments, the methodology adopted by him is called as :

- (1) Sequence annotation
 (2) Gene mapping
 (3) Expressed sequence tags
 (4) Bioinformatics

Q. 140. Which of the following occurs due to the presence of autosome linked dominant trait?

- (1) Sickle cell anaemia
 (2) Myotonic dystrophy
 (3) Haemophilia
 (4) Thalessemia

Q. 141. While explaining interspecific interaction of population, (+) sign is assigned for beneficial interaction, (–) sign is assigned for detrimental interaction and (0) for neutral interaction. Which of the following interactions can be assigned (+) for one species and (–) for another species involved in the interaction ?

- (1) predation (2) Amensalism
 (3) Commensalism (4) Competition

Q. 142. What is the role of large bundle sheath cells found around the vascular bundles in C_4 plants ?

- (1) To provide the site for photorespiratory pathway
 (2) To increase the number of chloroplast for the operation of Calvin cycle
 (3) To enable the plant to tolerate high temperature
 (4) To protect the vascular tissue from high light intensity

Q. 143. Match the plant with the kind of life cycle it exhibits

	List-I		List-II
(a)	<i>Spirogyra</i>	(i)	Dominant diploid sporophyte vascular plant, with highly reduced male or female gametophyte
(b)	Fern	(ii)	Dominant haploid free-living gametophyte
(c)	<i>Funaria</i>	(iii)	Dominant diploid sporophyte alternating with reduced gametophyte called prothallus

(d)	<i>Cycas</i>	(iv)	Dominant haploid leafy gametophyte alternating with partially dependent multicellular sporophyte
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Choose the **correct answer** from the options given below.

- (1) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
- (2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
- (3) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
- (4) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

Q. 144. Read the following statements on lipids and find out **correct** set on statements :

- (a) Lecithin found in the plasma membrane is a glycolipid
- (b) Saturated fatty acids possess one or more $c = c$ bonds
- (c) Gingly oil has lower melting point, hence remains as oil in winter
- (d) Lipids are generally insoluble in water but soluble in some organic solvents
- (e) when fatty acids is esterified with glycerol, monoglycerides are formed

Choose the **correct answer** from the options given below :

- (1) (a), (b) and (c) only
- (2) (a), (d) and (e) only
- (3) (c), (d) and (e) only
- (4) (a), (b) and (d) only

Q. 145. Transposons can be used during which one of the following ?

- (1) Polymerase Chain Reaction
- (2) Gene silencing
- (3) Autoradiography
- (4) Gene sequencing

Q. 146. Match **List-I** with **List-II**.

	List-I		List-II
(a)	Metacentric chromosome	(i)	Centromere situated close to the end forming one extremely short and one very long arms
(b)	Acrocentric chromosome	(ii)	Centromere at the terminal end
(c)	Sub-metacentric	(iii)	Centromere in the middle forming two equal arms of chromosomes

(d)	Telocentric chromosome	(iv)	Centromere slightly away from the middle forming one shorter arm and one longer arm
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Choose the **correct answer** from the options given below.

- (1) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
- (2) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
- (3) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
- (4) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

Q. 147. In the following palindromic base sequences of DNA, which one can be cut easily by particular restriction enzyme ?

- (1) 5' G A T A C T 3' ; 3' C T A T G A 5'
- (2) 5' G A A T T C 3' ; 3' C T T A A G 5'
- (3) 5' C T C A G T 3' ; 3' G A G T C A 5'
- (4) 5' G T A T T C 3' ; 3' C A T A A G 5'

Q. 148. The anatomy of springwood shows some peculiar features. Identity the **correct** set of statements about springwood :

- (a) It is also called as the earlywood
- (b) In spring season cambium produces xylem elements with narrow vessels
- (c) It is lighter in colour
- (d) The springwood along with autumn wood shows alternate concentric rings forming annual rings
- (e) It has lower density

Choose the correct answer from the options given below :

- (1) (a), (b), (d) and (e) Only
- (2) (a), (c), (d) and (e) Only
- (3) (a), (b) and (d) Only
- (4) (c), (d) and (e) Only

Q. 149. Which one of the following will accelerate phosphorus cycle ?

- (1) Burning of fossil fuels
- (2) Volcanic activity
- (3) Weathering of rocks
- (4) Rain fall and storms

Q. 150. Addition of more solutes in a given will :

- (1) raise its water potential

- (2) lower its water potential
- (3) make its water potential zero

- (4) not affect the water potential at all

ZOOLOGY

Section A

- Q. 151.** Breeding crops with higher levels of vitamins and minerals or higher proteins and healthier fats is called :
- (1) Bio-magnification
 - (2) Bio-remediation
 - (3) Bio-fortification
 - (4) Bio-accumulation
- Q. 152.** Regarding Meiosis, which of the statements is incorrect?
- (1) There are two stages in Meiosis, Meiosis-I and II
 - (2) DNA replication occurs in S phase of Meiosis-II
 - (3) Pairing of homologous chromosomes and recombination occurs in Meiosis-II
 - (4) Four haploid cells are formed at the end of Meiosis-II
- Q. 153.** Given below are two statements :
- Statement I :** Fatty acids and glycerols cannot be absorbed into the blood.
- Statement II :** Specialized lymphatic capillaries called lacteals carry chylomicrons into lymphatic vessels and ultimately into the blood.
- In the light of the above statements, choose the most appropriate answer from the options given below :
- (1) Both **Statement I** and **Statement II** are correct
 - (2) Both **Statement I** and **Statement II** are incorrect
 - (3) **Statement I** is correct but **Statement II** is incorrect
 - (4) **Statement I** is incorrect but **Statement II** is correct
- Q. 154.** At which stage of life the oogenesis process is initiated ?
- (1) Puberty
 - (2) Embryonic development stage
 - (3) Birth
 - (4) Adult
- Q. 155.** Detritivores breakdown detritus into smaller particles. This process is called :
- (1) Catabolism
 - (2) Fragmentation
 - (3) Humification
 - (4) Decomposition
- Q. 156.** If '8' *Drosophila* in a laboratory population of '80' died during a week, the death rate in the population is individuals per *Drosophila* per week.
- (1) 0.1
 - (2) 10
 - (3) 1.0
 - (4) zero
- Q. 157.** In which of the following animals, digestive tract has additional chambers like crop and gizzard ?
- (1) *Corvus*, *Columba*, *Chameleon*
 - (2) *Bufo*, *Balaenoptera*, *Bangarus*
 - (3) *Catla*, *Columba*, *Crocodilus*
 - (4) *Pavo*, *Psittacula*, *Corvus*
- Q. 158.** Which of the following statements with respect to Endoplasmic Reticulum is incorrect ?
- (1) RER has ribosomes attached to ER
 - (2) SER is devoid of ribosomes
 - (3) In prokaryotes only RER are present
 - (4) SER are the sites for lipid synthesis
- Q. 159.** Tegmina in cockroach, arises from :
- (1) Prothorax
 - (2) Mesothorax
 - (3) Metathorax
 - (4) Prothorax and Mesothorax
- Q. 160.** Which of the following is not the function of conducting part of respiratory system ?
- (1) It clears inhaled air from foreign particles
 - (2) Inhaled air is humidified
 - (3) Temperature of inhaled air is brought to body temperature
 - (4) Provides surface for diffusion of O₂ and CO₂
- Q. 161.** Under normal physiological conditions in human being every 100 mL of oxygenated blood can deliver mL of O₂ to the tissues.
- (1) 2 mL
 - (2) 5 mL
 - (3) 4 mL
 - (4) 10 mL
- Q. 162.** Select the **incorrect statement** with reference to mitosis :
- (1) All the chromosomes lie at the equator at metaphase.
 - (2) Spindle fibres attach to centromere of chromosomes.
 - (3) Chromosomes decondense at telophase.
 - (4) Splitting of centromere occurs at anaphase.
- Q. 163.** Which of the following is present between the adjacent bones of the vertebral column?
- (1) Intercalated discs
 - (2) Cartilage
 - (3) Areolar tissue
 - (4) Smooth muscle

Q. 164. If the length of a DNA molecule is 1.1 metres, what will be the approximate number of base pairs ?

- (1) 3.3×10^9 bp (2) 6.6×10^9 bp
(3) 3.3×10^6 bp (4) 6.6×10^6 bp

Q. 165. *In-situ* conservation refers to :

- (1) Protect and conserve the whole ecosystem
(2) Conserve only high risk species
(3) Conserve only endangered species
(4) Conserve only extinct species

Q. 166. Which of the following statements are true for spermatogenesis but do not hold true for Oogenesis?

- (a) It results in the formation of haploid gametes
(b) Differentiation of gamete occurs after the completion of meiosis
(c) Meiosis occurs continuously in a mitotically dividing stem cell population
(d) It is controlled by the Luteinising hormone (LH) and Follicle Stimulating Hormone (FSH) secreted by the anterior pituitary

(e) It is initiated at puberty

Choose the most appropriate answer from the options given below.

- (1) (c) and (e) only
(2) (b) and (c) only
(3) (b), (d) and (e) only
(4) (b), (c) and (e) only

Q. 167. Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) :

All vertebrates are chordates but all chordates are not vertebrates.

Reason (R) :

Notochord is replaced by vertebral column in the adult vertebrates.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(3) (A) is correct but (R) is not correct
(4) (A) is not correct but (R) is correct

Q. 168. Given below are two statements :

Statement I : Mycoplasma can pass through less than 1 micron filter size.

Statement II : Mycoplasma are bacteria with cell wall. In the light of the above statements, choose the most appropriate answer from the

options give below :

- (1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but Statement II is correct

Q. 169. Given below are two statements :

Statement I : Autoimmune disorder is a condition where body defense mechanism recognizes its own cells as foreign bodies.

Statement II : Rheumatoid arthritis is a condition where body does not attack self cells.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Statement I is correct but Statement II is incorrect
(4) Statement I is incorrect but Statement II is correct

Q. 170. In gene therapy of Adenosine Deaminase (ADA) deficiency, the patient requires periodic infusion of genetically engineered lymphocytes because :

- (1) Retroviral vector is introduced into these lymphocytes.
(2) Gene isolated from marrow cells producing ADA is introduced into cells at embryonic stages
(3) Lymphocytes from patient's blood are grown in culture, outside the body.
(4) Genetically engineered lymphocytes are not immortal cells.

Q. 171. Which of the following is a correct match for disease and its symptoms ?

- (1) Arthritis—Inflamed joints
(2) Tetany—high Ca^{2+} level causing rapid spasms.
(3) Myasthenia gravis—Genetic disorder resulting in weakening and paralysis of skeletal muscle.
(4) Muscular dystrophy—An autoimmune disorder causing progressive degeneration of skeletal muscle.

Q. 172. Given below are two statements :

Statement I :

The release of sperms into the seminiferous tubules is called spermiation.

Statements II :

Spermiogenesis is the process of formation of sperms from spermatogonia.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both Statement I and Statement II are correct
- (2) Both Statement I and Statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but Statement II is correct

Q. 173. Nitrogenous waste is excreted in the form of pellet or paste by :

- (1) *Ornithorhynchus*
- (2) *Salamandra*
- (3) *Hippocampus*
- (4) *Pavo*

Q. 174. Identify the microorganism which is responsible for the production of an immunosuppressive molecule cyclosporin A:

- (1) *Trichoderma polysporum*
- (2) *Clostridium butylicum*
- (3) *Aspergillus niger*
- (4) *Streptococcus cerevisiae*

Q. 175. Lippe's loop is a type of contraceptive used as :

- (1) Cervical barrier
- (2) Vault barrier
- (3) Non-medicated IUD
- (4) Copper releasing IUD

Q. 176. A dehydration reaction links two glucose molecules to produce maltose. If the formula for glucose is $C_6H_{12}O_6$ then what is the formula for maltose ?

- (1) $C_{12}H_{20}O_{10}$
- (2) $C_{12}H_{24}O_{12}$
- (3) $C_{12}H_{22}O_{11}$
- (4) $C_{12}H_{24}O_{11}$

Q. 177. Which of the following is not a connective tissue ?

- (1) Blood
- (2) Adipose tissue
- (3) Cartilage
- (4) Neuroglia

Q. 178. Which of the following functions is not performed by secretions from salivary glands?

- (1) Control bacterial population in mouth
- (2) Digestion of complex carbohydrates
- (3) Lubrication of oral cavity
- (4) Digestion of disaccharides

Q. 179. Given below are two statements :

Statement I : Restriction endonucleases

recognise specific sequence to cut DNA known as palindromic nucleotide sequence.

Statements II : Restriction endonucleases cut the DNA strand a little away from the centre of the palindromic site.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both Statement I and Statement II are correct
- (2) Both Statement I and Statement II are incorrect
- (3) Statement I is correct but Statement II is incorrect
- (4) Statement I is incorrect but Statement II is correct

Q. 180. Natural selection where more individuals acquire specific character value other than the mean character value, leads to:

- (1) Stabilising change
- (2) Directional change
- (3) Disruptive change
- (4) Random change

Q. 181. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Osteoporosis is characterised by decreased bone mass and increased chances of fractures.

Reason (R) : Common cause of Osteoporosis is increased levels of estrogen.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct
- (4) (A) is not correct but (R) is correct

Q. 182. In the taxonomic categories which hierarchical arrangement in ascending order is correct in case of animals ?

- (1) Kingdom, Phylum, Class, Order, Family, Genus, Species
- (2) Kingdom, Class, Phylum, Family, Order, Genus, Species
- (3) Kingdom, Order, Class, Phylum, Family, Genus, Species
- (4) Kingdom, Order, Phylum, Class, Family, Genus, Species

Q. 183. In an *E. coli* strain *i* gene gets mutated and its product can not bind the inducer molecule. If growth medium is provided with lactose, what will be the outcome ?

- (1) Only z gene will get transcribed
 (2) z, y, a genes will be transcribed
 (3) z, y, a genes will not be translated
 (4) RNA polymerase will bind the promoter region
- Q. 184.** Identify the asexual reproductive structure associated with *Penicillium* :
 (1) Zoospores (2) Conidia
 (3) Gemmules (4) Buds
- Q. 185.** Given below are two statements :
Statement I :
 The coagulum is formed of network of threads called thrombins.
Statement II :
 Spleen is the graveyard of erythrocytes.
 In the light of the above statements, choose the most appropriate answer from the options given below :
 (1) Both Statement I and Statement II are correct
 (2) Both Statement I and Statement II are incorrect
 (3) Statement I is correct but Statement II is incorrect
 (4) Statement I is incorrect but Statement II is correct
- Section B**
- Q. 186.** Match List-I with List-II.
- | List-I | List-II |
|------------------------|------------------------|
| (Biological Molecules) | (Biological functions) |
| (a) Glycogen | (i) Hormone |
| (b) Globulin | (ii) Biocatalyst |
| (c) Steroids | (iii) Antibody |
| (d) Thrombin | (iv) Storage product |
- Choose the correct answer from the options given below :
 (1) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)
 (2) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
 (3) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
 (4) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
- Q. 187.** Match List-I with List-II.
- | List-I | List-II |
|--------------------|-------------------------------------|
| (a) Bronchioles | (i) Dense Regular Connective Tissue |
| (b) Goblet cell | (ii) Loose Connective Tissue |
| (c) Tendons | (iii) Glandular Tissue |
| (d) Adipose Tissue | (iv) Ciliated Epithelium |
- Choose the correct answer from the options given below :
- (1) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
 (2) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
 (3) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
 (4) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
- Q. 188.** Which one of the following statements is correct ?
 (1) The atrio-ventricular node (AVN) generates an action potential to stimulate atrial contraction
 (2) The tricuspid and the bicuspid valves open due to the pressure exerted by the simultaneous contraction of the atria
 (3) Blood moves freely from atrium to the ventricle during joint diastole.
 (4) Increased ventricular pressure causes closing of the semilunar valves.
- Q. 189.** Which of the following are not the effect of Parathyroid hormone ?
 (a) Stimulates the process of bone resorption
 (b) Decreases Ca^{2+} level in blood
 (c) Reabsorption of Ca^{2+} by renal tubules
 (d) Decreases the absorption of Ca^{2+} from digested food
 (e) Increases metabolism of carbohydrates
 Choose the most appropriate answer from the options given below :
 (1) (a) and (c) only
 (2) (b), (d) and (e) only
 (3) (a) and (e) only
 (4) (b) and (c) only
- Q. 190.** Which of the following is a correct statement?
 (1) Cyanobacteria are a group of autotrophic organisms classified under Kingdom Monera.
 (2) Bacteria are exclusively heterotrophic organisms.
 (3) Slime moulds are saprophytic organisms classified under Kingdom Monera.
 (4) Mycoplasma have DNA, Ribosome and cell wall
- Q. 191.** Select the incorrect statement with respect to acquired immunity.
 (1) Primary response is produced when our body encounters a pathogen for the first time.
 (2) Anamnestic response is elicited on subsequent encounters with the same pathogen.
 (3) Anamnestic response is due to memory of first encounter.
 (4) Acquired immunity is non-specific type of defence present at the time of birth.
- Q. 192.** Match List-I with List-II with respect to methods of Contraception and their respective actions.
- | List-I | List-II |
|----------------|-------------|
| (a) Diaphragms | (i) Inhibit |

- | | | |
|-----------------------------|-------|--|
| | | ovulation and Implantation |
| (b) Contraceptive Pills | (ii) | Increase phagocytosis of sperm within Uterus |
| (c) Intra Uterine Devices | (iii) | Absence of Menstrual cycle and ovulation following parturition |
| (d) Lactational Amenorrhoea | (iv) | They cover the cervix blocking the entry of sperms |

Choose the correct answer from the options given below :

- (1) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)
 (2) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
 (3) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
 (4) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)
- Q. 193.** Statements related to human Insulin are given below. Which statement(s) is/are correct about genetically engineered Insulin?
- (a) Pro-hormone insulin contain extra stretch of C-peptide
 (b) A-peptide and B-peptide chains of insulin were produced separately in *E.coli*, extracted and combined by creating disulphide bond between them.
 (c) Insulin used for treatment of Diabetes was extracted from Cattles and Pigs.
 (d) Pro-hormone Insulin needs to be processed for converting into a mature and functional hormone.
 (e) Some patients develop allergic reactions to the foreign insulin.

Choose the most appropriate answer from the options given below :

- (1) (a), (b) and (d) only
 (2) (b) only
 (3) (c) and (d) only
 (4) (c), (d) and (e) only
- Q. 194.** The recombination frequency between the genes a & c is 5%, b & c is 15%, b & d is 9%, a & b is 20%, c & d is 24% and a & d is 29%. What will be the sequence of these genes on a linear chromosome ?
- (1) a, d, b, c
 (2) d, b, a, c
 (3) a, b, c, d
 (4) a, c, b, d
- Q. 195.** Ten *E. coli* cells with ^{15}N -dsDNA are incubated in medium containing ^{14}N nucleotide. After 60 minutes, how many *E. coli* cells will have DNA totally free from ^{15}N ?

- (1) 20 cells
 (2) 40 cells
 (3) 60 cells
 (4) 80 cells

Q. 196. Which of the following is not a desirable feature of a cloning vector ?

- (1) Presence of origin of replication
 (2) Presence of a marker gene
 (3) Presence of single restriction enzyme site
 (4) Presence of two or more recognition sites

Q. 197. Select the incorrect statement regarding synapses :

- (1) The membranes of presynaptic and postsynaptic neurons are in close proximity in an electrical synapse.
 (2) Electrical current can flow directly from one neuron into the other across the electrical synapse.
 (3) Chemical synapses use neurotransmitters
 (4) Impulse transmission across a chemical synapse is always faster than that across an electrical synapse.

Q. 198. Which of the following statements is not true?

- (1) Analogous structures are a result of convergent evolution
 (2) Sweet potato and potato is an example of analogy
 (3) Homology indicates common ancestry
 (4) Flippers of penguins and dolphins are a pair of homologous organs

Q. 199. If a colour blind female marries a man whose mother was also colour blind, what are the chances of her progeny having colour

- (1) 25% (2) 50%
 (3) 75% (4) 100%

Q. 200. Given below are two statements :

Statements I :

In a scrubber the exhaust from the thermal plant is passed through the electric wires to charge the dust particles.

Statements II :

Particular matter (PM 2.5) cannot be removed by scrubber but can be removed by an electrostatic precipitator.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both **Statement I** and **Statement II** are correct
 (2) Both **Statement I** and **Statement II** are incorrect
 (3) **Statement I** is correct but **Statement II** is incorrect
 (4) **Statement I** is incorrect but **Statement II** is correct

[illegible]

ANSWERS			
Q. No.	Answer Key	Topic's Name	Chapter's Name
		SECTION-A (PHYSICS)	
1	3	Work, Energy and Power	Work, Energy and Power
2	2	Electrostatic Potential and Capacitance	Electrostatics
3	3	Laws of Motion	Laws of Motion
4	3	Electronic Devices	Electronic Devices
5	2	Magnetic Effects of Current	Magnetic Effects of Current and Magnetism
6	3	Waves	Oscillations and Waves
7	2	Motion of System of Particles and Rigid Body	Motion of System of Particles and Rigid Body
8	4	Dual Nature of Matter and Radiation	Dual Nature of Matter and Radiation
9	2	Thermodynamics	Thermodynamics
10	1	Units and Measurement	Physical World and Measurement
11	3	Electrostatic Potential and Capacitance	Electrostatics
12	4	Current Electricity	Current Electricity
13	3	Nuclei	Atoms and Nuclei
14	4	Motion in a Straight Line	Kinematics
15	3	Electronic Devices	Electronic Devices
16	2	Motion of System of Particles and Rigid Body	Motion of System of Particles and Rigid Body
17	1	Dual Nature of Matter and Radiation	Dual Nature of Matter and Radiation
18	3	Units and Measurement	Physical World and Measurement
19	4	Electronic Devices	Electronic Devices
20	1	Work, Energy and Power	Work, Energy and Power
21	4	Wave Optics	Wave Optics
22	3	Alternating Current	Electromagnetic Induction and Alternating Current
23	1	Mechanical Properties of Fluids	Properties of Bulk Matter
24	3	Ray Optics and Optical Instruments	Optics
25	3	Motion in a Straight Line	Kinematics
26	2	Motion of System of Particles and Rigid Body	Motion of System of Particles and Rigid Body
27	2	Mechanical Properties of Fluids	Properties of Bulk Matter
28	4	Atoms	Atoms and Nuclei
29	2	Magnetic Effects of Current	Magnetic Effect of Current and Magnetism
30	2	Current Electricity	Current Electricity
31	3	Magnetic Effects of Current	Magnetic Effects of Current and Magnetism
32	4	Electromagnetic Waves	Electromagnetic Waves
33	3	Ray Optics and Optical Instruments	Optics
34	4	Electromagnetic Waves	Electromagnetic Waves
35	2	Gravitation	Gravitation

		SECTION-B (PHYSICS)	
36	1	Oscillations	Oscillations and Waves
37	3	Electronic Devices	Electronic Devices
38	2	Alternating Current	Electromagnetic Induction and Alternating Current
39	2	Electric Charges and Fields	Electrostatics
40	3	Electrostatic Potential and Capacitance	Electrostatics
41	3	Mechanical Properties of Solids	Properties of Bulk Matter
42	2	Ray Optics and Optical Instruments	Optics
43	4	Behavior of Perfect Gas and Kinetic Theory	Behavior of Perfect Gas and Kinetic Theory
44	3	Nuclei	Atoms and Nuclei
45	4	Units and Measurements	Physical World and Measurement
46	2	Current Electricity	Current Electricity
47	2	Concept of Vector and Motion in a Plane	Kinematics
48	2	Gravitation	Gravitation
49	3	Magnetics Effects of Current	Magnetic Effect of Current and Magnetism
50	3	Electromagnetic Induction	Electromagnetic Induction and Alternating Current
		SECTION-A (CHEMISTRY)	
51	3	Classification of solids	Solid state
52	4	Aromaticity	Organic Chemistry Some Basic Principles and Techniques
53	2	Molality	Some basic concepts of chemistry
54	1	IUPAC name of elements of periodic table	Classification of elements and periodicity in properties
55	1	Chlorination	Organic Chemistry Some Basic Principles and Techniques
56	3	Amines	Organic Compounds Containing Nitrogen
57	2	Properties of <i>f</i> -Block elements	<i>d</i> - and <i>f</i> -Block elements
58	4	Shapes of molecular orbitals	Structure of atom
59	3	Acidic strength of substituted phenols	Alcohols, Phenols and Ethers
60	4	Reactivity series	Electrochemistry
61	3	Hardy - Schulze rule	Surface Chemistry
62	4	Polymers	Polymers
63	4	Chemical reactions of carbonyl group	Aldehydes, ketones and carboxylic acids
64	2	Defects in solids	Solid state
65	3	Dalton's law of partial pressure	States of Matter : Gases and Liquids
66	1	Physical properties of aldehydes and ketones	Aldehydes, ketones and carboxylic acids
67	1	Preparation of carboxylic acids	Aldehydes, ketones and carboxylic acids
68	3	Enzymes	Surface Chemistry
69	2	Properties of <i>p</i> -block elements	<i>p</i> -Block elements
70	1	<i>p</i> -block elements	<i>p</i> -Block elements
71	2	Types of drugs	Chemistry in everyday life
72	3	Order of a chemical reaction	Chemical Kinetics

73	1	pH	Equilibrium
74	3	Quantitative estimation of nitrogen	Organic Chemistry Some Basic Principles and Techniques
75	4	Chemical bonding	Chemical Bonding and Molecular Structure
76	2	Work	Thermodynamics
77	2	Equilibrium	Equilibrium
78	1	Electrochemistry	Electrochemistry
79	1	Interhalogen compounds	p-block elements
80	4	Magnetic property	Chemical Bonding and Molecular Structure
81	3	Stereochemistry	Haloalkanes and haloarenes
82	2	Properties of <i>s</i> -block elements	<i>s</i> -Block Elements (Alkali and Alkaline Earth Metals)
83	4	Properties of <i>p</i> -block elements	<i>p</i> -block elements
84	4	Uses of <i>s</i> -block elements	<i>s</i> -Block Elements (Alkali and Alkaline Earth Metals)
85	4	IUPAC name of elements of complexes	Coordination compounds
SECTION-B (CHEMISTRY)			
86	1	Equilibrium	Equilibrium
87	2	Ores of metals	General Principles and Processes of Isolation of Elements
88	3	Lucas test for alcohols	Alcohols, Phenols and Ethers
89	1	Structure of atom	Structure of atom
90	4	Chemical reactions of alcohols	Alcohols, Phenols and Ethers
91	2	Aldol Condensation	Aldehydes, ketones and carboxylic acids
92	1	Ozonolysis	Organic Chemistry Some Basic Principles and Techniques
93	1	Chemical reactions of KMnO_4	<i>d</i> - and <i>f</i> -Block elements
94	4	Ideal gas equation	States of Matter : Gases and Liquids
95	2	First order reaction	Chemical Kinetics
96	1	IUPAC name of organic compounds	Organic Chemistry Some Basic Principles and Techniques
97	No correct Option	Nernst equation	Electrochemistry
98	3	Properties of complexes	Coordination compounds
99	1	Density of solids	Solid state
100	2	Pollutants	Environmental Chemistry
SECTION-A (BOTANY)			
101	3	Population and its Attributes	Organisms and populations
102	3	Ecosystem - Structure and Function, Productivity and Decomposition	Ecosystem
103	4	Anatomy of Dicotyledonous and Monocotyledonous plants	Anatomy of Flowering Plants
104	3	Long Distance Transport of Water and Transpiration	Transport in Plants
105	3	Phylum Annelida, Arthropoda, Mollusca, Echinodermata Mollusca, Echinodermata and Hemichordata	Animal Kingdom

106	2	The DNA, Search For Genetic Material RNA World	Molecular Basis of Inheritance
107	1	Secondary Growth	Anatomy of Flowering Plants
108	2	The Inflorescence and The Flower	Morphology of Flowering Plants
109	3	Principles of Biotechnology and Tools of Recombinant DNA Technology	Biotechnology : Principles & Processes
110	4	Differentiation, Dedifferentiation and Redifferentiation, Development and Plant Growth Regulators	Plant Growth and Development
111	2	Types of Classification and Algae	Plant Kingdom
112	4	Flower, Pre-Fertilisation : Structure and Events	Sexual Reproduction in Flowering Plants
113	4	Do Plants Breathe? Glycolysis	Respiration in Plants
114	3	Biodiversity Conservation	Biodiversity And Conservation
115	2	Use of ATP and NADPH	Photosynthesis in Higher Plants
116	3	Cell Cycle And Mitosis	Cell Cycle And Cell Division
117	3	Meiosis	Cell Cycle And Cell Division
118	3	Sex determination and Genetic Disorders	Principle of Inheritance and Variation
119	3	Differentiation, Dedifferentiation and Redifferentiation, Development and Plant Growth Regulators	Plant Growth and Development
120	1	Mendel's Laws of Inheritance and Chromosomal theory	Principle of Inheritance and Variation
121	4	Do Plants Breathe? Glycolysis	Respiration in Plants
122	3	Root, Stem and Leaf	Morphology of Flowering Plants
123	3	Types of Classification and Algae	Plant Kingdom
124	3	Pollution and its Types	Environmental Issues
125	1	Flower, Pre-Fertilisation : Structure and Events	Sexual Reproduction in Flowering Plants
126	2	Means of Transport and Plant-Water Relations	Transport in Plants
127	2	The Inflorescence and The Flower	Morphology of Flowering Plants
128	2	Process of Recombinant DNA Technology	Biotechnology : Principles & Processes
129	2	Plant Growth Regulators	Plant Growth and Development
130	3	Genetic Code, Translation Gene Expression And DNA Fingerprinting	Molecular Basis of Inheritance
131	1	Pigments involved in Photosynthesis, Light Reaction and The Electron Transport	Photosynthesis in Higher Plants
132	2	Mechanism of Absorption of Elements, Translocation of Solutes and Soil as Reservoir of Essential Elements	Mineral Nutrition
133	2	Biodiversity Conservation	Biodiversity And Conservation
134	1	Genetic Code, Translation Gene Expression And DNA Fingerprinting	Molecular Basis of Inheritance
135	2	Plant Growth Regulators	Plant Growth and Development
SECTION-B (BOTANY)			
136	2	Pollution and its Types	Environmental Issues
137	3	Mendel's Laws of Inheritance and Chromosomal theory	Principle of Inheritance and Variation
138	3	The Fruit, The Seed and The Families	Morphology of Flowering Plants
139	1	Genetic Code, Translation Gene Expression And DNA Fingerprinting	Molecular Basis of Inheritance

140	2	Sex determination and Genetic Disorders	Principle of Inheritance and Variation
141	1	Population and its Attributes	Organisms and populations
142	2	Pigments involved in Photosynthesis, Light Reaction and The Electron Transport	Photosynthesis in Higher Plants
143	2	Gymnosperms, Angiosperms and Plant Life Cycles	Plant Kingdom
144	3	Proteins, Polysaccharides, Nucleic Acids	Biomolecules
145	2	The DNA, Search For Genetic Material RNA World	Molecular Basis of Inheritance
146	1	Cell Cycle	Cell Cycle And Cell Division
147	2	Principles of Biotechnology and Tools of Recombinant DNA Technology	Biotechnology : Principles & Processes
148	2	Anatomy of Dicotyledonous and Monocotyledonous plants	Anatomy of Flowering Plants
149	3	Ecological Succession and Nutrient Cycling	Ecosystem
150	2	Means of Transport and Plant-Water Relations	Transport in Plants
SECTION-A (ZOOLOGY)			
151	3	Plant Breeding for Improved Food Quality	Strategies for enhancement in food production
152	2	Meiosis	Cell cycle and cell division
153	1	Absorption of Digested Products	Digestion and absorption
154	2	Oogenesis	Human reproduction
155	2	Decomposition	Ecosystem
156	1	Populations	Organisms and populations
157	4	Anatomy	Structural organisation in animals
158	3	The Endoplasmic Reticulum (ER)	Cell: the unit of life
159	2	Anatomy	Structural organisation in animals
160	4	Human Respiratory System	Breathing and exchange of gases
161	2	Transport of Oxygen	Breathing and exchange of gases
162	2	Metaphase	Cell cycle and cell division
163	2	Joints	Locomotion and movement
164	1	The DNA	Molecular basis of inheritance
165	1	How do we conserve Biodiversity?	Biodiversity and conservation
166	4	Oogenesis	Human reproduction
167	1	Phylum – Chordata	Animal kingdom
168	3	Eubacteria	Biological classification
169	3	Auto Immunity	Human health and disease
170	4	Gene Therapy	Biotechnology and its applications
171	1	Disorders Of Muscular And Skeletal System	Locomotion And Movement
172	3	The Male Reproductive System	Human Reproduction
173	4	Introduction To Excretion	Excretory Products And Their Elimination
174	1	Chemicals, Enzymes And Other Bioactive Molecules	Microbes In Human Welfare
175	3	Population Stabilisation And Birth Control	Reproductive Health
176	3	Nature Of Bond Linking Monomers In A Polymer	Biomolecules
177	4	Connective Tissue	Structural Organisation In Animals

178	4	Digestive Glands	Digestion And Absorption
179	1	Restriction Enzymes	Biotechnology : Principles And Processes
180	2	Hardy-Weinberg Principle	Evolution
181	3	Disorders Of Muscular And Skeletal System	Locomotion And Movement
182	1	Taxonomic Categories	The Living World
183	3	Lac Operon Model	Molecular Basis Of Inheritance
184	2	Asexual Reproduction	Reproduction In Organisms
185	4	Coagulation Of Blood	Body Fluids And Circulation
SECTION-B (ZOOLOGY)			
186	4	Biological Functions Of Biomolecules	Biomolecules
187	1	Animal Tissues	Structural Organisation In Animals
188	3	Human Circulatory System	Body Fluids And Circulation
189	2	Parathyroid Gland	Chemical Coordination And Integration
190	1	Kingdom Monera	Biological Classification
191	4	Acquired Immunity	Human Health And Disease
192	2	Population Stabilisation And Birth Control	Reproductive Health
193	2	Genetically Engineered Insulin	Biotechnology And Its Applications
194	4	Linkage And Recombination	Principles Of Inheritance And Variation
195	3	The Experimental Proof	Molecular Basis Of Inheritance
196	4	Cloning Vectors	Biotechnology : Principles And Processes
197	4	Transmission Of Impulses	Neural Control And Coordination
198	4	What Are The Evidences For Evolution?	Evolution
199	4	Mendelian Disorders	Principles Of Inheritance And Variation
200	2	Air Pollution And Its Control	Environmental Issues

NEET (UG)

17th July 2022 Paper

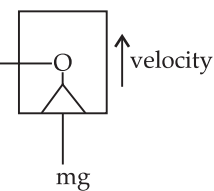
ANSWERS WITH EXPLANATION

PHYSICS

Section—A

1. Option (3) is correct.

Explanation:



$$\begin{aligned}
 m &= 2000 \text{ kg} \\
 T &= W + f \\
 &= mg + f \\
 &= (2000)(10) + 3000 \\
 &= 23000 \text{ N} \\
 \text{Power} &= \text{Force} \times \text{velocity} \\
 &= 23000 \times 1.5 \\
 &= 34500 \text{ W}
 \end{aligned}$$

2. Option (2) is correct.

Explanation: As,

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r},$$

or, $V \propto \frac{1}{r}$

Hence, lesser the radius/distance, more will be the potential.

3. Option (3) is correct.

Explanation: By law of conservation of momentum,

$$m \times 0 = \frac{2m}{5}(-v\hat{i}) + \frac{2}{5}m(v\hat{j}) + \frac{m}{5}(v')$$

$$\Rightarrow m(0) = (-2mv\hat{i}) + (2mv\hat{j}) + (mv')$$

$$\Rightarrow v' = \sqrt{(2v)^2 + (2v)^2}$$

$$\Rightarrow v' = \sqrt{8v^2}$$

$$\Rightarrow v' = 2\sqrt{2}v$$

4. Option (3) is correct.

Explanation: As temperature increases, the electrical resistance decreases for semi-conductors and insulators while increases

for conductors.

5. Option (2) is correct.

Explanation: $B = \mu_0 n I = \frac{\mu_0 N}{l} I$

$$\begin{aligned}
 \therefore B &= 4\pi \times 10^{-7} \times \frac{100}{1 \times 10^{-3}} \times 1 \\
 &= 12.56 \times 10^{-2} \text{ T}
 \end{aligned}$$

6. Option (3) is correct.

Explanation: $v \propto \sqrt{T}$

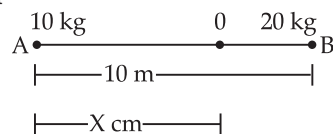
$$\frac{v_i}{v_f} = \sqrt{\frac{T_1}{T_2}}$$

$$\Rightarrow \frac{v_i}{v_f} = \sqrt{\frac{T}{2T}}$$

$$v_i : v_f = 1 : \sqrt{2}$$

7. Option (2) is correct.

Explanation:



$$\begin{aligned}
 X_{\text{cm}} &= \frac{M_1 \times M_2}{M_1 + M_2} \\
 &= \frac{10 \times 20}{10 + 20} \\
 &= \frac{200}{30} = \frac{20}{3} \text{ m}
 \end{aligned}$$

8. Option (4) is correct.

Explanation:

$$\text{As, } \lambda = \frac{h}{p}$$

i.e., $\lambda \propto \frac{1}{p}$

Hence, the graph will be hyperbolic.

9. Option (2) is correct.

Explanation: The given processes are;

1 $\rightarrow$ Isochoric

- 2 → Adiabatic
3 → Isothermal
4 → Isobaric

10. Option (1) is correct.

Explanation: Plane angle and solid angle are dimensionless, while having units.

11. Option (3) is correct.

Explanation: Electric field lines are always perpendicular to the equipotential surface.

12. Option (4) is correct.

Explanation: Given: $l = 10 \text{ m}$, $r = \frac{10^{-2}}{\pi} \text{ m}$,

$$R = 10 \Omega, E = 10 \text{ V/m}$$

$$\text{As, } J = \sigma E$$

$$\begin{aligned} \text{or, } J &= \frac{E}{\rho} = \frac{El}{RA} \\ &= \frac{10 \times 10}{10 \times \pi r^2} \\ &= \frac{10 \times 10}{10 \times 3.14 \times \left(\frac{10^{-2}}{\pi}\right)^2} \\ &= \frac{10 \times 10 \times (3.14)^2}{10 \times 3.14 \times 10^{-4}} \\ &= 3.14 \times 10^5 \text{ A/m}^2 \end{aligned}$$

13. Option (3) is correct.

Explanation: The represented or given reaction shows β^+ -decay.



14. Option (4) is correct.

Explanation: Slope of displacement-time graph gives velocity.

$$\text{Slope} = v = \frac{dx}{dt} = \tan \theta$$

$$\text{Hence, } \frac{v_1}{v_2} = \frac{\tan \theta_1}{\tan \theta_2} = \frac{\tan 30^\circ}{\tan 45^\circ}$$

$$\frac{v_1}{v_2} = \frac{1}{\sqrt{3}}$$

15. Option (3) is correct.

Explanation: In half-wave rectification, input frequency = output frequency

16. Option (2) is correct.

Explanation: $I_1 = \frac{Mr^2}{2}$ (about at centre)

$$I_2 = \frac{Mr^2}{4} \text{ (about at diameter)}$$

$$\text{As, } I = MK^2$$

$$\text{Hence, } K = \sqrt{\frac{I}{M}}$$

$$\begin{aligned} \frac{K_1}{K_2} &= \sqrt{\frac{I_1}{I_2}} \\ &= \sqrt{\frac{\frac{Mr^2}{2}}{\frac{Mr^2}{4}}} \\ &= \sqrt{2} : 1 \end{aligned}$$

17. Option (1) is correct.

Explanation: Apply Einstein's photoelectric equation

$$eV_s = h\nu - h\nu_0$$

$$\text{or, } \nu_0 = \frac{h\nu - eV_s}{h} \quad \dots(i)$$

From first condition,

$$\nu \rightarrow \nu \text{ and } V_s \rightarrow \frac{V_s}{2}$$

From equation (i)

$$\begin{aligned} \nu_0 &= \left(h\nu - e\frac{V_s}{2} \right) / h \\ \nu_0 &= \frac{2h\nu - eV_s}{2h} \quad \dots(ii) \end{aligned}$$

Again apply second condition,

$$\nu \rightarrow \frac{\nu}{2} \text{ and } V_s \rightarrow V_s$$

From equation (i),

$$\nu_0 = \frac{h\nu - 2eV_s}{2h} \quad \dots(iii)$$

Now, from equation (ii) and (iii)

$$2h\nu - eV_s = h\nu - 2eV_s$$

$$\text{or, } h\nu = -eV_s \quad \dots(iv)$$

Now from equation (i) and (iv)

$$\begin{aligned} \nu_0 h &= h\nu + h\nu \\ \nu_0 &= 2\nu \end{aligned}$$

18. Option (3) is correct.

Explanation: Dimensions of given options are as follows;

$$(1) \text{ Magnetic flux — } [ML^2T^{-2}A^{-1}]$$

$$(2) \text{ Self inductance — } [ML^2T^{-2}A^{-2}]$$

$$(3) \text{ Magnetic permeability — } [ML^2T^{-2}A^{-2}]$$

$$(4) \text{ Electric permittivity — } [M^{-1}L^{-3}T^4A^2]$$

Hence, the option (3) is correct.

19. Option (4) is correct.

Explanation: The potential drop will be equal in both circuits (a) and (c), as both the junctions are in forward biasing, and hence offers equal resistance.

20. Option (1) is correct.

Explanation: As, energy = power $\times$ time

$$\begin{aligned}\text{Hence, } E &= 100 \times 10^3 \times 1 \times 60 \times 60 \\ &= 360000 \times 10^3 \\ &= 36 \times 10^7 \text{ J}\end{aligned}$$

21. Option (4) is correct.

Explanation: In YDSE,

$$x = (n\lambda) \left(\frac{D}{d} \right)$$

According to question,

$$\begin{aligned}(n_1\lambda_1) \left(\frac{D}{d} \right) &= (n_2\lambda_2) \left(\frac{D}{d} \right) \\ &= 8 \times 600 \times 10^{-9} = n_2 \times 400 \times 10^{-9} \\ &= n_2 = \frac{8 \times 600 \times 10^{-9}}{400 \times 10^{-9}} = 12\end{aligned}$$

22. Option (3) is correct.

Explanation: Peak voltage is always $\sqrt{2}$ times of rms voltages in a.c.

$$V_0 = \sqrt{2} V_{\text{rms}}$$

23. Option (1) is correct.

Explanation: For soap bubble, $P - P_0 = \frac{4T}{R}$

As pressure is inverse in relation with radius, hence the pressure decreases as the soap bubble expands or radius increases.

24. Option (3) is correct.

Explanation: Given: $R_1 = R_2 = 20 \text{ cm} = 0.2 \text{ m}$, $\mu = 1.5$

$$\begin{aligned}\text{Power, } P &= \frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \\ P &= (1.5 - 1) \left\{ \frac{1}{0.2} - \left(\frac{1}{-0.2} \right) \right\} \\ &\quad \left[\text{for biconvex lens, } \begin{array}{l} R_1 = +ve \\ R_2 = -ve \end{array} \right] \\ P &= 0.5 \left(\frac{1}{0.2} + \frac{1}{0.2} \right) \\ P &= 0.5 \times \frac{2}{0.2} = 5\text{D}\end{aligned}$$

25. Option (3) is correct.

Explanation: As,

$$S_{\text{nth}} = u + \frac{a}{2} (2n - 1)$$

$$\text{Hence, } S_{1\text{st}} = \frac{a}{2} \{2(1) - 1\} = \frac{a}{2} \quad (1)$$

$$S_{2\text{nd}} = \frac{a}{2} \{2(2) - 1\} = \frac{a}{2} \quad (3)$$

$$S_{3\text{rd}} = \frac{a}{2} \{2(3) - 1\} = \frac{a}{2} \quad (5)$$

$$S_{4\text{th}} = \frac{a}{2} \{2(4) - 1\} = \frac{a}{2} \quad (7)$$

As the body is freely falling, the initial velocity is zero and $a = g$ in above each case.

Now, $S_{1\text{st}} : S_{2\text{nd}} : S_{3\text{rd}} : S_{4\text{th}} = 1 : 3 : 5 : 7$

26. Option (2) is correct.

Explanation: Given:

$$\omega_0 = 1200 \text{ rpm}$$

$$\omega = 3120 \text{ rpm}$$

$$t = 16 \text{ seconds}$$

$$\alpha = \frac{\omega - \omega_0}{t}$$

$$= \frac{3120 - 1200}{16}$$

$$= \frac{1920}{16} \text{ rpm}$$

$$= \frac{1920}{16} \times \frac{2\pi}{60} \text{ rad/s}^2$$

$$= 4\pi \text{ rad/s}^2$$

27. Option (2) is correct.

Explanation: Initially the velocity of the body falling in a viscous fluid increases but after some time it acquires a constant velocity known as terminal velocity. This is shown by point B.

28. Option (4) is correct.

Explanation: For first excited state, $n_1 = 2$

And for second excited state, $n_2 = 3$

$$\text{As, } T = -13.6 \frac{Z^2}{n^2}$$

$$T_1 = -13.6 \times \frac{(1)^2}{(2)^2}$$

$$T_2 = -13.6 \times \frac{(1)^2}{(3)^2}$$

$$T_1 : T_2 = \frac{1}{4} : \frac{1}{9} = 9 : 4$$

29. Option (2) is correct.

Explanation: Given:

$$a = 1\text{m}$$

$$R = 1\Omega$$

$$B = 0.5\text{T}$$

$$\text{Area of square} = (\text{side})^2 = a^2 = 1\text{ m}^2$$

$$\phi = BA \cos \theta$$

$$= 0.5 \times 1 \times \cos 0^\circ$$

$$= 0.5 \times 1 \times 1$$

$$= 0.5 \text{ Weber}$$

30. Option (2) is correct.

Explanation: In parallel combination, the potential remains same or constant.

$$\text{As, } P = \frac{V^2}{R}$$

$$P \propto \frac{1}{R}$$

$$\therefore \frac{P_1}{P_2} = \frac{R_2}{R_1} = \frac{200}{100}$$

$$P_1 : P_2 = 2 : 1$$

$$\text{And, } P = \frac{\text{Work done/Energy}}{\text{time taken}}$$

$$P \propto E \quad (\text{at same duration})$$

$$\therefore E_1 : E_2 = 2 : 1$$

31. Option (3) is correct.

Explanation: As, $\vec{d}\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \sin \theta}{r^2}$, so the

statement I is correct.

Statement-II is wrong as Biot Savart's law depends on current carrying element, which is also a vector quantity.

32. Option (4) is correct.

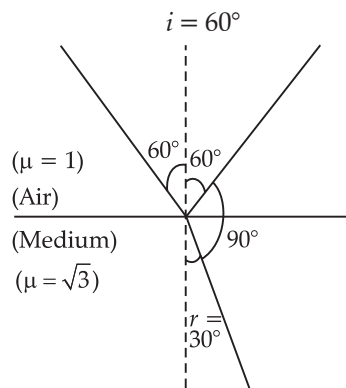
Explanation:

Waves	Wavelength (nearly)
AM Radio waves	10^2 m
Microwaves	10^{-2} m
Infrared Radiations	10^{-4} m
X-rays	$10^{-10}\text{ m} = 1\text{ \AA}$

33. Option (3) is correct.

Explanation: Given:

$$\mu = \sqrt{3}$$



By using Snell's law, we have;

$$\mu_1 \sin i = \mu_2 \sin r$$

$$1 \times \sin 60^\circ = \sqrt{3} \sin r$$

$$\sin r = \frac{\sqrt{3}/2}{\sqrt{3}} = \frac{1}{2}$$

$$r = 30^\circ$$

Hence, angle between reflected and refracted ray
 $= 60^\circ + 30^\circ = 90^\circ$

34. Option (4) is correct.

Explanation: As, $\mu = \sqrt{\epsilon_r \mu_r}$

$$\text{And, } \mu = \frac{c}{v}$$

$$\text{Hence, } v = \frac{c}{\mu} = \frac{c}{\sqrt{\epsilon_r \mu_r}}$$

35. Option (2) is correct.

Explanation: Gravitational Field Intensity (I_g)

$$= \frac{\text{Gravitational Force (F)}}{\text{Mass (M)}}$$

$$I_g = \frac{3}{60 \times 10^{-3}}$$

$$I_g = 50 \text{ N/kg}$$

Section—B

36. Option (1) is correct.

Explanation: $(n) (T_l) = (n+1) (T_s)$

$$\text{As, } T = 2\pi \sqrt{\frac{l}{g}}$$

$$\text{Hence, } (n) \left(2\pi \sqrt{\frac{1.21}{9.8}} \right) = (n+1) \left(2\pi \sqrt{\frac{1}{9.8}} \right)$$

$$\Rightarrow (n) (1.1) = (n+1) (1)$$

$$\Rightarrow 1.1n = n + 1$$

$$\Rightarrow 1.1n - n = 1$$

$$\Rightarrow n (1.1 - 1) = 1$$

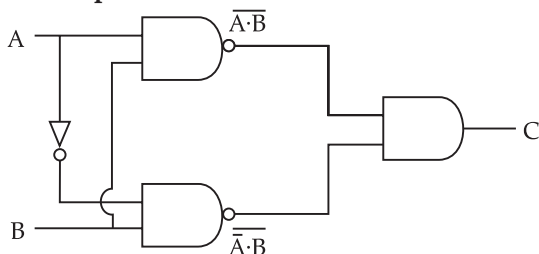
$$\Rightarrow 0.1n = 1$$

$$\Rightarrow n = \frac{1}{0.1} = 10$$

Now, Number of oscillation for shorter pendulum = $(n + 1) = 11$

37. **Option (3) is correct.**

Explanation:



$$C = \overline{A \cdot B + \bar{A} \cdot \bar{B}} \text{ (By using de-Morgan Theorem)}$$

$$C = \overline{B(A + \bar{A})} = \bar{B}$$

Hence, option (3) is most accurate one.

38. **Option (2) is correct.**

Explanation: Given: $L = 10 \text{ H}$, $C = 10 \mu\text{F}$,
 $R = 50 \Omega$

$$V = 200 \sin(100t)$$

Standard equation is $V = V_0 \sin \omega t$

Hence, $\omega = 100$

$$n = \frac{\omega}{2\pi} = \frac{100}{2\pi} = \frac{50}{\pi}$$

$$\begin{aligned} n_0 &= \frac{1}{2\pi\sqrt{LC}} \\ &= \frac{1}{2\pi\sqrt{10 \times 10 \times 10^{-6}}} \\ &= \frac{1}{2\pi} \times \sqrt{\frac{1}{10^{-4}}} \\ &= \frac{1}{2\pi} \times 10^2 = \frac{50}{\pi} \\ n &= n_0 = \frac{50}{\pi} \text{ Hz} \end{aligned}$$

39. **Option (2) is correct.**

Explanation: The provided figure is of an electric dipole.

And the electric field intensity for a dipole is always inverse of cube root of distance.

$$\text{Mathematically, } E = \frac{1}{4\pi\epsilon_0} \frac{p}{R^3}$$

$$\text{Hence, } E \propto \frac{1}{R^3}$$

40. **Option (3) is correct.**

Explanation: According to the question,

$$C_1 V_1 + C_2 V_2 = (C_1 + C_2) V$$

$$V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

$$V = \frac{C \times 100 + C \times 0}{C + C}$$

$$V = \frac{100C}{2C} = 50V$$

$$\text{Energy stored, } U = 2 \times \frac{1}{2} CV^2$$

[For given system]

$$\begin{aligned} U &= CV^2 \\ &= 900 \times 10^{-12} \times 50 \times 50 \\ &= 2250000 \times 10^{-12} \\ &= 2.25 \times 10^{-6} \text{ J} \end{aligned}$$

41. **Option (3) is correct.**

Explanation: In stretching a spring, its shape changes and hence the shear modulus. So, assertion (A) is correct.

Reason is incorrect as $Y_{\text{steel}} > Y_{\text{Copper}}$

42. **Option (2) is correct.**

Explanation: As, $\mu = \frac{c}{v}$

$$\text{Hence, } \mu \propto \frac{1}{v}$$

Now, Critical angle,

$$\begin{aligned} \sin i_c &= \frac{\mu_2}{\mu_1} = \frac{v_1}{v_2} \\ &= \frac{1.5 \times 10^8}{2 \times 10^8} = \frac{3}{4} \\ i_c &= \sin^{-1}\left(\frac{3}{4}\right) \\ i_c &= \sin^{-1}(0.750) \end{aligned}$$

43. **Option (4) is correct.**

Explanation:

$$\begin{aligned} \text{Required volume} &= \text{number of moles} \\ &\quad \times \text{standard volume} \\ &= \frac{\text{given mass}}{\text{molar mass}} \times 22.4 \text{ (L)} \\ &= \frac{4.5 \times 10^3}{18} \times 22.4 \times 10^{-3} \text{ (m}^3\text{)} \\ &= \frac{4.5 \times 22.4}{18} \\ &= 5.6 \text{ m}^3 \end{aligned}$$

44. Option (3) is correct.

Explanation: As, $R = R_0 A^{1/3}$

$$\text{Hence, } \frac{R_1}{R_2} = \frac{R_0(125)^{1/3}}{R_0(64)^{1/3}}$$

$$\frac{R_1}{R_2} = \left(\frac{125}{64}\right)^{1/3}$$

$$\frac{R_1}{R_2} = \frac{5}{4}$$

$$R_1 : R_2 = 5 : 4$$

45. Option (4) is correct.

Explanation: Area = length $\times$ breadth
 $= 55.3 \times 25$
 $= 1382.5 \text{ m}^2$

As least significant figure is two digits, hence resultant should be of two digits.

So, required answer should be

$$1400 = 14 \times 10^2 \text{ m}^2$$

46. Option (2) is correct.

Explanation: For precise measurement of unknown resistance, the resistances of arms P and Q should be approximately equal and small, as it maintains balancing.

47. Option (2) is correct.

Explanation: At highest point, only the horizontal component remains in consideration.

Hence, $u_x = u \cos \theta = 10 \cos 30^\circ$

$$10 \times \frac{\sqrt{3}}{2}$$

$$= 5\sqrt{3} \text{ ms}^{-1}$$

48. Option (2) is correct.

Explanation: The dimensional formula for required parts are as follow:

Gravitational constant — $[M^{-1}L^3T^{-2}]$

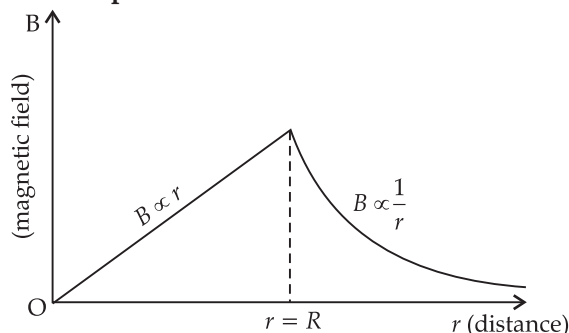
Gravitational potential energy — $[ML^2T^{-2}]$

Gravitational potential — $[L^2T^{-2}]$

Gravitational intensity — $[LT^{-2}]$

49. Option (3) is correct.

Explanation:



The graph clearly representing that given option (3) is correct.

50. Option (3) is correct.

Explanation: Given: $n = 1000$, $r = 10 \text{ m}$,
 $B = 2 \times 10^{-5} \text{ T}$, $R = 12.56 \Omega$ and $\omega = 2 \text{ rad/s}$

$$\begin{aligned} \text{As, } I_{\max} &= \frac{E_{\max}}{R} = \frac{NBA\omega}{R} \\ &= \frac{1000 \times 2 \times 10^{-5} \times \pi \times 10 \times 10 \times 2}{12.56} \\ &= \frac{2 \times 2 \times 3.14}{12.56} \times 10^0 [\because \text{area} = \pi r^2] \\ &= \frac{12.56}{12.56} \times 1 \quad [\because 10^0 = 1] \\ &= 1 \text{ A} \end{aligned}$$

□□□

CHEMISTRY

Section A

51. Option (3) is correct.

Explanation: In diamond, carbon atoms have tetrahedral arrangement, so hybridization is sp^3 .

The shape of graphite is trigonal planar and the hybridization possessed by it is sp^2 .

52. Option (4) is correct.

Explanation: c1ccccc1 does not follow Huckel rule as it has $8\pi e^-$ which violates $(4n + 2)\pi e^-$ rule.

53. Option (2) is correct.

Explanation:

$$\text{Molality} = \frac{\text{Moles of solute}}{\text{Weight of solvent (in kg)}}$$

$$\text{Weight} = \frac{\text{Moles of solute}}{\text{Molality}}$$

$$\text{Weight} = \frac{0.5}{1} \text{ kg}$$

$$\text{Weight} = 0.5 \times 1000 \text{ g}$$

$$\text{Weight} = 500 \text{ g}$$

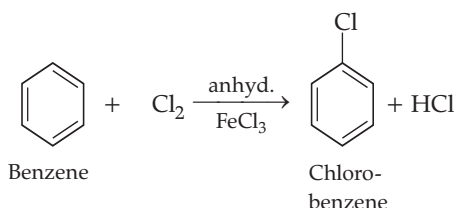
54. Option (1) is correct.

Explanation: As per IUPAC, notation for 1 is 'un' and g is 'enn' and the roots are put together in order of digits which make up the atomic number and 'ium' is added at the end.

So, 119 = un + un + enn + ium
= ununennium

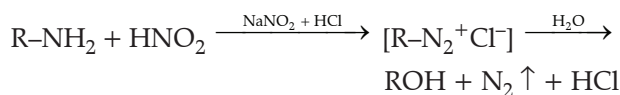
55. Option (1) is correct.

Explanation: Chlorobenzene is obtained by the chlorination of benzene.

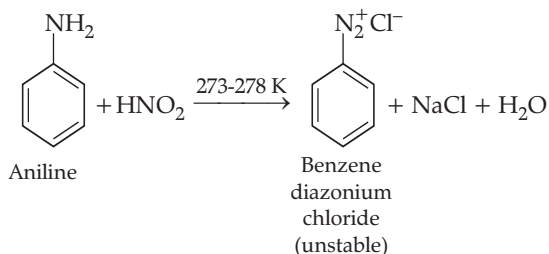


56. Option (3) is correct.

Explanation: Primary aliphatic amines reacts with nitrous acid to produce aliphatic diazonium salts which are unstable and liberates nitrogen gas and produces alcohols.



While, primary aromatic amines react with nitrous acid to form diazonium salts which are unstable.

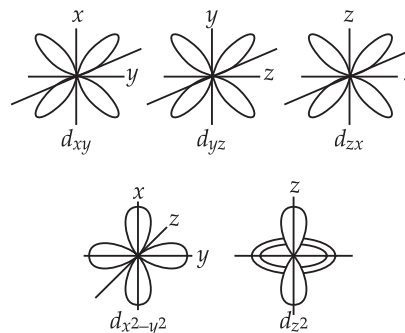


57. Option (2) is correct.

Explanation: The electronic configuration of Gadolinium (Gd) is [Xe] 4f⁷5d¹6s². In its Gd³⁺ state it acquires extra stability due to half-filled 4f sub-shell, i.e., [Xe] 4f⁷, and due to the high exchange enthalpy, it has low third ionization enthalpy.

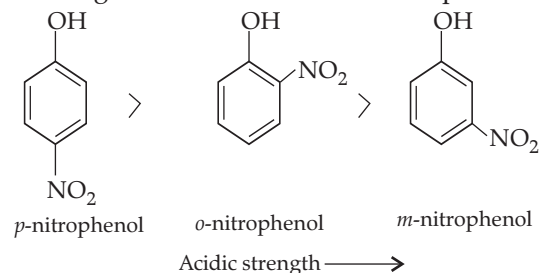
58. Option (4) is correct.

Explanation: The shapes of d_{xy} , d_{yz} , d_{zx} orbitals are similar to each other but the shapes of $d_{x^2-y^2}$ and d_{z^2} are different from each other.



59. Option (3) is correct.

Explanation: The acidic strength of monosubstituted nitrophenol is higher than phenol because of electron withdrawing group. But, *o*-nitrophenol, *m*-nitrophenol and *p*-nitrophenol have different acidic strength because of different distance of —NO₂ group from the —OH group and also at *ortho* and *para* positions, —NO₂ group withdraws electrons of the O—H bond towards itself by the stronger —R effect while the —NO₂ group of *meta* position withdraws electrons of the —OH bond by the weaker —I effect. So, acidic strength of *ortho* and *para*-nitrophenol is greater than that of *meta* nitrophenol.



60. Option (4) is correct.

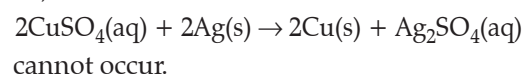
Explanation:

$$E_{\text{Cu}^{2+}/\text{Cu}}^0 = 0.39 \text{ V}, E_{\text{Zn}^{2+}/\text{Zn}}^0 = -0.76 \text{ V},$$

$$E_{\text{Fe}^{2+}/\text{Fe}}^0 = -0.44 \text{ V}, E_{\text{Ag}^{2+}/\text{Ag}}^0 = 0.80 \text{ V}$$

Considering the E^0 values, Ag cannot displace Cu from the solution.

So, the reaction :



61. Option (3) is correct.

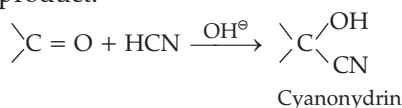
Explanation: According to Hardy Schulze Rule, Statement I is correct, since the greater the valence of the flocculating ion added, greater is its power to coagulate. But in statement II, Hardy-Schulze rule is not followed, hence statement II is incorrect.

- 62.
- Option (4) is correct.**

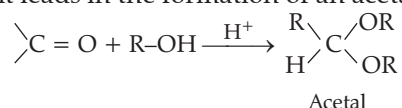
Explanation: Thermosetting polymers are not reusable.

- 63.
- Option (4) is correct.**

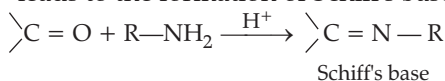
Explanation: When carbonyl group is reacted with HCN, cyanohydrin is formed as a product.



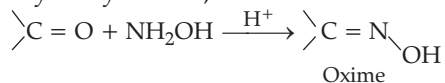
When carbonyl group is reacted with alcohol, it leads in the formation of an acetal.



On reaction with RNH_2 , carbonyl compound leads to the formation of Schiff's base.



When carbonyl compound is treated with hydroxyl amine, oxime is formed.



- 64.
- Option (2) is correct.**

Explanation: In point defects, electrical neutrality is not disturbed. Ether dislocation takes place or equal number of cation and anion gets missed.

- 65.
- Option (3) is correct.**

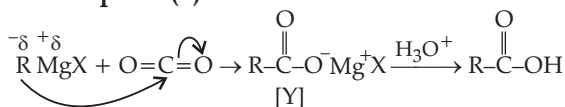
Explanation: According to Dalton's law of partial pressure, the total pressure exerted by a mixture of gases is equal to the sum of the partial pressure exerted by each individual gas in the mixture.

Mathematically, $p_{\text{total}} = p_1 + p_2 + p_3 + \dots$

So, option (3), i.e., the expression $p_i = x_i p$ is incorrect mathematical equation for Dalton's law of partial pressure.

- 66.
- Option (1) is correct.**

- 67.
- Option (1) is correct.**

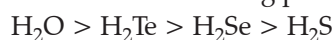


- 68.
- Option (3) is correct.**

Explanation: Enzymes are not polysaccharides, they are protein in nature.

- 69.
- Option (2) is correct.**

Explanation: Because of hydrogen bonding in water molecule, H_2O has highest boiling point. For the hydrides of S, Se and Te, boiling point increases with increase in molar mass. So, correct order of boiling point is:



- 70.
- Option (1) is correct.**

Explanation: MgH_2 is ionic in nature.

GeH_4 is electron precise as it has $8e^-$ without any lone pair.

B_2H_6 is an electron deficient compound as it has less than $8e^-$.

HF is electron rich hydride as it has 8 electrons with lone pair.

- 71.
- Option (2) is correct.**

Explanation: Cimetidine is an antacid.

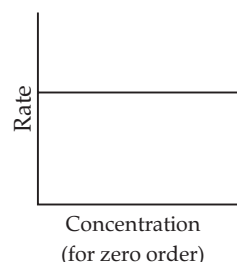
Seledane is an antihistamine.

Morphine is an analgesic.

Salvarsan is an antimicrobials.

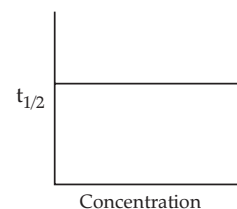
- 72.
- Option (3) is correct.**

Explanation: Since, rate of a zero order reaction does not depend on concentration, so for zero order reaction, $y = \text{rate}$ and $x = \text{concentration}$.



While in first order reaction, $t_{1/2} = \frac{0.693}{k}$, so

half life of reaction does not depend on concentration, so graph for first order reaction is:



- 73.
- Option (1) is correct.**

Explanation: Concentration of salt = 0.10 M

Concentration of acetic acid = 0.01 M

pK_a of acetic acid = 4.57

Since, $\text{pH} = pK_a + \log \frac{[\text{Salt}]}{[\text{Acid}]}$

$$\text{pH} = 4.57 + \log \left(\frac{0.1}{0.01} \right)$$

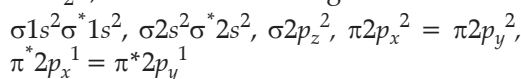
$$\text{pH} = 4.57 + \log \left(\frac{100}{10} \right)$$

$$\text{pH} = 4.57 + \log 10$$

$$\text{pH} = 4.57 + 1 \quad [\because \log 10 = 1]$$

$$\text{pH} = 5.5$$

In O_2^+ , the electronic configuration is:



Since, there is one unpaired electron in O_2^+ , so it is paramagnetic in nature.

81. Option (3) is correct.

Explanation: Enantiomers are non-superimposable mirror images of each other.

82. Option (2) is correct.

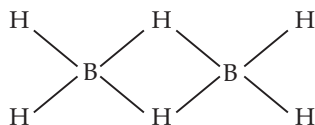
Explanation: Since, O_2^- is superoxide ion, so, its oxidation state is -1 .

So, oxidation number of KO_2 is:

$$x + (-1) = 0 \\ x = +1$$

83. Option (4) is correct.

Explanation:



In diborane (B_2H_6), hybridization of both the boron atoms is sp^3 .

84. Option (4) is correct.

Explanation: Lithium is used in electrochemical cells.

Sodium is used as coolant in fast breeder reactors.

KOH is used to absorb CO_2 gas.

Cs is used in photoelectric cells.

85. Option (4) is correct.

Explanation: IUPAC name of

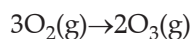


diaquasilver (I) dicyanidoargentate (I)

Section B

86. Option (1) is correct.

Explanation:



$$K_c = \frac{[O_3]^2}{[O_2]^3}$$

$$3 \times 10^{-59} = \frac{[O_3]^2}{(0.040)^3}$$

$$3 \times 10^{-59} = \frac{[O_3]^2}{(4 \times 10^{-2})^3}$$

$$[O_3]^2 = 3 \times 10^{-59} \times 64 \times 10^{-6}$$

$$[O_3]^2 = 192 \times 10^{-65}$$

$$[O_3]^2 = 19.2 \times 10^{-64}$$

$$[O_3] = \sqrt{19.2 \times 10^{-64}}$$

$$[O_3] = 4.38 \times 10^{-32} \text{ M}$$

87. Option (2) is correct.

Explanation: Haematite - Fe_2O_3

Magnetite - Fe_3O_4

Calamine - $ZnCO_3$

Kaolinite - $[Al_2(OH)_4Si_2O_5]$

88. Option (3) is correct.

Explanation: Tertiary alcohols are most reactive and immediately produce turbidity at room temperature on reaction with Lucas reagent.

89. Option (1) is correct.

Explanation:

Since, $r \propto \frac{n^2}{Z}$ (From Bohr's atomic model)

For third Bohr orbit of Li^{2+} , $n_{Li^{3+}} = 3$, $Z_{Li} = 3$

For second Bohr orbit of He^+ , $n_{He^+} = 2$,

$$Z_{He} = 2$$

$$(r_3)_{Li^{3+}} = \frac{(3)^2}{3}, (r_2)_{He^+} = \frac{(2)^2}{2}$$

$$\frac{(r_3)_{Li^{3+}}}{(r_2)_{He^+}} = \frac{(3)^2 \times 2}{3 \times (2)^2} = \frac{3}{2}$$

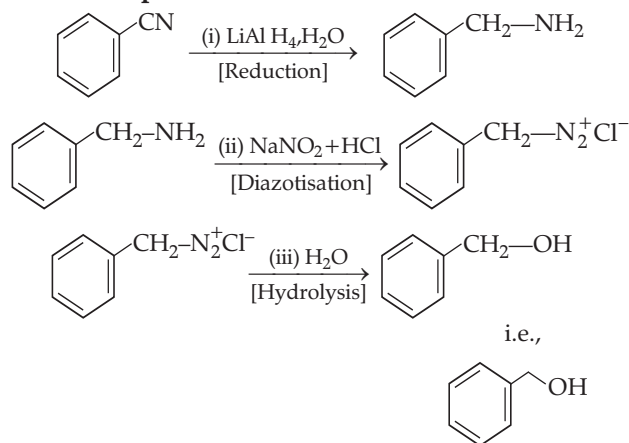
$$\frac{(r_3)_{Li^{3+}}}{105.8} = \frac{3}{2}$$

$$(r_3)_{Li^{3+}} = \frac{3}{2} \times 105.8$$

$$(r_3)_{Li^{3+}} = 158.7 \text{ pm}$$

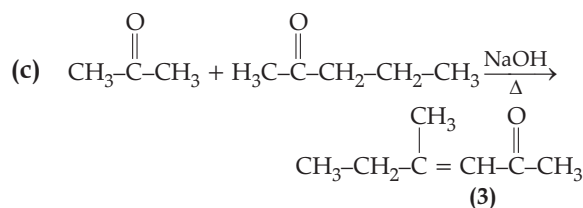
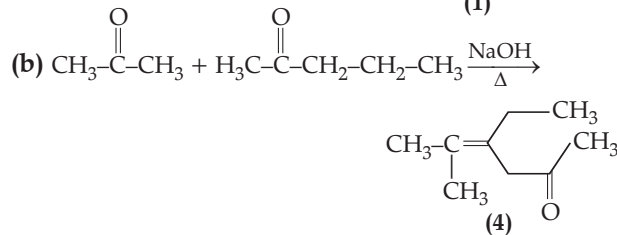
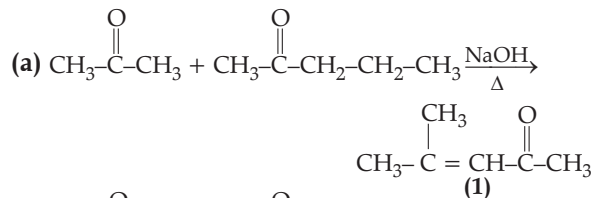
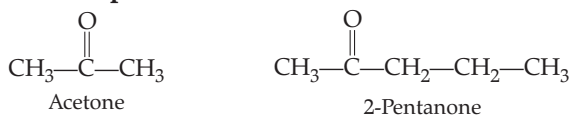
90. Option (4) is correct.

Explanation:



91. Option (2) is correct.

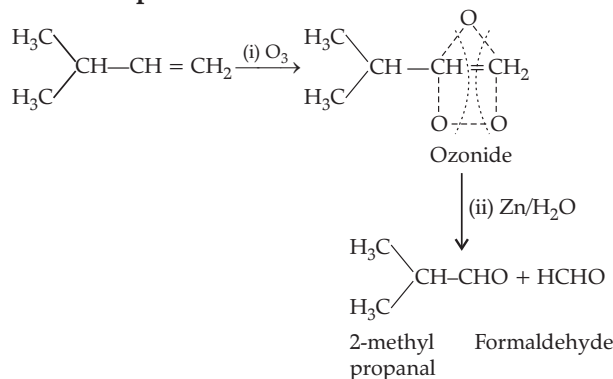
Explanation:



So, product (2) is not formed.

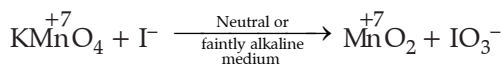
92. Option (1) is correct.

Explanation:



93. Option (1) is correct.

Explanation: Reaction of KMnO_4 with iodide ion in neutral or faintly alkaline medium:



So, the oxidation state of Mn changes from +7 to +4.

94. Option (4) is correct.

Explanation: $V = 10 \text{ L}$, $w_{\text{O}_2} = 64 \text{ g}$

$T = 27^\circ\text{C} = 27 + 273 \text{ K} = 300 \text{ K}$

1 mole of O_2 contains 32 g of O_2

So 64 g of O_2 will contain 2 moles of oxygen.

So, number of moles (n) = 2

$R = 0.0831 \text{ L bar K}^{-1} \text{ mol}^{-1}$

Applying ideal gas equation,

$$PV = nRT$$

$$P = \frac{nRT}{V} = \frac{2 \times 0.0831 \times 300}{10}$$

$$P = 4.9 \text{ bar}$$

95. Option (2) is correct.

Explanation: $A \rightarrow \text{Products}$ (given)

$$[A]_0 = 0.1 \text{ M}$$

$$[A] = 0.001 \text{ M}$$

$$t = 5 \text{ min}$$

For first order reaction,

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$$

$$k = \frac{2.303}{5} \log \left(\frac{0.1}{0.001} \right)$$

$$k = \frac{2.303}{5} \log (100)$$

$$k = \frac{2.303}{5} \log 10^2$$

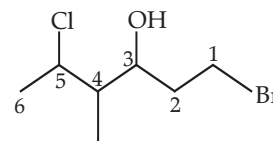
$$k = \frac{2.303}{5} \times 2 \log 10$$

$$k = \frac{2.303}{5} \times 2 = \frac{4.606}{5}$$

$$k = 0.9212 \text{ min}^{-1}$$

96. Option (1) is correct.

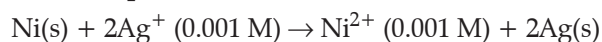
Explanation:



IUPAC name : 1-bromo-5-chloro-4 methyl hexan-3-ol

97. None of the option is correct.

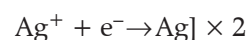
Explanation:



Oxidation half reaction:



Reduction half reaction:



Nernst equation:

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{2.303 RT}{nF} \log \frac{[\text{Ni}^{2+}]}{[\text{Ag}^+]^2}$$

$$E_{\text{cell}} = 10.5 - \frac{0.059}{2} \log \frac{(10^{-3})}{(10^{-3})^2}$$

$$E_{\text{cell}} = 10.5 - \frac{0.59}{2} \log 10^3$$

$$E_{\text{cell}} = 10.5 - \frac{0.59}{2} \times 3$$

$$E_{\text{cell}} = 10.5 - 0.885$$

$$E_{\text{cell}} = 10.4115 \text{ V}$$

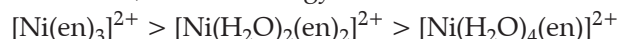
This will be the benefits, if attempted.

98. **Option (3) is correct.**

Explanation: More the strong field ligands present in the compound, greater is the splitting.

Since, en is a strong field ligand.

So, order of energy absorbed is:



(C)

(A)

(B)

99. **Option (1) is correct.**

Explanation: $a = 3.608 \times 10^{-8} \text{ cm}$ (given)

$$d = 8.92 \text{ g cm}^{-3}$$

For FCC unit cell, $Z = 4$

$$d = \frac{z \times M}{N_A \times a^3}$$

$$M = \frac{8.92 \times 6.022 \times 10^{23} \times (3.608 \times 10^{-8})^3}{4}$$

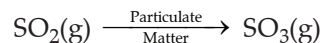
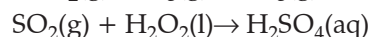
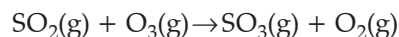
$$M = 630.73 \times 10^{-1}$$

$$M = 63.07 \text{ u}$$

$$M = 63.1 \text{ u}$$

100. **Option (2) is correct.**

Explanation: The pollution due to oxides of sulphur gets enhanced due to the presence of particulate matter, ozone and hydrogen peroxide.



□□□

BOTANY

Section A

101. **Option (3) is correct.**

Explanation: In predation, only a single species is negatively impacted.

102. **Option (3) is correct.**

Explanation: The process of degradation of detritus into simpler substances by the action of microbes is called decomposition. Decomposition rate is faster if the detritus is rich in nitrogen and water-soluble substances such as sugars, but not lignin and chitin.

103. **Option (4) is correct.**

Explanation: In monocotyledonous roots, xylem and phloem together constitute a vascular bundle. In monocots, the vascular bundles are collateral, conjoint and closed type.

104. **Option (3) is correct.**

Explanation: During apoplastic pathway, with the help of the intervening cells, water is transported from the root hair to the xylem. This movement beyond the cortex is blocked by the casparian strip of endodermal cells.

105. **Option (3) is correct.**

Explanation: All arthropods for e.g., lobsters, crabs, spiders, mites etc., consists a hard exoskeleton which is made up of a protein, chitin. This hard protective layer of arthropods give support for the attachment of arthropod's muscles.

106. **Option (2) is correct.**

Explanation: Euchromatin is loosely packed and enriched in genes, hence it is responsible for the active transcription. A typical nucleosome contains about 200 base pairs (bp) of the DNA helix. Heterochromatin is highly condensed and is transcriptionally silent.

107. **Option (1) is correct.**

Explanation: In old trees, the greater part of secondary xylem is dark brown and resistant to insect attack due to the deposition of secondary metabolites and organic compounds like tannins and resins in the central or innermost layer of the stem.

108. Option (2) is correct.

Explanation: The flowers of the plants such as bean, pea, *Cassia* and Gulmohar belonging to the family Fabaceae are zygomorphic i.e., having floral parts unequal in size or form.

109. Option (3) is correct.

Explanation: Gel electrophoresis is a technique of separation of DNA strands according to their size under the effect of an electric field across the gel. Bright orange coloured bands of DNA can be observed in the gel when exposed to UV light.

110. Option (4) is correct.

Explanation: Plasticity is an ability of an organism to change its phenotype in response to environmental condition. It is a characteristic feature in some plants such as cotton, coriander, buttercup, etc., which enables them to adapt to rapid environmental changes.

111. Option (2) is correct.

Explanation: Algae such as *Ulothrix* and *Volvox* belongs to the family chlorophyceae (green algae) and their reserved food material is in the form of starch.

112. Option (4) is correct.

Explanation: Most dominant pollinating agents among the animals, particularly among the insects are the "bees".

113. Option (4) is correct.

Explanation: In the process of glycolysis, four molecules of ATP and two molecules of NADPH are also formed, each produces 3 molecules of ATP which means total 6 ATP molecules are produced. Out of these 10ATP molecules, 2ATP molecules are initially utilised, one during the conversion of glucose to glucose 6-phosphate and the one during the conversion of fructose-6-phosphate to fructose 1, 6 bisphosphate. So, the net gain of ATP is eight molecules.

114. Option (3) is correct.

Explanation: The major causes of biodiversity loss are Habitat loss and Fragmentation, over-exploitation, Alien species invasion and co-extinction as explained in Rivet-popper hypothesis by Paul Ehrlich.

115. Option (2) is correct.

Explanation: The movement of ions across a semipermeable membrane bound structure, down their electrochemical gradient is known as chemiosmosis. ATP is produced

through chemiosmosis, which is the main molecule used by the cell as energy. Electron carriers like NADH and FADH donate electrons to the electron transport chain which cause conformational changes in the shapes of the protein molecules.

116. Option (3) is correct.

Explanation: Pairing of homologous chromosomes occurs during the prophase stage of Meiosis.

117. Option (3) is correct.

Explanation: Crossing over occurs during the pachytene stage of meiosis in which exchange of chromosomal material between maternal and paternal homologous chromosomes occur. Recombination nodules are multicomponent proteinaceous ellipsoids found associated with the synaptonemal complex during prophase-I of meiosis.

118. Option (3) is correct.

Explanation: XO type of sex determination is found in Grasshoppers, and some insects where females are homogametic and produce only single type of chromosome i.e., (XX). Males have only one X-chromosome (XO).

119. Option (3) is correct.

Explanation: Ethylene is a phytohormone responsible for inducing and promoting female flower development in cucumber.

120. Option (1) is correct.

Explanation: Mendel studied seven pairs of contrasting characters which include pea shape, pea colour, pod shape, pod colour, flower colour, plant size and position of flowers.

121. Option (4) is correct.

Explanation: Less than 7% of the energy is released from glucose during lactic acid fermentation. Not all of it is stored as ATP.

122. Option (3) is correct.

Explanation: In citrus and *Bougainvillea*, the thorns are developed from modified stem. Leaves are modified into many structures and the pointed structures are known as spines.

123. Option (3) is correct.

Explanation: Red algae belongs to the class Rhodophyceae in which they contain a red colour pigment - r-phycoerythrin in their body. Hydrocolloid carrageen can be obtained from Rhodophyceae only.

124. **Option (3) is correct.**

Explanation: Electrostatic precipitator is the device used for removing particulate matter present in the exhaust from a thermal power plant.

125. **Option (1) is correct.**

Explanation: Cleistogamy is a type of automatic self pollination of certain plants that can be seen especially in peanuts, peas etc. The advantage of such pollination is that, the flower is not dependent on pollinators but there is no chance for cross-pollination.

126. **Option (2) is correct.**

Explanation: The transportation of food material occurs with the help of plant tissue was identified by plant physiologists in "Girdling experiment". It showed that phloem is responsible for the translocation of food as it is present outside the xylem.

127. **Option (2) is correct.**

Explanation: Vexillary aestivation is the characteristic of the family Fabaceae. Hence, *Pisum sativum* shows vexillary aestivation. Stamens are said to be diadelphous when united into two bundles. As china rose has monadelphous stamens, pea plants have diadelphous stamens.

128. **Option (2) is correct.**

Explanation: Both the statements are correct but they do not show cause and effect.

129. **Option (2) is correct.**

Explanation: Ethylene is a simple, gaseous plant growth regulator which is responsible for inducing ripening in fruits and ageing tissues. It promote root growth and root hair formation to increase the absorption surface.

130. **Option (3) is correct.**

Explanation: Polymorphism is the presence of two or more variant forms of a specific DNA sequence occurring in different individuals or populations. It forms the basis of both genetic mapping and DNA fingerprinting.

131. **Option (1) is correct.**

Explanation: Phosphoenolpyruvate, found in the mesophyll cells is the primary CO_2 acceptor in C_4 plants. In C_4 plants, mesophyll cells lack RuBisCo-enzyme.

132. **Option (2) is correct.**

Explanation: *Frankia* produces nitrogen fixing nodules on the roots of *Alnus*.

133. **Option (2) is correct.**

Explanation: *Ex-situ* conservation is the conservation of species outside their natural habitat where they can be given special care and attention. For e.g., Botanical gardens, Zoos, Aquariums, Cryopreservation etc. Natural parks and wildlife sanctuaries are the *in-situ* conservation methods.

134. **Option (1) is correct.**

Explanation: As soon as the small sub-unit of ribosome encounters mRNA, the process of translation (mRNA to proteins) begins.

135. **Option (2) is correct.**

Explanation: (a) Manganese—Function in splitting of water during photosynthesis.

(b) Magnesium—Activates enzymes of respiration.

(c) Boron—Required for pollen germination.

(d) Iron—Activates the enzyme catalase.

Section B

136. **Option (2) is correct.**

Explanation: The engine in the CNG buses functions the same way as a gasoline engine. Natural gas is stored in a fuel tank, mostly at the back of the vehicle. The same diesel engine used in CNG buses is not tested as it makes the cost of conversion expensive.

137. **Option (3) is correct.**

Explanation: Genes that are located on different chromosomes assort independently. By determining the recombination frequencies, we can determine if the gene is assorting independently or not.

138. **Option (3) is correct.**

Explanation: In the given figure, i.e., T. S. of apple fruit, the part labelled 'C' is thalamus, which is a false fruit.

139. **Option (1) is correct.**

Explanation: Sequence annotation is the process of identifying the boundaries between genes and other features in a DNA sequence. This methodology is adopted by the geneticist for sequencing the whole genome of an organism.

140. **Option (2) is correct.**

Explanation: Myotonic dystrophy is the most common form of muscular dystrophy that begins at adulthood, characterised by progressive muscle weakness. It is caused due to the genetic changes and is inherited in an autosomal dominant manner.

141. Option (1) is correct.

Explanation: The interaction in which one organism consumes the body or parts of another organism is known as predation. For e.g., House cats killing mice, birds and other small insects.

142. Option (2) is correct.

Explanation: The large bundle sheath cells found around the vascular bundles in C_4 plants help to increase the number of chloroplast for the operation of Calvin cycle.

143. Option (2) is correct.

Explanation: (a) *Spirogyra*—Dominant haploid free-living gametophyte.

(b) *Fern*—dominant diploid sporophyte alternating with reduced gametophyte called prothallus.

(c) *Funaria*—Dominant haploid leafy gametophyte alternating with partially dependent multicellular sporophyte.

(d) *Cycas*—Dominant diploid sporophyte vascular plant with highly reduced male or female gametophyte.

144. Option (3) is correct.

Explanation: Lecithin is naturally found in body tissues. It is a phospholipid, a yellowish-brown fatty substance which contains glycerol, two fatty acids, phosphate group and choline. Saturated fatty acids have single bonds only. Unsaturated fatty acids can have one or more double bonds.

145. Option (2) is correct.

Explanation: Transposones can be used during Gene silencing methods. They are a

group of mobile genetic elements and can jump into different places of the genome, hence called jumping genes also.

146. Option (1) is correct.

Explanation: (a) **Metacentric chromosome**—Centromere in the middle forming two equal arms of chromosomes.

(b) **Telocentric chromosome**—Centromere situated close to the end forming one extremely short and one very long arm.

(c) **Sub-metacentric chromosome**—Centromere slightly away from the middle forming one shorter arm and one longer arm.

(d) **Telocentric chromosome**—Centromere at the terminal end.

147. Option (2) is correct.

Explanation: 5' G A A T T C 3'
3' C T T A A G 5'

In this palindromic sequence, EcoRI recognises and cuts at the restriction site and produces sticky ends.

148. Option (2) is correct.

Explanation: Cambium is very active during spring season and produces a large number of xylem elements which have vessels with wider cavities.

149. Option (3) is correct.

Explanation: Weathering of rocks accelerates the phosphorus cycle.

150. Option (2) is correct.

Explanation: Addition of more solutes in a given solution will lower its water potential and makes the solution saturated.

ZOOLOGY

Section A

151. Option (3) is correct.

Explanation: Bio-fortification is a process of breeding, in which nutritional quality of crop is enhanced with higher levels of vitamins and minerals or higher proteins and healthier fats.

152. Option (2) is correct.

Explanation: DNA replication occurs before meiosis I in S-phase of interphase, so, the statement II is incorrect. As DNA replication occurs only once during meiosis.

153. Option (1) is correct.

Explanation: Being insoluble in water, fatty acids and glycerols cannot be absorbed into the blood.

Lacteals are specialised lymphatic capillaries, which carry all dietary lipids or chylomicrons into lymphatic vessels and ultimately into the blood.

154. Option (2) is correct.

Explanation: Millions of gamete mother cells (oogonia) are formed within each fetal ovary and this process of oogenesis occurs during embryonic development stage.

155. **Option (2) is correct.**

Explanation: The process of breakdown of detritus into smaller particles by detritivores is known as Fragmentation.

156. **Option (1) is correct.**

Explanation: Death rate is expressed as the number of deaths in a given area during a given time per 1000 individuals. It can be calculated as death of individuals to that of total number of individual in a population.

So, Death rate of individuals per week

$$= \frac{8}{80} = 0.1.$$

157. **Option (4) is correct.**

Explanation: Crop and gizzard are additional chambers in digestive tract of birds. *Pavo* (Peacock), *Psittacula* (Parakeets), *Corvus* (Crow), all are birds, and possess crop and gizzard.

158. **Option (3) is correct.**

Explanation: Prokaryotes lack membrane bound organelles. So, in prokaryotes both RER and SER are absent. They do not have any kind of endoplasmic reticulum.

159. **Option (2) is correct.**

Explanation: Forewings of cockroach are called tegmina, they are emerged from mesothorax, while hind wings are emerged from metathorax.

160. **Option (4) is correct.**

Explanation: Atmospheric air, passes through nose → pharynx → larynx → trachea → bronchi → bronchioles, this system starting from nose to bronchioles, constitute conducting part of respiratory system, it filters, humidify and maintain the temperature of the gases entered the conducting system.

161. **Option (2) is correct.**

Explanation: Oxygen get bound to haemoglobin in the lung surface and get dissociated at the tissues. Every 100 mL of oxygenated blood can deliver around 5 mL of O₂ to the tissue under normal physiological conditions.

162. **Option (2) is correct.**

Explanation: During mitosis, spindle fibres attach to kinetochore of chromosome. Kinetochores are the protein, present around the centromere.

163. **Option (2) is correct.**

Explanation: White fibrous cartilage is present between the adjacent bones of vertebral column, which allows only a limited movement.

164. **Option (1) is correct.**

Explanation: The total length of double helix DNA = total number of base pairs × distance between two base pairs.

The length between two base pair is = 0.34 nm = 0.34×10^{-9} m

The length of given DNA molecule is = 1.1m.

1.1 = total number of base pairs × 0.34×10^{-9} m

$$\begin{aligned} \text{Total number of base pairs} &= \frac{1.1}{0.34 \times 10^{-9}} = \\ &= 3.23 \times 10^9 \\ 3.23 \times 10^9 &= \sim 3.3 \times 10^9. \end{aligned}$$

165. **Option (1) is correct.**

Explanation: *In-situ* means insite conservation, it is the protection and conservation of whole ecosystem in its natural habitat.

166. **Option (4) is correct.**

Explanation: Oogenesis occur at early embryonic stage, while spermatogenesis occurs at the stage of puberty.

Both spermatogenesis and oogenesis, results in the formation of haploid gametes.

167. **Option (1) is correct.**

Explanation: During embryonic development, members of vertebrata posses notochord, but the notochord is replaced by a cartilaginous or bony vertebral column in adult. Thus, all vertebrates are chordates but all chordates are not vertebrates.

168. **Option (3) is correct.**

Explanation: Mycoplasma are microscopic, they can penetrate 0.2 μm filter. They are categorised as bacteria, which lack cell wall.

169. **Option (3) is correct.**

Explanation: Rheumatoid arthritis is an autoimmune disease. In this disease, body's defence mechanism recognises its own cells as foreign bodies, and start attacking self cells.

170. **Option (4) is correct.**

Explanation: Patient suffering from adenosine deaminase deficiency require periodic infusion of genetically engineered lymphocytes because genetically engineered lymphocytes are not immortal.

171. **Option (1) is correct.**

Explanation:

Tetany—low level of Ca^{2+} causing rapid spasm.

Myasthenia gravis—Autoimmune disorder resulting in weakening and paralysis of skeletal muscle.

Muscular dystrophy—A genetic disorder causing degeneration of skeletal muscle.

172. **Option (3) is correct.**

Explanation: Spermiogenesis is release of sperms into seminiferous tubules. The process of transformation of spermatids into spermatozoa is called spermiogenesis.

173. **Option (4) is correct.**

Explanation: Reptiles, birds, lands snails and insects excrete nitrogenous waste as uric acid in the form of pellet or paste with a minimum loss of water. *Pavo* is a bird among the other options, which excretes in the form of pellet or paste.

174. **Option (1) is correct.**

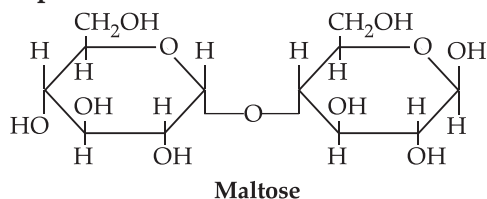
Explanation: *Trichoderma polysporum* is a fungus, which is responsible for the production of an immunosuppressive agent cyclosporin A.

175. **Option (3) is correct.**

Explanation: IUDs (Intrauterine devices) are method of birth control, inserted by doctors in uterus of females through vagina. Lippe's loop is an example of non-medicated IUD.

176. **Option (3) is correct.**

Explanation:



177. **Option (4) is correct.**

Explanation: Neuroglia are not a connective tissue, they are neural cells, which provide support to neurons.

178. **Option (4) is correct.**

Explanation: Secretions of salivary glands lubricate oral cavity, and salivary enzymes

performs partial digestion of complex carbohydrates, controls bacterial population in mouth, but do not perform the function of digestion of disaccharides.

179. **Option (1) is correct.**

Explanation: Restriction endonucleases recognise a specific palindromic nucleotide sequence in DNA. These endonucleases cut the DNA strand a little away from the centre of palindromic site.

180. **Option (2) is correct.**

Explanation: Natural selection can lead to stabilisation if more individuals acquire mean character value, and if more individual acquire value other than the mean character value, it leads to directional change.

181. **Option (3) is correct.**

Explanation: Osteoporosis is an age-related disorder, which is characterised by decrease bone mass and increased chances of fractures. Decreased level of estrogen is the most common cause of osteoporosis.

182. **Option (1) is correct.**

Explanation: "Kingdom, Phylum, class, order, Family, Genus, species." is correct ascending order of hierarchical arrangement of taxonomic categories.

183. **Option (3) is correct.**

Explanation: If the *i* gene of *E. coli* strain get mutated and its product can not bind the inducer molecule or lactose, then *i* gene will code for repressor, and due to mutation *z*, *y*, a gene will not be translated.

184. **Option (2) is correct.**

Explanation: Zoospores are asexual reproductive structure in algae. Gemmules are found in sponges.

Buds are asexual reproductive structure in *Hydra*.

185. **Option (4) is correct.**

Explanation: RBCs after completing their life span, which is of 120 days get destroyed in spleen. The coagulum or clot is formed of network of threads called fibrins.

Section B

186. **Option (4) is correct.**

Explanation: Glycogen is a storage product in animals. Globulins are proteins that can function as antibody. Many hormones are steroid in nature. Thrombin is an enzyme, that act as biocatalyst.

187. **Option (1) is correct.**

Explanation: Ciliated epithelium lines nasal cavity, bronchioles, etc.

Mucosal epithelium has goblet cells, which secrete mucus, and are glandular tissue.

Tendons are dense regular connective tissue.

Adipose tissue is a type of loose connective tissue, located mainly beneath the skin.

188. **Option (3) is correct.**

Explanation:

- Sino-atrial node (SAN) generates an action potential to stimulate atrial contraction.
- If atrial pressure is more than ventricular pressure, then it leads to closure of semilunar valves.

189. **Option (2) is correct.**

Explanation:

- Parathyroid hormone raises the level of calcium in the blood.
- It also influence calcium absorption from digested food.
- Thyroid hormones control metabolism of carbohydrate.

190. **Option (1) is correct.**

Explanation: Bacteria can be autotrophic or heterotrophic. Slime moulds are classified under kingdom Protista. Mycoplasma lack cell wall.

191. **Option (4) is correct.**

Explanation: Acquired immunity is pathogen specific, not present at the time of birth.

192. **Option (2) is correct.**

Explanation: Diaphragms cover the cervix blocking the entry of sperms.

Contraceptive pills inhibit ovulation and implantation. Intrauterine devices increases phagocytosis of sperm within Uterus.

Lactational amenorrhoea is a period of absence of menstrual cycle and ovulation following parturition.

193. **Option (2) is correct.**

Explanation: Pro-insulin does not contain extra stretch of C-peptide.

Genetically engineered insulin was not extracted from cattles and pigs, but it was artificially manufactured using *E.coli*.

194. **Option (4) is correct.**

Explanation: Recombination frequency is directly proportional to the distance between genes, So,

Let,

$$a \text{ \& } d = (X)$$

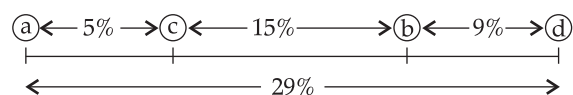
$$a \text{ \& } c = (Y)$$

$$c \text{ \& } b = (Z)$$

$$b \text{ \& } d = (A)$$

$$(X) = (Y) + (Z) + (A)$$

$$29 = 5 + 15 + 9$$



Hence, a, c, b, d is correct sequence.

195. **Option (3) is correct.**

Explanation: A new hybrid $^{15}\text{N} - ^{14}\text{N}$ is formed after 20 minutes. After 40 minutes equal hybrids of high density DNA and low density DNA will be formed.

After 60 minutes, all the DNA strands will be containing ^{14}N nucleotide.

196. **Option (4) is correct.**

Explanation: Presence of two or more recognition sites, will complicate gene cloning, hence, it is not a desirable feature of cloning vector.

197. **Option (4) is correct.**

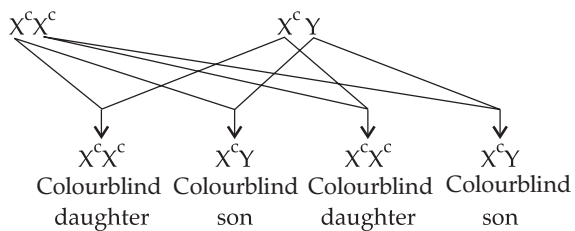
Explanation: Impulse transmission across electrical synapse is always faster than that of chemical synapse.

198. **Option (4) is correct.**

Explanation: Flippers of penguins and dolphins are a pair of analogous organs, as they perform same function but have different origin.

199. Option (4) is correct.

Explanation: If a colourblind female (X^cX^c) marries a man whose mother was also colourblind (X^cY)



100% Colourblind progeny.

200. Option (2) is correct.

Explanation: In a scrubber the exhaust is passed through a spray of water or lime.

Particulate matter (PM-2.5) cannot be removed by precipitator.

□□□

**SOLVED
PAPER**

**NEET (UG)
(Re-Examination)
4th September 2022**

**Code
U6**

Important Instructions :

- The test is of **3 hours 20 minutes** duration and Test Booklet contains **200** multiple choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**. 50 Questions in each subject are divided into two **Section (A and B)** as per details given below:
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos- 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** Questions in each subject (Question Nos- 36 to 50, 86 to 100, 136 to 150 and 80 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, the first ten questions answered by the candidate shall be evaluated.
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, one mark will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
- Use of Electronic/Manual Calculator is prohibited.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

PHYSICS

Section-A

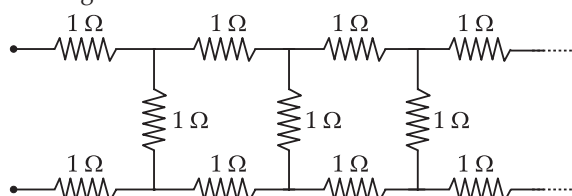
- Q. 1.** Identify the function which represents a non-periodic motion.
(1) $e^{-\omega t}$ (2) $\sin \omega t$
(3) $\sin \omega t + \cos \omega t$ (4) $\sin (\omega t + \pi/4)$
- Q. 2.** The magnetic field of a plane electromagnetic wave is given by $\vec{B} = 3 \times 10^{-8} \cos (1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{j}$, then the associated electric field will be:
(1) $3 \times 10^{-8} \cos (1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{i} \text{ V/m}$
(2) $3 \times 10^{-8} \sin (1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{i} \text{ V/m}$
(3) $9 \sin (1.6 \times 10^3 x - 48 \times 10^{10} t) \hat{k} \text{ V/m}$
(4) $9 \cos (1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{k} \text{ V/m}$
- Q. 3.** The incorrect statement about the property of a Zener diode is:
(1) Zener voltage remains constant at breakdown

- (2) It is designed to operate under reverse bias
(3) Depletion region formed is very wide
(4) p and n regions of zener diode are heavily doped
- Q. 4.** A cell of emf 4 V and internal resistance 0.5Ω is connected to a 7.5Ω external resistance. The terminal potential difference of the cell is:
(1) 3.75 V (2) 4.25 V
(3) 4 V (4) 0.375 V
- Q. 5.** Given below are two statements:
Statement-I:
In an a.c. circuit, the current through a capacitor leads the voltage across it.
Statement-II:
In a.c. circuits containing pure capacitance only, the phase difference between the current and the voltage is π .

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **statement-I** and **statement-II** are correct
- (2) Both **statement-I** and **statement-II** are incorrect
- (3) **Statement-I** is correct but **statement-II** is incorrect
- (4) **Statement-I** is incorrect but **statement-II** is correct

Q. 6. The equivalent resistance of the infinite network given below is:



- (1) 2Ω
- (2) $(1 + \sqrt{2}) \Omega$
- (3) $(1 + \sqrt{3}) \Omega$
- (4) $(1 + \sqrt{5}) \Omega$

Q. 7. A cricket ball is thrown by a player at a speed of 20 m/s in a direction 30° above the horizontal. The maximum height attained by the ball during its motion is ($g = 10 \text{ m/s}^2$):

- (1) 5 m
- (2) 10 m
- (3) 20 m
- (4) 25 m

Q. 8. A closely packed coil having 1000 turns has an average radius of 62.8 cm . If current carried by the wire of the coil is 1 A , the value of magnetic field produced at the centre of the coil will be (permeability of free space $= 4\pi \times 10^{-7} \text{ H/m}$) nearly:

- (1) 10^{-1} T
- (2) 10^{-2} T
- (3) 10^2 T
- (4) 10^{-3} T

Q. 9. An inductor of inductance 2 mH is connected to a 220 V , 50 Hz a.c. source. Let inductive reactance in the circuit is X_1 . If a 220 V d.c. source replaces the a.c. source in the circuit, then the inductive reactance in the circuit is X_2 . X_1 and X_2 respectively are:

- (1) 6.28Ω , zero
- (2) 6.28Ω , infinity
- (3) 0.628Ω , zero
- (4) 0.628Ω , infinity

Q. 10. During a cloudy day, a primary and a secondary rainbow may be created, then the:

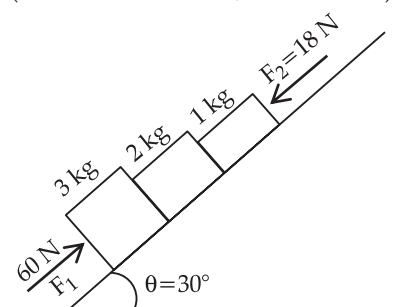
- (1) primary rainbow is due to double internal reflection and is formed above the secondary one.
- (2) primary rainbow is due to double internal reflection and is formed below the secondary one.
- (3) secondary rainbow is due to double internal reflection and is formed above the primary one.

(4) secondary rainbow is due to single internal reflection and is formed above the primary one.

Q. 11. The light rays having photons of energy 4.2 eV are falling on a metal surface having a work function of 2.2 eV . The stopping potential of the surface is:

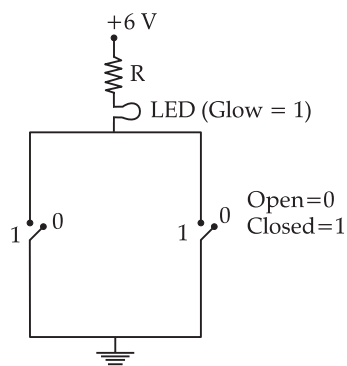
- (1) 2 eV
- (2) 2 V
- (3) 1.1 V
- (4) 6.4 V

Q. 12. In the diagram shown, the normal reaction force between 2 kg and 1 kg is (Given $g = 10 \text{ ms}^{-2}$) (Consider the surface, to be smooth):



- (1) 25 N
- (2) 39 N
- (3) 6 N
- (4) 10 N

Q. 13.



Identify the equivalent logic gate represented by the given circuit:

- (1) OR
- (2) NOR
- (3) AND
- (4) NAND

Q. 14. Two copper vessels A and B have the same base area but of different shapes. A takes twice the volume of water as that B requires to fill upto a particular common height. Then the correct statement among the following is:

- (1) Pressure on the base area of vessels A and B is same.
- (2) Pressure on the base area of vessels A and B is not same.
- (3) Both vessels A and B weigh the same.
- (4) Vessel B weighs twice that of A.

Q. 15. The distance between the two plates of a parallel plate capacitor is doubled and the area of each

plate is halved. If C is its initial capacitance, its final capacitance is equal to:

- (1) $2C$ (2) $C/2$
(3) $4C$ (4) $C/4$

Q. 16. The terminal velocity of a copper ball of radius 5 mm falling through a tank of oil at room temperature is 10 cm s^{-1} . If the viscosity of oil at room temperature is $0.9 \text{ kg m}^{-1} \text{ s}^{-1}$, the viscous drag force is:

- (1) $8.48 \times 10^{-3} \text{ N}$ (2) $8.48 \times 10^{-5} \text{ N}$
(3) $4.23 \times 10^{-3} \text{ N}$ (4) $4.23 \times 10^{-6} \text{ N}$

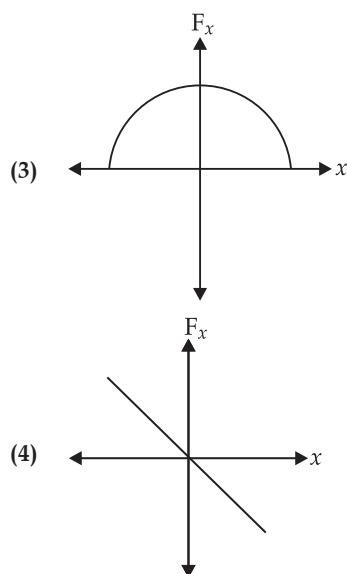
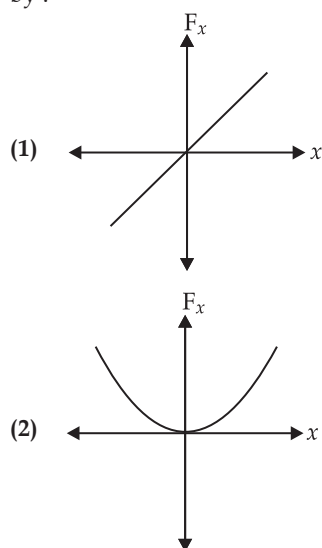
Q. 17. If $\vec{F} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{r} = 3\hat{i} + 2\hat{j} - 2\hat{k}$, then the scalar and vector products of $\vec{F}$ and $\vec{r}$ have the magnitudes respectively as:

- (1) $5, \sqrt{3}$ (2) $4, \sqrt{5}$
(3) $10, \sqrt{2}$ (4) $10, 2$

Q. 18. After passing through a polariser a linearly polarised light of intensity I is incident on an analyser making an angle of 30° with that of the polariser. The intensity of light emitted from the analyser will be :

- (1) $\frac{I}{2}$ (2) $\frac{I}{3}$
(3) $\frac{3I}{4}$ (4) $\frac{2I}{4}$

Q. 19. The restoring force of a spring with a block attached to the free end of the spring is represented by :



Q. 20. If the screen is moved away from the plane of the slits in a Young's double slit experiment, then the:

- (1) angular separation of the fringes increases
(2) angular separation of the fringes decreases
(3) linear separation of the fringes increases
(4) linear separation of the fringes decreases

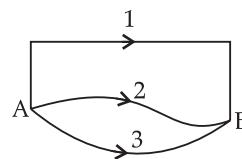
Q. 21. The effective capacitances of two capacitors are $3 \mu\text{F}$ and $16 \mu\text{F}$, when they are connected in series and parallel respectively. The capacitance of two capacitors are:

- (1) $10 \mu\text{F}, 6 \mu\text{F}$ (2) $8 \mu\text{F}, 8 \mu\text{F}$
(3) $12 \mu\text{F}, 4 \mu\text{F}$ (4) $1.2 \mu\text{F}, 1.8 \mu\text{F}$

Q. 22. The distance covered by a body of mass 5 g having linear momentum 0.3 kg m/s in 5 s is:

- (1) 300 m (2) 30 m
(3) 3 m (4) 0.3 m

Q. 23. A gravitational field is present in a region and a mass is shifted from A to B through different paths as shown. If W_1 , W_2 and W_3 represent the work done by the gravitational force along the respective paths, then



- (1) $W_1 = W_2 = W_3$
(2) $W_1 > W_2 > W_3$
(3) $W_1 > W_3 > W_2$
(4) $W_1 < W_2 < W_3$

Q. 24. The reciprocal of resistance is:

- (1) reactance (2) mobility
(3) conductivity (4) conductance

- Q. 25.** Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R),

Assertion (A):

When a fire cracker (rocket) explodes in mid air, its fragments fly in such a way that they continue moving in the same path, which the fire cracker would have followed, had it not exploded

Reason (R):

Explosion of cracker (rocket) occurs due to internal forces only and no external force acts for this explosion.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 (3) (A) is correct but (R) is not correct
 (4) (A) is not correct but (R) is correct

- Q. 26.** The threshold frequency of a photoelectric metal is ν_0 . If light of frequency $4\nu_0$ is incident on this metal, then the maximum kinetic energy of emitted electrons will be:

- (1) $h\nu_0$ (2) $2h\nu_0$
 (3) $3h\nu_0$ (4) $4h\nu_0$

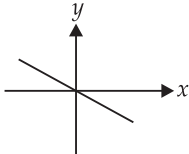
- Q. 27.** The ratio of the magnitude of the magnetic field and electric field intensity of a plane electromagnetic wave in free space of permeability μ_0 and permittivity ϵ_0 is (Given that c - velocity of light in free space):

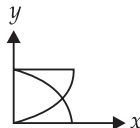
- (1) c (2) $\frac{1}{c}$
 (3) $\frac{c}{\sqrt{\mu_0\epsilon_0}}$ (4) $\frac{\sqrt{\mu_0\epsilon_0}}{c}$

- Q. 28.** The shape of the magnetic field lines due to an infinite long, straight current carrying conductor is:

- (1) a straight line (2) circular
 (3) elliptical (4) a plane

- Q. 29.** Match List - I with List - II:

	List - I (x-y graphs)		List - II (Situations)
(a)		(i)	Total mechanical energy is conserved
(b)		(ii)	Bob of a pendulum is oscillating under negligible air friction

(c)		(iii)	Restoring force of a spring
(d)		(iv)	Bob of a pendulum is oscillating along with air friction

Choose the **correct** answer from the options given below:

- (1) (a) - (iv), (b) - (i), (c) - (iii), (d) - (i)
 (2) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
 (3) (a) - (i), (b) - (iv), (c) - (iii), (d) - (i)
 (4) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)

- Q. 30.** Given below are two statements:

Statement I:

The law of radioactive decay states that the number of nuclei undergoing the decay per unit time is inversely proportional to the total number of nuclei in the sample.

Statement II:

The half life of a radionuclide is the sum of the life time of all nuclei, divided by the initial concentration of the nuclei at time $t = 0$.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both **Statement I** and **Statement II** are correct
 (2) Both **Statement I** and **Statement II** are incorrect
 (3) **Statement I** is correct but **Statement II** is incorrect
 (4) **Statement I** is incorrect but **Statement II** is correct

- Q. 31.** An ideal gas follows a process described by the equation $PV^2 = C$ from the initial (P_1, V_1, T_1) to final (P_2, V_2, T_2) thermodynamic states, where C is a constant. Then:

- (1) If $P_1 > P_2$ then $T_1 < T_2$
 (2) If $V_2 > V_1$ then $T_2 > T_1$
 (3) If $V_2 > V_1$ then $T_2 < T_1$
 (4) If $P_1 > P_2$ then $V_1 > V_2$

- Q. 32.** A standard filament lamp consumes 100 W when connected to 200 V a.c. mains supply. The peak current through the bulb will be :

- (1) 0.707 A (2) 1 A
 (3) 1.414 A (4) 2 A

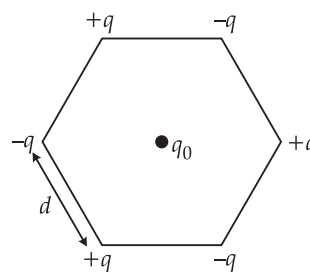
- Q. 33.** Let R_1 be the radius of the second stationary orbit and R_2 be the radius of the fourth stationary orbit of an electron in Bohr's model. The ratio $\frac{R_1}{R_2}$ is:

- (1) 0.25 (2) 0.5
 (3) 2 (4) 4

- Q. 34. The physical quantity that has the same dimensional formula as pressure is:
 (1) Force
 (2) Momentum
 (3) Young's modulus of elasticity
 (4) Coefficient of viscosity
- Q. 35. An energy of 484 J is spent in increasing the speed of a flywheel from 60 rpm to 360 rpm. The moment of inertia of the flywheel is:
 (1) 0.7 kg-m^2 (2) 3.22 kg-m^2
 (3) 30.8 kg-m^2 (4) 0.07 kg-m^2

Section-B

- Q. 36. The magnetic flux linked to a circular coil of radius R is:
 $\phi = 2t^3 + 4t^2 + 2t + 5 \text{ Wb}$
 The magnitude of induced emf in the coil at $t = 5 \text{ s}$ is:
 (1) 108 V (2) 197 V
 (3) 150 V (4) 192 V
- Q. 37. An astronomical refracting telescope is being used by an observer to observe planets in normal adjustment. The focal lengths of the objective and eye piece used in the construction of the telescope are 20 m and 2 cm respectively. Consider the following statements about the telescope:
 (a) The distance between the objective and eye piece is 20.02 m
 (b) The magnification of the telescope is $(-)$ 1000
 (c) The image of the planet is erect and diminished
 (d) The aperture of eye piece is smaller than that of objective
 The correct statements are:
 (1) (a), (b) and (c) (2) (b), (c) and (d)
 (3) (c), (d) and (a) (4) (a), (b) and (d)
- Q. 38. At any instant, two elements X_1 and X_2 have same number of radioactive atoms. If the decay constant of X_1 and X_2 are 10λ and λ respectively. Then the time when the ratio of their atoms becomes $\frac{1}{e}$ respectively will be:
 (1) $\frac{1}{11\lambda}$ (2) $\frac{1}{6\lambda}$
 (3) $\frac{1}{6\lambda}$ (4) $\frac{1}{5\lambda}$
- Q. 39. Six charges $+q, -q, +q, -q, +q$ and $-q$ are fixed at the corners of a hexagon of side d as shown in the figure. The work done in bringing a charge q_0 to the centre of the hexagon from infinity is (ϵ_0 - permittivity of free space):



- (1) Zero (2) $\frac{-q^2}{4\pi\epsilon_0 d}$
 (3) $\frac{-q^2}{4\pi\epsilon_0 d} \left(3 - \frac{1}{\sqrt{2}} \right)$ (4) $\frac{-q^2}{4\pi\epsilon_0 d} \left(6 - \frac{1}{\sqrt{2}} \right)$
- Q. 40. An organ pipe filled with a gas at 27°C resonates at 400 Hz in its fundamental mode. If it is filled with the same gas at 90°C , the resonance frequency at the same mode will be:
 (1) 420 Hz (2) 440 Hz
 (3) 484 Hz (4) 512 Hz
- Q. 41. The position-time ($x - t$) graph for positive acceleration is:
 (1)
- (2)
- (3)
- (4)
- Q. 42. The collector current in a common base amplifier using $n-p-n$ transistor is 24 mA. If 80% of the electrons released by the emitter is accepted by the collector, then the base current is numerically:
 (1) 6 mA and leaving the base
 (2) 3 mA and leaving the base
 (3) 6 mA and entering the base
 (4) 3 mA and entering the base

Q. 43. Three vessels of equal capacity have gases at the same temperature and pressure. The first vessel contains helium (monoatomic), the second contains fluorine (diatomic) and the third contains sulfur hexafluoride (polyatomic). The correct statement, among the following is:

- (1) All vessels contain unequal number of respective molecules
- (2) The root mean square speed of molecules is same in all three cases
- (3) The root mean square speed of helium is the largest
- (4) The root mean square speed of sulfur hexafluoride is the largest

Q. 44. In a gravitational field, the gravitational potential is given by, $V = \frac{K}{x}$ (J/kg). The gravitational field intensity at point (2, 0, 3) m is:

- (1) $+\frac{K}{2}$
- (2) $-\frac{K}{2}$
- (3) $-\frac{K}{4}$
- (4) $+\frac{K}{4}$

Q. 45. Two very long, straight, parallel conductors A and B carry current of 5 A and 10 A respectively and are at a distance of 10 cm from each other. The direction of current in two conductors is same. The force acting per unit length between two conductors is ($\mu_0 = 4\pi \times 10^{-7}$ SI unit):

- (1) $2 \times 10^{-4} \text{ Nm}^{-1}$ and is attractive
- (2) $2 \times 10^{-4} \text{ Nm}^{-1}$ and is repulsive
- (3) $1 \times 10^{-4} \text{ Nm}^{-1}$ and is attractive
- (4) $1 \times 10^{-4} \text{ Nm}^{-1}$ and is repulsive

Q. 46. The magnetic field on the axis of a circular loop of radius 100 cm carrying current $I = \sqrt{2}$ A, at point 1 m away from the centre of the loop is given by:

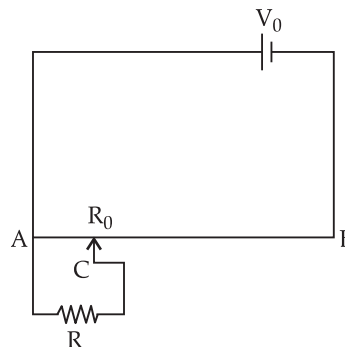
- (1) $3.14 \times 10^{-7} \text{ T}$
- (2) $6.28 \times 10^{-7} \text{ T}$
- (3) $3.14 \times 10^{-4} \text{ T}$
- (4) $6.28 \times 10^{-4} \text{ T}$

Q. 47. Two rods one made of copper and other made of steel of the same length and same cross sectional area are joined together. The thermal conductivity of copper and steel are $385 \text{ J s}^{-1} \text{ K}^{-1} \text{ m}^{-1}$ and $50 \text{ J s}^{-1} \text{ m}^{-1}$ respectively. The free ends of copper

and steel are held at 100°C and 0°C respectively. The temperature at the junction is, nearly:

- (1) 12°C
- (2) 50°C
- (3) 73°C
- (4) 88.5°C

Q. 48. The sliding contact C is at one fourth of the length of the potential wire (AB) from A as shown in the circuit diagram. If the resistance of the wire AB is R_0 , then the potential drop (V) across the resistor R is:



- (1) $\frac{4V_0R}{3R_0 + 16R}$
- (2) $\frac{4V_0R}{3R_0 + R}$
- (3) $\frac{2V_0R}{4R_0 + R}$
- (4) $\frac{2V_0R}{2R_0 + 3R}$

Q. 49. The ratio of coulomb's electrostatic force to the gravitational force between an electron and a proton separated by some distance is 2.4×10^{39} . The ratio of the proportionality constant, $K =$

$\frac{1}{4\pi\epsilon_0}$ to the Gravitational constant G is nearly

(Given that the charge of the proton and electron each = $1.6 \times 10^{-19} \text{ C}$, the mass of the electron = $9.11 \times 10^{-31} \text{ kg}$, the mass of the proton = $1.67 \times 10^{-27} \text{ kg}$):

- (1) 10^{20}
- (2) 10^{30}
- (3) 10^{40}
- (4) 10

Q. 50. The percentage error in the measurement of g is

(Given that $g = \frac{4\pi^2 L}{T^2}$, $L = (10 \pm 0.1) \text{ cm}$, $T = (100 \pm 1) \text{ s}$):

- (1) 2%
- (2) 5%
- (3) 3%
- (4) 7%

□□□

CHEMISTRY

Section A

Q. 51. The correct order of bond angles in the following compounds/species is :

- (1) $\text{H}_2\text{O} < \text{NH}_3 < \text{NH}_4^+ < \text{CO}_2$
- (2) $\text{H}_2\text{O} < \text{NH}_4^+ < \text{NH}_3 < \text{CO}_2$
- (3) $\text{H}_2\text{O} < \text{NH}_4^+ = \text{NH}_3 < \text{CO}_2$
- (4) $\text{CO}_2 < \text{NH}_3 < \text{H}_2\text{O} < \text{NH}_4^+$

Q. 52. K_H value for some gases at the same temperature 'T' are given:

gas	$K_H/\text{k bar}$
Ar	40.3
CO_2	1.67
HCHO	1.83×10^{-5}
CH_4	0.413

where K_H is Henry's Law constant in water. The order of their solubility in water is :

- (1) $\text{Ar} < \text{CO}_2 < \text{CH}_4 < \text{HCHO}$
- (2) $\text{Ar} < \text{CH}_4 < \text{CO}_2 < \text{HCHO}$
- (3) $\text{HCHO} < \text{CO}_2 < \text{CH}_4 < \text{Ar}$
- (4) $\text{HCHO} < \text{CH}_4 < \text{CO}_2 < \text{Ar}$

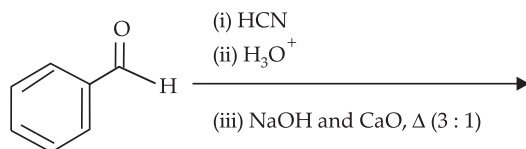
Q. 53. Which of the following reactions is a part of the large scale industrial preparation of nitric acid ?

- (1) $\text{NaNO}_3 + \text{H}_2\text{SO}_4 \xrightarrow[500\text{K}, 9\text{ bar}]{\text{Pt}} \text{NaHSO}_4 + \text{HNO}_3$
- (2) $4\text{NH}_3 + 5\text{O}_2 (\text{from air}) \xrightarrow[500\text{K}, 9\text{ bar}]{\text{Pt}} 4\text{NO} + 6\text{H}_2\text{O}$
- (3) $4\text{HPO}_3 + 2\text{N}_2\text{O}_5 \xrightarrow[500\text{K}, 9\text{ bar}]{\text{Pt}} 4\text{HNO}_3 + \text{P}_4\text{O}_{10}$
- (4) $\text{Cu}(\text{NO}_3)_2 + 2\text{NO}_2 + 2\text{H}_2\text{O} \xrightarrow[500\text{K}, 9\text{ bar}]{\text{Pt}} 4\text{HNO}_3 + \text{Cu}$

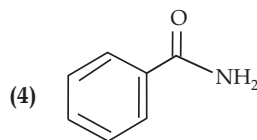
Q. 54. CaCl_2 and $\text{Ca}(\text{OCl})_2$ are components of :

- (1) gypsum
- (2) Portland cement
- (3) bleaching powder
- (4) lime water

Q. 55. The product formed from the following reaction sequence is :



- (1)
- (2)
- (3)



Q. 56. Match List-I with List-II :

List-I (Reaction)	List-II (Product formed)
(a) Gabriel synthesis	(i) Benzaldehyde
(b) Kolbe synthesis	(ii) Ethers
(c) Williamson synthesis	(iii) Primary amines
(d) Etard synthesis	(iv) Salicylic acid

Choose the correct answer from the options given below :

- (1) (a) - (iii), (b) - (i), (c) - (ii), (d) - (iv)
- (2) (a) - (ii), (b) - (iii), (c) - (i), (d) - (iv)
- (3) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)
- (4) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

Q. 57. Match List-I with List-II

List-I	List-II
(a) Sodium laurylsulphate	(i) Toilet soap
(b) Cetyltrimethyl ammonium chloride	(ii) Non-ionic detergent
(c) Sodium stearate	(iii) Anionic detergent
(d) Polyethyleneglycol stearate	(iv) Cationic detergent

Choose the correct answer from the options given below :

- (1) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)
- (2) (a) - (i), (b) - (iv), (c) - (ii), (d) - (iii)
- (3) (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)
- (4) (a) - (iii), (b) - (i), (c) - (ii), (d) - (iv)

Q. 58. Which of the following reactions is a decomposition redox reaction ?

- (1) $2\text{Pb}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{PbO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$
- (2) $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}(\text{g})$
- (3) $\text{Cl}_2(\text{g}) + 2\text{OH}(\text{aq}) \rightarrow \text{ClO}^-(\text{aq}) + \text{Cl}^-(\text{aq}) + 4\text{H}_2\text{O}(\text{l})$
- (4) $\text{P}_4(\text{s}) + 3\text{OH}(\text{aq}) + 3\text{H}_2\text{O}(\text{l}) \rightarrow \text{PH}_3(\text{g}) + 3\text{H}_2\text{PO}_2^-(\text{aq})$

Q. 59. If first ionization enthalpies of element X and Y are 419 kJ mol^{-1} and 590 kJ mol^{-1} , respectively and second ionization enthalpies of X and Y are 3069 kJ mol^{-1} and 1145 kJ mol^{-1} , respectively.

Then **correct** statement is :

- (1) X is an alkali metal and Y is an alkaline earth metal.
- (2) X is an alkaline earth metal and Y is an alkali metal.

- (3) Both X and Y are alkali metals.
 (4) Both X and Y are alkaline earth metals.
- Q. 60. Predict the order of reactivity of the following four isomers towards S_N2 reaction.
- (I) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$
 (II) $\text{CH}_3\text{CH}_2\text{CH}(\text{Cl})\text{CH}_3$
 (III) $(\text{CH}_3)_2\text{CHCH}_2\text{Cl}$
 (IV) $(\text{CH}_3)_3\text{CCl}$
- (1) (IV) > (III) > (II) > (I)
 (2) (I) > (II) > (III) > (IV)
 (3) (I) > (III) > (II) > (IV)
 (4) (IV) > (II) > (III) > (I)
- Q. 61. Match List-I with List-II :

List-I (Molecules)	List-II (Shape)
(a) NH_3	(i) Square pyramidal
(b) ClF_3	(ii) Trigonal bipyramidal
(c) PCl_5	(iii) Trigonal pyramidal
(d) BrF_5	(iv) T-shape

Choose the correct answer from the options given below :

- (1) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)
 (2) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)
 (3) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)
 (4) (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)
- Q. 62. Which among the following is a thermoplastic polymer ?
- (1) Bakelite
 (2) Polythene
 (3) Urea-formaldehyde resin
 (4) Melamine polymer

- Q. 63. Match List-I with List-II :

List-I (Compounds)	List-II (Molecular formula)
(a) Borax	(i) NaBO_2
(b) Kernite	(ii) $\text{Na}_2\text{B}_4\text{O}_7 \cdot 4\text{H}_2\text{O}$
(c) Orthoboric acid	(iii) H_3BO_3
(d) Borax bead	(iv) $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$

Choose the correct answer from the options given below :

- (1) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)
 (2) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
 (3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
 (4) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)
- Q. 64. Two half cell reactions are given below :
- $\text{CO}^{3+} + \text{e}^- \rightarrow \text{Co}^{2+}$, $E^\circ_{\text{Co}^{2+}/\text{Co}^{3+}} = -1.81 \text{ V}$
 $2\text{Al}^{3+} + 6\text{e}^- \rightarrow 2\text{Al}(\text{s})$, $E^\circ_{\text{Al}^{3+}/\text{Al}} = +1.66 \text{ V}$
 The standard EMF of a cell with feasible redox reaction will be :
- (1) +7.09 V (2) +0.15 V
 (3) +3.47 V (4) -3.47 V
- Q. 65. The element used for welding metals with high melting points is :
- (1) Cl_2 (2) H_2
 (3) Ne (4) He

- Q. 66. $\text{Na}_2\text{B}_4\text{O}_7 \xrightarrow{\text{heat}} \text{X} + \text{NaBO}_2$
 in the above reaction the product "X" is :

- (1) H_3BO_3 (2) B_2O_3
 (3) $\text{Na}_2\text{B}_2\text{O}_5$ (4) NaB_3O_5

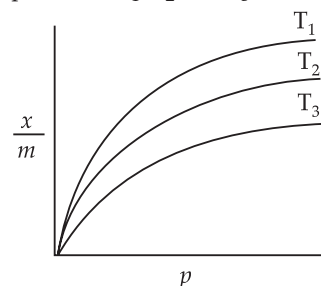
- Q. 67. The correct order of first ionization enthalpy for the given four element is :

- (1) $\text{C} < \text{N} < \text{F} < \text{O}$ (2) $\text{C} < \text{N} < \text{O} < \text{F}$
 (3) $\text{C} < \text{O} < \text{N} < \text{F}$ (4) $\text{C} < \text{F} < \text{N} < \text{O}$

- Q. 68. 0.01 M acetic acid solution is 1% ionised, then pH of this acetic acid solution is :

- (1) 3 (2) 2
 (3) 4 (4) 1

- Q. 69. Shown below are adsorption isotherms for a gas 'X' at temperatures T_1 , T_2 and T_3 :



p and $\frac{x}{m}$ represent pressure and extent of adsorption, respectively. The correct order of temperatures for the given adsorption is:

- (1) $T_1 > T_2 > T_3$ (2) $T_3 > T_2 > T_1$
 (3) $T_1 = T_2 = T_3$ (4) $T_1 = T_2 > T_3$

- Q. 70. The half life of a first order reaction is 2000 years. If the concentration after 8000 years is 0.02 M, then the initial concentration was :

- (1) 0.16 M (2) 0.32 M
 (3) 0.08 M (4) 0.04 M

- Q. 71. One mole of an ideal gas at 300 K is expanded isothermally from 1 L to 10 L volume. ΔU for this process is

(Use $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)

- (1) 1260 J (2) 2520 J
 (3) 5040 J (4) 0 J

- Q. 72. What is the hybridization shown by C_1 and C_2 carbons, respectively in the given compound ?



- (1) sp^2 and sp^3 (2) sp^2 and sp^2
 (3) sp^3 and sp^2 (4) sp^3 and sp^3

- Q. 73. The density of the solution is 2.15 g mL^{-1} , then mass of 2.5 mL solution in correct significant figures is

- (1) $5375 \times 10^{-3} \text{ g}$ (2) 5.4 g
 (3) 5.38 g (4) 53.75 g

- Q. 74. Flourine is a stronger oxidising agent than chlorine because :

- (1) F-F bond has a low enthalpy of dissociation.

- (2) Fluoride ion (F^-) has high hydration enthalpy
 (3) Electron gain enthalpy of fluorine is less negative than chlorine.
 (4) Fluorine has a very small size.

Choose the most appropriate answer from the options given :

- (1) (a) and (b) only (2) (a) and (c) only
 (3) (a) and (d) only (4) (b) and (c) only

Q. 75. Match List-I with List-II :

List-I (Complexes)	List-II (Types)
(a) $[Co(NH_3)_5NO_2]Cl_2$ and $[Co(NH_3)_5ONO]Cl_2$	(i) ionisation isomerism
(b) $[Cr(NH_3)_6][Co(CN)_6]$ and $[Cr(CN)_6][Co(NH_3)_6]$	(ii) coordination isomerism
(c) $[Co(NH_3)_5(SO_4)]Br$ and $[Co(NH_3)_5Br]SO_4$	(iii) linkage isomerism
(d) $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$	(iv) solvate isomerism

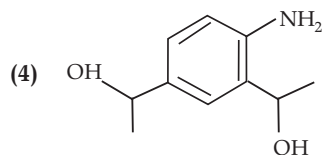
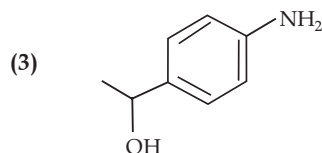
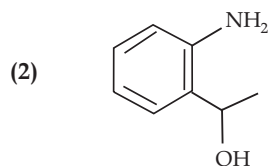
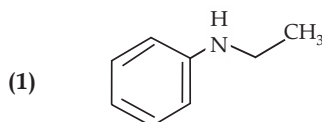
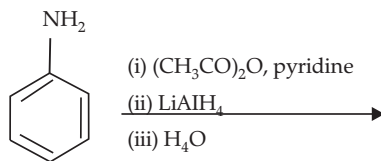
Choose the **correct answer** from the options given below :

- (1) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
 (2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
 (3) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)
 (4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

Q. 76. The incorrect statement about denaturation of proteins is

- (1) It results due to change of temperature and/or pH
 (2) It results in loss of biological activity of proteins.
 (3) A protein is formed from amino acids linked by peptide bonds.
 (4) Uncoiling of the helical structure takes place.

Q. 77. The product formed from the following reaction sequence is



Q. 78. Match List-I with List-II :

List-I (Defects)	List-II (Shown by)
(a) Frenkel defect	(i) non-ionic solids and density of the solid decreases
(b) Schottky defect	(ii) non-ionic solids and density of the solid increases
(c) Vacancy defect	(iii) ionic solids and density of the solid decreases
(d) Interstitial defect	(iv) ionic solids and density of the solid remains constant.

Choose the **correct answer** from the options given below :

- (1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
 (2) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
 (3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
 (4) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

Q. 79. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Chlorine is an electron withdrawing group but it is ortho, para directing in electrophilic aromatic substitution.

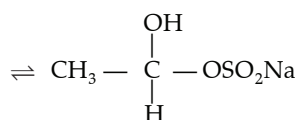
Reason (R) : Inductive effect of chlorine destabilises the intermediate carbocation formed during the

electrophilic substitution, however due to the more pronounced resonance effect, the halogen stabilises the carbocation at ortho and para positions.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct
- (4) (A) is not correct but (R) is correct

Q. 80. Which of the following reactions is not an example for nucleophilic addition-elimination reaction?



- (2) $\text{CH}_3\text{CHO} + \text{NH}_2\text{OH} \rightleftharpoons \text{CH}_3\text{CH} = \text{N}-\text{OH} + \text{H}_2\text{O}$
- (3) $\text{CH}_3\text{CHO} + \text{C}_6\text{H}_5\text{NHNH}_2 \rightleftharpoons \text{CH}_3\text{CH} = \text{N}-\text{NHC}_6\text{H}_5 + \text{H}_2\text{O}$
- (4) $\text{CH}_3\text{CHO} + \text{NH}_3 \rightleftharpoons \text{CH}_3\text{CH} = \text{NH} + \text{H}_2\text{O}$

Q. 81. Four gas cylinders containing He, N_2 , CO_2 and NH_3 gases separately are gradually cooled from a temperature of 500 K. Which gas will liquify first?

(Given T_c in K – He : 5.3, N_2 : 126, CO_2 : 304.1 and NH_3 : 405.5)

- (1) He
- (2) N_2
- (3) CO_2
- (4) NH_3

Q. 82. Decrease in size from left to right in actinoid series is greater and gradual than that in lanthanoid series due to:

- (1) 4f orbitals are penultimate
- (2) 4f orbitals have greater shielding effect
- (3) 5f orbitals have poor shielding effect
- (4) 5f orbitals have greater shielding effect

Q. 83. The decreasing order of boiling points of the following alkanes is:

- (a) Heptane
- (b) butane
- (c) 2-methylbutane
- (d) 2-methylpropane
- (e) hexane

Choose the **correct answer** from the options given below:

- (1) (a) > (c) > (e) > (d) > (b)
- (2) (c) > (d) > (a) > (e) > (b)
- (3) (a) > (e) > (b) > (c) > (d)
- (4) (a) > (e) > (c) > (b) > (d)

Q. 84. Match the reagents (**List-I**) with the product (**List-II**) obtained from phenol.

	List-I		List-II
(a)	(i) NaOH (ii) CO_2 (iii) H^+	(i)	Benzoquinone
(b)	(i) Aqueous NaOH + CHCl_3 (ii) H^+	(ii)	Benzene
(c)	Zn duct, Δ	(iii)	Salicyl aldehyde
(d)	$\text{Na}_2\text{Cr}_2\text{O}_7$, H_2SO_4	(iv)	Salicylic acid

Choose the **correct answer** from the options given below :

- (1) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
- (2) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
- (3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
- (4) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

Q. 85. Match List-I with List-II:

	List-I (quantum number)		List-II (Orbital)
(a)	$n = 2, l = 1$	(i)	2s
(b)	$n = 3, l = 2$	(ii)	3s
(c)	$n = 3, l = 0$	(iii)	2p
(d)	$n = 2, l = 0$	(iv)	3d

Choose the **correct answer** from the options given below :

- (1) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
- (2) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
- (3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
- (4) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

Section B

Q. 86. Which one of the following is not a calcination reaction?

- (1) $\text{ZnCO}_3 \xrightarrow{\Delta} \text{ZnO} + \text{CO}_2$
- (2) $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O} \xrightarrow{\Delta} \text{Fe}_2\text{O}_3 + x\text{H}_2\text{O}$
- (3) $\text{CaCO}_3 \cdot \text{MgCO}_3 \xrightarrow{\Delta} \text{CaO} + \text{MgO} + 2\text{CO}_2$
- (4) $\text{CaCO}_3 \cdot 2\text{HCl} \xrightarrow{\Delta} \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$

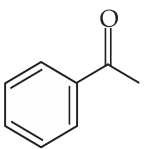
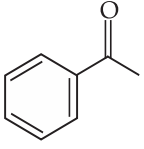
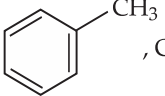
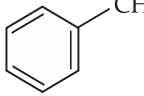
Q. 87. When electromagnetic radiation of wavelength 300 nm falls on the surface of a metal, electrons are emitted with the kinetic energy of $1.68 \times 10^5 \text{ J mol}^{-1}$. What is the minimum energy needed to remove an electron from the metal?

$$(h = 6.626 \times 10^{-34} \text{ Js}, c = 3 \times 10^8 \text{ ms}^{-1},$$

$$N_A = 6.022 \times 10^{23} \text{ mol}^{-1})$$

- (1) $2.31 \times 10^6 \text{ J mol}^{-1}$
- (2) $3.84 \times 10^4 \text{ J mol}^{-1}$
- (3) $3.84 \times 10^{-19} \text{ J mol}^{-1}$
- (4) $2.31 \times 10^5 \text{ J mol}^{-1}$

- Q. 88. For a chemical reaction
 $4A + 3B \rightarrow 6C + 9D$
 rate of formation of C is $6 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$ and
 rate of disappearance of A is $4 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$.
 The rate of reaction and amount of B consumed
 in interval of 10 seconds, respectively will be:
- (1) $1 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$ and $30 \times 10^{-2} \text{ mol L}^{-1}$
 - (2) $10 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$ and $10 \times 10^{-2} \text{ mol L}^{-1}$
 - (3) $1 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$ and $10 \times 10^{-2} \text{ mol L}^{-1}$
 - (4) $10 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$ and $30 \times 10^{-2} \text{ mol L}^{-1}$
- Q. 89. The incorrect method for the synthesis of alkenes is:
- (1) treatment of alkynes with Na in liquid NH_3
 - (2) heating alkyl halides with alcoholic KOH
 - (3) treating alkyl halides in aqueous KOH solution
 - (4) treating vicinal dihalides with Zn metal
- Q. 90. The incorrect method to synthesize benzaldehyde is:

- (1)  Cl , H_2 , Pd-BaSO₄
- (2)  OC_2H_5 , DIBAL-H, followed by H_2O
- (3)  CH_3 , CrO_2Cl_2 , followed by H_3O^+ in CS_2
- (4)  CH_3 , CH_3MgBr , followed by H_3O^+

- Q. 91. What fraction of Fe exists as Fe(III) in $\text{Fe}_{0.96}\text{O}$?
 (Consider $\text{Fe}_{0.96}$ to be made up of Fe(II) and Fe(III) only)
- (1) $\frac{1}{12}$ (2) 0.08
 - (3) $\frac{1}{16}$ (4) $\frac{1}{20}$

- Q. 92. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A):

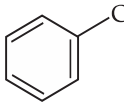
The metal carbon bond in metal carbonyls possesses both σ and π character.

Reason (R):

The ligand to metal bond is a π bond and metal to ligand bond is a σ bond.

In the light of the above statements, choose the

most appropriate answer from the options given below:

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
 - (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 - (3) (A) is correct but (R) is not correct.
 - (4) (A) is not correct but (R) is correct.
- Q. 93. Which one of the following reaction sequence is incorrect method to prepare phenol?
- (1) Aniline, $\text{NaNO}_2 + \text{HCl}$, H_2O , heating
 - (2) Cumene, O_2 , H_3O^+
 - (3) , NaOH, STP condition
 - (4) , oleum, NaOH, H_3O^+

- Q. 94. A vessel contains 3.2 g of dioxygen gas at STP (273.15 K and 1 atm pressure). The gas is now transferred to another vessel at constant temperature, where pressure becomes one third of the original pressure. The volume of new vessel in L is :

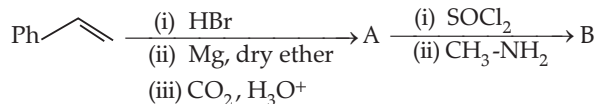
(Given - molar volume at STP is 22.4 L)

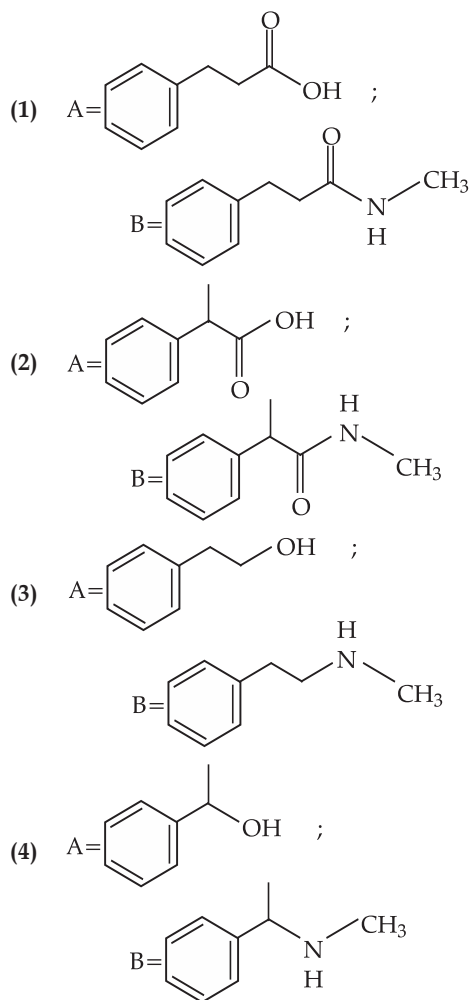
- (1) 6.72 (2) 2.24
 - (3) 22.4 (4) 67.2
- Q. 95. Match List-I with List-II:

	List-I		List-II
(a)	Biochemical oxygen demand	(i)	Oxidising mixture
(b)	Photochemical smog	(ii)	Polar stratospheric cloud
(c)	Classical smog	(iii)	organic matter in water
(d)	Ozone depletion	(iv)	reducing mixture

Choose the **correct answer** from the options given below:

- (1) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)
 - (2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
 - (3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
 - (4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
- Q. 96. The products A and B in the following reaction sequence are :





Q. 97. Given below are two statements:

Statement I:

Cr^{2+} is oxidising and Mn^{3+} is reducing in nature.

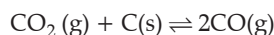
Statement II:

Sc^{3+} compounds are repelled by the applied magnetic field.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement I and Statement II are correct
- (2) Both Statement I and Statement II are incorrect
- (3) Statement I is correct but Statement II is incorrect
- (4) Statement I is incorrect but Statement II is correct

Q. 98. K_p for the following reaction is 3.0 at 1000 K.



What will be the value of K_c for the reaction at the same temperature?

(Given: $R = 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1}$)

- (1) 0.36
- (2) 3.6×10^{-2}
- (3) 3.6×10^{-3}
- (4) 3.6

Q. 99. Standard electrode potential for the cell with cell reaction

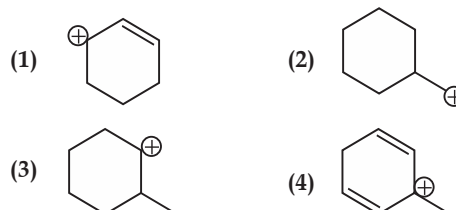


is 1.1 V. Calculate the standard Gibbs energy change for the cell reaction.

(Given $F = 96487 \text{ C mol}^{-1}$)

- (1) $-200.27 \text{ kJ mol}^{-1}$
- (2) $-212.27 \text{ kJ mol}^{-1}$
- (3) $-212.27 \text{ J mol}^{-1}$
- (4) $-200.27 \text{ J mol}^{-1}$

Q. 100. Which of the following is the most stable carbocation?



□□□

BOTANY

Section A

Q. 101. Which stage of meiosis can last for months or years in the oocytes of some vertebrates?

- (1) Leptotene
- (2) Pachytene
- (3) Diplotene
- (4) Diakinesis

Q. 102. When one CO_2 molecule is fixed as one molecule of triose phosphate, which of the following photochemically made, high energy chemical intermediates are used in the reduction phase?

- (1) 1 ATP + 1 NADPH
- (2) 1 ATP + 2 NADPH
- (3) 2 ATP + 1 NADPH

(4) 2 ATP + 2 NADPH

Q. 103. In *lac* operon, *z* gene codes for:

- (1) β -galactosidase
- (2) Permease
- (3) Repressor
- (4) Transacetylase

Q. 104. Initiation of lateral roots and vascular cambium during secondary growth takes place in cells of:

- (1) Epiblema
- (2) Cortex
- (3) Endodermis
- (4) Pericycle

Q. 105. Match List - I with List - II:

List - I	List - II
(a) Adenine	(i) Pigment

(b) Anthocyanin	(ii) Polysaccharide
(c) Chitin	(iii) Alkaloid
(d) Codeine	(iv) Purine

Choose the **correct answer** from the options given below :

- (1) (a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)
 (2) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
 (3) (a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)
 (4) (a) - (i), (b) - (iv), (c) - (iii), (d) - (ii)

Q. 106. Match List - I with List - II :

List - I	List - II
(a) In <i>lac</i> operon <i>i</i> gene codes for	(i) transacetylase
(b) In <i>lac</i> operon <i>z</i> gene codes for	(ii) permease
(c) In <i>lac</i> operon <i>y</i> gene codes for	(iii) β -galactosidase
(d) In <i>lac</i> operon <i>a</i> gene	(iv) Repressor codes for

Choose the **correct answer** from the options given below :

- (1) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)
 (2) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
 (3) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)
 (4) (a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)

Q. 107. Match List - I with List - II :

List - I	List - II
(a) <i>Chlamydomonas</i>	(i) Moss
(b) <i>Cycas</i>	(ii) Pteridophyte
(c) <i>Selaginella</i>	(iii) Alga
(d) <i>Sphagnum</i>	(iv) Gymnosperm

Choose the **correct answer** from the options given below :

- (1) (a) - (iii), (b) - (i), (c) - (ii), (d) - (iv)
 (2) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)
 (3) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)
 (4) (a) - (ii), (b) - (iii), (c) - (i), (d) - (iv)

Q. 108. Given below are two statements :

Statement I :

DNA polymerases catalyse polymerisation only in one direction, that is $5' \rightarrow 3'$

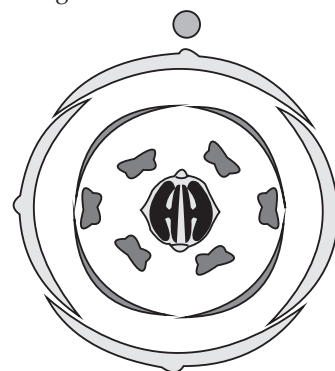
Statement II :

During replication of DNA, on one strand the replication is continuous while on other strand it is discontinuous.

In the light of the above statements, choose the **correct answer** from the options given below :

- (1) Both Statement I and Statement II are correct
 (2) Both Statement I and Statement II are incorrect
 (3) Statement I is correct but Statement II is incorrect
 (4) Statement I is incorrect but Statement II is correct

Q. 109. The Floral Diagram represents which one of the following families ?



- (1) Fabaceae (2) Brassicaceae
 (3) Solanaceae (4) Liliaceae

Q. 110. The pioneer species in a hydrarch succession are:

- (1) Free-floating angiosperms
 (2) Submerged rooted plants
 (3) Phytoplanktons
 (4) Filamentous algae

Q. 111. The number of time(s) decarboxylation of isocitrate occurs during single TCA cycle is :

- (1) One (2) Two
 (3) Three (4) Four

Q. 112. Given below are two statements :

Statement I :

Sickle cell anaemia and Haemophilia are autosomal dominant traits .

Statement II :

Sickle cell anaemia and Haemophilia are disorders of the blood.

In the light of the above statements, choose the **correct answer** from the options given below :

- (1) Both Statement I and Statement II are correct
 (2) Both Statement I and Statement II are incorrect
 (3) Statement I is correct but Statement II is incorrect
 (4) Statement I is incorrect but Statement II is correct

Q. 113. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) :

When a particular restriction enzyme cuts strand of DNA, overhanging stretches or sticky ends are formed.

Reason (R) :

Some restriction enzymes cut the strand of DNA a little away from the centre of palindromic site.

In the light of the above statements, choose the **correct answer** from the options given below :

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 (3) (A) is correct but (R) is not correct
 (4) (A) is not correct but (R) is correct

Q. 114. Give the correct descending order of organisms with reference to their estimated number found in Amazon forest.

- (a) Plants (b) Invertebrates
 (c) Fishes (d) Mammals
 (e) Birds

Choose the **correct answer** from the options given below :

- (1) (a) > (b) > (e) > (d) > (c)
 (2) (a) > (c) > (d) > (b) > (e)
 (3) (b) > (a) > (e) > (d) > (c)
 (4) (b) > (a) > (c) > (e) > (d)

Q. 115. Match List - I with List - II :

List - I	List - II
(a) Porins nodules	(i) Pink coloured
(b) <i>leg</i> haemoglobin	(ii) Lumen of thylakoid
(c) H ⁺ accumulation	(iii) Amphibolic pathway
(d) Respiration	(iv) Huge pores in outer membrane of mitochondria

Choose the correct answer from the options given below :

- (1) (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)
 (2) (a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)
 (3) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)
 (4) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

Q. 116. Which of the following growth regulators is an adenine derivative ?

- (1) Auxin (2) Cytokinin
 (3) Ethylene (4) Absciscic acid

Q. 117. The type of tissue commonly found in the fruit wall of nuts is :

- (1) Parenchyma (2) Collenchyma
 (3) Sclerenchyma (4) Sclereid

Q. 118. The species that come to appear in bare area are called :

- (1) Pioneer species
 (2) Invasive species
 (3) Competitive species
 (4) Species of seral community

Q. 119. In general the egg apparatus of embryo sac in angiosperm consists of :

- (1) One egg cell, two synergids, three antipodal cells, two Polar nuclei
 (2) One egg cell, two synergids, two antipodal cells, three Polar nuclei
 (3) One egg cell, three synergids, two antipodal cells, two Polar nuclei
 (4) One egg cell, two synergids, two antipodal cells, two Polar nuclei

Q. 120. Match List - I with List - II :

List - I	List - II
(a) Imbricate	(i) <i>Calotropis</i>
(b) Valvate	(ii) <i>Cassia</i>
(c) Vexillary	(iii) Cotton
(d) Twisted	(iv) Bean

Choose the **correct answer** from the options given below :

- (1) (a) - (ii), (b) - (i), (c) - (iii), (d) - (iv)
 (2) (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)
 (3) (a) - (ii), (b) - (iv), (c) - (iii), (d) - (i)
 (4) (a) - (i), (b) - (iii), (c) - (iv), (d) - (ii)

Q. 121. All successions irrespective of the habitat proceed to which type of climax community ?

- (1) Xeric (2) Mesic
 (3) Hydrophytic (4) Edaphic

Q. 122. Separation of DNA, fragments is done by a technique known as :

- (1) Polymerase Chain Reaction
 (2) Recombinant technology
 (3) Southern blotting
 (4) Gel electrophoresis

Q. 123. The phenomenon by which the undividing parenchyma cells start to divide mitotically during plant tissue culture is called as :

- (1) Differentiation (2) Dedifferentiation
 (3) Redifferentiation (4) Secondary growth

Q. 124. In meiosis, crossing over and exchange of genetic material between homologous chromosomes are catalyzed by the enzyme.

- (1) Phosphorylase (2) Recombinase
 (3) Transferase (4) Polymerase

Q. 125. The 5-C compound formed during TCA cycle is :

- (1) α -ketoglutaric acid
 (2) Oxalo succinic acid
 (3) Succinic acid
 (4) Fumaric acid

Q. 126. When a carrier protein facilitates the movement of two molecules across the membrane in same direction, it is called :

- (1) Uniport (2) Transport
 (3) Antiport (4) Symport

Q. 127. The World Summit on sustainable development held in 2002 in Johannesburg, South Africa pledged for:

- (1) A significant reduction in the current rate of biodiversity loss.
 (2) Declaration of more biodiversity hotspots.
 (3) Increase in agricultural production
 (4) Collection and preservation of seeds of different genetic strains of commercially important plants.
- Q. 128. Interfascicular cambium is present between :
 (1) Primary xylem and primary phloem
 (2) Pericycle and endodermis
 (3) Two vascular bundles
 (4) Secondary xylem and secondary phloem
- Q. 129. The ascent of xylem sap in plants is mainly accomplished by the :
 (1) size of the stomatal aperture
 (2) distribution of stomata on the upper and lower epidermis
 (3) cohesion and adhesion between water molecules
 (4) root pressure
- Q. 130. Which of the following statement is not correct?
 (1) Rhizome is a condensed form of stem
 (2) The apical bud in rhizome always remains above the ground
 (3) The rhizome is aerial with no distinct nodes and internodes
 (4) The rhizome is thick, prostrate and branched
- Q. 131. To ensure that only the desired pollens fall on the stigma in artificial hybridization process :
 (a) the female flower buds of plant producing unisexual flower need not be bagged.
 (b) there is no need to emasculate unisexual flowers of selected female parent
 (c) emasculated flowers are to be bagged immediately after cross pollination
 (d) emasculated flowers are to be bagged after removal of anthers
 (e) bisexual flowers, showing protogyny are never selected for cross
 Choose the **correct answer** from the options given below :
 (1) (a), (b) and (c) only
 (2) (b), (c) and (d) only
 (3) (b), (c) and (e) only
 (4) (a), (d) and (e) only
- Q. 132. The residual persistent part which forms the perisperm in the seeds of beet is :
 (1) Calyx (2) Endosperm
 (3) Nucellus (4) Integument
- Q. 133. The chromosomal theory of inheritance was proposed by :
 (1) Thomas Morgan (2) Sutton and Boveri
 (3) Gregor Mendel (4) Robert Brown
- Q. 134. Which of the following protects nitrogenase inside the root nodule of a leguminous plant?
 (1) Catalase
 (2) leg haemoglobin

- (3) Transaminase
 (4) Glutamate dehydrogenase
- Q. 135. The ability of plants to follow different pathways in response to environment leading to formation of different kinds of structures is called :
 (1) Redifferentiation (2) Development
 (3) Plasticity (4) Differentiation

Section B

- Q. 136. Which of the following pair represents free living nitrogen fixing aerobic bacteria?
 (1) *Rhizobium* and *Frankia*
 (2) *Azotobacter* and *Beijerinckia*
 (3) *Anabaena* and *Rhodospirillum*
 (4) *Pseudomonas* and *Thiobacillus*
- Q. 137. Match List-I with List-II :

List - I	List - II
(a) Sacred groves	(i) Alien species
(b) Zoological park	(ii) Release of large quantity of oxygen
(c) Nile perch	(iii) <i>Ex-situ</i> conservation
(d) Amazon forest	(iv) Khasi Hills in Meghalaya

Choose the **correct answer** from the options given below :

- (1) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)
 (2) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)
 (3) (a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)
 (4) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)
- Q. 138. Primary proteins are also called as polypeptides because :
 (1) They are linear chains
 (2) They are polymers of peptide monomers
 (3) Successive amino acids are joined by peptide bonds
 (4) They can assume many conformations
- Q. 139. Match List-I with List-II :

List - I	List - II
(a) Gene gun	(i) Replacement of a faulty gene by a normal healthy gene
(b) Gene therapy	(ii) Used for transfer of gene
(c) Gene cloning	(iii) Total DNA in the cells of an organism
(d) Genome	(iv) To obtain identical copies of a particular DNA molecule

Choose the **correct answer** from the options given below :

- (1) (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)
 (2) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)
 (3) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)
 (4) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

Q. 140. Match List-I with List-II :

List - I	List - II
(a) Bacteriophage $\phi \times 174$	(i) 48502 base pairs
(b) Bacteriophage lambda	(ii) 5386 nucleotides
(c) <i>Escherichia coli</i>	(iii) 3.3×10^9 base pairs
(d) Haploid content of human DNA	(iv) 4.6×10^6 base pairs

Choose the **correct answer** from the options given below:

- (1) (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)
 (2) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)
 (3) (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)
 (4) (a) - (i), (b) - (ii), (c) - (iv), (d) - (iii)

Q. 141. Which of the following can be expected if scientists succeed in introducing apomictic gene into hybrid varieties of crops ?

- (1) Polyembryony will be seen and each seed will produce many plantlets
 (2) Seeds of hybrid plants will show longer dormancy
 (3) Farmers can keep on using the seeds produced by the hybrids to raise new crop year after year
 (4) There will be segregation of the desired characters only in the progeny

Q. 142. Read the following statements and identify the characters related to the alga shown in the diagram:

- (a) It is a member of Chlorophyceae
 (b) Food is stored in the form of starch
 (c) It is a monoecious plant showing oogonium and antheridium
 (d) Food is stored in the form of laminarin or mannitol
 (e) It shows dominance of pigments chlorophyll a, c and Fucoxanthin.

Choose the **correct answer** from the options given below :



- (1) (a) and (b) only
 (2) (a), (b) and (c) only
 (2) (a), (c) and (d) only
 (4) (c), (d) and (e) only

Q. 143. Frugivorous birds are found in large numbers in tropical forests mainly because of :

- (1) lack of niche specialisation
 (2) higher annual rainfall
 (3) availability of fruits throughout the year
 (4) temperature conducive for their breeding

Q. 144. Which type of substance would face difficulty to pass through the cell membrane?

- (1) Substance with hydrophobic moiety
 (2) Substance with hydrophilic moiety
 (3) All substance irrespective of hydrophobic and hydrophilic moiety
 (4) Substance soluble in lipids

Q. 145. Identify the correct statements regarding chemiosmotic hypothesis :

- (a) Splitting of the water molecule takes place on the inner side of the membrane.
 (b) Protons accumulate within the lumen of the thylakoids.
 (c) Primary acceptor of electron transfers the electrons to an electron carrier.
 (d) NADP reductase enzyme is located on the stroma side of the membrane.
 (e) Protons increase in number in stroma.

Choose the **correct answer** from the options given below:

- (a) (a), (b) and (e) (2) (a), (b) and (d)
 (3) (b), (c) and (d) (4) (b), (c) and (e)

Q. 146. If a female individual is with small round head, furrowed tongue, partially open mouth and broad palm with characteristic palm crease. Also the physical, psychomotor and mental development is retarded. The karyotype analysis of such an individual will show :

- (1) 47 chromosomes with XXY sex chromosomes
 (2) 45 chromosomes with XO sex chromosomes
 (3) 47 chromosomes with XYY sex chromosomes
 (4) Trisomy of chromosome 21

Q. 147. Identify the **correct** sequence of events during Prophase I of meiosis :

- (a) Synapsis of homologous chromosomes
 (b) Chromosomes become gradually visible under microscope
 (c) Crossing over between non-sister chromatids of homologous chromosomes
 (d) Terminalisation of chiasmata
 (e) Dissolution of synaptonemal complex

Choose the **correct answer** from the options given below :

- (1) (a), (b), (c), (d), (e) (2) (b), (c), (d), (e), (a)
 (3) (b), (a), (c), (e), (d) (4) (a), (c), (d), (e), (b)

Q. 148. The enzyme (a) is needed for isolating genetic material from plant cells and enzyme (b) for isolating genetic material from fungus. Choose the correct pair of options from the following :

- (1) (a) Cellulase (b) Protease
 (2) (a) Cellulase (b) Chitinase
 (3) (a) Chitinase (b) Lipase
 (4) (a) Cellulase (b) Lipase

Q. 149. Match the List-I with List-II :

List - I	List - II
(a) Carbon dissolved in oceans	(i) 55 billion tons
(b) Annual fixation of carbon through photosynthesis	(ii) 71%
(c) PAR captured by plants	(iii) 4×10^3 kg
(d) Productivity of oceans	(iv) 2 to 10%

Choose the **correct answer** from the options given below :

- (1) (a) - (ii), (b) - (iv), (c) - (iii), (d) - (i)
 (2) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)
 (3) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)
 (4) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)

Q. 150. What is the expected percentage of F_2 progeny with yellow and inflated pod in dihybrid cross experiment involving pea plants with green coloured, inflated pod and yellow coloured constricted pod ?

- (1) 100% (2) 56.25%
 (3) 18.75% (4) 9%

ZOOLOGY

Section A

Q. 151. Pathogenic bacteria gain resistance to antibiotics due to changes in their :

- (1) Cosmids (2) Plasmids
 (3) Nucleus (4) Nucleoid

Q. 152. Milk of transgenic 'Cow Rosie' was nutritionally more balanced product for human babies than natural cow milk because it contained:

- (1) Human protein α -1-antitrypsin
 (2) Human alpha-lactalbumin
 (3) Human insulin-like growth factor
 (4) Human enzyme Adenosine Deaminase (ADA)

Q. 153. If the pH in lysosomes is increased to alkaline, what will be the outcome?

- (1) Hydrolytic enzymes will function more efficiently
 (2) Hydrolytic enzymes will become inactive
 (3) Lysosomal enzymes will be released into the cytoplasm
 (4) Lysosomal enzymes will be more active

Q. 154. Choose the incorrect enzymatic reaction :

- (1) Maltose $\xrightarrow{\text{Maltase}}$ Glucose + Galactose
 (2) Sucrose $\xrightarrow{\text{Sucrase}}$ Glucose + Fructose
 (3) Lactose $\xrightarrow{\text{Lactase}}$ Glucose + Galactose
 (4) Dipeptides $\xrightarrow{\text{Dipeptidases}}$ Amino acids

Q. 155. Which of the following reasons is mainly responsible for graft rejection in transplantation of organs ?

- (1) Inability of recipient to differentiate between 'self' and 'non-self' tissues/cells
 (2) Humoral immune response only
 (3) Auto-immune response
 (4) Cell-mediated response

Q. 156. If DNA contained sulphur instead of phosphorus and proteins contained phosphorus instead of sulfur, what would have been the outcome of Hershey and Chase experiment?

- (1) No radioactive sulfur in bacterial cells
 (2) Both radioactive sulfur and phosphorus in bacterial cells
 (3) Radioactive sulfur in bacterial cells
 (4) Radioactive phosphorus in bacterial cells

Q. 157. Two butterfly species are competing for the same nectar of a flower in a garden. To survive and coexist together, they may avoid competition in the same garden by:

- (1) feeding at the same time
 (2) choosing different foraging patterns
 (3) increasing time spent on attacking each other
 (4) predated on each other

Q. 158. Mad cow disease in cattle and Cr Jacob disease in humans are due to infection by _____.

- (1) Bacterium (2) Virus
 (3) Viroid (4) Prion

Q. 159. Which of the following is not an Intra Uterine device?

- (1) Progestogens (2) Multiload 375
 (3) Lippes loop (4) Progestasert

Q. 160. Match List-I with List-II :

List - I	List - II
(a) <i>Chlamydomonas</i>	(i) Conidia
(b) <i>Penicillium</i>	(ii) Zoospores
(c) <i>Hydra</i>	(iii) Gemmules
(d) <i>Sponge</i>	(iv) Buds

Choose the **correct answer** from the options given below:

- (1) (a) - (i), (b) - (iv), (c) - (iii), (d) - (ii)
 (2) (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)
 (3) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)
 (4) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

- Q. 161.** According to the sliding filament theory:
 (1) Actin and myosin filaments slide over each other to increase the length of the sarcomere.
 (2) Length of A-band does not change.
 (3) I-band increases in length
 (4) The actin filaments slide away from A-band resulting in shortening of sarcomere.
- Q. 162.** The amount of biomass or organic matter produced per unit area over a time period by plants during photosynthesis is called:
 (1) Secondary production
 (2) Primary production
 (3) Gross primary production
 (4) Net primary production
- Q. 163.** Given below are two statements:
Statement I :
 Amino acids have a property of ionizable nature of $-NH_2$ and $-COOH$ groups, hence have different structures at different pH.
Statement II :
 Amino acids can exist as Zwitterionic form at acidic and basic pH.
 In the light of the above statements, choose the **most appropriate answer** from the options given below:
 (1) Both Statement I and Statement II are correct
 (2) Both Statement I and Statement II are Incorrect
 (3) Statement I is correct but Statement II is incorrect
 (4) Statement I is incorrect but Statement II is correct
- Q. 164.** Which of the following types of epithelium is present in the bronchioles and Fallopian tubes?
 (1) Simple squamous epithelium
 (2) Simple columnar epithelium
 (3) Ciliated epithelium
 (4) Stratified squamous epithelium
- Q. 165.** Western Ghats have a large number of plants and animal species that are not found anywhere else. Which of the following term is used to notify such species?
 (1) Threatened species
 (2) Keystone species
 (3) Endemic species
 (4) Vulnerable species
- Q. 166.** Gout is a type of disorder which leads to:
 (1) Inflammation of joints due to accumulation of uric acid crystals
 (2) Weakening of bones due to decreased bone mass
 (3) Inflammation of joints due to cartilage degeneration
 (4) Weakening of bones due to low calcium level
- Q. 167.** Which of the following statements are correct with respect to vital capacity?

- (a) It includes ERV, TV and IRV
 (b) Total volume of air a person can inspire after a normal expiration
 (c) The maximum volume of air a person can breathe in after forced expiration
 (d) It includes ERV, RV and IRV.
 (e) The maximum volume of air a person can breath out after a forced inspiration.
 Choose the **most appropriate answer** from the options given below:

- (1) (b), (d) and (e) (2) (a), (c) and (d)
 (3) (a), (c) and (e) (4) (a) and (e)

- Q. 168.** A unique vascular connection between the digestive tract and liver is called _____.
 (1) Hepato-pancreatic system
 (2) Hepatic portal system
 (3) Renal portal system
 (4) Hepato-cystic system

- Q. 169.** Match **List-I** with **List-II** regarding the organs of Cockroach:

List - I	List - II
(a) Crop	(i) grinding the food particles
(b) Proventriculus	(ii) secretion of digestive juice
(c) Hepatic caecae	(iii) removal of nitrogenous waste
(d) Malpighian tubules	(iv) storage of food

Choose the **correct answer** from the options given below:

- (1) (a) – (iv), (b) – (i), (c) – (ii), (d) – (iii)
 (2) (a) – (iii), (b) – (ii), (c) – (i), (d) – (iv)
 (3) (a) – (ii), (b) – (iv), (c) – (i), (d) – (iii)
 (4) (a) – (i), (b) – (iv), (c) – (iii), (d) – (ii)

- Q. 170.** Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)** :

Assertion (A) :

Spirulina is a microbe that can be used for reducing environmental pollution.

Reason (R) :

Spirulina is a rich source of protein, carbohydrates, fats, minerals and vitamins.

In the light of the above statements, choose the **most appropriate answer** from the options given below:

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 (3) (A) is correct but (R) is not correct
 (4) (A) is not correct but (R) is correct

- Q. 171.** Panspermia, an idea that is still a favourite for some astronomers, means :
- (1) Creation of life from dead and decaying matter
 - (2) Creation of life from chemicals
 - (3) Origin of sperm in human testes
 - (4) Transfer of spores as unit of life from other planets of Earth
- Q. 172.** Arrange the components of mammary gland. (from proximal to distal)
- (a) Mammary duct
 - (b) Lactiferous duct
 - (c) Alveoli
 - (d) Mammary ampulla
 - (e) Mammary tubules
- Choose the most appropriate answer from the options given below :
- (1) (c) → (a) → (d) → (e) → (b)
 - (2) (b) → (c) → (e) → (d) → (a)
 - (3) (c) → (e) → (a) → (d) → (b)
 - (4) (e) → (c) → (d) → (b) → (a)
- Q. 173.** Select the **incorrect** match regarding the symbols used in Pedigree analysis:
- (1)  - Sex unspecified
 - (2)  - Affected individual
 - (3)  - Consanguineous mating
 - (4)  - Parent with male child affected with disease
- Q. 174.** Why CNG is considered better fuel than diesel?
- (a) It can not be adulterated
 - (b) It takes less time to fill the fuel tank
 - (c) It burns more efficiently
 - (d) It is cheaper
 - (e) It is less inflammable.
- Choose the **most appropriate** answer from the options given below
- (1) (a), (b), (c), (e) only
 - (2) (a), (c), (d) only
 - (3) (a), (b), (d), (e) only
 - (4) (c), (d), (e) only
- Q. 175.** Which of the following methods is not commonly used for introducing foreign DNA into the plant cell?
- (1) *Agrobacterium* mediated transformation
 - (2) Gene gun
 - (3) 'Disarmed pathogen' vectors
 - (4) Bacteriophages
- Q. 176.** Identify the region of human brain which has pneumotaxic centre that alters respiratory rate by reducing the duration of inspiration.
- (1) Medulla
 - (2) Pons
 - (3) Thalamus
 - (4) Cerebrum
- Q. 177.** How many secondary spermatocytes are required to form 400 million spermatozoa ?
- (1) 50 million
 - (2) 100 million
 - (3) 200 million
 - (4) 400 million
- Q. 178.** Choose the **correct** statement about a muscular tissue :
- (1) Skeletal muscle fibres are uninucleated and found in parallel bundles.
 - (2) Intercalated discs allow the cardiac muscle cells to contract as a unit.
 - (3) The walls of blood vessels are made up of columnar epithelium.
 - (4) Smooth muscles are multinucleated and involuntary.
- Q. 179.** Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
- Assertion (A) :**
FSH which interacts with membrane bound receptors does not enter the target cell.
- Reason (R) :**
Binding of FSH to its receptors generates second messenger (cyclic AMP) for its biochemical and physiological responses.
- In the light of the above statements, choose the **most appropriate** answer from the options given below ;
- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
 - (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 - (3) (A) is correct but (R) is not correct
 - (4) (A) is not correct but (R) is correct
- Q. 180.** Which of the following animals has three chambered heart ?
- (1) *Scoliodon*
 - (2) *Hippocampus*
 - (3) *Chelone*
 - (4) *Pteropus*
- Q. 181.** Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
- Assertion (A) :**
During pregnancy the level of thyroxine is increased in the maternal blood.
- Reason (R) :**
Pregnancy is characterised by metabolic changes in the mother.
- In the light of the above statements, choose the **most appropriate** answer from the options given below :
- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
 - (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 - (3) (A) is correct but (R) is not correct
 - (4) (A) is not correct but (R) is correct
- Q. 182.** Select the **Incorrect** statements with respect to Cyclostomes :
- (a) They lack scales and paired fins.
 - (b) They have circular mouth with Jaws.

(c) They bear 6-15 pairs of gills.
 (d) They migrate to deep sea for spawning.
 Choose **the most appropriate answer** from the options given below :

- (1) (a) and (b) only
- (2) (b) and (c) only
- (3) (b) and (d) only
- (4) (a) and (d) only

Q. 183. Role of enamel is to :

- (1) Connect crown of tooth with its root.
- (2) Masticate the food.
- (3) Form bolus.
- (4) Give basic shape to the teeth.

Q. 184. Choose the **correct** statements :

- (a) Bones support and protect softer tissues and organs
- (b) Weight bearing function is served by limb bones
- (c) Ligament is the site of production of blood cells.
- (d) Adipose tissue is specialised to store fats.
- (e) Tendons attach one bone to another.

Choose **the most appropriate answer** from the options given below :

- (1) (a), (b) and (d) only
- (2) (b), (c) and (e) only
- (3) (a), (c) and (d) only
- (4) (a), (b) and (e) only

Q. 185. Bivalent or Tetrad formation is a characteristic feature observed during :

- (1) Synaptonemal complex in zygotene stage
- (2) Chiasmata in Diplotene stage
- (3) Synaptonemal complex in Pachytene stage
- (4) Chiasmata in zygotene stage

Section B

Q. 186. Which of the following are true about the taxonomical aid 'key' ?

- (a) Keys are based on the similarities and dissimilarities.
- (b) Key is analytical in nature.
- (c) Keys are based on the contrasting characters in pair called couplet.
- (d) Same key can be used for all taxonomic categories.
- (e) Each statement in the key is called Lead.

Choose **the most appropriate answer** from the options given below :

- (1) (a), (b) and (c) only
- (2) (b), (c) and (d) only
- (3) (a), (b), (c) and (e) only
- (4) (a), (c), (d) and (e) only

Q. 187. A normal girl, whose mother is haemophilic marries a male with no ancestral history of haemophilia.

What will be the possible phenotypes of the offsprings ?

- (a) Haemophilic son and haemophilic daughter.
- (b) Haemophilic son and carrier daughter.
- (c) Normal daughter and normal son.
- (d) Normal son and haemophilic daughter.

Choose **the most appropriate answer** from the options given below :

- (1) (a) and (b) only
- (2) (b) and (c) only
- (3) (a) and (d) only
- (4) (b) and (d) only

Q. 188. IUDs are small objects made up of plastic or copper that are inserted in the uterine cavity. Which of the following statements are correct about IUDs ?

- (a) IUDs decrease phagocytosis of sperm within the uterus.
- (b) The released copper ions suppress the sperm motility.
- (c) IUDs do not make the cervix hostile to the sperm.
- (d) IUDs suppress the fertilization capacity of sperm.
- (e) The IUDs require surgical intervention for their insertion in the uterine cavity.

Choose **the most appropriate answer** from the options given below :

- (1) (a), (d) and (e) only
- (2) (b) and (c) only
- (3) (b) and (d) only
- (4) (d) only

Q. 189. Refer to the following statements for agarose-gel electrophoresis :

- (a) Agarose is a natural polymer obtained from sea-weed.
- (b) The separation of DNA molecules in agarose-gel electrophoresis depends on the size of DNA.
- (c) The DNA migrates from negatively-charged electrode to the positively-charged electrode
- (d) The DNA migrates from positively-charged electrode to the negatively-charged electrode.

Choose **the most appropriate answer** from the options given below :

- (1) (a) and (b) only
- (2) (a), (b) and (c) only
- (3) (a), (b) and (d) only
- (4) (b), (c) and (d) only

Q. 190. Match List - I with List - II

List - I	List - II
(a) Multipolar neuron	(i) Somatic neural system
(b) Bipolar neuron	(ii) Cerebral cortex
(c) Myelinated nerve fibre	(iii) Retina of Eye
(d) Unmyelinated nerve fibre	(iv) Spinal nerves

Choose the **correct answer** from the options given below :

- (1) (a) – (iii), (b) – (i), (c) – (iv), (d) – (ii)
 (2) (a) – (ii), (b) – (iv), (c) – (iii), (d) – (i)
 (3) (a) – (ii), (b) – (iii), (c) – (i), (d) – (iv)
 (4) (a) – (ii), (b) – (iii), (c) – (iv), (d) – (i)

Q. 191. Excretion in cockroach is performed by all, EXCEPT :

- (1) Urecoase glands (2) Malpighian tubules
 (3) Fat body (4) Hepatic caeca

Q. 192. Select the **correct** statement regarding mutation theory of evolution.

- (1) This theory was proposed by Alfred Wallace
 (2) Variations are small directional changes
 (3) Single step large mutation is a cause of speciation
 (4) Large differences due to mutations arise gradually in a population

Q. 193. Arrange the following formed elements in the decreasing order of their abundance in blood in humans :

- (a) Platelets (b) Neutrophils
 (c) Erythrocytes (d) Eosinophils
 (e) Monocytes

Choose the **most appropriate answer** from the options given below :

- (1) (c), (a), (b), (e), (d)
 (2) (c), (b), (a), (e), (d)
 (3) (d), (e), (b), (a), (c)
 (4) (a), (c), (b), (d), (e)

Q. 194. In the enzyme which catalyses the breakdown of: $\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{O}_2$ the prosthetic group is :

- (1) Nicotinamide adenine dinucleotide
 (2) Haem
 (3) Zinc
 (4) Niacin

Q. 195. Against the codon 5' UAC 3', what would be the sequence of anticodon on tRNA ?

- (1) 5' AUG 3' (2) 5' ATG 3'
 (3) 5' GTA 3' (4) 5' GUA 3'

Q. 196. Select the **correct** statements.

- (a) Angiotensin II activates the cortex of adrenal gland to release aldosterone.
 (b) Aldosterone leads to increase in blood pressure.
 (c) ANF acts as a check on renin-angiotensin mechanism.
 (d) ADH causes vasodilation.
 (e) Vasopressin is released from adenohypophysis.

Choose the **most appropriate answer** from the options given below :

- (1) (a), (b) and (e) only
 (2) (c), (d) and (e) only
 (3) (b), (c) and (d) only
 (4) (a), (b) and (c) only

Q. 197. With respect to metaphase, which of the following statements is incorrect ?

- (1) Complete disintegration of nuclear envelope takes place
 (2) Chromosomes are highly condensed
 (3) Metaphase chromosomes are made up of four sister chromatids held together by centromere
 (4) Chromosomes lie at the equator of the cell

Q. 198. Select the **incorrect** statement with respect to inbreeding of animals.

- (1) It is used for evolving pure lines in cattle.
 (2) It helps in accumulation of superior genes and elimination of less desirable genes.
 (3) It decreases homozygosity.
 (4) It exposes harmful recessive genes that are eliminated by selection.

Q. 199. Match List - I with List - II :

List - I	List - II
(a) Cellular barrier	(i) Interferons
(b) Cytokine barrier	(ii) Mucus
(c) Physical barrier	(iii) Neutrophils
(d) Physiological juice	(iv) HCl in gastric barrier

Choose the **correct answer** from the options given below :

- (1) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)
 (2) (a) - (ii), (b) - (iii), (c) - (i), (d) - (iv)
 (3) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)
 (4) (a) - (iii), (b) - (i), (c) - (ii), (d) - (iv)

Q. 200. If A and C make 30% and 20% of DNA, respectively, what will be the percentage composition of T and G ?

- (1) T : 20%, G : 30%
 (2) T : 30%, G : 20%
 (3) T : 30%, G : 30%
 (4) T : 20%, G : 20%

Booklet	Batch	Roll Number	Name	Test Date	Invigilator's Signature	Student's Signature	Test Center Code
(A) 10+1 (S)		0 0 0 0 0 0 0 0					0 0
(B) 10-2 (V)		1 1 1 1 1 1 1 1					1 1
(C) 10-3 (A)		2 2 2 2 2 2 2 2					2 2
(D) Crash (C)		3 3 3 3 3 3 3 3					3 3
(E)		4 4 4 4 4 4 4 4					4 4
(F) Paper		5 5 5 5 5 5 5 5					5 5
(G)		6 6 6 6 6 6 6 6					6 6
(H) Paper 1 (1)		7 7 7 7 7 7 7 7					7 7
(I) Paper 2 (2)		8 8 8 8 8 8 8 8					8 8
(J)		9 9 9 9 9 9 9 9					9 9

1 (a) (b) (c) (d)	6 (a) (b) (c) (d)	11 (a) (b) (c) (d)	16 (a) (b) (c) (d)	21 (a) (b) (c) (d)	26 (a) (b) (c) (d)
2 (a) (b) (c) (d)	7 (a) (b) (c) (d)	12 (a) (b) (c) (d)	17 (a) (b) (c) (d)	22 (a) (b) (c) (d)	27 (a) (b) (c) (d)
3 (a) (b) (c) (d)	8 (a) (b) (c) (d)	13 (a) (b) (c) (d)	18 (a) (b) (c) (d)	23 (a) (b) (c) (d)	28 (a) (b) (c) (d)
4 (a) (b) (c) (d)	9 (a) (b) (c) (d)	14 (a) (b) (c) (d)	19 (a) (b) (c) (d)	24 (a) (b) (c) (d)	29 (a) (b) (c) (d)
5 (a) (b) (c) (d)	10 (a) (b) (c) (d)	15 (a) (b) (c) (d)	20 (a) (b) (c) (d)	25 (a) (b) (c) (d)	30 (a) (b) (c) (d)
31 (a) (b) (c) (d)	36 (a) (b) (c) (d)	41 (a) (b) (c) (d)	46 (a) (b) (c) (d)	51 (a) (b) (c) (d)	56 (a) (b) (c) (d)
32 (a) (b) (c) (d)	37 (a) (b) (c) (d)	42 (a) (b) (c) (d)	47 (a) (b) (c) (d)	52 (a) (b) (c) (d)	57 (a) (b) (c) (d)
33 (a) (b) (c) (d)	38 (a) (b) (c) (d)	43 (a) (b) (c) (d)	48 (a) (b) (c) (d)	53 (a) (b) (c) (d)	58 (a) (b) (c) (d)
34 (a) (b) (c) (d)	39 (a) (b) (c) (d)	44 (a) (b) (c) (d)	49 (a) (b) (c) (d)	54 (a) (b) (c) (d)	59 (a) (b) (c) (d)
35 (a) (b) (c) (d)	40 (a) (b) (c) (d)	45 (a) (b) (c) (d)	50 (a) (b) (c) (d)	55 (a) (b) (c) (d)	60 (a) (b) (c) (d)
61 (a) (b) (c) (d)	66 (a) (b) (c) (d)	71 (a) (b) (c) (d)	76 (a) (b) (c) (d)	81 (a) (b) (c) (d)	86 (a) (b) (c) (d)
62 (a) (b) (c) (d)	67 (a) (b) (c) (d)	72 (a) (b) (c) (d)	77 (a) (b) (c) (d)	82 (a) (b) (c) (d)	87 (a) (b) (c) (d)
63 (a) (b) (c) (d)	68 (a) (b) (c) (d)	73 (a) (b) (c) (d)	78 (a) (b) (c) (d)	83 (a) (b) (c) (d)	88 (a) (b) (c) (d)
64 (a) (b) (c) (d)	69 (a) (b) (c) (d)	74 (a) (b) (c) (d)	79 (a) (b) (c) (d)	84 (a) (b) (c) (d)	89 (a) (b) (c) (d)
65 (a) (b) (c) (d)	70 (a) (b) (c) (d)	75 (a) (b) (c) (d)	80 (a) (b) (c) (d)	85 (a) (b) (c) (d)	90 (a) (b) (c) (d)
91 (a) (b) (c) (d)	96 (a) (b) (c) (d)	101 (a) (b) (c) (d)	106 (a) (b) (c) (d)	111 (a) (b) (c) (d)	116 (a) (b) (c) (d)
92 (a) (b) (c) (d)	97 (a) (b) (c) (d)	102 (a) (b) (c) (d)	107 (a) (b) (c) (d)	112 (a) (b) (c) (d)	117 (a) (b) (c) (d)
93 (a) (b) (c) (d)	98 (a) (b) (c) (d)	103 (a) (b) (c) (d)	108 (a) (b) (c) (d)	113 (a) (b) (c) (d)	118 (a) (b) (c) (d)
94 (a) (b) (c) (d)	99 (a) (b) (c) (d)	104 (a) (b) (c) (d)	109 (a) (b) (c) (d)	114 (a) (b) (c) (d)	119 (a) (b) (c) (d)
95 (a) (b) (c) (d)	100 (a) (b) (c) (d)	105 (a) (b) (c) (d)	110 (a) (b) (c) (d)	115 (a) (b) (c) (d)	120 (a) (b) (c) (d)
121 (a) (b) (c) (d)	126 (a) (b) (c) (d)	131 (a) (b) (c) (d)	136 (a) (b) (c) (d)	141 (a) (b) (c) (d)	146 (a) (b) (c) (d)
122 (a) (b) (c) (d)	127 (a) (b) (c) (d)	132 (a) (b) (c) (d)	137 (a) (b) (c) (d)	142 (a) (b) (c) (d)	147 (a) (b) (c) (d)
123 (a) (b) (c) (d)	128 (a) (b) (c) (d)	133 (a) (b) (c) (d)	138 (a) (b) (c) (d)	143 (a) (b) (c) (d)	148 (a) (b) (c) (d)
124 (a) (b) (c) (d)	129 (a) (b) (c) (d)	134 (a) (b) (c) (d)	139 (a) (b) (c) (d)	144 (a) (b) (c) (d)	149 (a) (b) (c) (d)
125 (a) (b) (c) (d)	130 (a) (b) (c) (d)	135 (a) (b) (c) (d)	140 (a) (b) (c) (d)	145 (a) (b) (c) (d)	150 (a) (b) (c) (d)
151 (a) (b) (c) (d)	156 (a) (b) (c) (d)	161 (a) (b) (c) (d)	166 (a) (b) (c) (d)	171 (a) (b) (c) (d)	176 (a) (b) (c) (d)
152 (a) (b) (c) (d)	157 (a) (b) (c) (d)	162 (a) (b) (c) (d)	167 (a) (b) (c) (d)	172 (a) (b) (c) (d)	177 (a) (b) (c) (d)
153 (a) (b) (c) (d)	158 (a) (b) (c) (d)	163 (a) (b) (c) (d)	168 (a) (b) (c) (d)	173 (a) (b) (c) (d)	178 (a) (b) (c) (d)
154 (a) (b) (c) (d)	159 (a) (b) (c) (d)	164 (a) (b) (c) (d)	169 (a) (b) (c) (d)	174 (a) (b) (c) (d)	179 (a) (b) (c) (d)
155 (a) (b) (c) (d)	160 (a) (b) (c) (d)	165 (a) (b) (c) (d)	170 (a) (b) (c) (d)	175 (a) (b) (c) (d)	180 (a) (b) (c) (d)
181 (a) (b) (c) (d)	186 (a) (b) (c) (d)	191 (a) (b) (c) (d)	196 (a) (b) (c) (d)		
182 (a) (b) (c) (d)	187 (a) (b) (c) (d)	192 (a) (b) (c) (d)	197 (a) (b) (c) (d)		
183 (a) (b) (c) (d)	188 (a) (b) (c) (d)	193 (a) (b) (c) (d)	198 (a) (b) (c) (d)		
184 (a) (b) (c) (d)	189 (a) (b) (c) (d)	194 (a) (b) (c) (d)	199 (a) (b) (c) (d)		
185 (a) (b) (c) (d)	190 (a) (b) (c) (d)	195 (a) (b) (c) (d)	200 (a) (b) (c) (d)		

ANSWERS			
Q. No.	Answer Key	Topic's Name	Chapter's Name
SECTION-A (PHYSICS)			
1	1	Electronic Devices	Electronic Devices
2	4	Electromagnetic Waves	Electromagnetic Waves
3	3	Electronic Devices	Electronic Devices
4	1	Current Electricity	Current Electricity
5	3	Alternating Currents	Electromagnetic Induction and Alternating Current
6	3	Current Electricity	Current Electricity
7	1	Motion in a Straight Line	Kinematics
8	4	Magnetic Effects of Current	Magnetic Effects of Current and Magnetism
9	3	Alternating Currents	Electromagnetic Induction and Alternating Current
10	3	Ray Optics and Optical Instruments	Optics
11	2	Dual Nature of Matter and Radiation	Dual Nature of Matter and Radiation
12	1	Laws of Motion	Laws of Motion
13	1	Electronic Devices	Electronic Devices
14	1	Mechanical Properties of Fluids	Properties of Bulk Matter
15	4	Electrostatic Potential and Capacitance	Electrostatics
16	1	Mechanical Properties of Fluids	Properties of Bulk Matter
17	3	Concept of Vector and Motion in a Plane	Kinematics
18	3	Ray Optics and Optical Instruments	Optics
19	4	Oscillations	Oscillations and Waves
20	3	Wave Optics	Optics
21	3	Electrostatic Potential and Capacitance	Electrostatics
22	1	Motion in a Straight Line	Kinematics
23	1	Gravitation	Gravitation
24	4	Current Electricity	Current Electricity
25	4	Motion of System of Particles and Rigid Body	Motion of System of Particles and Rigid Body
26	3	Dual Nature of Matter and Radiation	Dual Nature of Matter and Radiation
27	2	Electromagnetic Waves	Electromagnetic Waves
28	2	Magnetic Effects of Current	Magnetic Effects of Current and Magnetism
29	2	Motion in a Straight Line	Kinematics
30	2	Nuclei	Atoms and Nuclei
31	3	Behaviour of Perfect Gas and Kinetic theory	Behaviour of Perfect Gas and Kinetic theory
32	1	Alternating Currents	Electromagnetic Induction and Alternating Currents
33	1	Atoms	Atoms and Nuclei
34	3	Units and Measurements	Physical World and Measurement
35	1	Motion of System of Particles and Rigid Body	Motion of System of Particles and Rigid Body

SECTION-B (PHYSICS)			
36	4	Electromagnetic Induction	Electromagnetic Induction and Alternating Currents
37	4	Ray Optics and Optical Instruments	Optics
38	2	Nuclei	Atoms and Nuclei
39	1	Electrostatic Potential and Capacitance	Electrostatics
40	2	Waves	Oscillations and Waves
41	1	Motion in a Straight Line	Kinematics
42	3	Electronic Devices	Electronic Devices
43	3	Behaviour of Perfect Gas and Kinetic theory	Behaviour of Perfect Gas and Kinetic theory
44	3	Gravitation	Gravitation
45	3	Magnetic Effects of Current	Magnetic Effects of Current and Magnetism
46	1	Magnetic Effects of Current	Magnetic Effects of Current and Magnetism
47	4	Thermal Properties of Matter	Bulk Properties of Matter
48	1	Current Electricity	Current Electricity
49	1	Electrostatics and Gravitation	Electrostatics and Gravitation
50	3	Units and Measurements	Physical World and Measurement
SECTION-A (CHEMISTRY)			
51.	1	Bond angles	Chemical bonding and molecular structure
52.	1	Henry's law	Solutions
53.	2	Preparations of nitric acid	<i>p</i> -Block Elements
54.	3	Properties of alkaline earth metals	<i>s</i> -Block Elements (Alkali and Alkaline Earth Metals)
55.	2	Chemical reactions of aldehydes	Aldehydes, Ketones and Carboxylic Acids
56.	4	Chemical reactions of functional groups	Alcohols, Phenols and Ethers & Organic Compounds Containing Nitrogen
57.	3	Soaps & Detergents	Chemistry in Everyday Life
58.	1	Redox reactions	Redox reactions
59.	1	Properties of <i>s</i> -Block elements	<i>s</i> -Block Elements (Alkali and Alkaline Earth Metals)
60.	3	Nucleophilic substitution reactions	Haloalkanes and Haloarenes
61.	2	VSEPR Theory	Chemical bonding and molecular structure
62.	2	Thermoplastic polymers	Polymers
63.	1	Compounds of Boron	Some <i>p</i> -Block Elements Principles and Process
64.	3	Electrochemistry	Electrochemistry
65.	2	Applications of hydrogen	Hydrogen
66.	2	Compounds of Boron	Some <i>p</i> -Block Elements Principles and Process
67.	3	Properties of <i>p</i> -block elements	Some <i>p</i> -Block Elements Principles and Process
68.	3	pH	Equilibrium
69.	2	Adsorption isotherm	Surface Chemistry
70.	2	Chemical Kinetics	Chemical Kinetics
71.	4	Thermodynamics	Thermodynamics
72.	1	Hybridisation of organic compounds	Organic Chemistry: Some Basic Principles and Techniques
73.	2	Significant figures	Some Basic Concepts of Chemistry

74.	1	Properties of group-17 elements	Some <i>p</i> -Block Elements Principles and Process
75.	3	Isomerism in coordination compounds	Coordination Compounds
76.	3	Proteins	Biomolecules
77.	1	Chemical reactions of amines	Amines
78.	4	Defects in solids	Solid state
79.	1	Inductive effect	Organic Chemistry Some Basic Principles and Techniques
80.	1	Chemical reactions of aldehydes	Aldehydes, Ketones and Carboxylic Acids
81.	4	Critical temperature	States of Matter: Gases and Liquids
82.	3	Properties of actinoids	<i>d</i> and <i>f</i> -Block Elements
83.	4	Physical properties of alkanes	Hydrocarbons
84.	3	Phenols	Alcohols, Phenols and Ethers
85.	4	Quantum numbers	Structure of Atom
		SECTION-B (CHEMISTRY)	
86.	4	General Principles and Processes of Isolation of Elements	General Principles and Processes of Isolation of Elements
87.	4	Structure of Atom	Structure of Atom
88.	1	Rate of Chemical Reaction	Chemical Kinetics
89.	3	Synthesis of alkenes	Hydrocarbons
90.	4	Synthesis of benzaldehyde	Aldehydes, Ketones and Carboxylic Acids
91.	1	Solid State	Solid State
92.	3	Metal carbonyls	Coordination Compounds
93.	3	Preparation of phenol	Alcohols, Phenols and Ethers
94.	1	Boyle's law	States of Matter : Gases and Liquids
95.	3	Environmental Chemistry	Environmental Chemistry
96.	2	Chemical reactions of functional groups	Haloalkanes and Haloarenes & Aldehydes, ketones & carboxylic acids
97.	4	First transition series	<i>d</i> and <i>f</i> -Block Elements
98.	2	Equilibrium	Equilibrium
99.	2	Electrochemistry	Electrochemistry
100.	4	Reaction Intermediates	Organic Chemistry: Some Basic Principles and Techniques
		SECTION-A (BOTANY)	
101	3	Cell cycle and cell division	Cell cycle and cell division
102	1	Photosynthesis in higher plants	Photosynthesis in higher plants
103	1	Lac operon	Molecular basis of inheritance
104	4	Secondary growth	Anatomy of flowering plants
105	1	Secondary metabolites	Biomolecules
106	2	<i>Lac</i> operon	Molecular basis of inheritance
107	2	(i) Moss (ii) Pteridophyte (iii) Alga (iv) Gymnosperm	Plant kingdom
108	1	DNA Replication	Molecular basis of inheritance

109	2	Description of Some Important Families	Morphology of flowering plants
110	3	Successions	Ecosystem
111	2	Metabolic pathways	Respiration in plants
112	4	Genetic disorders	Principles of inheritance and variation
113	1	Restriction enzymes	Biotechnology: Principles and processes
114	4	Biodiversity	Biodiversity and its conservation
115	2	Mineral nutrition & respiration in plants	Mineral nutrition
116	2	Phytohormones	Plant growth and development
117	4	Plant tissues	Anatomy of flowering plants
118	1	Succession of plants	Ecosystem
119	1	Double fertilisation	Sexual reproduction in flowering plants
120	2	Description of some important families	Morphology of flowering plants
121	2	Successions	Ecosystem
122	4	Gel electrophoresis	Biotechnology: Principles and processes
123	2	Plant tissues	Anatomy of flowering plants
124	2	Cell cycle and cell division	Cell cycle and cell division
125	1	Metabolic pathways	Respiration in plants
126	4	Active and passive transport	Transport in plants
127	1	<i>Ex-situ</i> and <i>in-situ</i> conservation	Biodiversity and its conservation
128	3	Plant tissues	Anatomy of flowering plants
129	3	Transpiration	Transport in plants
130	3	Characteristics of flowering plants	Morphology of flowering plants
131	2	artificial hybridisation process	Sexual reproduction in flowering plants
132	3	Parts of a flowering plant	Sexual reproduction in flowering plants
133	2	Chromosomal theory of inheritance	Principles of inheritance and variation
134	2	Leguminous plants	Mineral nutrition
135	3	Growth and development	Plant growth and development
SECTION-B (BOTANY)			
136	2	Nitrogen fixing bacteria	Mineral nutrition
137	1	Ex-situ and in-situ conservation	Biodiversity and its conservation
138	3	Structure of protein	Biomolecules
139	1	Recombinant DNA technology	Biotechnology and its applications
140	3	The DNA	Molecular basis of inheritance
141	3	Apomictic gene	Sexual reproduction in flowering plants
142	2	Phaeophyceae	Plant kingdom
143	3	Species area relationship	Biodiversity and its conservation
144	2	Biomolecules	Biomolecules
145	2	Chemiosmotic hypothesis	Photosynthesis in higher plants
146	4	Down's syndrome	Principles of inheritance and variation
147	3	Cell cycle and cell division	Cell cycle and cell division
148	2	DNA isolation	Biotechnology: Principles and processes
149	3	Fixation in plants	Photosynthesis in higher plants

150	3	Mendel's breeding experiment	Principles of inheritance and variation
		SECTION-A (ZOOLOGY)	
151	2	Antibiotic resistance	Biotechnology: Principles and processes
152	2	Gene cloning	Biotechnology and its applications
153	2	Cell organelles	Cell -The unit of life
154	1	Enzymatic reactions	Biomolecules
155	4	Immune system	Human Health and Disease
156	3	Hershey and Chase experiment	Molecular basis of inheritance
157	2	Species interactions	Organisms and population
158	4	VIRUSES, VIROIDS, PRIONS AND LICHENS	Biological classification
159	1	Birth control methods	Reproductive health
160	2	Asexual reproduction	Reproduction in organisms
161	2	sliding filament theory	Locomotion and movement
162	2	Productivity	Ecosystem
163	1	Amino acids	Biomolecules
164	3	Types of epithelium	Structural organisation in animals
165	3	Endemic species	Biodiversity and its conservation
166	1	Disorders of muscular and skeletal system	Locomotion and movement
167	3	Vital capacity	Breathing and exchange of gases
168	2	Double circulation	Body fluids and circulation
169	1	Digestive system	Structural organisation in animals
170	2	Single cell protein	Strategies for enhancement in food production
171	4	Theory of spontaneous generation	Evolution
172	3	Components of mammary gland	Human reproduction
173	3	Pedigree analysis	Principles of inheritance and variation
174	2	Environmental issues	Environmental issues
175	4	Gene transfer methods	Biotechnology: Principles and processes
176	2	Respiratory rate	Breathing and exchange of gases
177	3	Spermatogenesis	Human reproduction
178	2	Muscular tissues	Structural organisation in animals
179	2	Hormones	Human reproduction
180	3	Structure of heart	Animal kingdom
181	1	Metabolic changes during pregnancy	Human reproduction
182	3	Cyclostomes	Animal kingdom
183	2	Role of enamel	Digestion and absorption
184	1	Tissues	Structural organisation in animals
185	1	Cell cycle and cell division	Cell cycle and cell division
		SECTION-B (ZOOLOGY)	
186	3	Taxonomical aid 'key'	The living world
187	2	Genetic disorders	Principles of inheritance and variation
188	3	Birth control methods	Reproductive health
189	2	Agarose-gel electrophoresis :	Biotechnology and its applications
190	4	Types of neurons	Neural control and co-ordination

191	4	Excretion in cockroach	Excretory products and their elimination
192	3	Mutation theory of evolution.	Evolution
193	1	Types of cells	Cell- The unit of life
194	2	Role of enzymes	Biomolecules
195	4	tRNA sequence	Molecular basis of inheritance
196	4	Renin-angiotensin mechanism.	Excretory products and their elimination
197	3	Cell cycle and cell division	Cell cycle and cell division
198	3	Inbreeding of animals	Strategies for enhancement in food production
199	4	Physical and physiological barriers	Human Health and Disease
200	2	Nitrogenous bases	Molecular basis of inheritance

NEET (UG)

(Re-examination)

4th September 2022 Paper

ANSWERS WITH EXPLANATION

PHYSICS

Section-A

1. Option (1) is correct.

Explanation: Only option (1) is representing non-periodic motion as it is an exponential function.

2. Option (4) is correct.

Explanation: Given, $B = 3 \times 10^{-8} \cos(1.6 \times 10^3 x + 48 \times 10^{10} t)$... (1)

$$B = B_0 \cos(kx + \omega t) \quad \dots (2)$$

On comparing equations (1) & (2), we get;

$$B_0 = 3 \times 10^{-8}$$

$$c = \frac{\omega}{k} = \frac{48 \times 10^{10}}{1.6 \times 10^3}$$

$$= 3 \times 10^8$$

And also, $c = \frac{E_0}{B_0}$

Hence, $E_0 = B_0 \times c$
 $= 3 \times 10^{-8} \times 3 \times 10^8$
 $= 9$

So, the required equation is:

$$E = 9 \cos(1.6 \times 10^3 x + 48 \times 10^{10} t)$$

3. Option (3) is correct.

Explanation: Zener diode is highly doped $p-n$ junction, and zener voltage is constant. Its depletion region is very thin.

4. Option (1) is correct.

Explanation: As, $i = \frac{E}{R+r}$

Hence, $i = \frac{4}{7.5+0.5} = \frac{4}{8} = \frac{1}{2}$

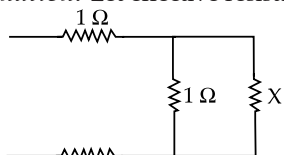
And, $V = E - ir$
 (where, the symbols have their usual meanings)
 $V = 4 - 0.5 \times 0.5$
 $V = 3.75 \text{ volt}$

5. Option (3) is correct.

Explanation: In pure capacitive circuits, the current leads the voltage by $\frac{\pi}{2}$ or 90° .

6. Option (3) is correct.

Explanation: Let effective resistance be X .



Then, $X = 1 + \frac{X}{X+1} + 1$

$$X = \frac{X+1+X+X+1}{X+1}$$

$$\frac{X(X+1)}{X^2-2X-2} = 0$$

$$X = \frac{2 \pm \sqrt{12}}{2}$$

$$X = 1 \pm \sqrt{3} \Omega$$

$$X = 1 + \sqrt{3} \Omega$$

(as R can't be negative)

7. Option (1) is correct.

Explanation: As, $H = \frac{u^2 \sin^2 \theta}{2g}$
 $= \frac{(20)^2 \sin^2 30^\circ}{2(10)} = 5\text{m}$

8. Option (4) is correct.

Explanation: As, magnetic field at centre of coil,

$$B = \frac{\mu_0 N i}{2R}$$

$$B = \frac{4\pi \times 10^{-7} \times 1000 \times 1}{2 \times 62.8 \times 10^{-2}}$$

$$= \frac{4 \times 3.14 \times 10^{-7} \times 10^3}{2 \times 62.8 \times 10^{-2}}$$

$$= 10^{-3} \text{ T}$$

9. Option (3) is correct.

Explanation: For AC circuits,

$$X_L = \omega L$$

$$X_L = 2\pi n L \quad (\because \omega = 2\pi n)$$

$$X_L = 0.628 \Omega$$

Hence, $X_1 = 0.628 \Omega$

For DC circuits, $\omega = 0$

$$X_L = \omega L$$

$$X_L = 0$$

Hence $X_2 = 0$

10. Option (3) is correct.

Explanation: Secondary rainbow is formed due to double internal reflection and is formed above the primary rainbow. The secondary rainbow is brighter than the primary rainbow.

11. Option (2) is correct.

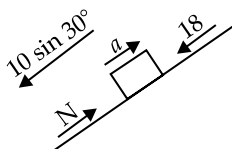
Explanation: By Einstein photoelectric equation, we have;

$$\begin{aligned} \text{K.E.} &= h\nu - \phi \\ eV_0 &= 4.2 \text{ eV} - 2.2 \text{ eV} \\ eV_0 &= 2 \text{ eV} = 2 \text{ volt} \end{aligned}$$

12. Option (1) is correct.**Explanation:** Total force along inclined plane,

$$\begin{aligned} f &= -F_1 + 6g\sin 30^\circ + F_2 \\ f &= -60 + 6g\sin 30^\circ + 18 \\ f &= -60 + 6 \times 10 \times \frac{1}{2} + 18 \\ f &= 12 \text{ N (downwards)} \end{aligned}$$

$$\text{acceleration, } a = \frac{f}{m} = \frac{12}{6} = 2 \text{ m/s}^2$$



As per question, the normal force between 2 kg and 1 kg is,

$$\begin{aligned} N - 18 - 10\sin 30^\circ &= ma \\ N - 18 - 10 \times \frac{1}{2} &= 1 \times 2 \\ N &= 2 + 23 \\ N &= 25 \text{ N} \end{aligned}$$

13. Option (1) II is correct.**Explanation:** As per question, the current flows only when the switch is ON.

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1

The truth table represents the OR Gate.

14. Option (1) is correct.**Explanation:** Pressure, $P = h\rho g$... (1)In above equation, ρ represents density of water, g represents acceleration due to gravity and h is the height of the vessel.

As both vessels have same height and containing water, then densities are also same. Hence, the pressure on the base area of vessels A and B is same.

15. Option (4) is correct.**Explanation:** The capacitance of parallel plate capacitor,

$$C = \frac{\epsilon_0 A}{d} \quad \dots (i)$$

(where, the symbols have their usual meanings.)

$$\text{Given, } d' = 2d \text{ \& } A' = \frac{A}{2}$$

$$\text{Then, } C' = \frac{\epsilon_0 \left(\frac{A}{2} \right)}{2d} = \frac{\epsilon_0 A}{4d} = \frac{C}{4}$$

16. Option (1) is correct.**Explanation:** By Stobal's law,

$$keF = 6\pi\eta rv$$

(where, the symbols have their usual meanings)

$$\begin{aligned} F &= 6 \times 3.14 \times 0.9 \times 5 \times 10^{-3} \times 10 \times 10^{-2} \\ F &= 84.78 \times 10^{-4} \\ &= 8.478 \times 10^{-3} \text{ N} = 8.48 \times 10^{-3} \text{ N} \end{aligned}$$

17. Option (3) is correct.

$$\text{Explanation: } \vec{F} = 2\hat{i} + \hat{j} - \hat{k}$$

$$\vec{r} = 3\hat{i} + 2\hat{j} - 2\hat{k} \quad (\text{given})$$

The scalar product of $\vec{F}$ and $\vec{r}$ is

$$\begin{aligned} \vec{F} \cdot \vec{r} &= (2)(3) + (1)(2) + (-1)(-2) \\ &= 6 + 2 + 2 = 10 \end{aligned}$$

The vector product of $\vec{F}$ and $\vec{r}$ is

$$\begin{aligned} \vec{F} \times \vec{r} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 3 & 2 & -2 \end{vmatrix} \\ &= \hat{i} [(1 \times -2) - (2 \times -1)] - \hat{j} [(2 \times -2) - (-1 \times 3)] + \hat{k} [(2 \times 2) - (3 \times 1)] \\ &= \hat{i} (0) - \hat{j} (-1) + \hat{k} (1) = \hat{j} + \hat{k} \end{aligned}$$

$$\text{Magnitude} = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$$

18. Option (3) is correct.**Explanation:** As per Malus law, $I = I_0 \cos^2 \phi$, where symbols have their usual meanings.

$$I' = I \cos^2 30^\circ$$

$$I' = I \left(\frac{\sqrt{3}}{2} \right)^2$$

$$I' = \frac{3I}{4}$$

19. Option (4) is correct.**Explanation:** Restoring force of spring,

$$F \propto -x$$

(where, x is the displacement)

Only graph (4) is representing the above equation.

20. Option (3) is correct.

$$\text{Explanation: As per YDSE, } x = \frac{\lambda D}{d},$$

(where, symbols have their usual meanings)

If the screen is moving away from the slits then D increases, and hence the fringe width (x) increases.

- 21.
- Option (3) is correct.**

Explanation:

$$C_p = C_1 + C_2 \quad \dots(1)$$

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} \quad \dots(2)$$

C_1 and C_2 are individual capacitance, C_s is the series capacitance & C_p is the parallel capacitance.

Hence, $16 = C_1 + C_2 \quad \dots(3)$

$$\frac{1}{3} = \frac{1}{C_1} + \frac{1}{C_2} \quad \dots(4)$$

By using equations (3) & (4), we have;

$$\frac{1}{3} = \frac{1}{C_1} + \frac{1}{(16 - C_1)}$$

$$\frac{1}{3} = \frac{16 - C_1 + C_1}{C_1 \times (16 - C_1)}$$

$$\Rightarrow \frac{1}{3} = \frac{16}{16C_1 - C_1^2}$$

$$\Rightarrow C_1^2 - 16C_1 + 48 = 0$$

$$\Rightarrow C_1 = \frac{16 \pm \sqrt{256 - 192}}{2}$$

[By Dharacharya rule, we have; $C = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$]

$$\Rightarrow C_1 = \frac{16 \pm \sqrt{64}}{2}$$

$$\Rightarrow C_1 = \frac{16 \pm \sqrt{8}}{2}$$

$$C_1 = 12 \text{ or } 4 \text{ (in } \mu\text{F)}$$

Hence, $C_2 = 4 \text{ or } 12 \text{ (in } \mu\text{F)}$

- 22.
- Option (1) is correct.**

Explanation: Given,

$$m = 5 \text{ g} = 0.005 \text{ kg},$$

$$P = mv = 0.3 \text{ kg m/s, and}$$

$$t = 5 \text{ s},$$

(where, symbols have their usual meanings.)

$$P = mv = 0.3$$

$$\text{Hence, } V = \frac{0.3}{m} = \frac{0.3}{0.005} = 60 \text{ m/s}$$

$$\text{Hence, the distance covered} = \text{speed} \times \text{time} \\ = 60 \times 5 = 300 \text{ m}$$

- 23.
- Option (1) is correct.**

Explanation: Since, gravitational force is a conservative force and hence the work done is independent of the paths covered between points A and B.

- 24.
- Option (4) is correct.**

Explanation: Conductance $\propto \frac{1}{\text{Resistance}}$

- 25.
- Option (4) is correct.**

Explanation: As no external force acts along

x -axis, hence centre of mass will move with same speed in x -direction before and after the explosion.

- 26.
- Option (3) is correct.**

Explanation: By using Einstein's photoelectric equation, we have,

$$KE = h\nu - \phi$$

$$KE = h\nu - h\nu_0$$

$$= h(4\nu_0) - h\nu_0$$

$$= 3h\nu_0$$

where, h = Planck's constant,

ν & ν_0 = frequencies, and

ϕ = work function.

- 27.
- Option (2) is correct.**

Explanation: As per question,

$$\frac{B_0}{E_0} = \frac{1}{c} = \sqrt{\mu_0 \epsilon_0}$$

(where, the symbols have their usual meanings.)

- 28.
- Option (2) is correct.**

Explanation: Straight current carrying wire produces circular magnetic field.



- 29.
- Option (2) is correct.**

Explanation: (a) Amplitude decreases continuously (damped oscillation)

(b) $a \propto -x$

(c) Amplitude is constant

(d) Total (K.E + P.E.) = constant

- 30.
- Option (2) is correct.**

Explanation: Both statements are incorrect.

As per law of disintegration, we have,

$$\frac{dN}{dt} = -\lambda N$$

(where, the symbols have their usual meanings)

Hence, $dN \propto N$, so the statement I is incorrect. $T_{1/2}$ is the time in which active number of nuclei becomes half, so the statement II is also incorrect.

- 31.
- Option (3) is correct.**

Explanation: As per question,

$$PV^2 = C \quad \dots(1)$$

And by ideal gas equation, we have;

$$PV = nRT \quad \dots(2)$$

$$\text{Hence, } \left(\frac{nRT}{V} \right) V^2 = C$$

$$\text{Hence, } TV = \text{constant}$$

$$\& \quad T \propto \frac{1}{V} \therefore V_2 > V_1 \text{ and } T_1 > T_2$$

32. Option (1) is correct.

Explanation: Power,

$$P = I_{\text{rms}} V_{\text{rms}}$$

$$I_{\text{rms}} = \frac{P}{V_{\text{rms}}} = \frac{100}{200} = \frac{1}{2}$$

$$I_0 = \sqrt{2} I_{\text{rms}}$$

$$I_0 = \sqrt{2} \times \frac{1}{2}$$

$$I_0 = \frac{1}{\sqrt{2}} = 0.707 \text{ A}$$

33. Option (1) is correct.

Explanation: As,

$$R_n = 0.529 \frac{n^2}{Z}$$

$$\text{Hence, } \frac{R_1}{R_2} = \frac{0.529 \frac{n_1^2}{Z_1}}{0.529 \frac{n_2^2}{Z_2}}$$

$$\frac{R_1}{R_2} = \frac{n_1^2}{n_2^2} = \frac{(2)^2}{(4)^2}$$

$$\therefore \frac{R_1}{R_2} = \frac{1}{4} = 0.25$$

(where, the symbols have their usual meanings)

34. Option (3) is correct.

Explanation: Dimensions of Pressure = $[\text{ML}^{-1}\text{T}^{-2}]$

Dimension of Force = $[\text{MLT}^{-2}]$

Dimension of Momentum = $[\text{MLT}^{-1}]$

Dimension of Young's modulus of elasticity = $[\text{ML}^{-1}\text{T}^{-2}]$

Dimension of coefficient of viscosity = $[\text{ML}^{-1}\text{T}^{-1}]$

35. Option (1) is correct.

Explanation: Given,

$$\omega_1 = 60 \text{ rpm} = 60 \times \frac{2\pi}{60} = 2\pi \text{ rad/s}$$

$$\omega_2 = 360 \text{ rpm} = 360 \times \frac{2\pi}{60} = 12\pi \text{ rad/s}$$

Change in kinetic energy,

$$\Delta \text{K.E} = \frac{1}{2} I(\omega_2^2 - \omega_1^2)$$

$$484 = \frac{1}{2} \times I(\omega_2 + \omega_1)(\omega_2 - \omega_1)$$

$$\Rightarrow I = \frac{484 \times 2}{14\pi \times 10\pi} = 0.7 \text{ kg.m}^2$$

(where, the symbols have their usual meanings)

Section-B

36. Option (4) is correct.

Explanation: Given :

$$\phi = 2t^3 + 4t^2 + 2t + 5, \text{ and}$$

$$\& \quad t = 5 \text{ s}$$

By Faraday's law of EMI, we have,

$$\text{emf, } = -\frac{d\phi}{dt}$$

(where, the symbols have their usual meanings)

$$e = \left| \frac{d\phi}{dt} \right| = \left| \frac{d}{dt}(2t^3 + 4t^2 + 2t + 5) \right|$$

$$= 6t^2 + 8t + 2$$

$$\text{At } t = 5 \text{ s,}$$

$$e = 6(5)^2 + 8(5) + 2$$

$$= 192 \text{ V}$$

37. Option (4) is correct.

Explanation: Given,

$$f_e = 2 \text{ cm, and}$$

$$f_o = 20 \text{ m} = 2000 \text{ cm}$$

For normal adjustment,

$$\text{Magnification power} = \frac{-f_o}{f_e} = \frac{-2000}{2}$$

$$= -1000$$

$$\text{Length of telescope} = f_o + f_e$$

$$= 2000 + 2$$

$$= 2002 \text{ cm}$$

$$= 20.02 \text{ m}$$

The image formed is inverted and magnified, and the aperture of objective is larger than eye piece of the telescope.

(f_o & f_e are focal lengths of objective & eye piece respectively.)

38. Option (2) is correct.

Explanation: By radioactivity law, we have,

$$N = N_0 e^{-\lambda t}$$

(where, the symbols have their usual meanings)

Hence, for two elements,

$$\frac{N_1}{N_2} = \frac{N_0 e^{-10\lambda t}}{N_0 e^{-\lambda t}} = \frac{1}{e}$$

$$\text{Hence, } e^{-9\lambda t} = e$$

$$t = \frac{1}{9\lambda}$$

39. Option (1) is correct.

Explanation:

Work done = Change in potential energy

$$W = U_f - U_i$$

$$\text{Hence, } W = 0$$

40. Option (2) is correct.

Explanation: As per question, we have;

frequency, $f = \frac{v}{4l}$

$$\text{Hence, } \frac{f_1}{f_2} = \frac{\frac{v_1}{4l}}{\frac{v_2}{4l}}$$

$$\Rightarrow \frac{400}{f_2} = \frac{v_1}{v_2}$$

$$\Rightarrow \frac{400}{f_2} = \sqrt{\frac{T_1}{T_2}} \quad [\because v \propto \sqrt{T}]$$

$$\Rightarrow \frac{400}{f_2} = \sqrt{\frac{300}{363}}$$

$$\Rightarrow f_2 = \sqrt{\frac{363}{300}} \times 400$$

$$\Rightarrow f_2 = 440 \text{ Hz}$$

41. Option (1) is correct.

Explanation: As $x = \frac{1}{2}at^2$, hence the graph

between x and t is upward parabola.

42. Option (3) is correct.

Explanation: As per question, we have,

$$I_c = 24 \text{ mA, and } \alpha = 0.8$$

$$\text{Now, } I_E = \frac{I_c}{\alpha} = \frac{24}{0.8} = 30 \text{ mA}$$

$$\text{As, } I_E = I_B + I_C$$

$$\text{Hence, } I_B = I_E - I_C = 6 \text{ mA}$$

(where, the symbols have their usual meanings)

43. Option (3) is correct.

$$\text{Explanation: As, } V_{\text{rms}} = \sqrt{\frac{3KT}{m}}$$

$$\text{Hence, } v_{\text{rms}} \propto \frac{1}{m}$$

So, lower the mass, higher/greater the velocity.

(where, the symbols have their usual meanings)

44. Option (3) is correct.

Explanation: Given,

$$V = \frac{-K}{x} (\text{J/kg})$$

$$\text{As, } E_x = \frac{-dv}{dx}$$

$$= K \frac{d(x-1)}{dx}$$

$$= \frac{-K}{x^2} = \frac{-K}{(2)^2} \quad (\text{at } x = 2)$$

$$= \frac{-K}{4}$$

45. Option (3) is correct.

Explanation: As per question,

$$\begin{aligned} \frac{F}{l} &= \frac{\mu_0}{4\pi} \times \frac{2I_1I_2}{r} \\ &= \frac{4\pi \times 10^{-7} \times 2 \times 5 \times 10}{2\pi \times 0.1} \\ &= 10^{-4} \text{ Nm}^{-1} \end{aligned}$$

The nature of the force is attractive.

(where, the symbols have their usual meanings)

46. Option (1) is correct.

Explanation: As per question, we have;

$$\begin{aligned} B &= B_0 \sin^3 \theta \\ &= \frac{\mu_0 I}{2\pi} \sin^3 (45^\circ) \\ &= \frac{4\pi \times 10^{-7} \times \sqrt{2}}{2 \times 1} \left(\frac{1}{\sqrt{2}} \right)^3 \\ &= 3.14 \times 10^{-7} \text{ T} \end{aligned}$$

47. Option (4) is correct.

Explanation: As per question, we have;

$$\begin{array}{ccc} 100^\circ\text{C} & \boxed{\text{Cu} \quad \text{Steel}} & 0^\circ\text{C} \\ & \theta = ? & \end{array}$$

$$\text{As } K = \frac{Qd}{A \Delta t},$$

where, K = thermal conductivity

Q = amount of heat transferred

d = distance between two planes

A = area of surface, and

ΔT = difference in Temp.

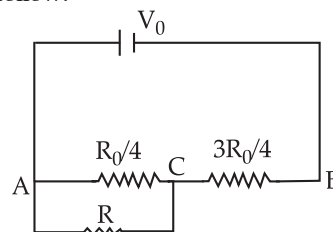
$$\begin{aligned} \frac{\Delta Q}{\Delta t} &= \frac{385 \times A \times (100^\circ\text{C} - \theta)}{l} \\ &= \frac{50 \times A \times (\theta - 0^\circ\text{C})}{l} \end{aligned}$$

$$\Rightarrow 385 (100^\circ\text{C} - \theta) = 50 (\theta - 0^\circ\text{C})$$

$$\Rightarrow \theta = 88.5^\circ\text{C}$$

48. Option (1) is correct.

Explanation: The given diagram can be redrawn as follow:



The potential across AC,

$$V_{AC} = \frac{R_{AC}}{R_{BC} + R_{AC}} \times V_0$$

$$= \frac{\frac{RR_0}{4R + R_0}}{\frac{3R_0}{4} + \frac{RR_0}{4R + R_0}} \times V_0$$

$$= \frac{4RV_0}{3R_0 + 16R}$$

49. Option (1) is correct.

Explanation: A.T.Q, we have,

$$\frac{F_e}{F_G} = \frac{\frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}}{\frac{Gm_1 m_2}{r^2}}$$

$$2.4 \times 10^{39} = \frac{K}{G} \times \frac{(1.6 \times 10^{-19})^2}{(9.1 \times 10^{-31} \times 1.67 \times 10^{-27})}$$

$$\left[\because K = \frac{1}{4\pi\epsilon_0} \right]$$

Hence,

$$\frac{K}{G} = \frac{2.4 \times 10^{39} \times 9.1 \times 10^{-31} \times 1.67 \times 10^{-27}}{1.6 \times 10^{-19} \times 1.6 \times 10^{-19}}$$

$$= 14.247 \times 10^{19}$$

$$= 1.4247 \times 10^{20} \approx 10^{20}$$

50. Option (3) is correct.

Explanation: Given:

$$g = \frac{4\pi^2 L}{T^2}$$

$$L = (10 \pm 0.1) \text{ cm, and}$$

$$T = (100 \pm 1) \text{ s}$$

Hence, error in measurement of g is

$$\frac{\Delta g}{g} = \frac{\Delta L}{L} + 2 \frac{\Delta T}{T}$$

$$= \frac{0.1}{10} \times 100 + 2 \times \frac{1}{100} \times 100$$

$$= 3\%$$

□□□

CHEMISTRY

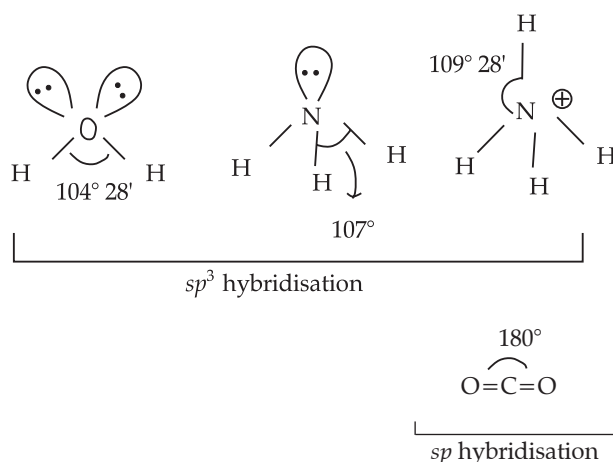
Section A

51. Option (1) is correct.

Explanation: According to VSEPR theory, bond angle decreases with increase in lone pair, for molecules of same hybridisation.

With increase in s -character, bond angle increase.

$\therefore$ order of bond angle =



52. Option (1) is correct.

Explanation: K_H = Henry Law constant of gas in water, related to solubility as:

$$\text{Solubility of a gas} \propto \frac{1}{K_H \text{ Value}}$$

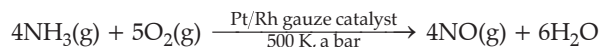
This means higher is the value of K_H , lower will be the solubility of gas in water.

$\therefore$ Solubility order of given gas in water

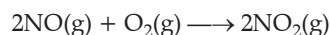
$$= HCHO > CH_4 > CO_2 > Ar$$

53. Option (2) is correct.

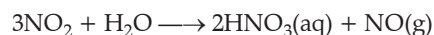
Explanation: This is the Ostwald process of preparing nitric acid on industrial scale.



$NO(g)$ combines with oxygen to give $NO_2(g)$.

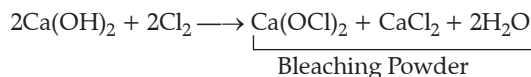


Nitrogen dioxide formed, dissolves in water to give HNO_3



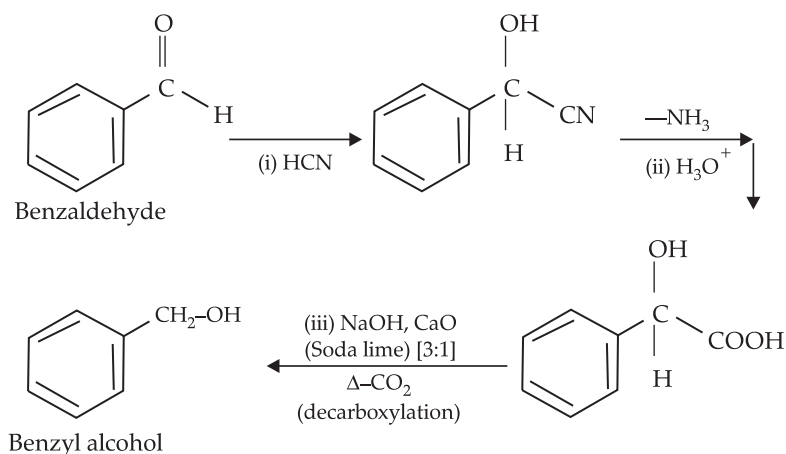
54. Option (3) is correct.

Explanation: Bleaching powder is prepared by passing Cl_2 gas through dry slaked lime.



55. Option (2) is correct.

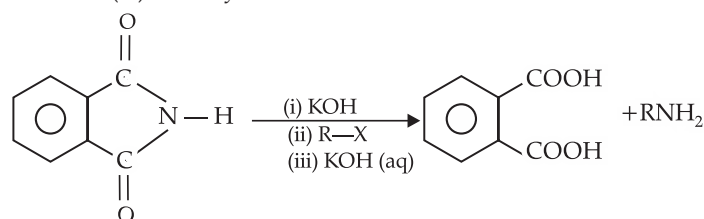
Explanation:



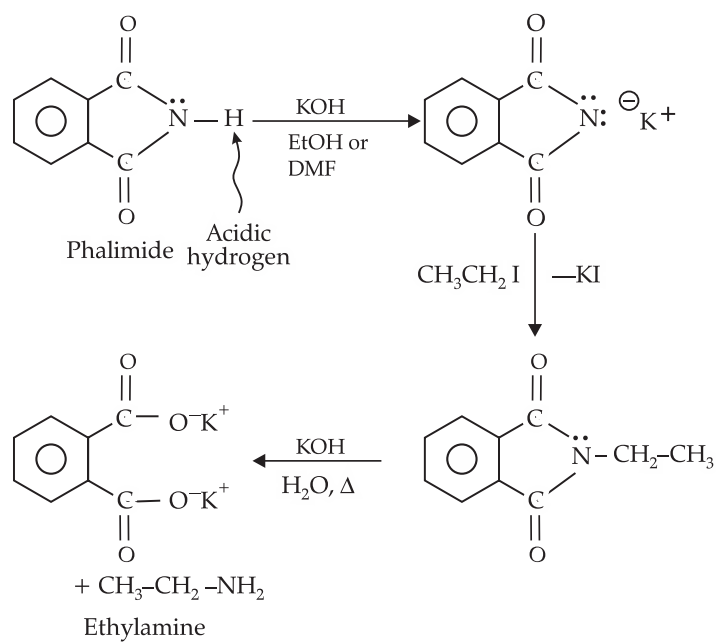
56. Option (4) is correct.

Explanation:

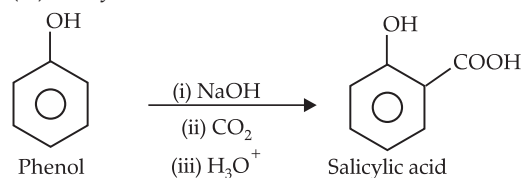
(a) Gabriel synthesis $\longrightarrow$ (iii) Primary amines



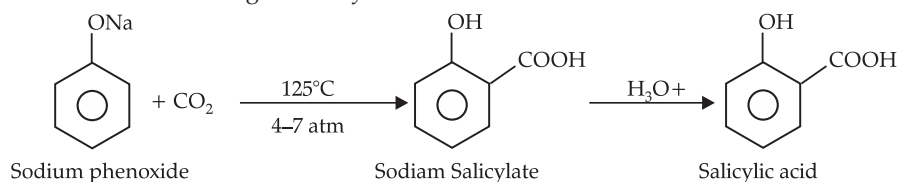
Phthalimide is alkylated with primary or unbranched secondary alkyl halides or sulphonate. The product thus obtained is hydrolysed to give primary amines.



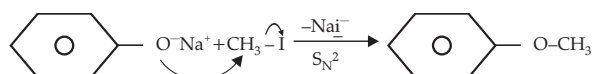
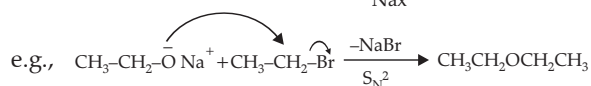
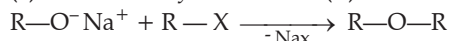
(b) Kolbe's synthesis $\longrightarrow$ (iv) Salicylic acid.



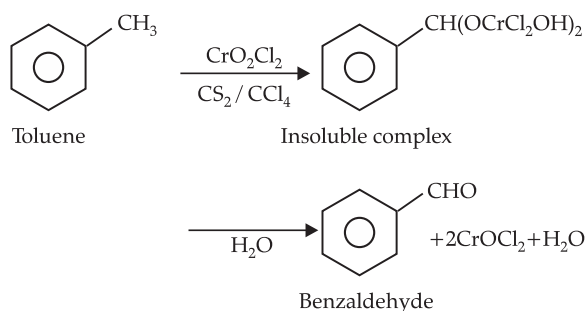
Heating of sodium phenoxide with CO_2 gas under pressure produces sodium salicylate as the major product, which on further acidification gives salicylic acid.



(c) Williamson synthesis $\longrightarrow$ (ii) Ethers



(d) Etard Reaction $\longrightarrow$ (i) Benzaldehyde



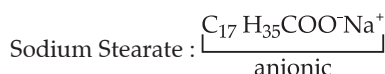
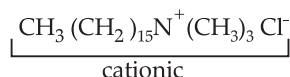
Treatment of toluene with chromyl chloride in an inert solvent like CS_2 or CCl_4 forms an insoluble complex $\text{C}_6\text{H}_5\text{CH}(\text{OCrCl}_2\text{OH})_2$ which further on hydrolysis produces benzaldehyde.

57. Option (3) is correct.

Explanation:



Cetyltrimethyl ammonium chloride :



61. Option (2) is correct.

Explanation:

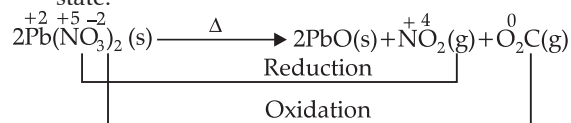
NH_3	ClF_3	PCl_5	BrF_5
bp = 3, lp = 1 Steric No. = 4 Hybrid $-\text{sp}^3$	bp = 3, lp = 2 Steric No. = 5 Hybrid $-\text{sp}^3\text{d}$	bp = 5, lp = 0 Steric No. = 5 Hybrid sp^3d	bp = 5, lp = 1 Steric No. = 6 Hybrid sp^3d^2
Trigonal pyramidal	T-shaped	Trigonal Bipyramidal	Square Pyramidal

(Component of bath soap or toilet soap)

Polyethyleneglycol stearate – Non ionic reagent.

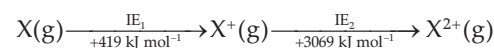
58. Option (1) is correct.

Explanation: Redox decomposition reaction is associated with breakdown of a large molecule into smaller molecule with change in oxidation state.



59. Option (1) is correct.

Explanation: The highest jump ion successive ionisation energy indicates a stable noble gas configuration.



The highest jump is observed in the IE_1 & IE_2 . It shows the attainment of stable noble gas configuration of X after losing one electron. Hence, X is an alkali metal.

The IE_1 of Y is $+590 \text{ kJ mol}^{-1}$ which is greater than IE_1 of X due to ns^2 configuration.

$\therefore$ X = Alkali metal and Y = Alkaline Earth metal

60. Option (3) is correct.

Explanation: The rate of reaction decreases with increase in stearu crowding around the substrate centre.

$$\therefore \text{Rate of } \text{S}_{\text{N}}^2 \propto \frac{1}{\text{Steric crowding}}$$

$\therefore$ Rate of reactivity order is : I > III > II > IV

62. Option (2) is correct.

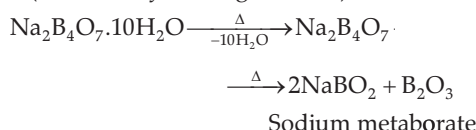
Explanation:

Thermoplastic polymer - Those plastics which can be remoulded again and again.**Thermosetting plastics** - Those plastics which becomes hard on cooling and permanently sets into a solid.(1) **Bakelite** - Thermosetting plastic polymer (A phenol - formaldehyde resin)(2) **Polythene** - Thermoplastic polymer.(3) **Urea - formaldehyde resin** - A thermosetting plastic polymer.(4) **Melamine polymer** - Thermosetting plastic polymer.

63. Option (1) is correct.

Explanation: (a) **Borax** - $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$ or $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$ (b) **Kernite** - $\text{Na}_2\text{B}_4\text{O}_7 \cdot 4\text{H}_2\text{O}$ (c) **Orthoboric acid** - H_3BO_3 (d) **Borax bead** - NaBO_2

(obtained by heating of borax)



64. Option (3) is correct.

Explanation: The cell reaction is written as :

$$E^\circ_{\text{Co}^{2+}/\text{Co}^{3+}} = -1.81 \text{ V (SOP)}$$

$$\therefore E^\circ_{\text{Co}^{3+}/\text{Co}^{2+}} = +1.81 \text{ V (SRP)}$$

$$E^\circ_{\text{Al}/\text{Al}^{3+}} = +1.66 \text{ V (SOP)}$$

$$\therefore E^\circ_{\text{Al}^{3+}/\text{Al}} = -1.66 \text{ V (SRP)}$$

Here E° is given in standard Reduction Potential

$$E^\circ_{\text{Co}^{3+}/\text{Co}} > E^\circ_{\text{Al}^{3+}/\text{Al}}$$

$\therefore \text{Co}^{3+}/\text{Co}$ will undergo reduction acting as cathode and Al^{3+}/Al will undergo oxidation acting as anode.

$$\therefore E^\circ_{\text{cell}} = E^\circ_{\text{cathode (Red.)}} - E^\circ_{\text{Anode (oxide)}}$$

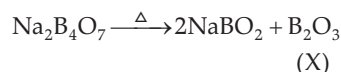
$$E^\circ_{\text{cell}} = 1.81 - (-1.66)$$

$$E^\circ_{\text{cell}} = +3.47 \text{ V}$$

65. Option (2) is correct.

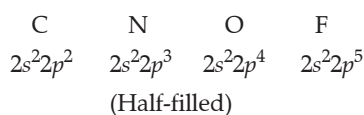
Explanation: Oxy-hydrogen flame produces a very high temperature which is used for welding and melting of metals with high melting point.

66. Option (2) is correct.



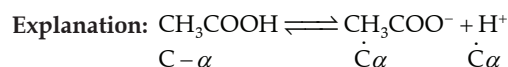
X = Boric anhydride

67. Option (3) is correct.

Explanation:Across the period Z_{eff} increases but nitrogen has half-filled orbital which makes it more stable.

$$\therefore \text{Order of } IE_1 = \text{C} < \text{O} < \text{N} < \text{F}$$

68. Option (3) is correct.

Ionisation of CH_3COOH is 1% in aqueous solution.

$$\text{So, } [\text{H}^+] = \frac{1}{100} \times \text{concentration of } \text{CH}_3\text{COOH}$$

$$= \frac{1}{100} \times 10^{-2} = 10^{-4}$$

$$\text{pH} = -\log [\text{H}^+] = -\log (1 \times 10^{-4})$$

$$\text{pH} = 4$$

69. Option (2) is correct.

Explanation: 'x' is mass of gas adsorbed on mass 'm' of the adsorbent at pressure 'p'. The curves in the given adsorption isotherm indicates that at a fixed pressure, there is decrease in physical adsorption with increase in temperature and vice versa.

So, the correct order of temperature is :

$$T_3 > T_2 > T_1$$

70. Option (2) is correct.

Explanation: Half life period ($t_{1/2}$) = 2000 years

$$k = \frac{0.693}{2000}$$

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]_t}$$

$$\frac{0.693}{2000} = \frac{2.303}{8000} \log \frac{[A]_0}{0.02}$$

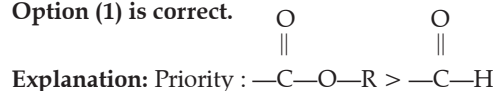
$$\therefore [A]_0 = 0.32 \text{ M}$$

71. Option (4) is correct.

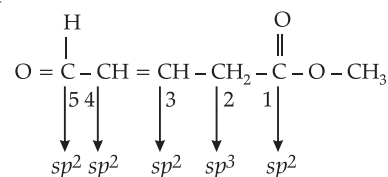
Explanation: In an isothermal expansion or compression process, the temperature of system remains constant throughout the process. Since, for an ideal gas, U depends only on temperature, it follows that

$$dU = 0$$

72. Option (1) is correct.

**Explanation:** Priority : $-\text{C}-\text{O}-\text{R} > -\text{C}-\text{H}$

$\therefore$ The priority of numbering goes to alkoxy carbonyl group.



73. **Option (2) is correct.**

Explanation: $\text{Density} = \frac{\text{Mass}}{\text{Volume}}$

Density = 2.15 g/mL, Volume of solution = 2.5 mL

$$\therefore 2.15 = \frac{\text{Mass}}{2.5} \Rightarrow \text{Mass} = 5.375 \text{ g}$$

The correct significant figure should have the same number of significant digits as the number with least number of significant digits.

$$\therefore \text{Mass} = 5.4 \text{ g}$$

74. **Option (1) is correct.**

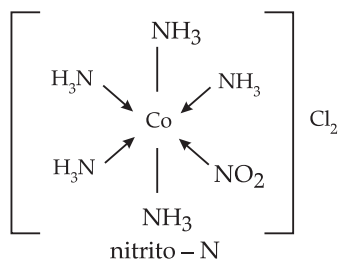
Explanation: Even though fluorine has lower electron gain enthalpy than chlorine, it is a strong oxidising agent. The correct reason that can be attributed for this behaviour is:

- (i) Small size of fluorine, makes it suitable for its high hydration enthalpy.
- (ii) The bond dissociation enthalpy of fluorine is lower than chlorine due to high lone pair lone pair repulsion.

75. **Option (3) is correct.**

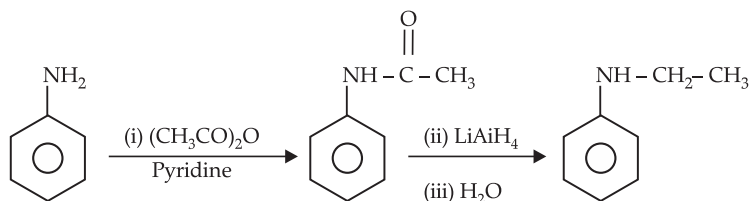
Explanation:

(a) $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_5\text{ONO}]\text{Cl}_2$ – linkage Isomerism due to difference in binding site.



77. **Option (1) is correct.**

Explanation:



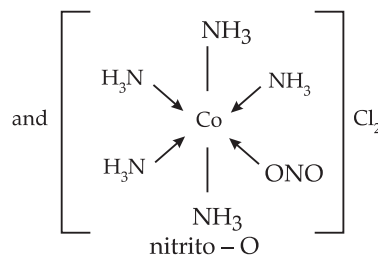
78. **Option (4) is correct.**

Explanation: (a) **Frenkel Defect**—formed in ionic compounds where an ion is missing from its lattice site and occupies interstitial site. Compounds suffering from such defects does not have any change in their density.

(b) **Schottky defect**—Observed in ionic compounds where equal number of ions are found missing

79. **Option (1) is correct.**

Explanation: Chlorine is *ortho* directing in electrophilic aromatic substitution instead of its electron withdrawing nature because inductive effect of chlorine destabilises the intermediate carbocation which is formed during



(b) $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$ and

$[\text{Cr}(\text{CN})_6][\text{Co}(\text{NH}_3)_6]$

– It is an example of coordination Isomerism due to exchange of ligands between the two coordination spheres.

(c) $[\text{Co}(\text{NH}_3)_5(\text{SO}_4)]\text{Br}$ and $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$

– It is an example of Ionisation Isomerism arising due to formation of different ions due to ionisation. The case arises due to exchange of ions between the coordination sphere and ionisation sphere.

(d) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ and $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$

– It is an example of solvate isomerism arising due to difference in H_2O molecules acting as potential ligand and as water of crystallisation.

76. **Option (3) is correct.**

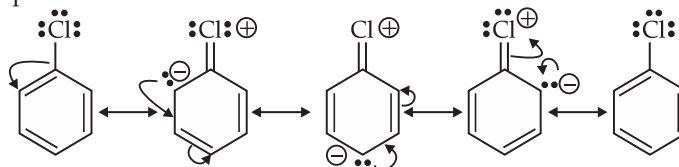
Explanation: Enzymes act at of optimum temperature and pH. Tertiary structure is responsible for biological activities of proteins. With increase in temperature weather H-bond breaks and denaturation occurs resulting in uncoiling of helical structure.

from their lattice site. Such compounds have lower density.

(c) **Vacancy defect**—Observed in non-ionic solids where either an anion or cation is missing from their sites. Density of such solids are lower.

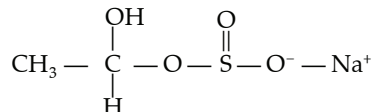
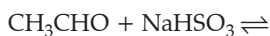
(d) **Interstitial defects**—Observed in non-ionic solids where an extra cation may get into interstitial site, increasing the density of solid.

the electrophilic substitution. But because of more strong resonance effect, the halogen stabilises the carbocation at *ortho*- and *para*- position.



80. Option (1) is correct.

Explanation: Nucleophilic addition – elimination reaction is associated with elimination of H_2O .

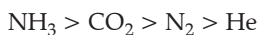


Here the above reaction is an addition reaction.

81. Option (4) is correct.

Explanation: Higher is the critical temperature, easier is the liquefaction of gas.

The order of liquefaction of gases will be:



Hence, among the given gases NH_3 has the highest critical temperature. Therefore, it will be liquefied first.

82. Option (3) is correct.

Explanation: More is the number of poor shielders, greater will be Z_{eff} and hence, smaller is the size.

Actinoids consists of more no. of poor shielders (3d, 4d, 5d, 4f). Therefore, size continuously decreases left to right.

83. Option (4) is correct.

Explanation: With increase in molecular mass, van Der Waal force increases and hence boiling point also increases.

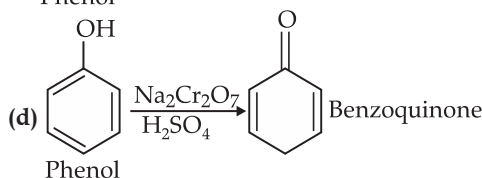
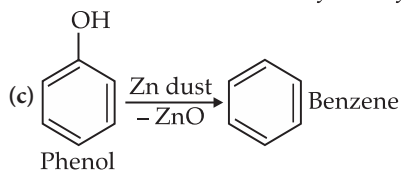
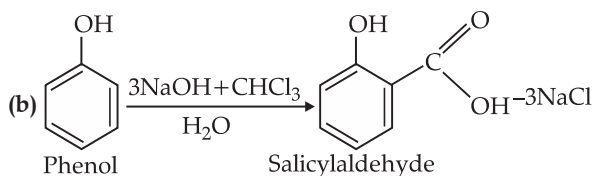
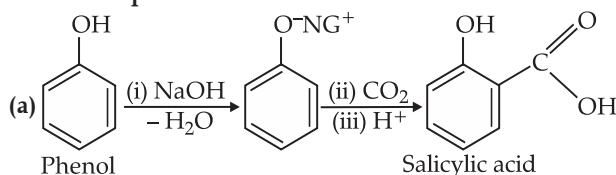
$\therefore$ BP : Heptane > Hexane > 2-Methyl butane

with the increase in branching, surface area decreases of molecules of same molecular mass which results in decrease in boiling point.

$\therefore$ BP : butane > α -methyl propane.

84. Option (3) is correct.

Explanation:



85. Option (4) is correct.

Explanation: $n = 1$ $l = 0$ (s) $n = 4$ $l = 0$ (s)

$n = 2$ $l = 0$ (s) 1 (p)

1 (p) 2 (d)

$n = 3$ $l = 0$ (s) 3 (f)

1 (p)

2 (d)

Principle QN (n)	Azimuthal QN (l)	Possible values ($n - 1$)	Subshell
$n = 1$	0		s
$n = 2$	0		s
	1		p
$n = 3$	0		s
	1		p
	2		d
$n = 4$	0		s
	1		p
	2		d
	3		f

So, for $n = 2, l = 1$, orbital is 2p

for $n = 3, l = 2$, orbital is 3d

for $n = 3, l = 0$, orbital is 3s

for $n = 2, l = 0$, orbital is 2s

86. Option (4) is correct.

Explanation: Calcination process involves heating of ore in absence of air below melting point. But here reaction taking place is a displacement reaction.

87. Option (4) is correct.

Explanation: Since, the striking photon has energy equal to $h\nu$ and the minimum energy required to eject the electron is $h\nu_0$ (work function W_0), the difference in energy ($h\nu - h\nu_0$) is transferred as the kinetic energy of the photoelectron.

$$h\nu = h\nu_0 + \text{KE}$$

$h\nu_0$ = minimum energy required to remove electron)

Energy of a 300 nm photon is given by

$$\Rightarrow E = \frac{hc}{\lambda}$$

$$= \frac{6.626 \times 10^{-34} \text{ J s} \times 3 \times 10^8 \text{ ms}^{-1}}{300 \times 10^{-9} \text{ m}}$$

$$= 6.626 \times 10^{-19} \text{ J}$$

Energy of one mole of photons

$$= 6.626 \times 10^{-19} \text{ J} \times 6.022 \times 10^{23} \text{ mol}^{-1}$$

$$= 3.99 \times 10^5 \text{ J mol}^{-1}$$

The minimum energy needed to remove one mole of electrons = $(3.99 - 1.68) \times 10^5 \text{ J mol}^{-1}$

$$= 2.31 \times 10^5 \text{ J mol}^{-1}$$

88. Option (1) is correct.

Explanation: $4A + 3B \longrightarrow 6C + 9D$

$$\text{Rate of reaction} = -\frac{1}{4} \frac{d[A]}{dt} = -\frac{1}{3} \frac{d[B]}{dt}$$

$$= \frac{1}{6} \frac{d[C]}{dt} = \frac{1}{9} \frac{d[D]}{dt}$$

Rate of reaction = Rate of formation of C.

$$\therefore r = \frac{1}{6} \frac{d[C]}{dt} = \frac{1}{6} \times 6 \times 10^{-2}$$

$$= 1 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$$

Also, rate of reaction = Rate of disappearance of B

$$r = -\frac{1}{3} \frac{d[B]}{dt} = \frac{d[B]}{dt} = 3 \times r$$

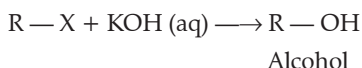
$$= 3 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$$

Consumption of B in 10 seconds = $3 \times 10^{-2} \times 10$

$$= 30 \times 10^{-2} \text{ mol L}^{-1}$$

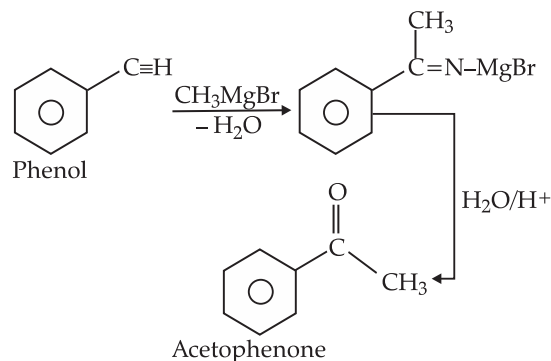
89. Option (3) is correct.

Explanation: Alkyl halide in aqueous KOH results in formation of alcohol.



90. Option (4) is correct.

Explanation:



91. Option (1) is correct.

Explanation: Let's assume Fe in $\text{Fe}^{2+} = x$

Let's assume Fe in $\text{Fe}^{3+} = (0.96 - x)$

Using the charge conservation formula for $\text{Fe}_{0.96}\text{O}_{1.00}$,

We get: $(x)(+2) + (0.96 - x)(+3) - 2 = 0$

$$= 2x + 2.88 - 3x - 2 = 0$$

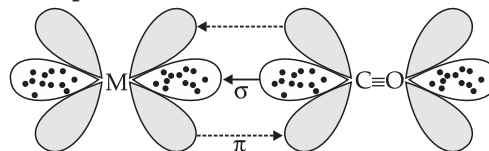
$$x = 0.88$$

$$\text{So, number of ions of } \text{Fe}^{3+} = 0.96 - x = 0.96 - 0.88 = 0.08$$

$$\text{Fraction of } \text{Fe}^{3+} \text{ ion} = \frac{0.08}{0.96} = \frac{1}{12}$$

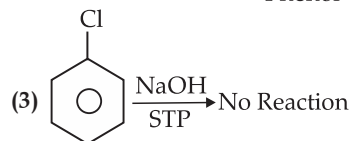
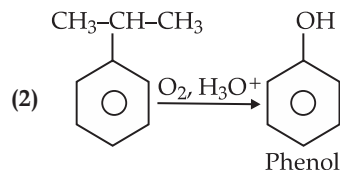
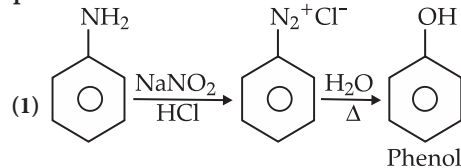
92. Option (3) is correct.

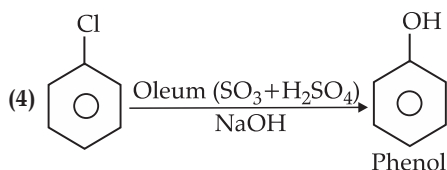
Explanation:



93. Option (3) is correct.

Explanation:





94. Option (1) is correct.

Explanation:

$$\text{No. of moles of O}_2 = \frac{\text{Given weight of O}_2}{\text{Mol. weight of O}_2} = \frac{3.2}{32} = 0.1$$

$$\text{Molar vol. at STP} = 0.1 \times 22.4 = 2.24 \text{ L}$$

$$P_1 = 1 \text{ atm}, V_1 = 2.24 \text{ L}, P_2 = \frac{1}{3} \text{ atm}, V_2 = ?$$

According to Boyle's law : $P_1 V_1 = P_2 V_2$

$$\therefore 1 \times 2.24 = \frac{1}{3} \times V_2$$

$$V_2 = 2.24 \times 3 = 6.72 \text{ L}$$

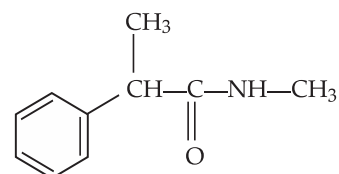
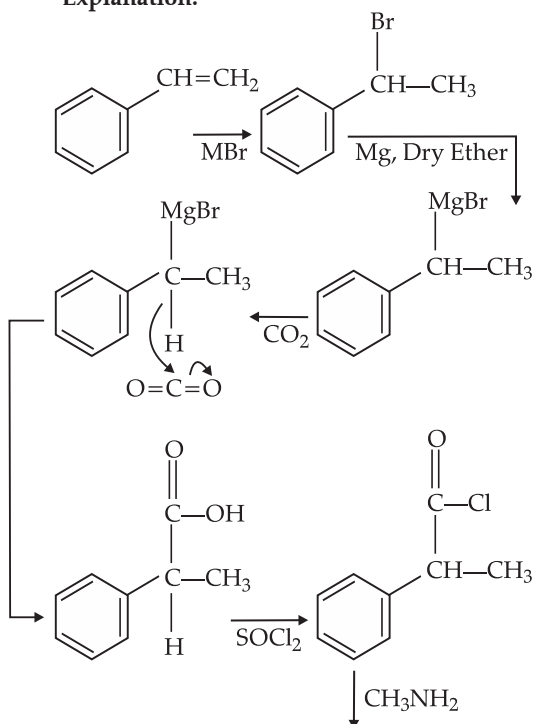
95. Option (3) is correct.

Explanation:

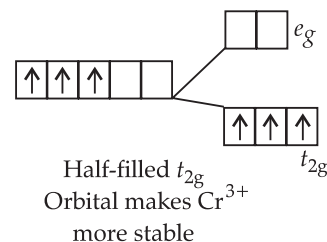
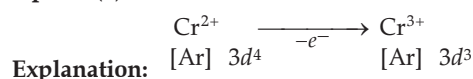
- (a) Biological oxygen demand (BOD) – Organic matter in water
- (b) Photochemical smog – Oxidising in nature (occurs in warm, dry and sunny climate)
- (c) Classical smog – Reducing in nature (occurs in cool and humid climate)
- (d) Ozone layer depletion – Polar stratospheric cloud

96. Option (2) is correct.

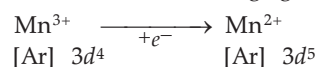
Explanation:



97. Option (4) is correct.



Therefore, Cr^{2+} acts as a reducing agent.

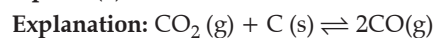


Half filled d -orbital (more stable)

Therefore, Mn^{3+} acts as an oxidising agent.

Sc^{3+} — $[\text{Ar}]$ All elements are paired making it diamagnetic. Hence Sc^{3+} compounds are repelled when placed in an external magnetic field.

98. Option (2) is correct.



$$K_p = K_c (\text{RT})^{\Delta n_g}$$

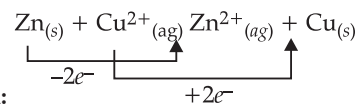
$$K_p = 3.0, R = 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1}$$

$$\Delta n_g = 2 - 1 = 1$$

$$\therefore 3.0 = K_c (0.083 \times 1000)^1$$

$$K_c = \frac{3}{83} = 3.6 \times 10^{-2}$$

99. Option (2) is correct.



Explanation:

$$E_{\text{Cell}} = 1.1 \text{ V}, n = 2, f = 96487 \text{ C mol}^{-1}$$

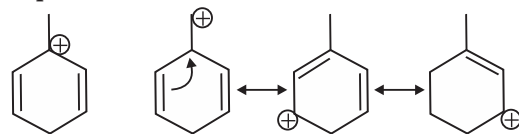
$$\Delta G^\circ = -n F E_{\text{Cell}}$$

$$= -2 \times 96487 \times 1.1 \text{ J}$$

$$= -212.27 \text{ kJ}$$

100. Option (4) is correct.

Explanation:



– No. of α -H = 3

– 3° carbocation

– Resonance stabilised

BOTANY

Section A

101. Option (3) is correct.

Explanation: Diplotene (a stage of Prophase-I of Meiosis) which can delay for months or years in oocyte of some vertebrates such as humans. In human female, cycle the cell arrest during embryonic stage and continue after puberty. It is completed at the time of ovulation during menstruation.

102. Option (1) is correct.

Explanation: During C_3 cycle (calvin cycle) 3 phases continue, namely **carboxylation**, **Reduction & regeneration**. During reduction or fixation phase, each CO_2 molecule require one mole of ATP and one mole of NADPH.

103. Option (1) is correct.

lac operon is an example of polycistronic operon in which 3 gene are present namely Z, Y and A.

Z — β -galactosidase

Y — Permease

A — Transacetylase

104. Option (4) is correct.

Explanation: In Dicot root, the vascular bundles are **radial**, in which vascular cambium is purely secondary and arise from **pericycle** with **lateral roots** as well. Lateral roots thus have **endogenous** origin.

105. Option (1) is correct.

Explanation:

Adenine	— A nitrogenous purine base
Anthocyanin	— Purple coloured pigment in vacuole
Chitin	— Polymer of aminated sugar i.e., N-acetyl-D-glucosamine, present in the exoskeleton of insects, molluscs, nematodes, cell wall of fungi etc.
Codeine	— Alkaloid (acts on CNS to relieve mild to moderate pain)

106. Option (2) is correct.

Explanation: i-gene in *lac* operon code for repressor while z codes for β -galactosidase, y for permease and a is for Transacetylase. All these genes are a part of single structure called *lac operon*.

107. Option (2) is correct.

Explanation:

<i>Chlamydomonas</i>	— A green algae
<i>Cycas</i>	— Fossil gymnosperm
<i>Selaginella</i>	— Pteridophyte, Microphyllous
<i>Sphagnum</i>	— Moss, Peat Mass/cotton mass.

108. Option (1) is correct.

Explanation: DNA polymerase can read template DNA in only one direction i.e., $3' \rightarrow 5'$ thus DNA replication can continue only in $5' \rightarrow 3'$ direction, thus it can continue only on one strand of DNA. The DNA stand on which replication is continuous is called **leading strand** whereas the strand on which replication is discontinuous is said to be **lagging strand**.

109. Option (2) is correct.

Explanation: Above mentioned floral diagram is of the family **Brassicaceae**. It is indicated by two unique characters—(1) Tetrameric flower i.e., (four petalled cross-shaped) and (2) Tetradynamous condition of stamens (two long and two short).

110. Option (3) is correct.

Explanation: Hydrarch succession is a series of communities in aquatic habitat. It is started with **phytoplankton** as pioneer species. Both Hydrarch and Xerarch are ended in nearly same type of condition called **Mesarch**.

111. Option (2) is correct.

Explanation: Isocitrate (Isocitric acid) undergoes two times decarboxylation during TCA cycle. During first decarboxylation, it forms α -Ketoglutaric acid and in the second time, it forms succinyl Co-A

112. Option (4) is correct.

Explanation: Sick cell anemia is autosomal recessive trait while haemophilia is X-linked recessive trait.

Sickle cell anaemia is blood disorder while haemophilia is the disorder of blood clotting.

113. Option (1) is correct.

Explanation: With the help of most restriction enzymes sticky ends in DNA can be created. The flanking region in DNA can be formed due to cutting of DNA strand little away from restriction site.

114. Option (4) is correct.

Explanation: Amazon rain forest contains following number of species in respective group.

Invertebrates	125000
Plants	40000
Fishes	3000
Birds	1300
Mammals	427

115. Option (2) is correct.

Explanation:

Porins	Huge pores in outer membrane of Mitochondria, Chloroplast & some bacteria.
Leg Haemoglobin	pink coloured pigment, O_2 scavenger

H ⁺ accumulation	Lumen of thylakoid (Inside chloroplast)
Respiration	Amphibolic (catabolic + Anabolic) pathway.

116. **Option (2) is correct.**

Explanation: Cytokinin is a cell division inducing hormone derived from **Adenine**, a purine type nitrogenous base.

→ Cytokinin was first obtained from autoclaved Herring sperm DNA.

117. **Option (4) is correct.**

Explanation: In fruit wall of several nuts, various types of sclereid tissues are present. These are called **stone cells**. These are dead cells with obliterated lumen.

118. **Option (1) is correct.**

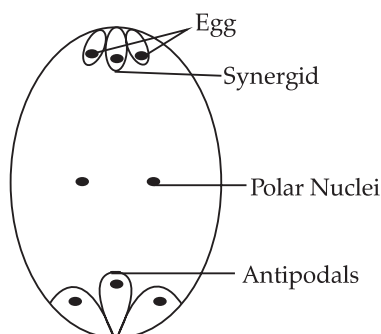
Explanation: During process of succession, the species which occupy bare are first time is called **Pioneer** species. It is the species with least adaptation and highest environmental resistance.

119. **Option (1) is correct.**

Explanation: A typical embryo sac is 8 nucleate, 7 celled structure formed in most angiosperms.

It has a 3 celled egg apparatus in which one egg cell and two synergids are present.

Note: The answer has been provided with respect to the embryo sac, not egg apparatus.



120. **Option (2) is correct.**

Explanation:

Imbricate	Gulmohar, <i>Cassia</i> etc (Fabaceae) <i>Calotropis</i> (Asclepiadaceae)
Valvate	Beans (Fabaceae or leguminosae)
Vexillary	
Twisted	Cotton (Malvaceae)

121. **Option (2) is correct.**

Explanation: All succession, either started from water body or sand, leads to the Mesic condition.

122. **Option (4) is correct.**

Explanation: With the help of Gel electrophoresis technique, separation of DNA fragments, based on their size (mol. wt) can take place. DNA fragments

can be separated on Agarose gel with the help of electrode and can be visualised with the help of ethidium bromide under UV-light.

123. **Option (2) is correct.**

Explanation: The process of conversion of a **Non-dividing** (permanent cell) to a dividing cell is called **de-differentiation**. This is an important process for secondary growth which takes place in plants.

124. **Option (2) is correct.**

Explanation: During Meiosis-I, the stage **Prophase I** occurs in which various substages are present. Of these substages, the most important is **Pachytene stage** in which exchange of chromosomal segments takes place between non-sister chromatids of homologous chromosome, with the help of enzyme recombinase.

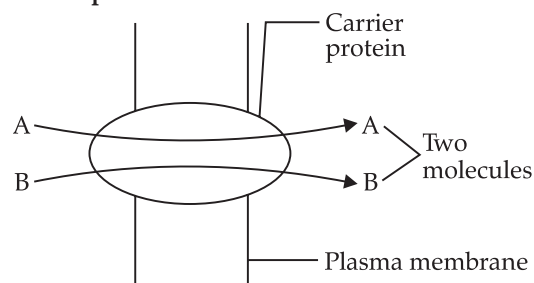
125. **Option (1) is correct.**

Explanation: During TCA cycle, after first decarboxylation of oxalosuccinic acid, α -Ketoglutaric acid 5-C compound (α -KGA) is formed.

α -KGA undergoes further decarboxylation to form a 4 carbon compound i.e., succinyl-CO-A.

126. **Option (4) is correct.**

Explanation:



Above condition is called **Symport** in which the two molecules are moving in the same direction. This is a type of **Facilitated diffusion** of non-Polar substance from plasma membrane.

127. **Option (1) is correct.**

Explanation: World summit on sustainable development takes place to ensure significant reduction in the current rate of Biodiversity loss. This summit was held in 2002 in Johannesburg, South Africa.

128. **Option (3) is correct.**

Explanation: Inter fascicular cambium is present between two vascular bundles and formed from de differentiation of permanent tissues in medullary rays. Interfascicular cambium is present in between primary xylem and primary phloem.

129. **Option (3) is correct.**

Explanation: Ascent of sap in xylem is mainly due to transpiration pull which is supported by physical properties of water such as cohesion and adhesion between water molecules. Cohesion is also called **Tensile strength of water molecule**.

- 130.
- Option (3) is correct.**

Explanation: The Rhizome is underground stem modification in which nodes and internodes are distinct. The nodes contain several buds from which new plants arise. It is prostrate and branched. e.g. **Sugarcane, Banana**, etc.

- 131.
- Option (2) is correct.**

Explanation: During the process of artificial pollination in pea plant, emasculation is only needed in Bisexual plant, selected as female parent. After removing stamens from such flower and transferring selected pollens, it needs to be bagged to prevent non-desired pollen to come in contact with this flower.

- 132.
- Option (3) is correct.**

Explanation: Persistent nucellus in beet and pepper is called **Perisperm**. Nucellus is diploid sporogenous tissue which is generally degraded, but present in above mentioned seeds.

- 133.
- Option (2) is correct.**

Explanation: Chromosomal theory of inheritance was proposed by **Sutton** and **Boveri** in 1902. They both connect the idea of Mendelian factor to the new concept of chromosome and find very close similarity between these two.

- 134.
- Option (2) is correct.**

Explanation: **Leg-Haemoglobin** is a pink O_2 -scavenger-pigment present in root nodules of leguminous plants. It helps in protection of **Nitrogenase** from the O_2 , as it is O_2 -sensitive enzyme.

- 135.
- Option (3) is correct.**

Explanation: Phenotypic plasticity is the process through which plant follow different pathways in response to different environmental conditions. For example Heterophylly in **Cotton, Coriander** and **Larkspur**.

Section B

- 136.
- Option (2) is correct.**

Explanation: **Azotobacter** and **Beijerinckia** are free living, N_2 fixing, Aerobic bacteria. **Rhizobium** and **Frankia** are associated bacteria. **Pseudomonas** and **Thiobacillus** are denitrifying bacteria while **Rhodospirillum** is **anaerobic** non, N_2 Fixing bacteria.

- 137.
- Option (1) is correct.**

Explanation:

Sacred grooves Zoological park	Khasi Hill in Meghalaya <i>Ex-situ</i> (off site) conservation Alien species—causes loss of 200 species of unique Assemblage of Chichilid Fishes
Nile perch	
Amazon forest	Lungs of planet Earth, release about 20% of total oxygen.

- 138.
- Option (3) is correct.**

Explanation: Primary structure in proteins are continuous, in which successive amino acids are added with the help of peptide bond formation.

It is simplest yet generally non-Functional structure of proteins.

- 139.
- Option (1) is correct.**

Explanation:

Gene gun	Particle gun, used for transfer of gene.
Gene therapy	Replacement of non-working gene by a working gene.
Gene cloning	To obtain identical copies of a particular DNA molecule.
Genome	Total DNA in the cells of an organism in Haploid state (n)

- 140.
- Option (3) is correct.**

Explanation:

Organism	Genome size
Bacteriophage $\phi \times 174$	5386 Nucleotides (SS)
Bacteriophage lambda	48502 bp
<i>E. Coli</i>	4.6×10^6 bp
Haploid content of Human DNA	3.3×10^9 bp

- 141.
- Option (3) is correct.**

Explanation: It Apomictic gene are introduced into hybrid crop. Varieties, the seed formed in them without fertilisation, leads to continuation of desired combination of characters. For unlimited generations, thus Farmers can keep their seeds for next year.

- 142.
- Option (2) is correct.**

Explanation: The given figure is of *Chara*, a green algae of family **chlorophyceae**, pigments are **chl a**, **b** and **carotene**. Stored food material is in the form of starch. It is monoecious and contain **Oogonium** (female sex organ) and **Antheridium** (male sex organ).

- 143.
- Option (3) is correct.**

Explanation: Fruit eating birds (Frugivorous) are present in tropical forest as food is available in throughout the year. For example, fig plant support such birds in their survival. Thus, Fig is also called **Keystone** species in TRF.

- 144.
- Option (2) is correct.**

Explanation: As plasma membrane is made up to phospholipid, it only allow free transfer of **Fat soluble substances** (Hydrophobic) it does not allow Hydrophilic substance to pass from it.

- 145.
- Option (2) is correct.**

Explanation: Chemi-osmotic hypothesis of ATP formation was given by **P. A. Mitcheli**. According to which proton flow causes ATP formation. Primary acceptor of electrons, i.e., Phaeophytin transfers its electron to **proton carrier** that is Plastoquinone, non an e^- carrier.

- 146.
- Option (4) is correct.**

Explanation: The symptoms mentioned in the question are of down's syndrome which is caused by trisomy of chromosome 21. Total chromosome number turn to be 47. It is also called **Mongoloid Idiocy**.

- 147.
- Option (3) is correct.**

Explanation: The order of events during prophase I of Meiosis-I is as follows:

Leptotene → Thickening of chromosome

Zygotene → Synapsis of chromosome

Pachytene → Crossing over

Diplotene → Dissolution of synaptonemal complex

Diakinesis → Terminalisation of chiasmata.

- 148.
- Option (2) is correct.**

Explanation: Cellulase can degrade cellulosic cell wall of plant cell while chitinase is used for degrading **Chitin** (Fungal cellulase) for obtaining DNA from cell.

- 149.
- Option (3) is correct.**

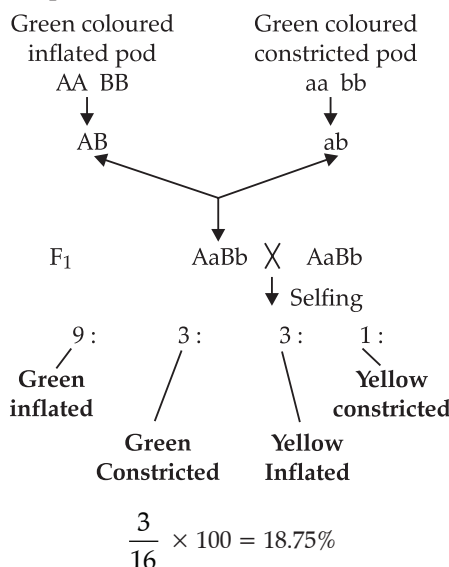
Explanation:

- Carbon dissolved in oceans = About 71%
- Annual fixation of CO₂ through photosynthesis is = 4×10^{13} kg
- PAR captured by plants = 2 – 10%

- Productivity of oceans = 55 billion tonnes

- 150.
- Option (3) is correct.**

Explanation:



ZOOLOGY

Section A

- 151.
- Option (2) is correct.**

Explanation: Plasmids are extra chromosomal DNA in bacterial cytoplasm which provide additional feature to the bacteria such as antibiotic resistance, N₂-Fixation, fertility, etc.

Plasmid may contain resistance genes for several antibiotics.

- 152.
- Option (2) is correct.**

Explanation: Transgenic cow **Rosie** produced milk rich in Human protein **α-Lactalbumin** which is easily digestible to babies with lactose intolerance.

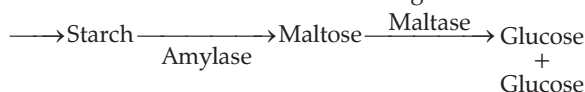
→ Such proteins which are produced in other organisms through DNA recombinant technology are called **recombinant proteins**.

- 153.
- Option (2) is correct.**

Explanation: Lysosomal enzymes are generally **Hydrolytic** enzymes which are functional in Acidic pH only. When pH is turned to basic, medium these enzymes will become inactive.

- 154.
- Option (1) is correct.**

Explanation: **Maltase** leads to breakdown of disaccharide maltose into two glucose subunits.



Salivary Pancreatic

- 155.
- Option (4) is correct.**

Explanation: During organ transplantation, the graft rejection takes place due to cell mediated response, in which T-cell controls the identification of **Self** from **Non-self**.

This rejection can be prevented by **Immuno-**

suppressant agent i.e., Cyclosporin-A.

- 156.
- Option (3) is correct.**

Explanation: If **Sulphur** and **Phosphorus** change their presence in DNA and protein respectively the result of **Hershey** and **Chase** experiment will be reversed. Bacteria with radioactive sulphur formed at the base of Test Tube.

- 157.
- Option (2) is correct.**

Explanation: Two competing species can co-exist for the same type of resource by changing their foraging pattern. These butterflies can obtain nectar at different times and chase different floral parts to avoid competition.

- 158.
- Option (4) is correct.**

Explanation: Mad cow disease (Bovine spongiform encephalopathy) and Cr. Jacob disease in humans are caused by misfolded, Infectious protein particle called **Prion**. These are proteins which attack neural tissues, generally.

- 159.
- Option (1) is correct.**

Explanation: Progestogens are pills containing progesterone. These are **NOT IUDs** as they can be taken in the form of daily/weekly or monthly pills.

- 160.
- Option (2) is correct.**

Explanation:

<i>Chlamydomonas</i>	Zoospore
<i>Penicillium</i>	Conidia
<i>Hydra</i>	Budding (External budding)
Sponges	Gemmule (internal budding)

These all are methods of Asexual reproduction.

161. Option (2) is correct.

Explanation: According to **Sliding filament theory**, thin. Filament slide over **Thick filament** (mainly myosin). That is why the length of A band does not change as A band is made up of thick filament.

162. Option (2) is correct.

Explanation: Productivity is the total amount of biomass or organic matter produced per unit area. Over a period of time by plants. This is also called **Primary production**.

→ Productivity of consumer is called Secondary production.

163. Option (1) is correct.

Explanation: Amino acids can exist in zwitterionic form which contains both positive and negative charges together, thus called **Amphoteric molecule**. These forms of amino acids, are having both moieties.

164. Option (3) is correct.

Explanation: Epithelium of Bronchiole and Fallopian tube is lined by the cilia for the movement of attached substances in specific direction.

→ Egg movement towards uterus is due to these surface outgrowths.

165. Option (3) is correct.

Explanation: Western ghats have large number of endemic species which are absent in all other parts of the world. These consist of amphibians and large number of flowering plants. Western-ghats is one of biodiversity hotspots in India.

166. Option (1) is correct.

Explanation: Accumulation of uric acid crystals leads to inflammation of joints which is commonly called as **Gout**. It is a metabolic disorder in which Uric acid degrading enzyme remains inactive or absent.

167. Option (3) is correct.

Explanation: Vital capacity does not include **RV** as this cannot be measure of by using respirometer. Vital capacity is the maximum inspiration after force full expiration.

168. Option (2) is correct.

Explanation: Organ to organ transportation of any substance/nutrients is called **portal system**, occurs via special veins called **Portal veins**. Transfer of material from digestive tract to liver is via **Hepato-Portal veins**.

169. Option (1) is correct.

Explanation:

Crop	— Storage chamber for food.
Pro. ventriculus (Gizzard)	→ Have Chitinous/Cuticular teeth for grinding the food particular.
Hepatic caecae	→ Gastric caeca — Secrete digestive juice

Malpighian Tubule	— Main excretory structure for the removal of nitrogenous waste
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170. Option (2) is correct.

Explanation: *Spirulina* (a BGA) can be used as an agent reduce environmental pollution. It absorb heavy metals and other harmful substances from environment thus reduce pollution. It can also be used as SCP (Single cell protein) as source of protein, carbohydrate, fat, mineral & vitamins.

171. Option (4) is correct.

Explanation: Panspermia is a hypothesis which explains the origin of life on earth due to transfer of spores from other planet of earth. The theory faces high level of criticism as environment between two planets cannot support Life.

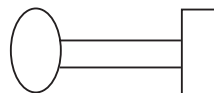
172. Option (3) is correct.

Explanation: Mammary glands have various structures arranged in defined order—from proximal to distal they shows—

Alveoli → Mammary tubules → Mammary duct → Mammary ampulla → Lactiferous duct

173. Option (3) is correct.

Explanation: In consanguine marriage/mating occurs between two close relatives and can be represented by two lateral lines between male and female as follows:



This lead to expression of numerous harmful genes.

174. Option (2) is correct.

Explanation: CNG (Compressed Natural Gas) is better fuel than diesel because it cannot be adulterated, It burns more efficiently and It is cheaper than diesel.

→ Its distribution is difficult.

175. Option (4) is correct.

Explanation: Bacteriophage can be used in introducing DNA into animal and bacterial cell, not plant cell.

Gene Gun, Agrobacterium mediated Transfer and Disarmed pathogen are used as vector for plant cells.

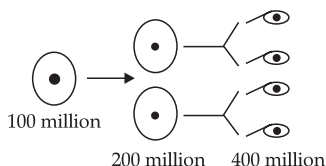
176. Option (2) is correct.

Explanation: Pneumotaxic centre present in the pons region of brain controls the respiratory rhythm centre. Single from this centre can control the duration of inspiration and expiration thus control and alter respiratory rate.

177. Option (3) is correct.

Explanation: Spermatogenesis is the formation of spermatozoa in males. Each secondary spermatocyte undergo Meiosis II and form two spermatids.

→ Thus 200 million spermatocytes are required to produce 400 million spermatozoa.



178. **Option (2) is correct.**

Explanation: Skeletal muscle fibres are multinucleated and bundled in a parallel manner.

Communication junction (Gap junction) in the intercalated disc of cardiac muscles allow the cardiac muscle cells to contract as a unit.

Blood vessel lining contains simple squamous epithelium.

179. **Option (2) is correct.**

Explanation: Because FSH (Follicle Stimulating Hormone) is a fat soluble hormone, it cannot cross the plasma membrane for entering into target cell, but generate secondary messenger (cAMP) which in turn regulate cellular metabolism.

180. **Option (3) is correct.**

Explanation: 3 chambered heart is present in *Chelone* (Turtle), as Reptiles have 3 chambered heart.

Pteropus is fire fox (a mammal) with 4 chambered heart.

→ *Scoliodon* → 2 chambered Heart.

181. **Option (1) is correct.**

Explanation: During pregnancy, thyroxine level increases in maternal blood because metabolic rate in body increases during pregnancy. As thyroxine maintain Basal Metabolic rate (BMR) to generate heat.

182. **Option (3) is correct.**

Explanation: Cyclostomes are jawless thus option with jawed behaviour is not correct.

They normally live in marine water but for spawning they move to fresh water.

→ They are scaleless and contain paired fins.

183. **Option (2) is correct.**

Explanation: Enamel is the hardest part of the body which protect the soft portion of teeth. It is a nonliving part made up of minerals and salt.

It provide surface for mastication/chewing.

184. **Option (1) is correct.**

Explanation: Bone protect inner visceral parts and limb bone helps weight bearing.

→ Adipose tissue store fat.

→ Ligament joins bone to bone and tendons attach muscles to the bones, bone marrow is the site where blood cells are produced not ligament.

185. **Option (1) is correct.**

Explanation: Bivalent or Tetrad formation is characteristic feature of **Zygotene** stage of

prophase I of **Meiosis I**, during which 4 chromatids can be observed on a single centromere.

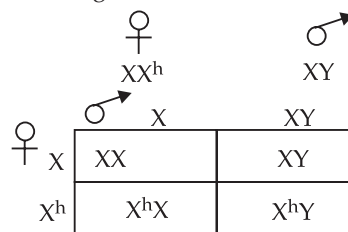
Section B

186. **Option (3) is correct.**

Explanation: Taxonomic keys are specialised taxonomic aid which help in studying taxonomy and identifying organisms. Each category have different set of keys for its segregation. Each key count aim **couplet** and **lead**.

187. **Option (2) is correct.**

Explanation: The girl will be carrier (Heterozygous) for Hemophilia while the male have no history of **Haemophilia**. This can be explained with the help of following cross:



188. **Option (3) is correct.**

Explanation: IUDs suppress the fertilisation capacity of sperm. If also reduces sperm motility. copper releasing IUDs also reduces sperm motility and fertilization capacity. But IUDs increase phagocytosis of sperms. They also make uterus hostile.

189. **Option (2) is correct.**

Explanation: Agarose gel electrophoresis is the process of separation of DNA fragments based on their size. As DNA is — vely charged, it moves from negative charged electrode to tve charged electrode inside the Gel Agarose gel shows **sieving effect**.

190. **Option (4) is correct.**

Explanation:

Multipolar neuron	Cerebral cortex, Brain
Bipolar neuron	Retina of eye
Myclinated nerve fibre	Peripheral Nerves e.g., Cranial Nerve
Unmyelinate nerve fibre	Nerve innervations to voluntary muscles.

191. **Option (4) is correct.**

Explanation: Excretion in cockroach should not be performed by Hepatic caeca as it is glandular in nature which secrete digestive Juice. It is least concerned with excretion.

192. **Option (3) is correct.**

Explanation: The concept of evolution through mutation was given by Hugo de vries.

→ Mutations are sudden aberration/change that arise in the population thus they are discontinuous in nature.

→ Large single step changes are called Saltation which lead to speciation.

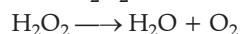
193. **Option (1) is correct.**

Explanation: Number or abundance of blood cells can be arranged in decreasing order as follows:

RBCs > Platelets > Neutrophils > Monocytes > Eosinophils > Platelets

194. **Option (2) is correct.**

Explanation: The enzyme which catalyse the breakdown of H_2O_2 as :



is called **Catalase** which have prosthetic group **Haem**. This enzyme is activated by iron. The breakdown occurs inside peroxisome during photorespiration.

195. **Option (4) is correct.**

Explanation: 5' UAC 3' code on mRNA can be complemented by 5' GUA 3' on anticodon loop of tRNA.

Codon anticodon pairing between mRNA and rRNA is the key of translation process.

196. **Option (4) is correct.**

Explanation: ADH (Antidiuretic Hormone) called **Vasopressin**. Which is a Vasoconstrictor.

Vasopressin is released from **Posterior Pituitary** not Adenohypophysis.

197. **Option (3) is correct.**

Explanation: Chromosome with 4 sister chromatids are not present during **Metaphase**.

Metaphasic chromosome consist of 2 chromatids attached together by a centromere. This is the thickest and shortest chromosome with prominent structure.

198. **Option (3) is correct.**

Explanation: Inbreeding always increases homozygosity thus leads to loss of superior characters.

In breeding can be synonymously used as continuous selling through which purelines can be generated for stability of characters.

199. **Option (4) is correct.**

Explanation:

Skin and mucous membrane	— Physical barriers
HCl is gastric juice	— Physiological barrier
Phagocytic cells	— Cellular barrier
Interferons	— Cytokine barriers.

200. **Option (2) is correct.**

Explanation:

A → 30% = T → 30%

C → 20% = G → 20%

According to Chargaff's rule, Adenine (A) pairs with Thymine (T) and Cytosine (C) pairs with Guanine (G). So, The ds DNA have equal percentage of A and T or G and C.



SOLVED PAPER 2021

Time : 3 Hours

Max. Marks : 720

Important Instructions :

The test is of 3 hours duration and test contains 200 questions. Each question carries 4 marks.

Number of Questions to be attempted out of 200 Questions	180 Questions
Total Marks	720 Marks
Number of questions in each subject and section	- Physics: Section A – 35, Section B – 15 - Chemistry: Section A – 35, Section B – 15 - Zoology: Section A – 35, Section B – 15 - Botany: Section A – 35, Section B – 15
Internal Choice in NEET Question Paper	Yes; Section B in each subject attempt only 10 questions out of 15 questions
Marking Scheme	Each correct answer +4 marks. For each incorrect answer –1 mark

PHYSICS

Section A

- Q. 1. In a potentiometer circuit a cell of EMF 1.5 V gives balance point at 36 cm length of wire. If another cell of EMF 2.5 V replaces the first cell, then at what length of the wire, the balance point occurs ?

- (a) 62 cm (b) 60 cm
(c) 21.6 cm (d) 64 cm

- Q. 2. Column-I gives certain physical terms associated with flow of current through a metallic conductor.

Column-II gives some mathematical relations involving electrical quantities. Match Column-I and Column-II with appropriate relations.

Column-I

- (A) Drift Velocity
(B) Electrical Resistivity
(C) Relaxation Period
(D) Current Density

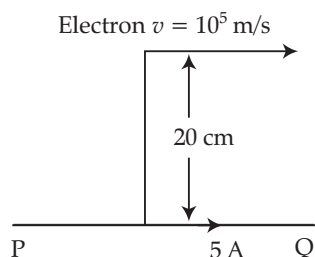
Column-II

- (P) $\frac{m}{ne^2\rho}$
(Q) neV_d
(R) $\frac{eE}{m}\tau$
(S) $\frac{E}{J}$

- (a) (A) - (R), (B) - (Q), (C) - (S), (D) - (P)
(b) (A) - (R), (B) - (S), (C) - (P), (D) - (Q)
(c) (A) - (R), (B) - (S), (C) - (Q), (D) - (P)
(d) (A) - (R), (B) - (P), (C) - (S), (D) - (Q)

- Q. 3. An infinitely long straight conductor carries a current of 5A as shown. An electron is moving with a speed of 10^5 m/s parallel to

the conductor. The perpendicular distance between the electron and the conductor is 20 cm at an instant. Calculate the magnitude of the force experienced by the electron at that instant.

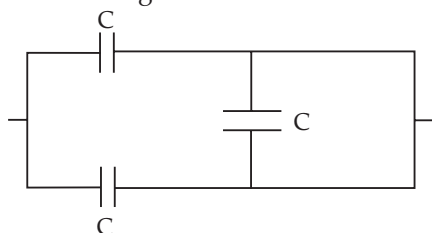


- (a) $8 \times 10^{-20} \text{ N}$ (b) $4 \times 10^{-20} \text{ N}$
 (c) $8\pi \times 10^{-20} \text{ N}$ (d) $4\pi \times 10^{-20} \text{ N}$
- Q. 4.** Polar molecules are the molecules
 (a) Having a permanent electric dipole moment.
 (b) Having zero dipole moment.
 (c) Acquire a dipole moment only in the presence of electric field due to displacement of charges.
 (d) Acquire a dipole moment only when magnetic field is absent.
- Q. 5.** A cup of coffee cools from 90°C to 80°C in t minutes, when the room temperature is 20°C . The time taken by a similar cup of coffee to cool from 80°C to 60°C at a room temperature same at 20°C is
 (a) $\frac{5}{13}t$ (b) $\frac{13}{10}t$
 (c) $\frac{13}{5}t$ (d) $\frac{10}{13}t$
- Q. 6.** Consider the following statements (A) and (B) and identify the correct answer.
 (A) A zener diode is connected in reverse bias, when used as a voltage regulator.
 (B) The potential barrier of p-n junction lies between 0.1 V to 0.3 V.
 (a) (A) is incorrect but (B) is correct.
 (b) (A) and (B) both are correct.
 (c) (A) and (B) both are incorrect
 (d) (A) is correct and (B) is incorrect.
- Q. 7.** A nucleus with mass number 240 breaks into two fragments each of mass number 120, the binding energy per nucleon of unfragmented nuclei is 7.6 MeV while that of fragments is 8.5 MeV. The total gain in the Binding Energy in the process is
 (a) 216 MeV (b) 0.9 MeV
 (c) 9.4 MeV (d) 804 MeV
- Q. 8.** Two charged spherical conductors of radius R_1 and R_2 are connected by a wire. Then the ratio of surface charge densities of the spheres (σ_1/σ_2) is
 (a) $\frac{R_1^2}{R_2^2}$ (b) $\frac{R_1}{R_2}$
 (c) $\frac{R_2}{R_1}$ (d) $\sqrt{\frac{R_1}{R_2}}$
- Q. 9.** A body is executing simple harmonic motion with frequency ' n ', the frequency of its potential energy is
 (a) $4n$ (b) n
 (c) $2n$ (d) $3n$
- Q. 10.** The half-life of a radioactive nuclide is 100 hours. The fraction of original activity that will remain after 150 hours would be
 (a) $\frac{2}{3\sqrt{2}}$ (b) $\frac{1}{2}$
 (c) $\frac{1}{2\sqrt{2}}$ (d) $\frac{2}{3}$
- Q. 11.** For a plane electromagnetic wave propagating in x -direction, which one of the following combination gives the correct possible directions for electric field (E) and magnetic field (B) respectively ?
 (a) $-\hat{j} + \hat{k}, -\hat{j} + \hat{k}$ (b) $\hat{j} + \hat{k}, \hat{j} + \hat{k}$
 (c) $-\hat{j} + \hat{k}, -\hat{j} - \hat{k}$ (d) $\hat{j} + \hat{k}, -\hat{j} - \hat{k}$
- Q. 12.** A parallel plate capacitor has a uniform electric field 'E' in the space between the plates. If the distance between the plates is ' d ' and the area of each plate is ' A ', the energy stored in the capacitor is
 (ϵ_0 = permittivity of free space)
 (a) $\frac{E^2 Ad}{\epsilon_0}$ (b) $\frac{1}{2} \epsilon_0 E^2$
 (c) $\epsilon_0 EAd$ (d) $\frac{1}{2} \epsilon_0 E^2 Ad$
- Q. 13.** If force [F], acceleration [A] and time [T] are chosen as the fundamental physical quantities. Find the dimensions of energy.
 (a) $[F][A^{-1}][T]$ (b) $[F][A][T]$
 (c) $[F][A][T^2]$ (d) $[F][A][T^{-1}]$

- Q. 14.** A particle is released from height S from the surface of the Earth. At a certain height its kinetic energy is three times its potential energy. The height from the surface of earth and the speed of the particle at that instant are respectively

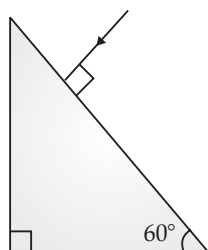
(a) $\frac{S}{4}, \sqrt{\frac{3gS}{2}}$ (b) $\frac{S}{4}, \frac{3gS}{2}$
 (c) $\frac{S}{4}, \sqrt{\frac{3gS}{2}}$ (d) $\frac{S}{2}, \sqrt{\frac{3gS}{2}}$

- Q. 15.** The equivalent capacitance of the combination shown in the figure is



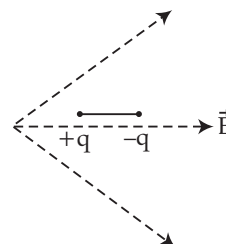
- (a) $\frac{3C}{2}$ (b) $3C$
 (c) $2C$ (d) $\frac{C}{2}$
- Q. 16.** If E and G respectively denote energy and gravitational constant, then $\frac{E}{G}$ has the dimensions of
- (a) $[M^2] [L^{-2}] [T^{-1}]$ (b) $[M^2] [L^{-1}] [T^0]$
 (c) $[M] [L^{-1}] [T^{-1}]$ (d) $[M] [L^0] [T^0]$
- Q. 17.** A lens of large focal length and large aperture is best suited as an objective of an astronomical telescope since

- (a) a large aperture contributes to the quality and visibility of the images.
 (b) a large area of the objective ensures better light gathering power.
 (c) a large aperture provides a better resolution.
 (d) all of the above
- Q. 18.** Find the value of the angle of emergence from the prism. Refractive index of the glass is $\sqrt{3}$



- (a) 90° (b) 60°
 (c) 30° (d) 45°

- Q. 19.** A dipole is placed in an electric field as shown. In which direction will it move ?



- (a) Towards the right as its potential energy will increase.
 (b) Towards the left as its potential energy will increase.
 (c) Towards the right as its potential energy will decrease.
 (d) Towards the left as its potential energy will decrease.
- Q. 20.** A screw gauge gives the following readings when used to measure the diameter of a wire
 Main scale reading : 0 mm
 Circular scale reading : 52 divisions
 Given that 1 mm on main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is
- (a) 0.052 cm (b) 0.52 cm
 (c) 0.026 cm (d) 0.26 cm

- Q. 21.** A small block slides down on a smooth inclined plane, starting from rest at time $t = 0$. Let S_n be the distance travelled by the block in the interval $t = n - 1$ to $t = n$. Then, the ratio $\frac{S_n}{S_{n+1}}$
- (a) $\frac{2n}{2n-1}$ (b) $\frac{2n-1}{2n}$
 (c) $\frac{2n-1}{2n+1}$ (d) $\frac{2n+1}{2n+1}$

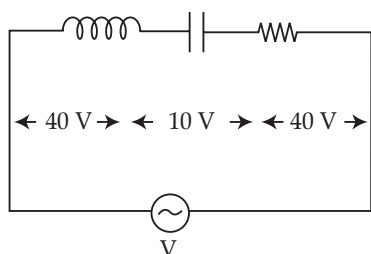
- Q. 22.** A convex lens 'A' of focal length 20 cm and a concave lens 'B' of focal length 5 cm are kept along the same axis with a distance ' d ' between them. If a parallel beam of light falling on 'A' leaves 'B' as a parallel beam, then the distance ' d ' in cm will be
- (a) 30 (b) 25
 (c) 15 (d) 50

- Q. 23.** The escape velocity from the Earth's surface is V . The escape velocity from the surface of another planet having a radius, four times that of Earth and same mass density is

(a) $4V$ (b) V
(c) $2V$ (d) $3V$

- Q. 24.** An inductor of inductance L , a capacitor of capacitance C and a resistor of resistance ' R ' are connected in series to an ac source of potential difference ' V ' volts as shown in figure.

Potential difference across L , C and R is $40V$, $10V$ and $40V$, respectively. The amplitude of current flowing through LCR series circuit is $10\sqrt{2}A$. The impedance of the circuit is

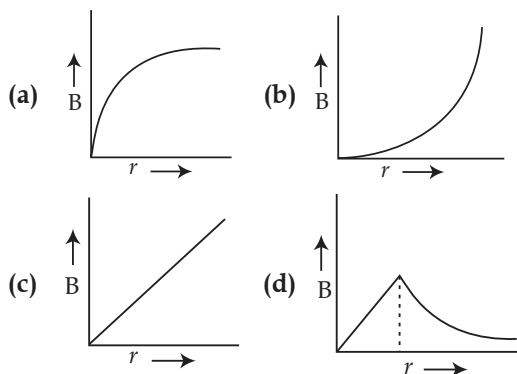


(a) 5Ω (b) $4\sqrt{2}\Omega$
(c) $\frac{5}{\sqrt{2}}\Omega$ (d) 4Ω

- Q. 25.** A spring is stretched by 5 cm by a force 10 N . The time period of the oscillations when a mass of 2 kg is suspended by it is

(a) 0.628 s (b) 0.0628 s
(c) 6.28 s (d) 3.14 s

- Q. 26.** A thick current carrying cable of radius ' R ' carries current ' I ' uniformly distributed across its cross section. The variation of magnetic field $B(r)$ due to the cable with the distance ' r ' from the axis of the cable is represented by



- Q. 27.** A radioactive nucleus A_ZX undergoes spontaneous decay in the sequence

${}^A_ZX \rightarrow {}^{z-1}_{-1}B \rightarrow {}^{z-3}_{-3}C \rightarrow {}^{z-2}_{-2}D$, where Z is the atomic number of element X . The possible decay particles in the sequence are

(a) β^- , α , β^+ (b) α , β^- , β^+
(c) α , β^+ , β^- (d) β^+ , α , β^-

- Q. 28.** The velocity of a small ball of mass M and density d , when dropped in a container filled with glycerine becomes constant after some time. If the density of glycerine is $\frac{d}{2}$, then the viscous force acting on the ball will be

(a) $2Mg$ (b) $\frac{Mg}{2}$
(c) Mg (d) $\frac{3}{2}Mg$

- Q. 29.** Water falls from a height of 60 m at the rate of 15 kg/s to operate a turbine. The losses due to frictional force are 10% of the input energy. How much power is generated by the turbine?

($g = 10\text{ m/s}^2$)

(a) 7.0 kW (b) 10.2 kW
(c) 8.1 kW (d) 12.3 kW

- Q. 30.** A capacitor of capacitance ' C ', is connected across an ac source of voltage V , given by $V = V_0\sin\omega t$

The displacement current between the plates of the capacitor, would then be given by

(a) $I_d = V_0\omega C\sin\omega t$ (b) $I_d = V_0\omega C\cos\omega t$
(c) $I_d = \frac{V_0}{\omega C}\cos\omega t$ (d) $I_d = \frac{V_0}{\omega C}\sin\omega t$

- Q. 31.** The number of photons per second on an average emitted by the source of monochromatic light of wavelength 600 nm , when it delivers the power of $3.3 \times 10^{-3}\text{ watt}$ will be ($h = 6.6 \times 10^{-34}\text{ J s}$)

(a) 10^{15} (b) 10^{18}
(c) 10^{17} (d) 10^{16}

- Q. 32.** The electron concentration in an n -type semiconductor is the same as hole concentration in a p -type semiconductor. An external field (electric) is applied across each of them. Compare the currents in them.

- (a) No current will flow in p -type, current will only flow in n -type
 (b) Current in n -type = current in p -type
 (c) Current in p -type > current in n -type
 (d) Current in n -type > current in p -type.
- Q. 33.** The effective resistance of a parallel connection that consists of four wires of equal length, equal area of cross-section and same material is 0.25Ω . What will be the effective resistance if they are connected in series ?
- (a) 4Ω (b) 0.25Ω
 (c) 0.5Ω (d) 1Ω
- Q. 34.** Match Column - I and Column - II and choose the correct match from the given choices.

Column - I	Column - II
(A) Root mean square speed of gas molecules	(P) $\frac{1}{3} nmv^{-2}$
(B) Pressure exerted by ideal gas	(Q) $\sqrt{\frac{3RT}{M}}$
(C) Average kinetic energy of a molecule	(R) $\frac{5}{2} RT$
(D) Total internal energy of 1 mole of a diatomic gas	(S) $\frac{3}{2} k_B T$
(a) (A) - (R), (B) - (Q), (C) - (P), (D) - (S)	
(b) (A) - (R), (B) - (P), (C) - (S), (D) - (Q)	
(c) (A) - (Q), (B) - (R), (C) - (S), (D) - (P)	
(d) (A) - (Q), (B) - (P), (C) - (S), (D) - (R)	

- Q. 35.** An electromagnetic wave of wavelength ' λ ' is incident on a photosensitive surface of negligible work function. If ' m ' mass is of photoelectron emitted from the surface has de-Broglie wavelength λ_d , then

(a) $\lambda = \left(\frac{2h}{mc} \right) \lambda_d^2$ (b) $\lambda = \left(\frac{2m}{hc} \right) \lambda_d^2$
 (c) $\lambda_d = \left(\frac{2mc}{h} \right) \lambda^2$ (d) $\lambda = \left(\frac{2mc}{h} \right) \lambda_d^2$

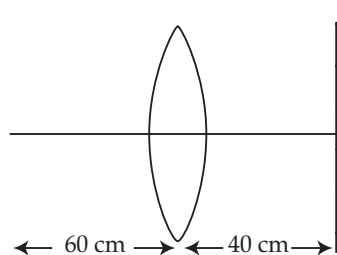
Section B

- Q. 36.** From a circular ring of mass ' M ' and radius ' R ' an arc corresponding to a 90° sector is removed. The moment of inertia of the remaining part of the ring about an axis

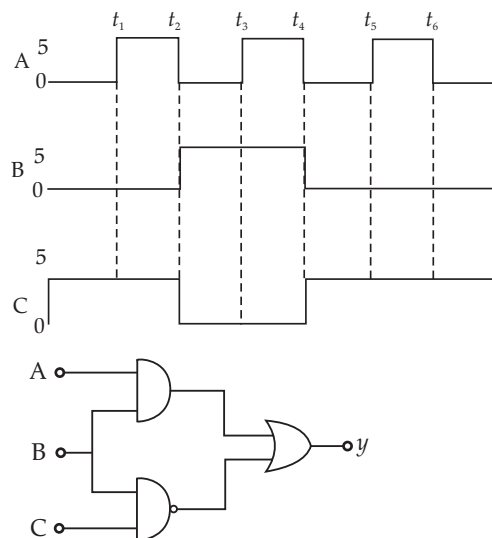
passing through the centre of the ring and perpendicular to the plane of the ring is ' K ' times ' MR^2 '. Then the value of ' K ' is

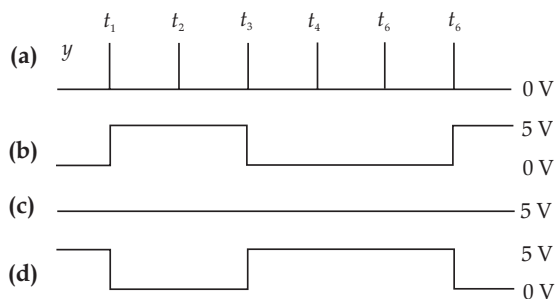
- (a) $\frac{1}{8}$ (b) $\frac{3}{4}$
 (c) $\frac{7}{8}$ (d) $\frac{1}{4}$

- Q. 37.** A point object is placed at a distance of 60 cm from a convex lens of focal length 30 cm. If a plane mirror were put perpendicular to the principal axis of the lens and at a distance of 40 cm from it, the final image would be formed at a distance of



- (a) 20 cm from the plane mirror, it would be a virtual image
 (b) 20 cm from the lens, it would be a real image
 (c) 30 cm from the lens, it would be a real image
 (d) 30 cm from the plane mirror, it would be a virtual image
- Q. 38.** For the given circuit, the input digital signals are applied at the terminals A, B and C. What would be the output at the terminal y ?

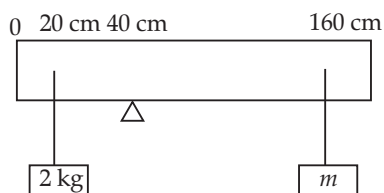




- Q. 39.** A car starts from rest and accelerates at 5 m/s^2 . At $t = 4 \text{ s}$, a ball is dropped out of a window by a person sitting in the car. What is the velocity and acceleration of the ball at $t = 6 \text{ s}$?

(Take $g = 10 \text{ m/s}^2$)

- (a) $20\sqrt{2} \text{ m/s}$, 10 m/s^2
 (b) 20 m/s , 5 m/s^2
 (c) 20 m/s , 0
 (d) $20\sqrt{2} \text{ m/s}$, 0
- Q. 40.** A uniform rod of length 200 cm and mass 500 g is balanced on a wedge placed at 40 cm mark. A mass of 2 kg is suspended from the rod at 20 cm and another unknown mass ' m ' is suspended from the rod at 160 cm mark as shown in the figure. Find the value of ' m ' such that the rod is in equilibrium. ($g = 10 \text{ m/s}^2$)

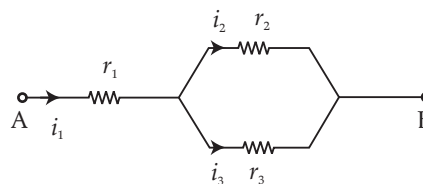


- (a) $\frac{1}{12} \text{ kg}$ (b) $\frac{1}{2} \text{ kg}$
 (c) $\frac{1}{3} \text{ kg}$ (d) $\frac{1}{6} \text{ kg}$
- Q. 41.** Two conducting circular loops of radii R_1 and R_2 are placed in the same plane with their centres coinciding. If $R_1 \gg R_2$, the mutual inductance M between them will be directly proportional to

- (a) $\frac{R_2^2}{R_1}$ (b) $\frac{R_1}{R_2}$
 (c) $\frac{R_2}{R_1}$ (d) $\frac{R_1^2}{R_2}$

- Q. 42.** Three resistors having resistances r_1 , r_2 and r_3 are connected as shown in the given circuit.

The ratio $\frac{i_3}{i_1}$ of currents in terms of resistances used in the circuit is



- (a) $\frac{r_2}{r_1 + r_3}$ (b) $\frac{r_1}{r_2 + r_3}$
 (c) $\frac{r_2}{r_2 + r_3}$ (d) $\frac{r_1}{r_1 + r_2}$
- Q. 43.** A particle of mass ' m ' is projected with a velocity $v = kV_e$ ($k < 1$) from the surface of the earth.

(V_e = escape velocity)

The maximum height above the surface reached by the particle is

- (a) $\frac{Rk^2}{1 - k^2}$ (b) $R \left(\frac{k}{1 - k} \right)^2$
 (c) $R \left(\frac{k}{1 + k} \right)^2$ (d) $\frac{R^2 k}{1 + k}$
- Q. 44.** A uniform conducting wire of length $12a$ and resistance ' R ' is wound up as a current carrying coil in the shape of,

- (i) an equilateral triangle of side ' a '.
 (ii) a square of side ' a '.

The magnetic dipole moments of the coil in each case respectively are

- (a) $4 Ia^2$ and $3 Ia^2$ (b) $\sqrt{3} Ia^2$ and $3 Ia^2$
 (c) $3 Ia^2$ and Ia^2 (d) $3 Ia^2$ and $4 Ia^2$
- Q. 45.** A step down transformer connected to an ac mains supply of 220 V is made to operate at 11 V , 44 W lamp. Ignoring power losses in the transformer, what is the current in the primary circuit?

- (a) 4 A (b) 0.2 A
 (c) 0.4 A (d) 2 A

- Q. 46.** A series LCR circuit containing 5.0 H inductor, $80 \mu\text{F}$ capacitor and 40Ω resistor is connected to 230 V variable frequency ac source. The angular frequencies of the source at which power transferred to the circuit is half the power at the resonant angular frequency are likely to be

- (a) 42 rad/s and 58 rad/s
 (b) 25 rad/s and 75 rad/s
 (c) 50 rad/s and 25 rad/s
 (d) 46 rad/s and 54 rad/s
- Q. 47. Twenty seven drops of same size are charged at 200 V each. They combine to form a bigger drop. Calculate the potential of the bigger drop.
- (a) 1980 V (b) 660 V
 (c) 1320 V (d) 1520 V
- Q. 48. A ball of mass 0.15 kg is dropped from a height 10 m, strikes the ground and rebounds to the same height.
- The magnitude of impulse imparted to the ball is ($g = 10 \text{ m/s}^2$) nearly
- (a) 1.4 kg m/s (b) 0 kg m/s
 (c) 4.2 kg m/s (d) 2.1 kg m/s
- Q. 49. In the product

$$\vec{F} = q(\vec{v} \times \vec{B})$$

$$= q\vec{v} \times (\vec{B}_i + \vec{B}_j + \vec{B}_k)$$

For $q = 1$ and $\vec{v} = 2\hat{i} + 4\hat{j} + 6\hat{k}$ and

$$\vec{F} = 4\hat{i} - 20\hat{j} + 12\hat{k}$$

What will be the complete expression for $\vec{B}$?

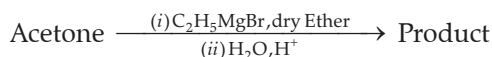
- (a) $6\hat{i} + 6\hat{j} - 8\hat{k}$ (b) $-8\hat{i} - 8\hat{j} - 6\hat{k}$
 (c) $-6\hat{i} - 6\hat{j} - 8\hat{k}$ (d) $8\hat{i} + 8\hat{j} - 6\hat{k}$
- Q. 50. A particle moving in a circle of radius R with a uniform speed takes a time T to complete one revolution. If this particle were projected with the same speed at an angle ' θ ' to the horizontal, the maximum height attained by it equals 4R. The angle of projection, θ , is then given by

- (a) $\theta = \sin^{-1} \left(\frac{2gT^2}{\pi^2 R} \right)^{1/2}$
 (b) $\theta = \cos^{-1} \left(\frac{gT^2}{\pi^2 R} \right)^{1/2}$
 (c) $\theta = \cos^{-1} \left(\frac{\pi^2 R}{gT^2} \right)^{1/2}$
 (d) $\theta = \sin^{-1} \left(\frac{\pi^2 R}{gT^2} \right)^{1/2}$

CHEMISTRY

Section A

- Q. 51. The molar conductance of NaCl, HCl and CH_3COONa at infinite dilution are 126.45, 426.16 and $91.0 \text{ S cm}^2 \text{ mol}^{-1}$ respectively. The molar conductance of CH_3COOH at infinite dilution is.
- Choose the right option for your answer.
- (a) $540.48 \text{ S cm}^2 \text{ mol}^{-1}$
 (b) $201.28 \text{ S cm}^2 \text{ mol}^{-1}$
 (c) $390.71 \text{ S cm}^2 \text{ mol}^{-1}$
 (d) $698.28 \text{ S cm}^2 \text{ mol}^{-1}$
- Q. 52. What is the IUPAC name of the organic compound formed in the following chemical reaction ?



- (a) 2-methylbutan-2-ol
 (b) 2-methylpropan-2-ol

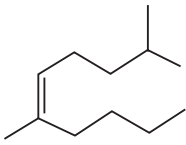
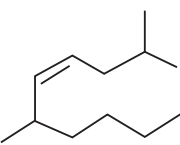
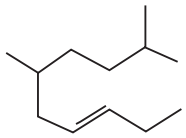
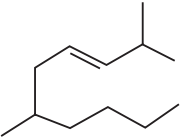
- (c) pentan-2-ol
 (d) pentan-3-ol

- Q. 53. The following solutions were prepared by dissolving 10 g of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) in 250 ml of water (P_1), 10 g of urea ($\text{CH}_4\text{N}_2\text{O}$) in 250 ml of water (P_2) and 10 g of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) in 250 ml of water (P_3).

The right option for the decreasing order of osmotic pressure of these solutions is

- (a) $P_3 > P_1 > P_2$ (b) $P_2 > P_1 > P_3$
 (c) $P_1 > P_2 > P_3$ (d) $P_2 > P_3 > P_1$
- Q. 54. The right option for the statement "Tyndall effect is exhibited by", is
- (a) urea solution. (b) NaCl solution.
 (c) glucose solution. (d) starch solution.
- Q. 55. The maximum temperature that can be achieved in blast furnace is

- (a) Upto 5000 K (b) Upto 1200 K
 (c) Upto 2200 K (d) Upto 1900 K

- Q. 56.** The correct option for the number of body centred unit cells in all 14 types of Bravais lattice unit cells is
 (a) 3 (b) 7
 (c) 5 (d) 2
- Q. 57.** An organic compound contains 78% (by wt.) carbon and remaining percentage of hydrogen. The right option for the empirical formula of this compound is [Atomic wt. of C is 12, H is 1]
 (a) CH_4 (b) CH
 (c) CH_2 (d) CH_3
- Q. 58.** Zr ($Z=40$) and Hf ($Z=72$) have similar atomic and ionic radii because of
 (a) having similar chemical properties.
 (b) belonging to same group.
 (c) diagonal relationship.
 (d) lanthanoid contraction.
- Q. 59.** Ethylene diaminetetraacetate (EDTA) ion is
 (a) tridentate ligand with three "N" donor atoms.
 (b) hexadentate ligand with four "O" and two "N" donor atoms.
 (c) unidentate ligand.
 (d) bidentate ligand with two "N" donor atoms.
- Q. 60.** The correct structure of 2, 6-Dimethyl-dec-4-ene is
 (a)  (b) 
 (c)  (d) 
- Q. 61.** The RBC deficiency is deficiency disease of
 (a) Vitamin B_2 (b) Vitamin B_{12}
 (c) Vitamin B_6 (d) Vitamin B_1
- Q. 62.** BF_3 is planar and electron deficient compound. Hybridization and number of electrons around the central atom, respectively are
 (a) sp^2 and 8 (b) sp^3 and 4
 (c) sp^3 and 6 (d) sp^2 and 6
- Q. 63.** Dihedral angle of least stable conformer of ethane is
 (a) 0° (b) 120°
 (c) 180° (d) 60°
- Q. 64.** Among the following alkaline earth metal halides, one which is covalent and soluble in organic solvents is
 (a) beryllium chloride.
 (b) calcium chloride.
 (c) strontium chloride.
 (d) magnesium chloride.
- Q. 65.** Which one of the following polymers is prepared by addition polymerisation ?
 (a) Dacron (b) Teflon
 (c) Nylon-66 (d) Novolac
- Q. 66.** The structures of beryllium chloride in solid state and vapour phase, are
 (a) chain in both.
 (b) chain and dimer, respectively.
 (c) linear in both.
 (d) dimer and Linear, respectively.
- Q. 67.** Which one of the following methods can be used to obtain highly pure metal which is liquid at room temperature ?
 (a) Zone refining (b) Electrolysis
 (c) Chromatography (d) Distillation
- Q. 68.** Which one among the following is the correct option for right relationship between C_P and C_V for one mole of ideal gas ?
 (a) $C_V = RC_P$ (b) $C_P + C_V = R$
 (c) $C_P - C_V = R$ (d) $C_P = RC_V$
- Q. 69.** Tritium, a radioactive isotope of hydrogen, emits which of the following particles ?
 (a) Neutron (n) (b) Beta (β^-)
 (c) Alpha (α) (d) Gamma (γ)
- Q. 70.** The major product formed in dehydrohalogenation reaction of 2-Bromo pentane is Pent-2-ene. This product formation is based on ?
 (a) Huckel's Rule (b) Saytzeff's Rule
 (c) Hund's Rule (d) Hofmann Rule

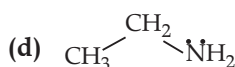
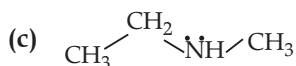
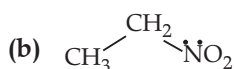
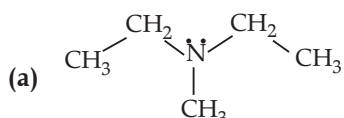
Q. 71. A particular station of All India Radio, New Delhi broadcasts on a frequency of 1,368 kHz (kilohertz). The wavelength of the electromagnetic radiation emitted by the transmitter is [speed of light $c = 3.0 \times 10^8 \text{ ms}^{-1}$]

- (a) 21.92 cm (b) 219.3 m
(c) 219.2 m (d) 2192 m

Q. 72. Right option for the number of tetrahedral and octahedral voids in hexagonal primitive unit cell are

- (a) 12, 6 (b) 8, 4
(c) 6, 12 (d) 2, 1

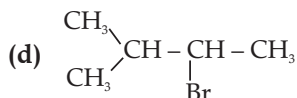
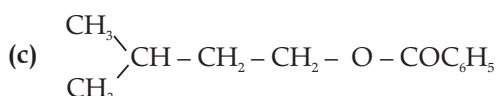
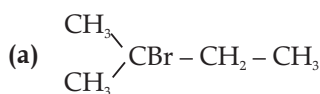
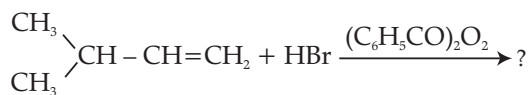
Q. 73. Identify the compound that will react with Hinsberg's reagent to give a solid which dissolves in alkali.



Q. 74. The pK_a of dimethylamine and pK_a of acetic acid are 3.27 and 4.77 respectively at T (K). The correct option for the pH of dimethylammonium acetate solution is

- (a) 6.25 (b) 8.50
(c) 5.50 (d) 7.75

Q. 75. The major product of the following chemical reaction is



Q. 76. Match List-I with List-II.

List-I**List-II**

- | | |
|--------------------|---------------------------|
| (a) PCl_5 | (i) Square pyramidal |
| (b) SF_6 | (ii) Trigonal planar |
| (c) BrF_5 | (iii) Octahedral |
| (d) BF_3 | (iv) Trigonal bipyramidal |

Choose the correct answer from the options given below.

- (a) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
(b) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
(c) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
(d) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

Q. 77. The incorrect statement among the following is

- (a) actinoids are highly reactive metals, especially when finely divided.
(b) actinoid contraction is greater than lanthanoid contraction.
(c) most of the trivalent Lanthanoid ions are colorless in the solid state.
(d) lanthanoids are good conductors of heat and electricity.

Q. 78. **Statement I :** Acid strength increases in the order given as $\text{HF} \ll \text{HCl} \ll \text{HBr} \ll \text{HI}$.

Statement II : As the size of the elements F, Cl, Br, I increases down the group, the bond strength of HF, HCl, HBr and HI decreases and so the acid strength increases.

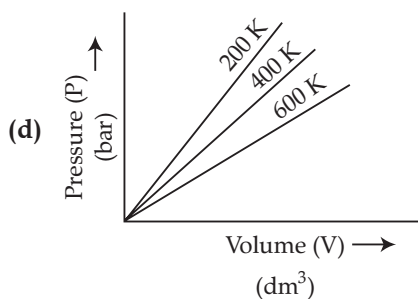
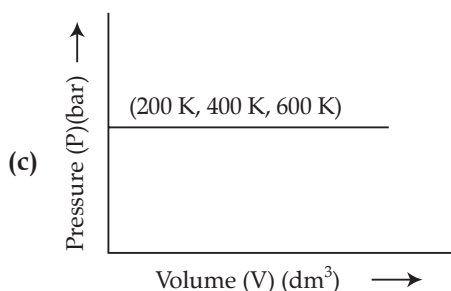
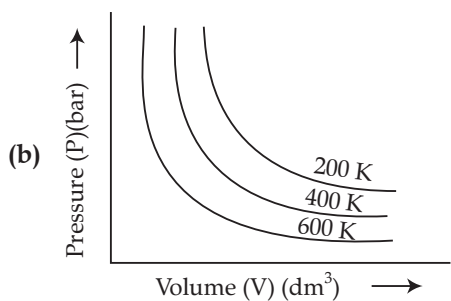
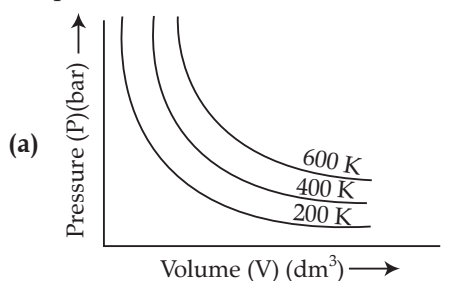
In the light of the above statements, choose the correct answer from the options given below.

- (a) Statement I is incorrect but Statement II is true.
(b) Both statement I and Statement II are true.
(c) Both Statement I and Statement II are false.
(d) Statement I is correct but statement II is false.

Q. 79. Noble gases are named because of their inertness towards reactivity. Identify an incorrect statement about them.

- (a) Noble gases have large positive values of electron gain enthalpy.
(b) Noble gases are sparingly soluble in water.

- (c) Noble gases have very high melting and boiling points.
 (d) Noble gases have weak dispersion forces.
- Q. 80. The compound which shows metamerism is
- (a) $C_4H_{10}O$ (b) C_5H_{12}
 (c) C_3H_8O (d) C_3H_6O
- Q. 81. Choose the correct option for graphical representation of Boyle's law, which shows a graph of pressure vs. volume of a gas at different temperatures



- Q. 82. The correct sequence of bond enthalpy of 'C—X' bond is
- (a) $CH_3-Cl > CH_3-F > CH_3-Br > CH_3-I$
 (b) $CH_3-F < CH_3-Cl < CH_3-Br < CH_3-I$

- (c) $CH_3-F > CH_3-Cl > CH_3-Br > CH_3-I$
 (d) $CH_3-F < CH_3-Cl < CH_3-Br < CH_3-I$

Q. 83. Given below are two statements :

Statement I :

Aspirin and Paracetamol belong to the class of narcotic analgesics.

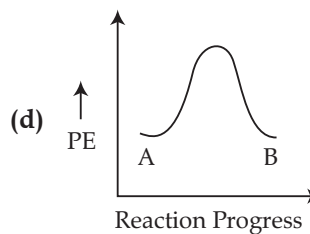
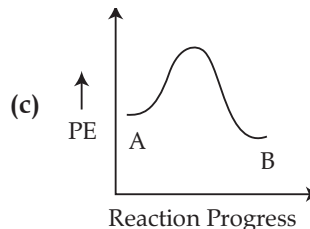
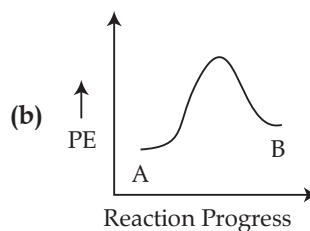
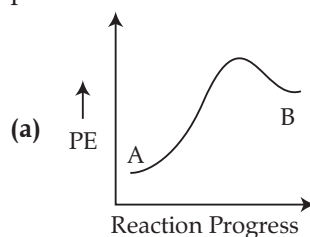
Statement II :

Morphine and Heroin are non-narcotic analgesics.

In the light of the above statements, choose the correct answer from the options given below.

- (a) Statement I is incorrect but Statement II is true.
 (b) Both Statement I and Statement II are true.
 (c) Both Statement I and Statement II are false.
 (d) Statement I is correct but Statement II is false.

Q. 84. For a reaction $A \rightarrow B$, enthalpy of reaction is -4.2 kJ mol^{-1} and enthalpy of activation is 9.6 kJ mol^{-1} . The correct potential energy profile for the reaction is shown in option.



Q. 85. Which of the following reactions is the metal displacement reaction? Choose the right option.

- (a) $2\text{Pb}(\text{NO}_3)_2 \rightarrow 2\text{PbO} + 4\text{NO}_2 + \text{O}_2 \uparrow$
 (b) $2\text{KClO}_3 \xrightarrow{\Delta} 2\text{KCl} + 3\text{O}_2$
 (c) $\text{Cr}_2\text{O}_3 + 2\text{Al} \xrightarrow{\Delta} \text{Al}_2\text{O}_3 + 2\text{Cr}$
 (d) $\text{Fe} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2 \uparrow$

Section B

Q. 86. The slope of Arrhenius plot $\left(\ln k v/s \frac{1}{T} \right)$ of first order reaction is $-5 \times 10^3 \text{K}$. The value of E_a of the reaction is. Choose the correct option for your answer.

[Given $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$]

- (a) -83 kJ mol^{-1} (b) 41.5 kJ mol^{-1}
 (c) 83.0 kJ mol^{-1} (d) 166 kJ mol^{-1}

Q. 87. Match List-I with List-II.

- | List-I | List-II |
|--|-----------------------------|
| (a) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$ | (i) Acid rain |
| (b) $\text{HOCl}(\text{g}) \xrightarrow{h\nu} \text{OH} \cdot + \text{Cl} \cdot$ | (ii) Smog |
| (c) $\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{H}_2\text{O} + \text{CO}_2$ | (iii) Ozone depletion |
| (d) $\text{NO}_2(\text{g}) \xrightarrow{h\nu} \text{NO}(\text{g}) + \text{O}(\text{g})$ | (iv) Tropospheric pollution |

Choose the correct answer from the options given below.

- (a) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)
 (b) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
 (c) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
 (d) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

Q. 88. Match List-I with List-II

- | List-I | List-II |
|--|------------------------------------|
| (a)  $\xrightarrow[\text{CuCl}]{\text{CO, HCl, Anhyd. AlCl}_3}$ | (i) Hell-Volhard-Zelinsky reaction |

- (b) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 + \text{NaOX} \longrightarrow$ (ii) Gattermann Koch reaction
 (c) $\text{R}-\text{CH}_2-\text{OH} + \text{R COOH} \xrightarrow{\text{Conc. H}_2\text{SO}_4}$ (iii) Haloform reaction
 (d) $\text{R}-\text{CH}_2\text{COOH} \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) X}_2/\text{Red P}}$ (iv) Esterification

Choose the correct answer from the options given below.

- (a) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)
 (b) (a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)
 (c) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)
 (d) (a) - (i), (b) - (iv), (c) - (iii), (d) - (ii)

Q. 89. From the following pairs of ions which one is not an iso-electronic pair?

- (a) $\text{Fe}^{2+}, \text{Mn}^{2+}$ (b) $\text{O}^{2-}, \text{F}^-$
 (c) $\text{Na}^+, \text{Mg}^{2+}$ (d) $\text{Mn}^{2+}, \text{Fe}^{3+}$

Q. 90. The molar conductivity of 0.007 M acetic acid is $20 \text{ S cm}^2 \text{ mol}^{-1}$. What is the dissociation constant of acetic acid? Choose the correct option.

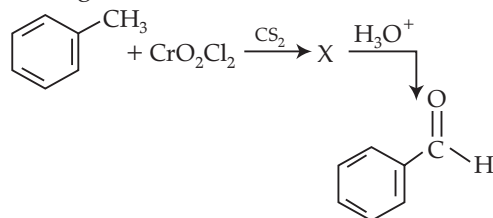
$$\left[\begin{array}{l} \Lambda_{\text{H}^+}^\circ = 350 \text{ S cm}^2 \text{ mol}^{-1} \\ \Lambda_{\text{CH}_3\text{COO}^-}^\circ = 50 \text{ S cm}^2 \text{ mol}^{-1} \end{array} \right]$$

- (a) $2.50 \times 10^{-5} \text{ mol L}^{-1}$
 (b) $1.75 \times 10^{-4} \text{ mol L}^{-1}$
 (c) $2.50 \times 10^{-4} \text{ mol L}^{-1}$
 (d) $1.75 \times 10^{-5} \text{ mol L}^{-1}$

Q. 91. In which one of the following arrangements the given sequence is not strictly according to the properties indicated against it?

- (a) $\text{CO}_2 < \text{SiO}_2 < \text{SnO}_2 < \text{PbO}_2$: Increasing oxidizing power
 (b) $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$: Increasing acidic strength
 (c) $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$: Increasing pK_a values
 (d) $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3$: Increasing acidic character

- Q. 92.** The intermediate compound 'X' in the following chemical reaction is



- (a)
- (b)
- (c)
- (d)

- Q. 93.** Which of the following molecules is non-polar in nature ?

- (a) NO_2 (b) POCl_3
(c) CH_2O (d) SbCl_5

- Q. 94.** Match List-I with List-II.

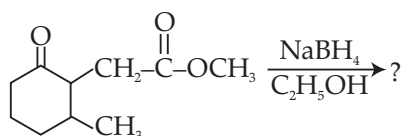
- | List-I | List-II |
|--|---------------|
| (a) $[\text{Fe}(\text{CN})_6]^{3-}$ | (i) 5.92 BM |
| (b) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ | (ii) 0 BM |
| (c) $[\text{Fe}(\text{CN})_6]^{4-}$ | (iii) 4.90 BM |
| (d) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ | (iv) 1.73 BM |

Choose the correct answer from the options given below.

- (a) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
(b) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
(c) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
(d) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)
- Q. 95.** $\text{CH}_3\text{CH}_2\text{COO}^-\text{Na}^+ \xrightarrow[\text{Heat}]{\text{NaOH, ?}} \text{CH}_3\text{CH}_3 + \text{Na}_2\text{CO}_3$

Consider the above reaction and identify the missing reagent/chemical.

- (a) DIBAL-H (b) B_2H_6
(c) Red Phosphorus (d) CaO
- Q. 96.** The product formed in the following chemical reaction is



- (a)
- (b)
- (c)
- (d)

- Q. 97.** The correct option for the value of vapour pressure of a solution at 45°C with benzene to octane in molar ratio 3 : 2 is

[At 45°C vapour pressure of benzene is 280 mm of Hg and that of octane is 420 mm of Hg. Assume Ideal gas]

- (a) 350 mm of Hg (b) 160 mm of Hg
(c) 168 mm of Hg (d) 336 mm of Hg

- Q. 98.** For irreversible expansion of an ideal gas under isothermal condition, the correct option is

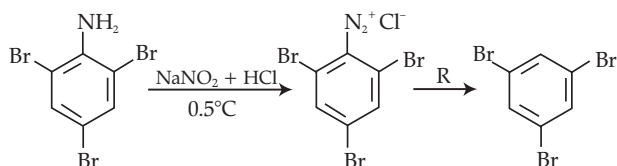
- (a) $\Delta U \neq 0, \Delta S_{\text{total}} = 0$
(b) $\Delta U = 0, \Delta S_{\text{total}} = 0$
(c) $\Delta U \neq 0, \Delta S_{\text{total}} \neq 0$
(d) $\Delta U = 0, \Delta S_{\text{total}} \neq 0$

- Q. 99.** Choose the correct option for the total pressure (in atm.) in a mixture of 4 g O_2 and 2 g H_2 confined in a total volume of one litre at 0°C is

[Given $R = 0.082 \text{ L atm mol}^{-1}\text{K}^{-1}$, $T = 273 \text{ K}$]

- (a) 26.02 (b) 2.518
(c) 2.602 (d) 25.18

- Q. 100.** The reagent 'R' in the given sequence of chemical reaction is



- (a) CuCN/KCN (b) H_2O
(c) $\text{CH}_3\text{CH}_2\text{OH}$ (d) HI

BOTANY

Section A

- Q. 101.** The production of gametes by the parents, formation of zygotes, the F_1 and F_2 plants, can be understood from a diagram called
 (a) Net square. (b) Bullet square.
 (c) Punch square. (d) Punnett square.
- Q. 102.** Gemmae are present in
 (a) some Liverworts.
 (b) mosses.
 (c) pteridophytes.
 (d) some Gymnosperms.
- Q. 103.** The factor that leads to Founder effect in a population is
 (a) genetic drift.
 (b) natural selection.
 (c) genetic recombination.
 (d) mutation.
- Q. 104.** Plants follow different pathways in response to environment or phases of life to form different kinds of structures. This ability is called
 (a) maturity. (b) elasticity.
 (c) flexibility. (d) plasticity.
- Q. 105.** Match List-I with List-II.

	List-I		List-II
(a)	Cells with active cell division capacity	(i)	Vascular tissues
(b)	Tissues having all cells similar in structure and function	(ii)	Meristematic tissues
(c)	Tissue having different types of cells	(iii)	Scleritis
(d)	Dead cells with highly thickened walls and narrow lumen	(iv)	Simple tissue

Select the correct answer from the options given below.

- (a) (b) (c) (d)
 (a) (iii) (ii) (iv) (i)
 (b) (ii) (iv) (i) (iii)
 (c) (iv) (iii) (ii) (i)
 (d) (i) (ii) (iii) (iv)
- Q. 106.** The term used for transfer of pollen grains

from anthers of one plant to stigma of a different plant which, during pollination, brings genetically different types of pollen grains to stigma, is

- (a) Cleistogamy (b) Xenogamy
 (c) Geitonogamy (d) Chasmogamy
- Q. 107.** Which of the following is a correct sequence of steps in a PCR (Polymerase Chain Reaction)
 (a) Annealing, Denaturation, Extension
 (b) Denaturation, Annealing, Extension
 (c) Denaturation, Extension, Annealing
 (d) Extension, Denaturation, Annealing
- Q. 108.** Match List-I with List-II.

	List-I		List-II
(a)	Lenticels	(i)	Phellogen
(b)	Cork cambium	(ii)	Suberin deposition
(c)	Secondary cortex	(iii)	Exchange of gases
(d)	Cork	(iv)	Phelloderm

Choose the correct answer from the options given below.

- (a) (b) (c) (d)
 (a) (iv) (ii) (i) (iii)
 (b) (iv) (i) (iii) (ii)
 (c) (iii) (i) (iv) (ii)
 (d) (ii) (iii) (iv) (i)
- Q. 109.** Inspite of interspecific competition in nature, which mechanism the competing species might have evolved for their survival ?
 (a) Predation
 (b) Resource partitioning
 (c) Competitive release
 (d) Mutualism
- Q. 110.** DNA strands on a gel stained with ethidium bromide when viewed under UV radiation, appear as
 (a) bright blue bands.
 (b) yellow bands.
 (c) bright orange bands.
 (d) dark red bands.
- Q. 111.** Which of the following algae produce Carrageen ?
 (a) Blue-green algae (b) Green algae
 (c) Brown algae (d) Red algae

- Q. 112.** Diadelphous stamens are found in
 (a) China rose and citrus.
 (b) China rose.
 (c) Citrus.
 (d) Pea.
- Q. 113.** The plant hormone used to destroy weeds in a field
 (a) IBA (b) IAA (c) NAA (d) 2, 4-D
- Q. 114.** Which of the following plants is monoecious ?
 (a) *Cycas circinalis*
 (b) *Carica papaya*
 (c) *Chara*
 (d) *Marchantia polymorpha*
- Q. 115.** Which of the following are not secondary metabolites in plants ?
 (a) Rubber, gums
 (b) Morphine, codeine
 (c) Amino acids, glucose
 (d) Vinblastin, curcumin
- Q. 116.** Complete the flow chart on central dogma.
 (a) $\text{DNA} \xrightarrow{(b)} \text{mRNA} \xrightarrow{(c)} (d)$
 (a) (a)-Transduction; (b)-Translation; (c)-Replication; (d)-Protein
 (b) (a)-Replication; (b)-Transcription; (c)-Transduction; (d)-Protein
 (c) (a)-Translation; (b)-Replication; (c)-Transcription; (d)-Transduction
 (d) (a)-Replication; (b)-Transcription; (c)-Translation; (d)-Protein
- Q. 117.** The site of perception of light in plants during photoperiodism is
 (a) leaf. (b) shoot apex.
 (c) stem. (d) axillary bud.
- Q. 118.** In the equation $GPP - R = NPP$ R represents
 (a) Respiration losses
 (b) Radiant energy
 (c) Retardation factor
 (d) Environmental factor
- Q. 119.** Match List-I with List-II.

List-I		List-II	
(a)	Cristae	(i)	Primary construction in chromosome

(b)	Thylakoids	(ii)	Disc-shaped sacs in Golgi apparatus
(c)	Centromere	(iii)	Infoldings in mitochondria
(d)	Cisternae	(iv)	Flattened membranous sacs in stroma of plastids

Choose the correct answer from the options given below.

- (a) (b) (c) (d)
 (a) (ii) (iii) (iv) (i)
 (b) (iv) (iii) (ii) (i)
 (c) (i) (iv) (iii) (ii)
 (d) (iii) (iv) (i) (ii)
- Q. 120.** Which of the following algae contains mannitol as reserve food material ?
 (a) *Ulothrix* (b) *Ectocarpus*
 (c) *Gracilaria* (d) *Volvox*
- Q. 121.** Which of the following is an incorrect statement ?
 (a) Nuclear pores act as passages for proteins and RNA molecules in both directions between nucleus and cytoplasm.
 (b) Mature sieve tube elements possess a conspicuous nucleus and usual cytoplasmic organelles.
 (c) Microbodies are present both in plant and animal cells.
 (d) The perinuclear space forms a barrier between the materials present inside the nucleus and that of the cytoplasm.
- Q. 122.** Mutations in plant cells can be induced by
 (a) zeatin. (b) kinetin.
 (c) infrared rays. (d) gamma rays.
- Q. 123.** The first stable product of CO_2 fixation in Sorghum is
 (a) phosphoglyceric acid.
 (b) pyruvic acid.
 (c) oxaloacetic acid.
 (d) succinic acid.
- Q. 124.** Match List-I with List-II.

List-I		List-II	
(a)	Cohesion	(i)	More attraction in liquid phase

(b)	Adhesion	(ii)	Mutual attractor among water molecules
(c)	Surface tension	(iii)	Water loss in liquid phase
(d)	Guttation	(iv)	Attraction towards polar surfaces

Choose the correct answer from the options given below.

- (a) (b) (c) (d)
- (a) (ii) (i) (iv) (iii)
 (b) (ii) (iv) (i) (iii)
 (c) (iv) (iii) (ii) (i)
 (d) (iii) (i) (iv) (ii)
- Q. 125.** During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out
 (a) Polysaccharides (b) RNA
 (c) DNA (d) Histones
- Q. 126.** Amensalism can be represented as
 (a) Species A (+); Species B (0)
 (b) Species A (-); Species B (0)
 (c) Species A (+); Species B (+)
 (d) Species A (-); Species B (-)
- Q. 127.** Which of the following stages of meiosis involves division of centromere ?
 (a) Telophase II
 (b) Metaphase I
 (c) Metaphase II
 (d) Anaphase II
- Q. 128.** When gene targeting involving gene amplification is attempted in an individual's tissue to treat disease, it is known as
 (a) safety testing.
 (b) biopiracy.
 (c) gene therapy.
 (d) molecular diagnosis.
- Q. 129.** The amount of nutrients, such as carbon, nitrogen, phosphorus and calcium present in the soil at any given time, is referred as
 (a) standing crop.
 (b) climax.
 (c) climax community.
 (d) standing state.
- Q. 130.** Which of the following statements is not correct ?

- (a) Pyramid of numbers in a grassland ecosystem is upright.
 (b) Pyramid of biomass in sea is generally inverted.
 (c) Pyramid of biomass in sea is generally upright.
 (d) Pyramid of energy is always upright.

Q. 131. A typical angiosperm embryo sac at maturity is

- (a) 8-nucleate and 8-celled
 (b) 8-nucleate and 7-celled
 (c) 7-nucleate and 8-celled
 (d) 7-nucleate and 7-celled

Q. 132. When the centromere is situated in the middle of two equal arms of chromosomes, the chromosome is referred as

- (a) acrocentric.
 (b) metacentric.
 (c) telocentric.
 (d) sub-metacentric.

Q. 133. Genera like *Selaginella* and *Salvinia* produce two kinds of spores. Such plants are known as

- (a) Heterosporous (b) Homosorus
 (c) Heterosorus (d) Homosporous

Q. 134. Match List-I with List-II

List-I		List-II	
(a)	Protoplast fusion	(i)	Totipotency
(b)	Plant tissue culture	(ii)	Pomato
(c)	Meristem culture	(iii)	Somaclones
(d)	Micro-propagation	(iv)	Virus free plants

Choose the correct answer from the options given below.

- (a) (b) (c) (d)
- (a) (iv) (iii) (ii) (i)
 (b) (iii) (iv) (ii) (i)
 (c) (ii) (i) (iv) (iii)
 (d) (iii) (iv) (i) (ii)

Q. 135. Which of the following is not an application of PCR (Polymerase Chain Reaction) ?

- (a) Detection of gene mutation
 (b) Molecular diagnosis
 (c) Gene amplification
 (d) Purification of isolated protein

Section B

- Q. 136.** Identify the correct statement.
- Split gene arrangement is characteristic of prokaryotes.
 - In capping, methyl guanosine triphosphate is added to the 3' end of hnRNA.
 - RNA polymerase binds with Rho factor to terminate the process of transcription in bacteria.
 - The coding strand in a transcription unit is copied to an mRNA.
- Q. 137.** Now a days it is possible to detect the mutated gene causing cancer by allowing radioactive probe to hybridise its complementary DNA in a clone of cells, followed by its detection using autoradiography because
- mutated gene does not appear on photographic film as the probe has complementarity with it.
 - mutated gene partially appears on a photographic film.
 - mutated gene completely and clearly appears on a photographic film.
 - mutated gene does not appear on a photographic film as the probe has no complementarity with it.
- Q. 138.** What is the role of RNA polymerase III in the process of transcription in eukaryotes ?
- Transcribes only snRNAs
 - Transcribes rRNAs (28S, 18S and 5.8S)
 - Transcribes tRNA, 5s rRNA and snRNA
 - Transcribes precursor of mRNA
- Q. 139.** Select the correct pair.
- | | |
|--|--|
| (a) Loose parenchyma | – Spongy cell rupturing the parenchyma epidermis and forming a lens shaped opening in bark |
| (b) Large colorless empty cells in the epidermis of grass leaves | – Subsidiary cells |
| (c) In dicot leaves, vascular bundles are surrounded by large thick-walled cells | – Conjunctive tissue |
| (d) Cells of medullary rays | – Interfascicular cambium |
- that form part of cambial ring
- Q. 140.** In the exponential growth equation $N_t = N_0 e^{rt}$, e represents
- The base of geometric logarithms
 - The base of number logarithms
 - The base of exponential logarithms
 - The base of natural logarithms
- Q. 141.** Which of the following statements is incorrect ?
- Oxidation-reduction reactions produce proton gradient in respiration.
 - During aerobic respiration, role of oxygen is limited to the terminal stage.
 - In ETC (Electron Transport Chain), one molecule of $\text{NADH} + \text{H}^+$ gives rise to 2 ATP molecules, and one FADH_2 gives rise to 3 ATP molecules.
 - ATP is synthesized through complex V.
- Q. 142.** Which of the following statements is incorrect ?
- Cyclic photophosphorylation involves both PS I and PS II.
 - Both ATP and $\text{NADPH} + \text{H}^+$ are synthesized during non-cyclic photophosphorylation.
 - Stroma lamellae have PS I only and lack NADP reductase.
 - Grana lamellae have both PS I and PS II.
- Q. 143.** Which of the following statements is correct ?
- Some of the organisms can fix atmospheric nitrogen in specialized cells called sheath cells.
 - Fusion of two cells is called Karyogamy.
 - Fusion of protoplasts between two motile or non-motile gametes is called plasmogamy.
 - Organisms that depend on living plants are called saprophytes.
- Q. 144.** DNA fingerprinting involves identifying differences in some specific regions in DNA sequence, called as
- Polymorphic DNA.
 - Satellite DNA.
 - Repetitive DNA.
 - Single nucleotides.
- Q. 145.** Match List-I with List-II.
- | | List-I | | List-II |
|-----|------------------------|-------|----------------------|
| (a) | Protein | (i) | C = C double bonds |
| (b) | Unsaturated fatty acid | (ii) | Phosphodiester bonds |
| (c) | Nucleic acid | (iii) | Glycosidic bonds |
| (d) | Polysaccharide | (iv) | Peptide bonds |

Choose the correct answer from the options given below.

- (a) (b) (c) (d)
 (a) (iv) (iii) (i) (ii)
 (b) (iv) (i) (ii) (iii)
 (c) (i) (iv) (iii) (ii)
 (d) (ii) (i) (iv) (iii)

Q. 146. Match Column - I with Column - II:

Column - I		Column- II	
(a)	<i>Nitrococcus</i>	(i)	Denitrification
(b)	<i>Rhizobium</i>	(ii)	Conversion of ammonia to nitrite
(c)	<i>Thiobacillus</i>	(iii)	Conversion of nitrite to nitrate
(d)	<i>Nitrobacter</i>	(iv)	Conversion of atmospheric nitrogen to ammonia

Choose the correct answer from options given below.

- (a) (b) (c) (d)
 (a) (iv) (iii) (ii) (i)
 (b) (ii) (iv) (i) (iii)
 (c) (i) (ii) (iii) (iv)
 (d) (iii) (i) (iv) (ii)

Q. 147. In some members of which of the following pairs of families, pollen grains retain their viability for months after release ?

- (a) Rosaceae ; Leguminosae
 (b) Poaceae ; Rosaceae
 (c) Poaceae ; Leguminosae
 (d) Poaceae ; Solanaceae

Q. 148. Plasmid pBR322 has PstI restriction enzyme site within gene *amp^R* that confers ampicillin resistance.

If this enzyme is used for inserting a gene for β -galactoside production and the recombinant plasmid is inserted in an *E. coli* strain

- (a) It will be able to produce a novel protein

with dual ability.

- (b) It will not be able to confer ampicillin resistance to the host cell.
 (c) The transformed cells will have the ability to resist ampicillin as well as produce β -galactoside.
 (d) It will lead to lysis of host cell.

Q. 149. Match Column-I with Column-II

Column-I	Column-II
(a) $\% \frac{K_5}{C_1 + 2 + (2)A_{(3)} + 1G_1}$	(i) Brassicaceae
(b) $\oplus \frac{K_5}{C_5} \frac{A_5}{G_{(2)}}$	(ii) Liliaceae
(c) $\oplus \frac{P_{(3+3)}}{A_3 + 3} \frac{G_{(3)}}{G_{(3)}}$	(iii) Fabaceae
(d) $\oplus \frac{K_2 + 2}{C_4} \frac{A_2 - 4}{G_{(2)}}$	(iv) Solanaceae

Select the correct answer from the options given below.

- (a) (b) (c) (d)
 (a) (iv) (ii) (i) (iii)
 (b) (iii) (iv) (ii) (i)
 (c) (i) (ii) (iii) (iv)
 (d) (ii) (iii) (iv) (i)

Q. 150. Match List-I with List-II.

List-I	List-II
(a) S phase	(i) Proteins are synthesized
(b) G ₂ phase	(ii) Inactive phase
(c) Quiescent stage	(iii) Interval between mitosis and initiation of DNA replication
(d) G ₁ phase	(iv) DNA replication

Choose the correct answer from the options given below.

- (a) (b) (c) (d)
 (a) (ii) (iv) (iii) (i)
 (b) (iii) (ii) (i) (iv)
 (c) (iv) (ii) (iii) (i)
 (d) (iv) (i) (ii) (iii)

ZOOLOGY

Section A

Q. 151. Which of the following RNAs is not required for the synthesis of protein ?

- (a) siRNA (b) mRNA

- (c) tRNA (d) rRNA

Q. 152. Succus entericus is referred to as

- (a) chyme. (b) pancreatic juice.
 (c) intestinal juice. (d) gastric juice.

Q. 153. Which is the "Only enzyme" that has "Capability" to catalyse Initiation, Elongation and Termination in the process of transcription in prokaryotes ?

- (a) DNase
- (b) DNA dependent DNA polymerase
- (c) DNA dependent RNA polymerase
- (d) DNA Ligase

Q. 154. Which of the following characteristics is incorrect with respect to cockroach ?

- (a) 10th abdominal segment in both sexes, bears a pair of anal cerci.
- (b) A ring of gastric caeca is present at the junction of midgut and hind gut.
- (c) Hypopharynx lies within the cavity enclosed by the mouth parts.
- (d) In females, 7th-9th sterna together form a genital pouch.

Q. 155. Match List - I with List - II

List-I		List-II	
(a)	Metamerism	(i)	Coelenterata
(b)	Canal system	(ii)	Ctenophora
(c)	Comb plates	(iii)	Annelida
(d)	Cnidoblasts	(iv)	Porifera

Choose the correct answer from the options given below.

- (a) (iv) (i) (ii) (iii)
- (b) (iv) (iii) (i) (ii)
- (c) (iii) (iv) (i) (ii)
- (d) (iii) (iv) (ii) (i)

Q. 156. Which of the following is not an objective of Biofortification in crops ?

- (a) Improve micronutrient and mineral content
- (b) Improve protein content
- (c) Improve resistance to diseases
- (d) Improve vitamin content

Q. 157. Identify the incorrect pair

- (a) Drugs – Ricin
- (b) Alkaloids – Codeine
- (c) Toxin – Abrin
- (d) Lectins – Concanavalin A

Q. 158. Match List-I with List-II

List-I		List-II	
(a)	<i>Aspergillus niger</i>	(i)	Acetic Acid
(b)	<i>Acetobacter aceti</i>	(ii)	Lactic Acid
(c)	<i>Clostridium butylicum</i>	(iii)	Citric Acid
(d)	<i>Lactobacillus</i>	(iv)	Butyric Acid

Choose the correct answer from the options given below.

- (a) (iv) (ii) (i) (iii)
- (b) (iii) (i) (iv) (ii)
- (c) (i) (ii) (iii) (iv)
- (d) (ii) (iii) (i) (iv)

Q. 159. With regard to insulin choose correct options.

- (a) C-peptide is not present in mature insulin.
- (b) The insulin produced by rDNA technology has cpeptide.
- (c) The pro-insulin has C-peptide
- (d) A-peptide and B-peptide of insulin are interconnected by disulphide bridges.

Choose the correct answer from the options given below.

- (a) (b), (d) only
- (b) (a), (d) only
- (c) (b), (c) only
- (d) (a), (c), (d) only

Q. 160. Select the favourable conditions required for the formation of oxyhaemoglobin at the alveoli.

- (a) Low pO_2 , low pCO_2 , more H^+ , higher temperature
- (b) High pO_2 , low pCO_2 , less H^+ , lower temperature
- (c) Low pO_2 , high pCO_2 , more H^+ , higher temperature
- (d) High pO_2 , high pCO_2 , less H^+ , higher temperature

Q. 161. Match the following:

List-I		List-II	
(a)	<i>Physalia</i>	(i)	Pearl oyster
(b)	<i>Limulus</i>	(ii)	Portuguese Man of War
(c)	<i>Ancylostoma</i>	(iii)	Living fossil
(d)	<i>Pinctada</i>	(iv)	Hookworm

Choose the correct answer from the options given below.

- (a) (b) (c) (d)
- (a) (i) (iv) (iii) (ii)
- (b) (ii) (iii) (i) (iv)
- (c) (iv) (i) (iii) (ii)
- (d) (ii) (iii) (iv) (i)
- Q. 162.** The fruit fly has 8 chromosomes (2n) in each cell.
During interphase of Mitosis if the number of chromosomes at G₁ phase is 8, what would be the number of chromosomes after S phase ?
(a) 32 (b) 8 (c) 16 (d) 4
- Q. 163.** Which one of the following organisms bears hollow and pneumatic long bones ?
(a) *Ornithorhynchus* (b) *Neophron*
(c) *Hemidactylus* (d) *Macropus*
- Q. 164.** Dobson units are used to measure thickness of
(a) Troposphere. (b) CFCs.
(c) Stratosphere. (d) Ozone.
- Q. 165.** Which enzyme is responsible for the conversion of inactive fibrinogens to fibrins ?
(a) Thrombokinas (b) Thrombin
(c) Renin (d) Epinephrine
- Q. 166.** For effective treatment of the disease, early diagnosis and understanding its pathophysiology is very important. Which of the following molecular diagnostic techniques is very useful for early detection ?
(a) Hybridization Technique
(b) Western Blotting Technique
(c) Southern Blotting Technique
(d) ELISA Technique
- Q. 167.** The organelles that are included in the endomembrane system are
(a) Golgi complex, Endoplasmic reticulum, Mitochondria and Lysosomes.
(b) Endoplasmic reticulum, Mitochondria, Ribosomes and Lysosomes.
(c) Endoplasmic reticulum, Golgi complex, Lysosomes and Vacuoles.
(d) Golgi complex, Mitochondria, Ribosomes and Lysosomes.
- Q. 168.** If Adenine makes 30% of the DNA molecule, what will be the percentage of Thymine, Guanine and Cytosine in it ?
(a) T : 20 ; G : 25 ; C : 25
(b) T : 20 ; G : 30 ; C : 20

(c) T : 20 ; G : 20 ; C : 30

(d) T : 30 ; G : 20 ; C : 20

- Q. 169.** A specific recognition sequence identified by endonucleases to make cuts at specific positions within the DNA is
(a) poly(A) tail sequences.
(b) degenerate primer sequence.
(c) okazaki sequences.
(d) palindromic nucleotide sequences.
- Q. 170.** Which of the following statements wrongly represents the nature of smooth muscle ?
(a) These muscles are present in the wall of blood vessels.
(b) These muscle have no striations.
(c) They are involuntary muscles.
(d) Communication among the cells is performed by intercalated discs.

- Q. 171.** Match List-I with List-II.

List-I		List-II	
(a)	Vaults	(i)	Entry of sperm through Cervix is blocked
(b)	IUDs	(ii)	Removal of Vas deferens
(c)	Vasectomy	(iii)	Phagocytosis of sperms within the Uterus
(d)	Tubectomy	(iv)	Removal of fallopian tube

Choose the correct answer from the options given below

- (a) (b) (c) (d)
- (a) (iii) (i) (iv) (ii)
- (b) (iv) (ii) (i) (iii)
- (c) (i) (iii) (ii) (iv)
- (d) (ii) (iv) (iii) (i)
- Q. 172.** During the process of gene amplification using PCR, if very high temperature is not maintained in the beginning, then which of the following steps of PCR will be affected first ?
(a) Ligation (b) Annealing
(c) Extension (d) Denaturation
- Q. 173.** In a cross between a male and female, both heterozygous for sickle cell anaemia gene, what percentage of the progeny will be diseased ?
(a) 100% (b) 50%
(c) 75% (d) 25%

Q. 174. Read the following statements.

- (a) Metagenesis is observed in Helminths.
- (b) Echinoderms are triploblastic and coelomate animals.
- (c) Round worms have organ-system level of body organization.
- (d) Comb plates present in ctenophores help in digestion.
- (e) Water vascular system is characteristic of Echinoderms.

Choose the correct answer from the options given below.

- (a) (b), (c) and (e) are correct
- (b) (c), (d) and (e) are correct
- (c) (a), (b) and (c) are correct
- (d) (a), (d) and (e) are correct

Q. 175. Receptors for sperm binding in mammals are present on

- (a) zona pellucida.
- (b) corona radiata.
- (c) vitelline membrane.
- (d) perivitelline space.

Q. 176. Sphincter of oddi is present at

- (a) junction of jejunum and duodenum.
- (b) ileo-caecal junction.
- (c) junction of hepato-pancreatic duct and duodenum.
- (d) gastro-oesophageal junction.

Q. 177. Chronic auto immune disorder affecting neuro muscular junction leading to fatigue, weakening and paralysis of skeletal muscle is called as

- (a) gout.
- (b) arthritis.
- (c) muscular dystrophy.
- (d) myasthenia gravis.

Q. 178. Persons with 'AB' blood group are called as "Universal recipients". This is due to

- (a) absence of antibodies, anti-A and anti-B, in plasma.
- (b) absence of antigens A and B on the surface of RBCs.
- (c) absence of antigens A and B in plasma.
- (d) presence of antibodies, anti-A and anti-B, on RBCs.

Q. 179. The centriole undergoes duplication during

- (a) G₂ phase (b) S-phase
- (c) Prophase (d) Metaphase

Q. 180. Erythropoietin hormone which stimulates R.B.C. formation is produced by

- (a) juxtaglomerular cells of the kidney.
- (b) alpha cells of pancreas.
- (c) the cells of rostral adenohypophysis.
- (d) the cells of bone marrow.

Q. 181. Which one of the following belongs to the family Muscidae ?

- (a) House fly (b) Fire fly
- (c) Grasshopper (d) Cockroach

Q. 182. Veneral diseases can spread through

- (a) Using sterile needles
- (b) Transfusion of blood from infected person
- (c) Infected mother to foetus
- (d) Kissing
- (e) Inheritance

Choose the correct answer from the option given below

- (a) (a) and (c) only
- (b) (a), (b) and (c) only
- (c) (b), (c) and (d) only
- (d) (b) and (c) only

Q. 183. The partial pressures (in mm Hg) of oxygen (O₂) and carbon dioxide (CO₂) at alveoli (the site of diffusion) are

- (a) pO₂ = 159 and pCO₂ = 0.3
- (b) pO₂ = 104 and pCO₂ = 40
- (c) pO₂ = 40 and pCO₂ = 45
- (d) pO₂ = 95 and pCO₂ = 40

Q. 184. Which one of the following is an example of hormone releasing IUD ?

- (a) Multiload 375 (b) CuT
- (c) LNG 20 (d) Cu 7

Q. 185. Which stage of meiotic prophase shows terminalisation of chiasmata as its distinctive feature ?

- (a) Pachytene (b) Leptotene
- (c) Zygotene (d) Diakinesis

Section B

Q. 186. Which of these is not an important component of initiation of parturition in humans ?

- (a) Release of Prolactin
- (b) Increase in estrogen and progesterone ratio
- (c) Synthesis of prostaglandins
- (d) Release of Oxytocin

Q. 187. Following are the statements about prostomium of earthworm.

- (a) It serves as a covering for mouth.
- (b) It helps to open cracks in the soil into which it can crawl.
- (c) It is one of the sensory structures.
- (d) It is the first body segment.

Choose the correct answer from the options given below.

- (a) (b) and (c) are correct
- (b) (a), (b) and (c) are correct
- (c) (a), (b) and (d) are correct
- (d) (a), (b), (c) and (d) are correct

Q. 188. The Adenosine deaminase deficiency results into

- (a) Addison's disease
- (b) Dysfunction of Immune system
- (c) Parkinson's disease
- (d) Digestive disorder

Q. 189. Match List-I with List-II

List-I		List-II	
(a)	Adaptive radiation	(i)	Selection of resistant varieties due to excessive use of herbicides and pesticides
(b)	Convergent evolution	(ii)	Bones of forelimbs in Man and Whale
(c)	Divergent evolution	(iii)	Wings of Butterfly and Bird
(d)	Evolution by anthropogenic action	(iv)	Darwin Finches

Choose the correct answer from the options given below.

- (a) (b) (c) (d)
- (a) (i) (iv) (iii) (ii)
- (b) (iv) (iii) (ii) (i)
- (c) (iii) (ii) (i) (iv)
- (d) (ii) (i) (iv) (iii)

Q. 190. Match List-I with List-II

List-I		List-II	
(a)	Allen's Rule	(i)	Kangaroo rat
(b)	Physiological adaptation	(ii)	Desert lizard
(c)	Behavioural adaptation	(iii)	Marine fish at depth
(d)	Biochemical adaptation	(iv)	Polar seal

Choose the correct answer from the options given below.

- (a) (b) (c) (d)
- (a) (iv) (iii) (ii) (i)
- (b) (iv) (ii) (iii) (i)
- (c) (iv) (i) (iii) (ii)
- (d) (iv) (i) (ii) (iii)

Q. 191. Which one of the following statements about Histones is wrong ?

- (a) Histones carry positive charge in the side chain.
- (b) Histones are organized to form a unit of 8 molecules.
- (c) The pH of histones is slightly acidic.
- (d) Histones are rich in amino acids - Lysine and Arginine.

Q. 192. Assertion (A): A person goes to high altitude and experiences 'altitude sickness' with symptoms like breathing difficulty and heart palpitations.

Reason (R): Due to low atmospheric pressure at high altitude, the body does not get sufficient oxygen.

In the light of the above statements, choose the correct answer from the options given below

- (a) (A) is false but (R) is true
- (b) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (c) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (d) (A) is true but (R) is false

Q. 193. Which of the following is not a step in Multiple Ovulation Embryo Transfer Technology (MOET) ?

- (a) Fertilized eggs are transferred to surrogate mothers at 8-32 cell stage.
- (b) Cow is administered hormone having LH like activity for super ovulation.
- (c) Cow yields about 6-8 eggs at a time.
- (d) Cow is fertilized by artificial insemination.

Q. 194. Match List-I with List-II

List-I		List-II	
(a)	Filariasis	(i)	<i>Haemophilus influenzae</i>
(b)	Amoebiasis	(ii)	<i>Trichophyton</i>
(c)	Pneumonia	(iii)	<i>Wuchereria bancrofti</i>
(d)	Ringworm	(iv)	<i>Entamoeba histolytica</i>

Choose the correct answer from the options given below

- (a) (b) (c) (d)
 (a) (ii) (iii) (i) (iv)
 (b) (iv) (i) (iii) (ii)
 (c) (iii) (iv) (i) (ii)
 (d) (i) (ii) (iv) (iii)

Q. 195. Following are the statements with reference to 'lipids'.

- (a) Lipids having only single bonds are called unsaturated fatty acids.
 (b) Lecithin is a phospholipid.
 (c) Trihydroxy propane is glycerol.
 (d) Palmitic acid has 20 carbon atoms including carboxyl carbon.
 (e) Arachidonic acid has 16 carbon atoms.

Choose the correct answer from the options given below.

- (a) (b) and (e) only (b) (a) and (b) only
 (c) (c) and (d) only (d) (b) and (c) only

Q. 196. Which of the following secretes the hormone, relaxin, during the later phase of pregnancy ?

- (a) Uterus (b) Graafian follicle
 (c) Corpus luteum (d) Foetus

Q. 197. Identify the types of cell junctions that help to stop the leakage of the substances across a tissue and facilitation of communication with neighbouring cells via rapid transfer of ions and molecules.

- (a) Adhering junctions and Gap junctions, respectively.
 (b) Gap junctions and Adhering junctions, respectively.
 (c) Tight junctions and Gap junctions, respectively.

- (d) Adhering junctions and Tight junctions, respectively.

Q. 198. Match List-I with List-II

List-I		List-II	
(a)	Scapula	(i)	Cartilaginous joints
(b)	Cranium	(ii)	Flat bone
(c)	Sternum	(iii)	Fibrous joints
(d)	Vertebral column	(iv)	Triangular flat bone

Choose the correct answer from the options given below

- (a) (b) (c) (d)
 (a) (iv) (iii) (ii) (i)
 (b) (i) (iii) (ii) (iv)
 (c) (ii) (iii) (iv) (i)
 (d) (iv) (ii) (iii) (i)

Q. 199. During muscular contraction which of the following events occur ?

- (a) 'H' zone disappears
 (b) 'A' band widens
 (c) 'I' band reduces in width
 (d) Myosine hydrolyzes ATP, releasing the ADP and Pi.
 (e) Z-lines attached to actins are pulled inwards.

Choose the correct answer from the options given below

- (a) (b), (d), (e), (a) only
 (b) (a), (c), (d), (e) only
 (c) (a), (b), (c), (d) only
 (d) (b), (c), (d), (e) only

Q. 200. **Statement I:** The codon 'AUG' codes for methionine and phenylalanine.

Statement II: 'AAA' and 'AAG' both codons code for the amino acid lysine.

In the light of the above statements, choose the correct answer from the options given below.

- (a) Statement I is incorrect but Statement II is true.
 (b) Both Statement I and Statement II are true.
 (c) Both Statement I and Statement II are false.
 (d) Statement I is correct but Statement II is false.

Booklet	Batch	Roll Number	Name	Test Date	Invigilator's Signature	Student's Signature	Test Center Code
(A) 10+1 (S)		0 0 0 0 0 0 0 0					0 0
(B) 10-2 (V)		1 1 1 1 1 1 1 1					1 1
(C) 10-3 (A)		2 2 2 2 2 2 2 2					2 2
(D) Crash (C)		3 3 3 3 3 3 3 3					3 3
(E)		4 4 4 4 4 4 4 4					4 4
(F) Paper		5 5 5 5 5 5 5 5					5 5
(G)		6 6 6 6 6 6 6 6					6 6
(H) Paper 1 (1)		7 7 7 7 7 7 7 7					7 7
(I) Paper 2 (2)		8 8 8 8 8 8 8 8					8 8
(J)		9 9 9 9 9 9 9 9					9 9

The OMR Sheet will be computer checked. Fill the circles completely and dark enough for proper detection. Use ballpen (black or blue) for marking.

Avoid Improper Marking

Partially Filled

Lightly Filled

Certified that all the entries in this section have been properly filled by the student

Test Center Code

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1 (a) (b) (c) (d)	6 (a) (b) (c) (d)	11 (a) (b) (c) (d)	16 (a) (b) (c) (d)	21 (a) (b) (c) (d)	26 (a) (b) (c) (d)
2 (a) (b) (c) (d)	7 (a) (b) (c) (d)	12 (a) (b) (c) (d)	17 (a) (b) (c) (d)	22 (a) (b) (c) (d)	27 (a) (b) (c) (d)
3 (a) (b) (c) (d)	8 (a) (b) (c) (d)	13 (a) (b) (c) (d)	18 (a) (b) (c) (d)	23 (a) (b) (c) (d)	28 (a) (b) (c) (d)
4 (a) (b) (c) (d)	9 (a) (b) (c) (d)	14 (a) (b) (c) (d)	19 (a) (b) (c) (d)	24 (a) (b) (c) (d)	29 (a) (b) (c) (d)
5 (a) (b) (c) (d)	10 (a) (b) (c) (d)	15 (a) (b) (c) (d)	20 (a) (b) (c) (d)	25 (a) (b) (c) (d)	30 (a) (b) (c) (d)
31 (a) (b) (c) (d)	36 (a) (b) (c) (d)	41 (a) (b) (c) (d)	46 (a) (b) (c) (d)	51 (a) (b) (c) (d)	56 (a) (b) (c) (d)
32 (a) (b) (c) (d)	37 (a) (b) (c) (d)	42 (a) (b) (c) (d)	47 (a) (b) (c) (d)	52 (a) (b) (c) (d)	57 (a) (b) (c) (d)
33 (a) (b) (c) (d)	38 (a) (b) (c) (d)	43 (a) (b) (c) (d)	48 (a) (b) (c) (d)	53 (a) (b) (c) (d)	58 (a) (b) (c) (d)
34 (a) (b) (c) (d)	39 (a) (b) (c) (d)	44 (a) (b) (c) (d)	49 (a) (b) (c) (d)	54 (a) (b) (c) (d)	59 (a) (b) (c) (d)
35 (a) (b) (c) (d)	40 (a) (b) (c) (d)	45 (a) (b) (c) (d)	50 (a) (b) (c) (d)	55 (a) (b) (c) (d)	60 (a) (b) (c) (d)
61 (a) (b) (c) (d)	66 (a) (b) (c) (d)	71 (a) (b) (c) (d)	76 (a) (b) (c) (d)	81 (a) (b) (c) (d)	86 (a) (b) (c) (d)
62 (a) (b) (c) (d)	67 (a) (b) (c) (d)	72 (a) (b) (c) (d)	77 (a) (b) (c) (d)	82 (a) (b) (c) (d)	87 (a) (b) (c) (d)
63 (a) (b) (c) (d)	68 (a) (b) (c) (d)	73 (a) (b) (c) (d)	78 (a) (b) (c) (d)	83 (a) (b) (c) (d)	88 (a) (b) (c) (d)
64 (a) (b) (c) (d)	69 (a) (b) (c) (d)	74 (a) (b) (c) (d)	79 (a) (b) (c) (d)	84 (a) (b) (c) (d)	89 (a) (b) (c) (d)
65 (a) (b) (c) (d)	70 (a) (b) (c) (d)	75 (a) (b) (c) (d)	80 (a) (b) (c) (d)	85 (a) (b) (c) (d)	90 (a) (b) (c) (d)
91 (a) (b) (c) (d)	96 (a) (b) (c) (d)	101 (a) (b) (c) (d)	106 (a) (b) (c) (d)	111 (a) (b) (c) (d)	116 (a) (b) (c) (d)
92 (a) (b) (c) (d)	97 (a) (b) (c) (d)	102 (a) (b) (c) (d)	107 (a) (b) (c) (d)	112 (a) (b) (c) (d)	117 (a) (b) (c) (d)
93 (a) (b) (c) (d)	98 (a) (b) (c) (d)	103 (a) (b) (c) (d)	108 (a) (b) (c) (d)	113 (a) (b) (c) (d)	118 (a) (b) (c) (d)
94 (a) (b) (c) (d)	99 (a) (b) (c) (d)	104 (a) (b) (c) (d)	109 (a) (b) (c) (d)	114 (a) (b) (c) (d)	119 (a) (b) (c) (d)
95 (a) (b) (c) (d)	100 (a) (b) (c) (d)	105 (a) (b) (c) (d)	110 (a) (b) (c) (d)	115 (a) (b) (c) (d)	120 (a) (b) (c) (d)
121 (a) (b) (c) (d)	126 (a) (b) (c) (d)	131 (a) (b) (c) (d)	136 (a) (b) (c) (d)	141 (a) (b) (c) (d)	146 (a) (b) (c) (d)
122 (a) (b) (c) (d)	127 (a) (b) (c) (d)	132 (a) (b) (c) (d)	137 (a) (b) (c) (d)	142 (a) (b) (c) (d)	147 (a) (b) (c) (d)
123 (a) (b) (c) (d)	128 (a) (b) (c) (d)	133 (a) (b) (c) (d)	138 (a) (b) (c) (d)	143 (a) (b) (c) (d)	148 (a) (b) (c) (d)
124 (a) (b) (c) (d)	129 (a) (b) (c) (d)	134 (a) (b) (c) (d)	139 (a) (b) (c) (d)	144 (a) (b) (c) (d)	149 (a) (b) (c) (d)
125 (a) (b) (c) (d)	130 (a) (b) (c) (d)	135 (a) (b) (c) (d)	140 (a) (b) (c) (d)	145 (a) (b) (c) (d)	150 (a) (b) (c) (d)
151 (a) (b) (c) (d)	156 (a) (b) (c) (d)	161 (a) (b) (c) (d)	166 (a) (b) (c) (d)	171 (a) (b) (c) (d)	176 (a) (b) (c) (d)
152 (a) (b) (c) (d)	157 (a) (b) (c) (d)	162 (a) (b) (c) (d)	167 (a) (b) (c) (d)	172 (a) (b) (c) (d)	177 (a) (b) (c) (d)
153 (a) (b) (c) (d)	158 (a) (b) (c) (d)	163 (a) (b) (c) (d)	168 (a) (b) (c) (d)	173 (a) (b) (c) (d)	178 (a) (b) (c) (d)
154 (a) (b) (c) (d)	159 (a) (b) (c) (d)	164 (a) (b) (c) (d)	169 (a) (b) (c) (d)	174 (a) (b) (c) (d)	179 (a) (b) (c) (d)
155 (a) (b) (c) (d)	160 (a) (b) (c) (d)	165 (a) (b) (c) (d)	170 (a) (b) (c) (d)	175 (a) (b) (c) (d)	180 (a) (b) (c) (d)
181 (a) (b) (c) (d)	186 (a) (b) (c) (d)	191 (a) (b) (c) (d)	196 (a) (b) (c) (d)		
182 (a) (b) (c) (d)	187 (a) (b) (c) (d)	192 (a) (b) (c) (d)	197 (a) (b) (c) (d)		
183 (a) (b) (c) (d)	188 (a) (b) (c) (d)	193 (a) (b) (c) (d)	198 (a) (b) (c) (d)		
184 (a) (b) (c) (d)	189 (a) (b) (c) (d)	194 (a) (b) (c) (d)	199 (a) (b) (c) (d)		
185 (a) (b) (c) (d)	190 (a) (b) (c) (d)	195 (a) (b) (c) (d)	200 (a) (b) (c) (d)		

ANSWER KEY

1	(b)	41	(a)	81	(a)	121	(b)	161	(d)
2	(b)	42	(c)	82	(c)	122	(d)	162	(b)
3	(a)	43	(a)	83	(c)	123	(c)	163	(b)
4	(a)	44	(b)	84	(c)	124	(b)	164	(d)
5	(c)	45	(b)	85	(c)	125	(c)	165	(b)
6	(d)	46	(d)	86	(b)	126	(b)	166	(d)
7	(a)	47	(a)	87	(d)	127	(d)	167	(c)
8	(c)	48	(c)	88	(a)	128	(c)	168	(d)
9	(c)	49	(c)	89	(a)	129	(d)	169	(d)
10	(c)	50	(a)	90	(d)	130	(c)	170	(d)
11	(c)	51	(c)	91	(c)	131	(b)	171	(c)
12	(d)	52	(a)	92	(b)	132	(b)	172	(d)
13	(c)	53	(b)	93	(d)	133	(a)	173	(d)
14	(a)	54	(d)	94	(a)	134	(c)	174	(a)
15	(c)	55	(c)	95	(d)	135	(d)	175	(a)
16	(b)	56	(a)	96	(a)	136	(c)	176	(c)
17	(d)	57	(d)	97	(d)	137	(d)	177	(d)
18	(b)	58	(d)	98	(d)	138	(c)	178	(a)
19	(c)	59	(b)	99	(d)	139	(d)	179	(b)
20	(a)	60	(b)	100	(c)	140	(d)	180	(a)
21	(c)	61	(b)	101	(d)	141	(c)	181	(a)
22	(c)	62	(d)	102	(a)	142	(a)	182	(d)
23	(a)	63	(a)	103	(a)	143	(c)	183	(b)
24	(a)	64	(a)	104	(d)	144	(c)	184	(c)
25	(a)	65	(b)	105	(b)	145	(b)	185	(d)
26	(d)	66	(b)	106	(b)	146	(b)	186	(a)
27	(d)	67	(d)	107	(b)	147	(a)	187	(b)
28	(b)	68	(c)	108	(c)	148	(b)	188	(b)
29	(c)	69	(b)	109	(b)	149	(b)	189	(b)
30	(b)	70	(b)	110	(c)	150	(d)	190	(d)
31	(d)	71	(b)	111	(d)	151	(a)	191	(c)
32	(d)	72	(a)	112	(d)	152	(c)	192	(b)
33	(a)	73	(d)	113	(d)	153	(c)	193	(b)
34	(d)	74	(d)	114	(c)	154	(b)	194	(c)
35	(d)	75	(b)	115	(c)	155	(d)	195	(d)
36	(b)	76	(b)	116	(d)	156	(c)	196	(c)
37	(a)	77	(c)	117	(a)	157	(a)	197	(c)
38	(c)	78	(b)	118	(a)	158	(b)	198	(a)
39	(a)	79	(c)	119	(d)	159	(d)	199	(b)
40	(a)	80	(a)	120	(b)	160	(b)	200	(a)



SOLVED PAPER 2021

ANSWERS WITH EXPLANATIONS

PHYSICS

1. Option (b) is correct.

Given,

EMF of cell (E_1) = 1.5 V

Balance point (l_1) = 36 cm

EMF of replaced cell (E_2) = 2.5 V

New balance point (l_2) = ?

Now, from the application of potentiometer

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

Now, substituting the values,

$$\frac{1.5}{2.5} = \frac{36}{l_2}$$

$$\Rightarrow l_2 = 36 \times \frac{2.5}{1.5}$$

$$\Rightarrow l_2 = (12 \times 5) \text{ cm}$$

$$\Rightarrow l_2 = 60 \text{ cm}$$

Hence, the new balancing point of potentiometer will be 60 cm.

2. Option (b) is correct.

Drift velocity (V_d) = ?

Suppose, applied electric field at the two end of conductor is E

$$\frac{\vec{E}}{\rho}$$

Force on electron = eE

$$\text{Acceleration (a)} = \frac{F}{m}$$

$$\Rightarrow a = \frac{eE}{m}$$

Now, $V = u + at$

Initial velocity, $u = 0$

$$\Rightarrow V_d = at$$

$$\Rightarrow V_d = \frac{eE}{m} t$$

Electrical resistivity = ?

$$R = \rho \cdot \frac{l}{A}$$

$$V = IR$$

$$\Rightarrow E = \frac{V}{l}$$

$$\Rightarrow E \cdot l = I \times \rho \frac{l}{A}$$

$$\Rightarrow E \cdot \frac{l}{l} = \left(\frac{I}{A} \right) \rho \quad \left(\because J = \frac{I}{A} \right)$$

$$\Rightarrow E = J\rho$$

$$\Rightarrow \rho = \frac{E}{J}$$

Relaxation period

$$V_d = \frac{eE\tau}{m}$$

We know, $I = neAV_d$

$$\Rightarrow I = neA \left(\frac{eE\tau}{m} \right)$$

$$\Rightarrow I = \frac{ne^2 A E \tau}{m}$$

$$\Rightarrow I = \frac{ne^2 A \tau}{m} \left(\frac{V}{l} \right) \quad \left(\because E = \frac{V}{l} \right)$$

$$\Rightarrow \frac{m \cdot l}{ne^2 A \tau} = \frac{V}{I}$$

$$\Rightarrow \frac{ml}{ne^2 A \tau} = R \quad (\because V = IR)$$

$$\Rightarrow \frac{ml}{ne^2 A \tau} = \rho \cdot \frac{l}{A} \quad \left(\because R = \frac{\rho l}{A} \right)$$

$$\Rightarrow \rho = \frac{m}{ne^2 \tau}$$

$$\text{Current density (J)} = \frac{I}{A}$$

$$\Rightarrow J = \frac{neAV_d}{A}$$

$$\Rightarrow J = neV_d$$

Hence, the correct option will be option *b*.

A $\rightarrow$ R

B $\rightarrow$ S

C $\rightarrow$ P

D $\rightarrow$ Q

3. **Option (a) is correct.**

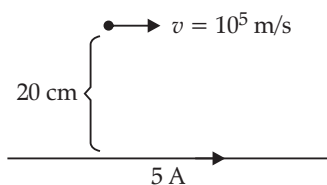
Given,

Current in conductor (I) = 5 A

Speed of electron (v) = 10^5 m/s

Perpendicular distance between electron and conductor = 20 cm

Force on electron = ?



Magnetic field due to current carrying conductor

$$B = \frac{\mu_0 i}{2\pi a}$$

So, force on charge

$$F = evB$$

$$\Rightarrow F = 1.6 \times 10^{-19} \times 10^5 \times \frac{\mu_0 \times 5}{2\pi \times 0.2}$$

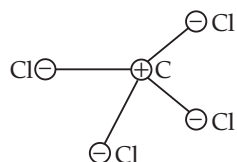
$$\Rightarrow F = 1.6 \times 10^{-19} \times 10^5 \times \left(\frac{2\mu_0}{4\pi} \right) \times \frac{5}{0.2}$$

$$\Rightarrow F = 1.6 \times 10^{-19} \times 10^{-7} \times 10^5 \times 5 \times 10 \text{ N}$$

$$\Rightarrow F = 8.0 \times 10^{-20} \text{ N}$$

4. **Option (a) is correct.**

It is a kind of molecule, in which, the one end of the molecule is slightly positive, while other end of the molecule is slightly negative.



Hence, molecules have a permanent electric dipole moment of their own.

5. **Option (c) is correct.**

Given,

Initial temperature of the cup of coffee (T_1) = 90°C

Final temperature (T_2) = 80°C

Time taken = t min

Room temperature (T) = 20°C

Time taken to cool a similar cup of coffee from 80°C to 60°C = ?

Time taken = t_1 min

Room temperature (T) = 20°C

Now, applying Newton's cooling law,

$$K \left(\frac{T_1 + T_2}{2} - T \right) = \frac{T_1 - T_2}{t}$$

Now, substituting the values

$$K \left(\frac{90 + 80}{2} - 20 \right) = \frac{90 - 80}{t}$$

$$\Rightarrow K \left(\frac{170}{2} - 20 \right) = \frac{10}{t}$$

$$\Rightarrow K(85 - 20) = \frac{10}{t}$$

$$\Rightarrow K(65) = \frac{10}{t} \quad \dots(i)$$

For second case

$$K \left(\frac{80 + 60}{2} - 20 \right) = \frac{80 - 60}{t_1}$$

$$K \left(\frac{140}{2} - 20 \right) = \frac{20}{t_1}$$

$$\Rightarrow K(70 - 20) = \frac{20}{t_1}$$

$$\Rightarrow K(50) = \frac{20}{t_1} \quad \dots(ii)$$

Now, from equation (i) and (ii)

$$\Rightarrow \frac{K \times 65}{50K} = \frac{\frac{10}{t}}{\frac{20}{t_1}}$$

$$\Rightarrow \frac{13}{10} = \frac{10 \times t_1}{20t}$$

$$\Rightarrow t_1 = \frac{13 \times 20}{10 \times 10} = \frac{26t}{10}$$

$$\Rightarrow t_1 = \frac{13}{5}t$$

6. **Option (d) is correct.**

When we are connecting zener diode in reverse bias, and after the breakdown, voltage across the zener diode becomes constant.

Hence, zener diode works as a voltage regulator.

Hence, statement A is correct.

We know that the potential barrier across silicon diode is 0.7 V.

Hence, statement B is incorrect.

7. Option (a) is correct.

Given,

Mass number of nucleus (m) = 240

The masses of two fragments are

$$m_1 = m_2 = 120$$

Binding energy per nucleon of unfragmented nuclei = 7.6 MeV

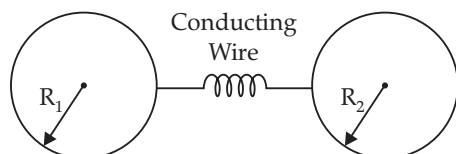
BE of fragmented nuclei = 8.5 MeV

Total gain in Binding energy = ?

$$\begin{aligned} \text{Total gain in Binding energy} &= \text{BE of products} \\ &\quad - \text{BE of reactants} \\ &= (120 + 120) \times 8.5 - 240 \times 7.6 \\ &= 240 \times 8.5 - 240 \times 7.6 \\ &= 216 \text{ MeV} \end{aligned}$$

Hence, gain in binding energy = 216 MeV

8. Option (c) is correct.



After connecting through the conducting wire, the potential on both the sphere will become same.

$$\begin{aligned} \frac{Q_1}{4\pi\epsilon_0 R_1} &= \frac{Q_2}{4\pi\epsilon_0 R_2} \\ \frac{Q_1 R_1}{4\pi\epsilon_0 R_1^2} &= \frac{Q_2 R_2}{4\pi\epsilon_0 R_2^2} \\ \Rightarrow \frac{\sigma_1 R_1}{\epsilon_0} &= \sigma_2 \cdot \frac{R_2}{\epsilon_0} \end{aligned}$$

$$\text{So, } \frac{\sigma_1}{\sigma_2} = \frac{R_2}{R_1}$$

9. Option (c) is correct.

Given,

Frequency of simple harmonic motion = n

The frequency of its potential energy = ?

The equation of displacement of SHM

$$x = A \sin(\omega t + \phi) \quad \dots(i)$$

The potential energy of the particle executing

$$\text{SHM}(u) = \frac{1}{2} Kx^2$$

$$\Rightarrow u = \frac{1}{2} K [A \sin(\omega t + \phi)]^2 \quad \dots(ii)$$

From equation (i) and (ii)

$$\omega = 2\pi n$$

$$n = \omega/2\pi \quad \dots(iii)$$

$$\text{And } u = \frac{K}{2} A^2 \sin^2(\omega t + \phi)$$

$$\Rightarrow \sin^2 \theta = \frac{1 - \cos \theta}{2} = \frac{1}{2} - \frac{1}{2} \cos \theta$$

$$\Rightarrow u = \frac{KA^2}{2 \times 2} (1 - \cos 2(\omega t + \phi))$$

Hence, frequency of potential energy

$$f_1 = \frac{2\omega}{2\pi} \quad \dots(iv)$$

Now, divide equation (iv) from equation (iii)

$$\begin{aligned} \frac{f_1}{n} &= \frac{\frac{2\omega}{2\pi}}{\frac{\omega}{2\pi}} \\ &= \frac{f_1}{n} = 2 \end{aligned}$$

$$\Rightarrow f_1 = 2n$$

Hence, the required frequency of potential energy = $2n$

10. Option (c) is correct.

Given,

Half life of radioactive nuclide

$$(t_{1/2}) = 100 \text{ hour}$$

Fraction of original activity after 150 hour is = ?

$$t = 150 \text{ hour}$$

The activity of radioactive substance

$$A = A_0 \left(\frac{1}{2} \right)^{t/t_{1/2}}$$

Now, substituting the values,

$$\frac{A}{A_0} = \left(\frac{1}{2} \right)^{\frac{150}{100}}$$

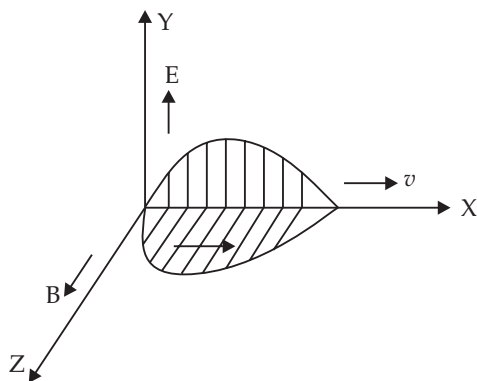
$$\Rightarrow A = A_0 \left(\frac{1}{2} \right)^{\frac{3}{2}}$$

$$\Rightarrow \frac{A}{A_0} = \frac{1}{2\sqrt{2}}$$

Hence, the required fraction of original activity = $\frac{1}{2\sqrt{2}}$

11. Option (c) is correct.

Given,



Direction of propagation of electromagnetic wave = +X - direction

From the given graph of the propagation of electromagnetic wave,

We can analyze that direction of propagation of wave, direction of electric field and direction of magnetic fields are perpendicular to each other.

$$\vec{E} \cdot \vec{B} = 0$$

$$\text{So, } (-\hat{j} + \hat{k}) \cdot (-\hat{j} - \hat{k}) = 0$$

$$\Rightarrow +1 - 1 = 0$$

12. Option (d) is correct.

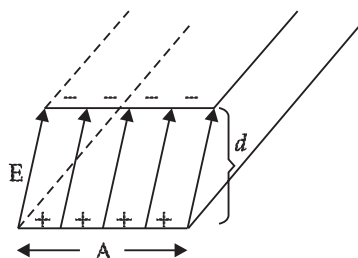
Given,

Electric field between the parallel plate capacitor = E

Distance between the plates = d

Area of each plate = A

Energy stored in the capacitor = ?



The developed potential between the plates (V) = $\vec{E} \cdot d$

$$\text{Capacitance} = \frac{\epsilon_0 A}{d}$$

Hence, stored potential energy

$$u = \frac{1}{2} CV^2$$

$$\Rightarrow u = \frac{1}{2} \left(\frac{\epsilon_0 A}{d} \right) (\vec{E} \cdot d)^2$$

$$\Rightarrow u = \frac{1}{2} \frac{\epsilon_0 A}{d} E^2 \cdot d^2$$

$$\Rightarrow u = \frac{1}{2} \epsilon_0 A d E^2$$

13. Option (c) is correct.

Given,

Chosen fundamental quantities are,

Force = F

Acceleration = A

Time = T

Dimensions of energy = ?

$$E \propto F^a A^b T^c$$

$$\Rightarrow [E] = [F^a][A^b][T^c]$$

Now,

$$\Rightarrow [E] = [ML^2T^{-2}]$$

$$[A] = [M^{\circ}LT^{-2}]$$

$$[T] = [M^{\circ}L^{\circ}T^1]$$

Substituting the values,

$$[ML^2T^{-2}] = [MLT^{-2}]^a [T]^b [T]^c$$

$$[ML^2T^{-2}] = [M^a L^{a+b} T^{-2a-2b+c}]$$

Now, comparing the exponents,

$$a = 1 \quad \dots(i)$$

$$a + b = 2 \quad \dots(ii)$$

$$-2a - 2b + c = -2 \quad \dots(iii)$$

From equation (i), (ii) and (iii)

$$b = 2 - 1$$

$$b = 1$$

$$\text{And, } C = -2 + 2a + 2b$$

$$= -2 + 2 + 2$$

$$C = +2$$

$$\text{Hence, } [E] = [FAT^2]$$

14. Option (a) is correct.

Given,

Height from which particle released = S

Let at height h,

Kinetic energy = 3 Potential energy

Initially, potential energy = mgS.

At height h,

$$mgh + \frac{1}{2}mv^2 = mgS$$

$$gh + \frac{v^2}{2} = gS \quad \dots(i)$$

From the given condition,

$$\frac{1}{2}mv^2 = 3mgh$$

$$\Rightarrow v^2 = 6gh \quad \dots(ii)$$

From equation (i) and (ii)

$$gh + \frac{6gh}{2} = gS$$

$$\Rightarrow h + 3h = S$$

$$\Rightarrow 4h = S$$

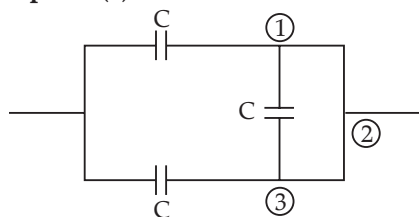
$$\Rightarrow h = \frac{S}{4}$$

Now, substituting the value of h in equation (ii)

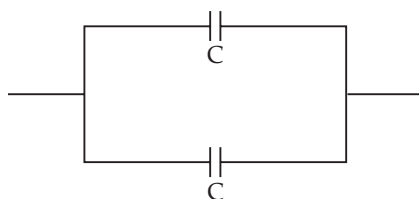
$$v^2 = 6g \times \left(\frac{S}{4}\right)$$

$$\Rightarrow v = \sqrt{6g \frac{S}{4}} = \frac{\sqrt{3gS}}{2}$$

15. Option (c) is correct.



In the above circuit diagram, point (1) and (3) are at same potential, so it will disappear from circuit



$$C_{eq} = C + C$$

$$= 2C$$

Hence, the required equivalent capacitance of the circuit = $2C$

16. Option (b) is correct.

Given,

E = Energy

G = Gravitational constant

$$\frac{E}{G} = ?$$

Dimensional formula of energy

$$(E) = [ML^2T^{-2}]$$

Dimensional formula of gravitational constant

$$= \frac{[MLT^{-2}L^2]}{[M^2]}$$

$$= [M^{-1}L^3T^{-2}]$$

$$\text{Hence, } \frac{E}{G} = \frac{[ML^2T^{-2}]}{[M^{-1}L^3T^{-2}]}$$

$$= [M^2L^{-1}T^0]$$

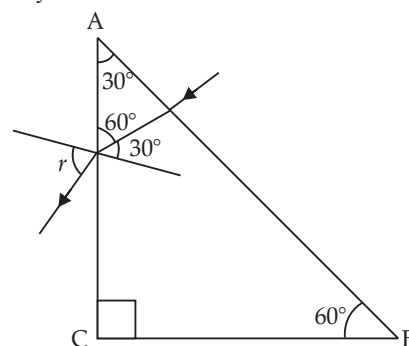
17. Option (d) is correct.

In the astronomical telescope, the focal length of the objective lens and aperture is quite large. Due to which the light gathering power of telescope used to increase, and the resolving power or the ability to observe two object distinctly gets increase.

By using larger diameters, the fainter objects can be observed so, the larger focal length of the objective lens contributes in the better quality and visibility of images.

18. Option (b) is correct.

As we know, when any ray of light incident normally on any surface, then it will not face any deviation.



So, angle of incident on the surface $AC(i) = 30^\circ$

Now, applying Snell's law,

$$\text{Refractive index } (\mu) = \frac{\sin i}{\sin r}$$

$$\Rightarrow \frac{\sin 30^\circ}{\sin r} = \sqrt{3}$$

$$\Rightarrow \sin r = \frac{1}{\sqrt{3}}$$

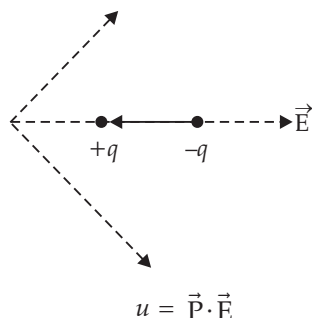
$$\Rightarrow \sin r = \frac{\sqrt{3}}{2} = \sin(60^\circ)$$

$$r = 60^\circ$$

Hence, the required deviation in the ray $\phi = 60^\circ$

19. Option (c) is correct.

Potential energy of electric dipole



Angle between electric field and electric dipole = 180°

$$u = -PE \cos 180^\circ$$

$$u = -(-PE)$$

$$\Rightarrow u = +PE$$

So, when we will move towards right, the strength of electric potential will decrease.
Net force and net torque on dipole will be zero.

20. Option (a) is correct.

Given,

Main scale reading = 0 mm

Circular scale reading = 52 divisions

1 mm on main scale = 100 divisions on circular scale

$$\text{Pitch } (p) = 1 \text{ mm}$$

$$n = 100$$

$$\text{Least count (LC)} = \frac{p}{n}$$

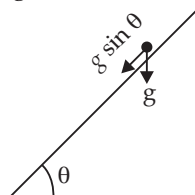
$$\text{LC} = \frac{1}{100} \\ = 0.01 \text{ mm}$$

Hence, diameter of the wire

$$= \text{MSR} + \text{CSR} \times \text{LC} \\ = 0 + (52 \times 0.01) \\ = 0.052 \text{ cm}$$

21. Option (c) is correct.

Let the angle of inclination be θ .



Distance covered in the duration n^{th} second is S_n

Let distance travelled during $(n+1)^{\text{th}}$ second = S_{n+1}

$$S_n = \frac{g}{2} \sin \theta (n^2) - \frac{g}{2} \sin \theta (n-1)^2$$

$$\Rightarrow S_n = \frac{g}{2} \sin \theta (n - n + 1)(n + n - 1)$$

$$\Rightarrow S_n = \frac{g}{2} \sin \theta (2n - 1) \quad \dots(i)$$

Distance travel during $(n+1)^{\text{th}}$ second

$$S_{n+1} = \frac{\sin \theta g}{2} (n+1)^2 - \frac{1}{2} g \sin \theta (n)^2$$

$$\Rightarrow S_{n+1} = \frac{1}{2} g \sin \theta (n+1 - n)(n+1 + n)$$

$$S_{n+1} = \frac{1}{2} g \sin \theta (1)(2n+1) \quad \dots(ii)$$

From equation (i) and (ii)

$$\Rightarrow \frac{S_n}{S_{n+1}} = \frac{\frac{1}{2} g \sin \theta (2n-1)}{\frac{1}{2} g \sin \theta (2n+1)}$$

$$\Rightarrow \frac{S_n}{S_{n+1}} = \frac{(2n-1)}{(2n+1)}$$

22. Option (c) is correct.

Given,

Focal length of the convex lens A,

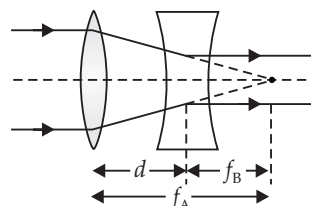
$$(f_A) = 20 \text{ cm}$$

Focal length of concave lens (B) is

$$(f_B) = 5 \text{ cm}$$

Distance between the lens = d

The emergent ray is parallel to X-axis.



As we know, when any parallel beam of light incident on convex lens, then it will meet at focal point.

If a beam of light gets parallel to principal axis after the refraction, then it means that object is at focal point of the concave lens.

Hence,

$$d = f_A - f_B \\ = 20 \text{ cm} - 5 \text{ cm} \\ = 15 \text{ cm}$$

Hence, the required distance between the lens is 15 cm.

23. Option (a) is correct.

Given,

Escape velocity from Earth's surface = $V_e = V$

Escape velocity from another planet = ?

Let the radius of earth = R

Radius of other planet = $4R$

Let, Escape velocity of other planet be V_o

Escape velocity on the surface of other planet

$$V_o = \sqrt{\frac{GM_o \times 2}{4R}}$$

Mass densities are same,

$$\sigma m_e = \sigma m_o$$

$$M_e = \sigma m_e \times V$$

$$= \sigma m_e \times \frac{4\pi}{3} R^3$$

$$M_o = \sigma m_o \times \frac{4\pi}{3} (4R)^3$$

$$= \sigma m_o \times \frac{4\pi}{3} R^3 \times 64$$

$$\begin{aligned} \text{Now, } \frac{V_e}{V_o} &= \frac{\sqrt{\frac{GM_e \times 2}{R}}}{\sqrt{\frac{GM_o \times 2}{4R}}} \\ &= \sqrt{\frac{\sigma m_e \times \frac{4\pi}{3} R^3 \times 1}{\sigma m_o \times \frac{4\pi}{3} R^3 \times 64}} \end{aligned}$$

$$\frac{V_e}{V_o} = \sqrt{\frac{1}{16}}$$

$$\Rightarrow V_o = 4 V_e$$

$$\Rightarrow V_o = 4 V$$

Hence, the required escape velocity on other planet's surface will be $4 V$.

24. Option (a) is correct.

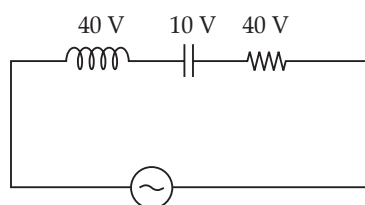
Given,

Inductance of the inductor = L

Capacitance of the capacitor = C

Resistance of the resistor = R

Potential difference = V



$$I_0 = 10\sqrt{2}A$$

$$V_{rms} = \sqrt{V_R^2 + (V_L - V_C)^2}$$

Now, substituting the values,

$$V_{rms} = \sqrt{40^2 + (40 - 10)^2}$$

$$= \sqrt{40^2 + 30^2}$$

$$= \sqrt{1600 + 900}$$

$$= \sqrt{2500} = 50 V$$

$$I_{rms} = \frac{I_0}{\sqrt{2}} = \frac{10\sqrt{2}}{\sqrt{2}}$$

$$= 10$$

$$V_{rms} = I_{rms} \times Z$$

$$\Rightarrow Z = \frac{V_{rms}}{I_{rms}}$$

$$\Rightarrow Z = \frac{50}{10}$$

$$Z = 5 \Omega$$

Hence, the required impedance of the circuit = 5Ω .

25. Option (a) is correct.

Given,

Stretching in spring (X) = 5 cm

Force (F) = 10 N

Mass (m) = 2 kg

The time period of oscillation of spring

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$\Rightarrow T = 2\pi\sqrt{\frac{2}{k}} \quad \dots(i)$$

From force balance equation,

$$kx = F$$

$$\Rightarrow k = \frac{F}{x}$$

$$\Rightarrow k = \frac{10}{5 \times 10^{-2}} \text{ N/m}$$

$$\Rightarrow k = 200 \text{ N/m}$$

Now put the value of k in equation (i)

$$\Rightarrow T = 2\pi\sqrt{\frac{2}{200}}$$

$$T = \frac{2\pi}{10}$$

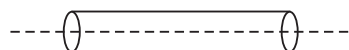
$$\Rightarrow T = \frac{2 \times 3.14}{10} \text{ s}$$

$$\Rightarrow T = \frac{6.28}{10} \text{ s}$$

$$\Rightarrow T = 0.628 \text{ s}$$

26. **Option (d) is correct.**

Radius of cable = R



Current = I

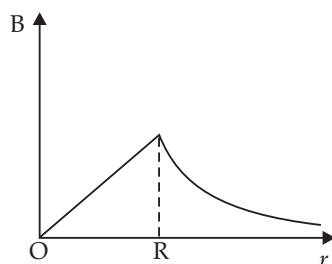
$B(r) = ?$

From ampere's circuital law

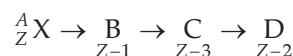
$$B = \frac{\mu_0 I}{2\pi R^2} \cdot r \quad (\text{For } r < R)$$

$$B = \frac{\mu_0 I}{2\pi r} \quad (\text{For } r \geq R)$$

Hence,



27. **Option (d) is correct.**

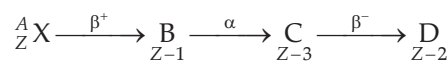


β^+ decreased atomic number by 1.

α decreased atomic number by 2.

β^- increased atomic number by 1.

Now,



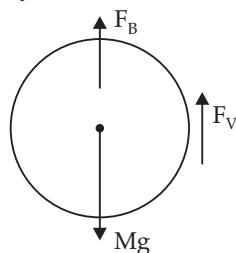
Hence, from the above reaction we can conclude that the possible decay particles in the sequence are β^+ , α , β^-

28. **Option (b) is correct.**

Given,

Mass of ball = M

Density of ball = d



$$\text{Density of glycerine} = \frac{d}{2}$$

Viscous force = ?

Now, applying force balance equation,

$$Mg = F_B + F_V$$

$$\Rightarrow dVg = \frac{d}{2} Vg + F_V$$

$$\Rightarrow F_V = dVg - \frac{d}{2} Vg$$

$$\Rightarrow F_V = \frac{d}{2} Vg$$

$$dV = M$$

$$\text{Hence, } F_V = \frac{M}{2} g$$

29. **Option (c) is correct.**

Given,

Height = 60 m

$$\text{Rate of water fall } \left(\frac{dm}{dt} \right) = 15 \text{ kg/s}$$

Losses due to fractional force = 10%

Power generated by the turbine = ?

Power = Rate of change of energy

$$p = \frac{mgh}{dt}$$

$$g = 10 \text{ m/s}^2$$

$$\Rightarrow p = 15 \times 10.0 \times 60$$

$$\Rightarrow p = 9000 \text{ J/s}$$

But, here fractional loss = 10%

So, generated power,

$$p_1 = 90 \times \frac{9000}{100} = 8100 \text{ W}$$

$$p_1 = 8.1 \text{ kW}$$

30. **Option (b) is correct.**

Capacitance of capacitor = C

AC source of voltage = V

$$V = V_0 \sin(\omega t)$$

Displacement current

$$I_d = C \cdot \frac{dV}{dt} \quad \dots(i)$$

$$\text{Now, } \frac{dV}{dt} = V_0 \omega \cos(\omega t)$$

$$\text{So, } I_d = V_0 \omega \cos(\omega t)$$

31. Option (d) is correct.

Given,

Wavelength of monochromatic light

$$(\lambda) = 600 \text{ nm}$$

Delivered power (p) = $3.3 \times 10^{-3} \text{ W}$

$$h = 6.6 \times 10^{-34} \text{ Js}$$

$$\text{Power of photon (P)} = \frac{E}{t}$$

$$\Rightarrow p = \frac{n hc}{t \lambda}$$

$$\text{Now, } \frac{n}{t} = \frac{p \times \lambda}{hc}$$

$$\Rightarrow \frac{n}{t} = \frac{600 \times 10^{-9} \times 3.3 \times 10^{-3} \text{ Wm}}{6.6 \times 10^{-34} \times 3 \times 10^8}$$

$$\Rightarrow \frac{n}{t} = \frac{1980 \times 10^{-12}}{19.8 \times 10^{-26}}$$

$$\Rightarrow \frac{n}{t} = 10^{16}$$

Hence, the total number of photon per second be 10^{16} .

32. Option (d) is correct.

Let the cross-sectional area be A ,Number of hole be n and drift velocity be V_d

$$i = ne A V_d \quad \dots(i)$$

$$V_d = \mu E$$

$$\text{For hole, } V_d = \mu_h E$$

$$\text{For electron } V_d = \mu_e E$$

$$\text{Hence, } \frac{I_n}{I_p} = \frac{neA\mu_e E}{neA\mu_h E}$$

$$\Rightarrow \frac{I_n}{I_p} = \frac{\mu_e}{\mu_h}$$

As we know, electron density > Hole density

$$\mu_e > \mu_h$$

$$\text{So, } I_n > I_p$$

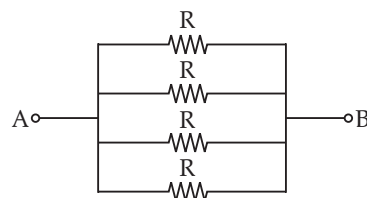
Hence, current in n -type > Current in p -type.

33. Option (a) is correct.

Given,

Let the length of wire be l and cross-section of the wire be A

Each wire is made-up of same material.

Resistance of wire = 0.25Ω 

$$\frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R}$$

$$\Rightarrow \frac{1}{R_{eq}} = \frac{4}{R}$$

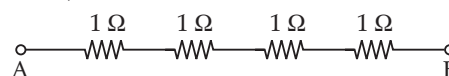
$$\Rightarrow R_{eq} = \frac{R}{4}$$

Now, substituting the values,

$$\Rightarrow 0.25 = \frac{R}{4}$$

$$\Rightarrow R = 0.25 \times 4 = 1 \Omega$$

Now, if each wire are connected in series.



$$R_{eq} = 1 + 1 + 1 + 1 \\ = 4 \Omega$$

34. Option (d) is correct.

Root mean square speed of gas molecule

$$V_{rms} = \sqrt{\frac{3RT}{M}}$$

$$\text{Pressure exerted by ideal gas} = \frac{1}{3} nm V^2$$

Average kinetic energy of molecule

$$= \frac{3}{2} k_B T$$

$$\text{Total internal energy of gas (u)} = \frac{n}{2} f RT$$

Hence,

$$n = 1$$

$$f = 5$$

$$u = \frac{5}{2} RT$$

$$(A) \rightarrow (Q)$$

$$(B) \rightarrow (P)$$

$$(C) \rightarrow (S)$$

$$(D) \rightarrow (R)$$

35. Option (d) is correct.

Given,

Wavelength of electromagnetic wave = λ Mass of photoelectron = m

de-broglie wavelength = λ_d

Now, applying photoelectric equation,

$$KE = \frac{-hc}{\lambda} + \phi_0 \quad \dots(i)$$

Kinetic energy be maximum,

If $\phi_0 = 0$

So, $KE = \frac{hc}{\lambda} = \frac{p^2}{2m}$

$$\Rightarrow p^2 = \frac{2hcm}{\lambda}$$

$$\Rightarrow p = \sqrt{\frac{2hcm}{\lambda}}$$

Now for de-broglie wavelength,

$$\lambda_d = \frac{h}{p} = \frac{h}{\sqrt{\frac{2hcm}{\lambda}}}$$

$$\lambda_d = \sqrt{\frac{h\lambda}{2mc}}$$

$$\lambda_d^2 = \frac{h\lambda}{2mc}$$

$$\lambda = \lambda_d^2 \left(\frac{2mc}{h} \right)$$

36. Option (b) is correct.

Given,

Mass of circular ring = M

Radius of Ring = R

Arc angle = 90°

$$I = kMR^2$$

$$K = ?$$

$$P = \frac{M}{2\pi R}$$

$$dM = p \cdot dl \quad \dots(i)$$

$$d\theta = \frac{\text{arc}}{\text{Radius}}$$

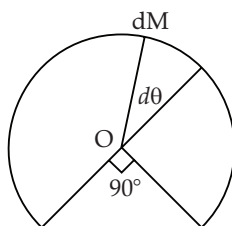
$$dl = R \cdot d\theta$$

So, $dM = \frac{M}{2\pi R} \cdot R \cdot d\theta \quad \dots(ii)$

$$dI = dM \cdot R^2$$

$$\Rightarrow dI = \frac{M}{2\pi R} \cdot R \cdot d\theta \cdot R^2$$

$$\Rightarrow dI = \frac{M}{2\pi} R^2 d\theta$$



Now, taking the integration of both side

$$\begin{aligned} \int_0^1 dI &= \int_0^{\frac{3\pi}{2}} \frac{M}{2\pi} R^2 \cdot d\theta \\ &= \frac{M}{2\pi} R^2 d\theta \Big|_0^{\frac{3\pi}{2}} \\ &= \frac{M}{2\pi} R^2 \frac{3 \times \pi}{2} \\ &= \frac{3}{4} MR^2 \end{aligned}$$

Hence, the value of $K = \frac{3}{4}$

37. Option (a) is correct.

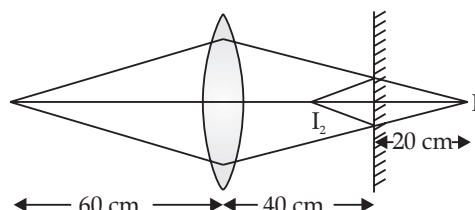
Given,

Distance, between object and convex lens

$$u = 60 \text{ cm}$$

Focal length of convex (f) = 30 cm

Distance between plane mirror and convex lens (d) = 40 cm



Applying lens law

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{30} + \frac{1}{(-60)}$$

$$\Rightarrow \frac{1}{v_1} = \frac{1}{60}$$

$$\Rightarrow v = 60 \text{ cm}$$

I_1 , will work as a object for plane mirror which will produce image at 20 cm in front of it.

Now, again this image will work as a object.

So, $u_1 = 20 \text{ cm}$

$$\frac{1}{v_1} - \frac{1}{u_1} = \frac{1}{f}$$

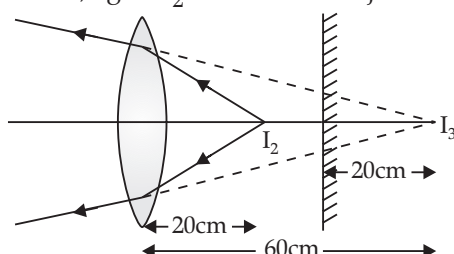
$$\Rightarrow \frac{1}{v_1} = \frac{1}{f} + \frac{1}{u_1}$$

$$\Rightarrow \frac{1}{v_1} = \frac{1}{30} + \frac{1}{(-20)} = \frac{2-3}{60}$$

$$\Rightarrow \frac{1}{v_1} = -\frac{1}{60}$$

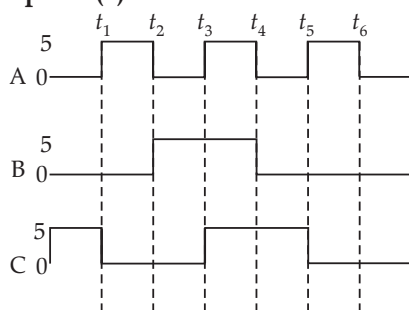
$$v_1 = -60 \text{ cm}$$

Now, again I_2 will work as object



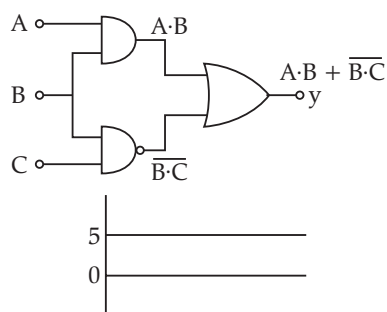
Hence, final virtual image will be formed at 60 cm from lens or we can say 20 cm behind mirror.

38. Option (c) is correct.



Now, from the above diagram creating the truth table

	A	B	C	A.B	$\overline{B \cdot C}$	$y = A \cdot B + \overline{B \cdot C}$
$0 \rightarrow t_1$	0	0	1	0	1	1
$t_1 \rightarrow t_2$	1	0	1	0	1	1
$t_2 \rightarrow t_3$	0	1	0	0	1	1
$t_3 \rightarrow t_4$	1	1	0	1	1	1
$t_4 \rightarrow t_5$	0	0	1	0	1	1
$t_5 \rightarrow t_6$	1	0	1	0	1	1
$t_6 \rightarrow t_7$	0	0	1	0	1	1



39. Option (a) is correct.

Given,

Initial speed of car (u) = 0

Acceleration of car (a) = 5 m/s²

$$t_1 = 4 \text{ s}$$

$$t_2 = 6 \text{ s}$$

Velocity gained by car at 4 s

$$V_x = u + at$$

$$\Rightarrow V_x = 0 + 5 \times 4$$

$$\Rightarrow V_x = 20 \text{ m/s}$$

So, ball also will acquire 20 m/s in horizontal direction.

Velocity gained by car at 6th s

$$V_y = u + gt$$

$$\Rightarrow V_y = 0 + 10 \times (6 - 4)$$

$$\Rightarrow V_y = 10 \times 2 = 20 \text{ m/s}$$

Hence, net velocity acquired by car

$$\text{or } V = \sqrt{V_x^2 + V_y^2}$$

$$\Rightarrow V = \sqrt{20^2 + 20^2}$$

$$\Rightarrow V = \sqrt{20^2 \times 2}$$

$$\Rightarrow V = 20\sqrt{2} \text{ m/s}$$

Acceleration on the ball = $g = 10 \text{ m/s}^2$

40. Option (a) is correct.

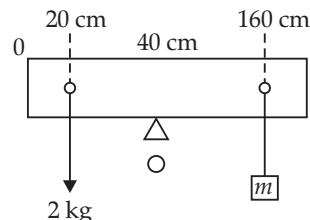
Given,

Length of rod = 200 cm

Mass = 500 g

$$= \frac{1}{2} \text{ kg}$$

Balance marks = 40 cm



Now, taking the torque, about point 'O'.

$$\Rightarrow 2 \times 20 \times 10^{-2} = m \times (160 - 40) \times 10^{-2} + 0.500 \times 60 \times 10^{-2}$$

$$\Rightarrow 40 \times 10^{-2} = 1.2m + 0.3$$

$$\Rightarrow 0.4 = 1.2m + 0.3$$

$$\Rightarrow 1.2m = 0.4 - 0.3$$

$$\Rightarrow 1.2m = 0.1$$

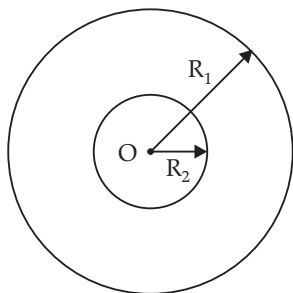
$$\Rightarrow m = \frac{1}{12} \text{ kg}$$

41. Option (a) is correct.

Given,

Radii of the circular loops are R_1 and R_2

$$R_1 \gg R_2$$

Mutual inductance = M Let current in coil = i 

Magnetic field at centre of the loop

$$B = \frac{\mu_0 i}{2\pi R_1}$$

So, magnetic flux through the inner coil

$$\phi = B \cdot A$$

$$\Rightarrow \phi = \frac{\mu_0 i}{2\pi R_1} \times \pi R_2^2 \quad \dots(i)$$

$$\text{But, magnetic flux} = M_i \quad \dots(ii)$$

So, from equation (i) and (ii)

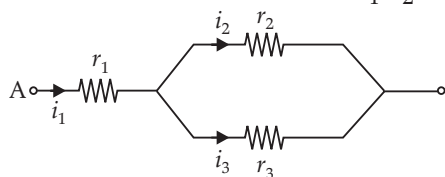
$$\frac{\mu_0 i}{2\pi R_1} \times \pi R_2^2 = M_i$$

$$\Rightarrow M = \frac{\mu_0 R_2^2}{2R_1}$$

$$\text{or } M \propto \frac{R_2^2}{R_1}$$

42. Option (c) is correct.

Given,

Resistance of the resistor are r_1 , r_2 , and r_3 .

$$i_1 = i_2 + i_3$$

$$V_2 = V_3 \quad (\because \text{Parallel connection})$$

$$\Rightarrow i_2 r_2 = i_3 r_3$$

$$\Rightarrow i_2 = i_3 \times \frac{r_3}{r_2} \quad \dots(ii)$$

Now, from equation (i) and (ii)

$$i_1 = i_3 \times \frac{r_3}{r_2} + i_3$$

$$\Rightarrow i_1 = \frac{i_3 r_3 + i_3 r_2}{r_2}$$

$$\Rightarrow i_1 = i_3 \frac{(r_3 + r_2)}{r_2}$$

$$\Rightarrow \frac{i_3}{i_1} = \frac{r_2}{r_2 + r_3}$$

43. Option (a) is correct.

Given,

Mass of particle = m Velocity of projection (v) = kV_e

$$(k < 1)$$

Now, applying the conservation of energy,

$$\frac{1}{2}mv^2 - \frac{GmM}{R} = -\frac{GmM}{(R+h)}$$

$$\Rightarrow \frac{v^2}{2} = \frac{GM}{R} - \frac{GM}{(R+h)}$$

$$\Rightarrow \frac{v^2}{2} = GM \left(\frac{1}{R} - \frac{1}{R+h} \right)$$

$$\Rightarrow \frac{v^2}{2} = GM \left(\frac{R+h-R}{R(R+h)} \right)$$

$$\Rightarrow \frac{v^2}{2} = GM \left(\frac{h}{R(R+h)} \right)$$

$$\Rightarrow \frac{(kV_e)^2}{2} = GM \left(\frac{h}{R(R+h)} \right) \quad \dots(i)$$

Since, escape velocity

$$V_e = \sqrt{\frac{2GM}{R}}$$

Now, substituting the value of V_e in equation (1)

$$\frac{k^2}{2} \left(\frac{2GM}{R} \right) = \frac{2GMh}{R(R+h)}$$

$$\Rightarrow k^2 = \frac{h}{R+h}$$

$$\Rightarrow \frac{R+h}{h} = \frac{1}{k^2}$$

$$\Rightarrow \frac{R}{h} + 1 = \frac{1}{k^2}$$

$$\Rightarrow \frac{R}{h} = \frac{1}{k^2} - 1 = \frac{1-k^2}{k^2}$$

$$\Rightarrow h = R \left(\frac{k^2}{1-k^2} \right)$$

44. Option (b) is correct.

Given,

Length of conducting wire = $12a$ Resistance of the wire = R Let current in each shape = I (i) Side of equilateral triangle = a (ii) Side of square coil = a

Magnetic dipole for triangular coil

$$M = NIA$$

Number of turns in triangular coil

$$N = \frac{12a}{3a} = 4$$

$$I = \frac{V}{R}$$

$$A = \frac{\sqrt{3}}{4} a^2$$

$$\text{So, } M_1 = 4 \times \frac{\sqrt{3}}{4} a^2 \times \frac{V}{R}$$

$$M_1 = \sqrt{3} a^2 I$$

Magnetic dipole for square coil

$$M_2 = N_2 I A_2$$

$$= \left(\frac{12a}{4a} \right) \times \frac{V}{R} \times a^2$$

$$M_2 = 3Ia^2$$

Hence, the required magnetic dipole moments are

$$M_1 = \sqrt{3} Ia^2$$

and

$$M_2 = 3Ia^2$$

45. Option (b) is correct.

Given,

The potential of AC mains supply

$$V = 220 \text{ V}$$

Operational rating,

$$V_0 = 11 \text{ V}$$

$$P_0 = 44 \text{ W}$$

As there is no power loss, so

Input power = Output power

$$\Rightarrow V_p I_p = V_s I_s = P_o$$

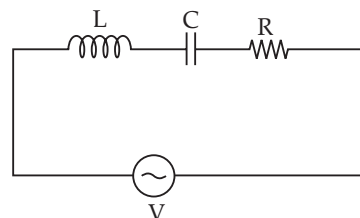
$$\Rightarrow 220 \times I_p = 44$$

$$\Rightarrow I_p = \frac{44}{220} \text{ A}$$

$$\Rightarrow I_p = \frac{2}{10} = 0.2 \text{ A}$$

46. Option (d) is correct.

Given,

Inductance of inductor (L) = 5.0 H Capacitance of capacitor (C) = $80 \mu\text{F}$ Resistance of resistor (R) = 40Ω Terminal voltage (V) = 230 V 

Resonance frequency

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{5.0 \times 80 \times 10^{-6}}}$$

$$\Rightarrow \omega_0 = \frac{1}{20 \times 10^{-3}} \text{ rad/s}$$

$$\Rightarrow \omega_0 = 50 \text{ rad/s}$$

As per question,

Half power frequencies are given as

$$\omega = \omega_0 \pm \frac{R}{2L}$$

$$\omega_L = 50 - \frac{40}{2 \times 5}$$

$$\Rightarrow \omega_L = (50 - 4) \text{ rad/s}$$

$$\Rightarrow \omega_L = 46 \text{ rad/s}$$

$$\omega_H = 50 + \frac{40}{2 \times 5}$$

$$= 50 + 40$$

$$= 54 \text{ rad/s}$$

47. Option (a) is correct.

Given,

Total number of drops = 27 Potential of drops = 200 V

When all the drops will combine, then the new potential of the drop = ?

Total volume always be conserve.

$$\frac{4}{3} \pi R^3 = 27 \times \frac{4}{3} \pi r^3$$

$$R = 3r$$

For smaller drop

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

$$\Rightarrow 200 = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \quad \dots(i)$$

Let charge on one drop = q

For larger drop,

$$V_1 = \frac{1}{4\pi\epsilon_0} \frac{Q}{R} \quad \dots(ii)$$

and

$$Q = 27q$$

$$\Rightarrow V_1 = \frac{1}{4\pi\epsilon_0} \frac{27q}{3r}$$

$$\Rightarrow V_1 = \frac{27}{3} \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r} \right)$$

$$\Rightarrow V_1 = 9 \times 200 \text{ V}$$

$$\Rightarrow V_1 = 1800 \text{ V}$$

Note: Data given in the question is incorrect. Due to which answer is not matching with option.

If we will take $V = 220 \text{ V}$

$$\text{So, } V_1 = 9 \times 220 \text{ V} \\ = 1980 \text{ V}$$

48. Option (c) is correct.

Given,

Mass of ball (m) = 0.15 kg

Height from which ball dropped = 10 m

Gravitational acceleration (g) = 10 m/s²

Magnitude of impulse imparted to the ball = ?

Velocity, at which ball is going to hit the ground (v) = ?

$$v^2 = u^2 + 2gh$$

$$\Rightarrow v = \sqrt{0 + 2 \times 10 \times 10}$$

$$v = 10\sqrt{2} \text{ m/s}$$

As it is given that, ball rebounds with same speed.

So, Magnitude of imparted impulse

$$= P_f - P_i \\ = m\vec{v}(\hat{j}) - m\vec{v}(-\hat{j})$$

$$|L| = 2 \times 0.15 \times 10\sqrt{2} \text{ kg m/s}$$

$$\Rightarrow |L| = 0.3 \times 10\sqrt{3} \text{ kg m/s}$$

$$\Rightarrow |L| = 30\sqrt{3} \text{ kg m/s}$$

$$\Rightarrow |L| = 4.2$$

$$\Rightarrow |L| = 4.2 \text{ kg m/s}$$

49. Option (c) is correct.

Given,

$$\vec{F} = q(\vec{V} \times \vec{B})$$

$$\Rightarrow \vec{F} = q\vec{V} \times (\vec{B}\hat{i} + \vec{B}\hat{j} + \vec{B}\hat{k})$$

$$\vec{F} = 2\hat{i} + 4\hat{j} + 6\hat{k}$$

$$q = 1$$

$$\vec{F} = 4\hat{i} - 20\hat{j} + 12\hat{k}$$

$$\text{Now } \vec{V} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 4 & 6 \\ B & B & B_0 \end{vmatrix}$$

$$= \hat{i}(4B_0 - 6B) - \hat{j}(2B_0 - 6B)$$

$$\text{Now, } + \hat{k}(2B - 4B)$$

$$4\hat{i} - 20\hat{j} + 12\hat{k} = \begin{bmatrix} (4B_0 - 6B)\hat{i} - \hat{j}(2B_0 - 6B) \\ + (-2B)\hat{k} \end{bmatrix}$$

Comparing both side,

$$4 = 4B_0 - 6B \quad \dots(i)$$

$$-20 = (2B_0 - 6B) \quad \dots(ii)$$

$$12 = -2B \quad \dots(iii)$$

Solving equation (i), (ii) and (iii)

$$B = -6$$

$$-20 = -B_0 - 36$$

$$\Rightarrow B_0 = \frac{-36 + 20}{2} = -8$$

$$\text{Hence, } \vec{B} = -6\hat{i} - 6\hat{j} - 8\hat{k}$$

50. Option (a) is correct.

Given,

Radius of circle = R

Time period of revolution = T

Now, again particle were projected with same speed = v

Angle of projection = θ

Maximum height = 4R

$$\theta = ?$$

For circular motion,

$$V = \frac{2\pi R}{T}$$

Now, for projectile motion,

$$H_{\max} = \frac{v^2 \sin^2 \theta}{2g}$$

$$\Rightarrow 4R = \left(2 \frac{\pi R}{T} \right)^2 \frac{\sin^2 \theta}{2g}$$

$$\Rightarrow 4R = \frac{4\pi^2 R^2}{T^2} \frac{\sin^2 \theta}{2g}$$

$$\Rightarrow \sin^2 \theta = \frac{2gT^2}{\pi^2 R}$$

$$\sin \theta = \sqrt{\frac{2gT^2}{\pi^2 R}}$$

$$\Rightarrow \theta = \sin^{-1} \left(\sqrt{\frac{2gT^2}{\pi^2 R}} \right)^{1/2}$$

CHEMISTRY

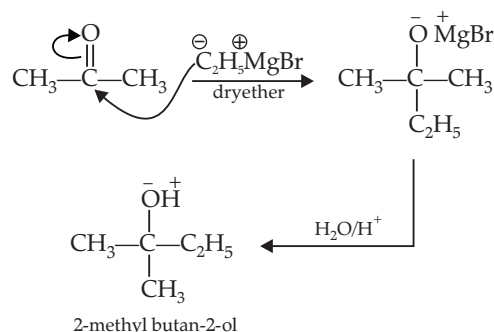
51. Option (c) is correct.

From Kohlrausch's law, Λ_m^0 of weak acid can be calculated.

$$\begin{aligned}\Lambda_m^0(\text{CH}_3\text{COOH}) &= \Lambda_m^0(\text{CH}_3\text{COONa}) \\ &\quad + \Lambda_m^0(\text{HCl}) - \Lambda_m^0(\text{NaCl}) \\ &= 91.0 \text{ S cm}^2 \text{ mol}^{-1} + 426.16 \\ &\quad \text{S cm}^2 \text{ mol}^{-1} - 126.45 \\ &\quad \text{S cm}^2 \text{ mol}^{-1} \\ &= 390.71 \text{ S cm}^2 \text{ mol}^{-1}\end{aligned}$$

52. Option (a) is correct.

$\text{C}_2\text{H}_5\text{MgBr}$ is a Grignard reagent in which C_2H_5 acts as a nucleophile and gives nucleophilic addition reaction with ketone to form tertiary alcohol as a major product.



53. Option (b) is correct.

First, we find molarity of the solution from the formula:

$$M = \frac{\text{weight of solute}}{\text{molar mass}} \times \frac{1}{V_{(L)}}$$

$$\text{For glucose} \Rightarrow M = \frac{10\text{gm}}{180\text{gm/mol}} \times \frac{1}{0.250\text{L}} = 0.22 \text{ M}$$

$$\text{For urea} \Rightarrow M = \frac{10\text{gm}}{60\text{gm/mol}} \times \frac{1}{0.250\text{L}} = 0.66 \text{ M}$$

$$\text{For sucrose} \Rightarrow M = \frac{10\text{gm}}{342\text{gm/mol}} \times \frac{1}{0.250} = 0.12 \text{ M}$$

From the formula $\pi = CRT$ here C is molar concentration of the solution.

At a given temperature $\pi \propto C$, higher the concentration, higher will be osmotic pressure.

So, from molarity of the given solution, osmotic pressure of the solution will be $P_2 > P_1 > P_3$

54. Option (d) is correct.

Tyndall effect is the property of colloidal solution in which scattering of light takes place through colloidal particles. Here urea, NaCl and glucose solution are true solution which does not exhibit Tyndall Effect, therefore, correct answer is starch solution.

55. Option (c) is correct.

Blast furnace is the furnace where heating of metal oxide takes place at higher temperature to obtain metal.

Maximum temperature obtained in blast furnace is upto 2200 K. which is present in combustion zone.

56. Option (a) is correct.

In body centred unit cell, atom are arranged in corner & body centre. Body centred unit cell is present in 3 crystal system i.e., cubic tetragonal and orthorhombic crystal system.

57. Option (d) is correct.

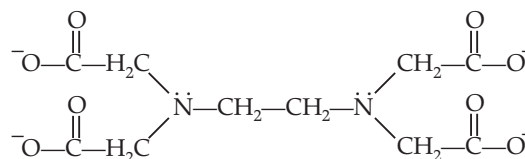
Element	Mass per-centage (w/w)	Number of mole	Mole Ratio
C	78%	$78/12 = 6.5$	$6.5/6.5 = 1$
H	22%	$22/1 = 22$	$22/6.5 = 3.38 \approx 3$

On the basis of above calculation, the empirical formula is CH_3 .

58. Option (d) is correct.

Due to Lanthanoid contraction, the atomic radii of Zr ($z = 40$) and Hf ($z = 72$) are nearly same. Because of poor screening effect by $4f$ electrons, which lead to the lanthanoid contraction. This is the reason of Zn and Hf have almost equal atomic size.

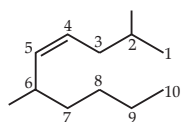
59. Option (b) is correct.



EDTA ion is a hexadentate ligand having four O-donor atoms and two N-donor atoms.

Nitrogen atom contains lone pair of electrons while oxygen carries negative charge which can be donated to metal or metal ion.

60. **Option (b) is correct.**



2,6 - dimethyl - dec - 4 - ene

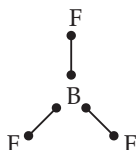
From IUPAC name it is clear that, main chain contains 10 carbon atoms and a double bond at position 4. Similarly it contains 2 methyl substituent at carbon 2 and 6.

61. **Option (b) is correct.**

Vitamin	Deficiency diseases
B ₂ (Riboflavin)	Burning sensation of the skin, Digestive disorder and cheilosis.
B ₁₂	Pernicious anaemia (RBC deficient in haemoglobin)
B ₆ (Pyridoxine)	Convulsions
B ₁ (Thiamine)	Beri beri (loss of appetite, retarded growth)

62. **Option (d) is correct.**

B(z = 5)	1s ²	2s ²	2p ¹
Ground state	$\boxed{\uparrow\downarrow}$	$\boxed{1}$	$\boxed{}\boxed{}\boxed{}$
Excited state	$\boxed{1}$	$\boxed{1}\boxed{1}\boxed{1}$	$\boxed{}\boxed{}\boxed{}$
		$\underbrace{\hspace{1.5cm}}_{sp^2}$	

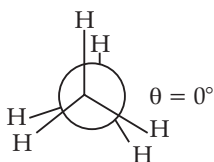


Here, Boron atom is sp^2 hybridized and it contains 6 electrons around the central atom after the formation of bond with boron atom.

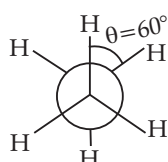
63. **Option (a) is correct.**

Conformers are obtained by the free rotation of carbon-carbon single bond. Ethane has two conformer (1) Eclipsed (2) Staggered.

In eclipsed structure, dihedral angle is 0° and it is least stable while in staggered structure, dihedral angle is 60° and it is stable.



Eclipsed ($\theta = 0^\circ$)



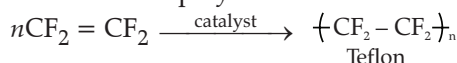
Staggered ($\theta = 60^\circ$)

64. **Option (a) is correct.**

All given options belong to the chloride of alkaline earth metals, all chloride of alkaline earth metals are ionic in nature except Be because due to high polarizability of small size Be^{2+} ion, it forms covalent halide which is soluble in organic solvent.

65. **Option (b) is correct.**

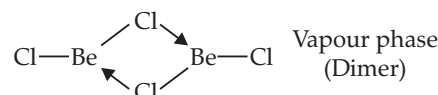
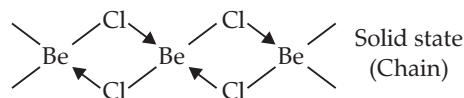
Addition polymers are mainly formed by unsaturated compound and its derivatives. Here, Teflon is prepared by addition polymerisation reaction whereas Dacron, Novolac and Nylon - 66 are prepared by condensation polymerisation.



Addition polymer

66. **Option (b) is correct.**

$BeCl_2$ is an electron-deficient compound which through coordinate bond tries to complete its octet. $BeCl_2$ is solid state and in vapour phase forms chain and dimer respectively to complete its octet.



$BeCl_2$ exist as $(BeCl_2)_n$ polymeric chain in solid form, while $BeCl_2$ exist as dimer $(BeCl_2)_2$ in vapour phase.

67. **Option (d) is correct.**

Distillation is the process which is generally used for the purification of metals having low boiling point. E.g. Zn, Hg etc.

68. **Option (c) is correct.**

$$\text{Here } q_V = C_V \Delta T = \Delta U$$

$$q_P = C_P \Delta T = \Delta H$$

ΔH = Enthalpy change

ΔU = Internal energy change

$$\text{We know } \Delta H = \Delta U + \Delta(PV)$$

$$= \Delta U + \Delta(RT)$$

$$\Delta H = \Delta U + R \Delta T$$

$$\text{Or } C_P \Delta T = C_V \Delta T + R \Delta T$$

$$C_P = C_V + R$$

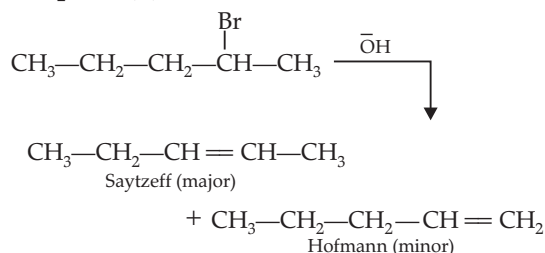
$$C_P - C_V = R$$

69. Option (b) is correct.

Out of three isotopes of Hydrogen : Protium, Deuterium and Tritium, only Tritium is radioactive and due to radioactivity, it emits β particles.

Tritium emits electrons through beta decay: It is a beta emitting radioactive isotope of hydrogen.

70. Option (b) is correct.



Saytzeff product is that product which contains highly substituted alkene while in Hofmann product less substituted alkene is formed during dehydrohalogenation of 2-bromopentane.

71. Option (b) is correct.

$$\text{Energy} = \frac{hc}{\lambda} = h\nu$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$\nu = 1368 \text{ kHz} = 1368 \times 10^3 \text{ s}^{-1}$$

$$\lambda = \frac{c}{\nu} = \frac{3 \times 10^8 \text{ m/s}}{1368 \times 10^3 \text{ s}^{-1}} = 219.3 \text{ m}$$

72. Option (a) is correct.

A hexagonal primitive unit cell contains 6 atoms in the crystal structure i.e., $n = 6$ we know that, for hcp,

The number of octahedral voids will be equal to the number of atoms in the crystal structure.

$$\text{No. of octahedral voids} = 6$$

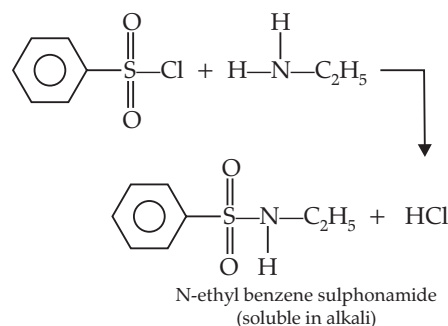
The number of tetrahedral voids will be equal to two times the number of atoms in the crystal structure.

$$\text{No. of tetrahedral voids} = 2 \times 6 = 12$$

73. Option (d) is correct.

Hinsberg reagent is Benzene Sulphonyl chloride $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$ which is used to distinguish between primary, secondary and tertiary amine.

Primary amines react with Hinsberg reagent to form N-ethyl benzene Sulphonamide which is soluble in alkali, while 2° and 3° amines do not get dissolve in alkali.



74. Option (d) is correct.

Dimethyl ammonium acetate

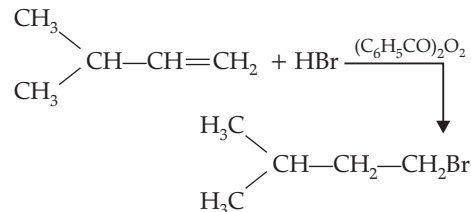
$((\text{CH}_3)_2\text{NH COOCH}_3)$ is a salt of weak acid and weak base whose pH can be calculated as

$$\text{pH} = \frac{1}{2}(\text{p}k_w + \text{p}k_a - \text{p}k_b)$$

$$= \frac{1}{2}(14 + 4.77 - 3.27)$$

$$\text{pH} = 7.75$$

75. Option (b) is correct.

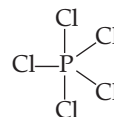


Here Anti-Markonikov's rule or peroxide effect takes place where Br^- attach to that carbon of alkene which contains higher number of C-atoms.

76. Option (b) is correct.

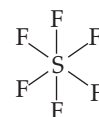
(a) PCl_5 – (iv) Trigonal bipyramidal

sp^3d hybridised



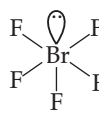
(b) SF_6 – (iii) Octahedral

sp^3d^2 hybridised



(c) BrF_5 – (i) Square pyramidal

sp^3d^2 hybridised



(d) BF_3 – (ii) Trigonal planar

sp^2 hybridised



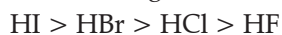
77. Option (c) is correct.

Most of the trivalent Lanthanoid ions are coloured in both their solid state as well as in aqueous solution. Option (c) is incorrect statement rest option are correct.

78. Option (b) is correct.

In a periodic table, among haloacids of halogen acidic nature increases when we move from top to bottom because the bond strength of $H-X$ decrease due to increase in bond length of HX . Both statement (I) and (II) are correct.

Acidic strength of HX

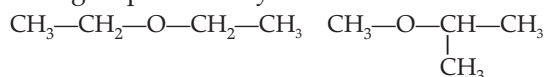
**79. Option (c) is correct.**

Noble gases have weak dispersion force or London dispersion force due to which the boiling point and melting point of noble gas are low. So statement (c) is incorrect about noble gas.

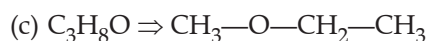
80. Option (a) is correct.

Metamerism is a type of position isomerism which is shown by divalent species, like ether, thioether, ketone, 2° amine etc.

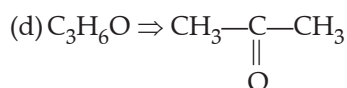
(a) $C_4H_{10}O$ will have different alkyl group attached with polyvalent functional group that's why show metamerism.



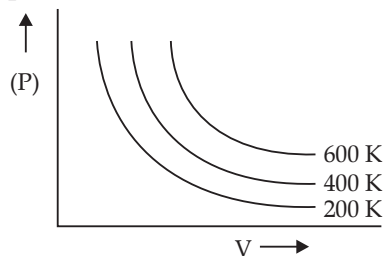
(b) No, polyvalent functional group in C_5H_{12} so can not show metamerism.



Only one arrangement possible so cannot show metamerism.



Only one arrangement possible, so cannot show metamerism.

81. Option (a) is correct.

According to Boyle's law at constant temperature $P \propto \frac{1}{V}$ or $PV = K$

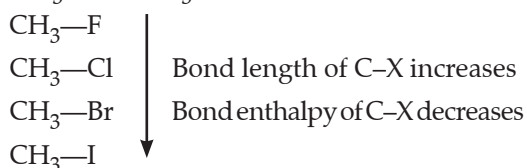
The value of PV will increase with temperature.

82. Option (c) is correct.

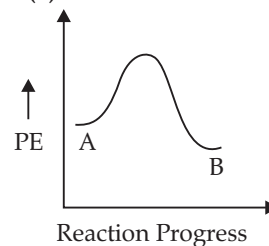
In a periodic table, on moving down the group, size of halogen atom increases from

F to I. Hence bond lengths from $C-F$ to $C-I$ increases.

$\therefore$ Bond dissociation enthalpy decreases from CH_3-F to CH_3-I decreases.

**83. Option (c) is correct.**

Both Aspirin and Paracetamol belong to non-narcotic analgesic while Morphine and Heroin belong to narcotic analgesic. Both statement (I) and (II) are false.

84. Option (c) is correct.

$$\begin{aligned} \text{From } \Delta H_{\text{rxn}} &= (E_a)_f - (E_a)_b \\ -4.2 &= 9.6 - (E_a)_b \\ (E_a)_b &= 9.6 + 4.2 = 13.8 \text{ kJ/mol.} \end{aligned}$$

Here $(E_a)_b$ is highly positive and the reaction is exothermic in nature. So possible graph is (c).

85. Option (c) is correct.

$Cr_2O_3 + Al \rightarrow Al_2O_3 + 2Cr$ represent metal displacement reaction, Both reaction (a) and (b) are decomposition reaction while reaction (d) does not represent metal displacement reaction.

86. Option (b) is correct.

From Arrhenius equation $K = Ae^{-E_a/RT}$
on taking $\log_{10}$ both side

$$\log K = \log A - \frac{E_a}{R} \times \frac{1}{2.303 T}$$

$$\text{Slope of } \log K \text{ v/s } \frac{1}{T} \Rightarrow m = \frac{-E_a}{R}$$

$$-5 \times 10^3 = \frac{-E_a}{R}$$

$$E_a = 5 \times 10^3 \times 8.314 \text{ J/mol}$$

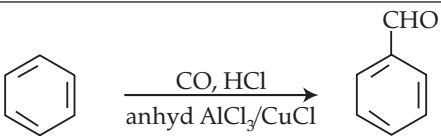
$$= 41.57 \times 10^3 \text{ J/mol}$$

$$E_a = 41.5 \text{ kJ/mol}$$

87. Option (d) is correct.

(a)	$2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$	(iv)	Tropospheric pollution
(b)	$\text{HOCl}(\text{g}) \xrightarrow{h\nu} \text{Cl} + \dot{\text{O}}\text{H}$	(iii)	Ozone depletion
(c)	$\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{H}_2\text{O} + \text{CO}_2$	(i)	Acid rain
(d)	$\text{NO}_2(\text{g}) \xrightarrow{h\nu} \text{NO}(\text{g}) + \text{O}(\text{g})$	(ii)	Smog

88. Option (a) is correct.

(a)		(ii)	Gattermann Koch reaction
(b)	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 + \text{NaOX} \longrightarrow \text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CX}_3$	(iii)	Haloform reaction
(c)	$\text{R}-\text{CH}_2-\text{OH} + \text{R}'\text{COOH} \xrightarrow[\text{H}_2\text{SO}_4]{\text{Conc.}} \text{R}-\text{CH}_2-\text{O}-\overset{\text{O}}{\parallel}{\text{C}}-\text{R}'$	(iv)	Esterification
(d)	$\text{R}-\text{CH}_2\text{COOH} \xrightarrow[\text{H}_2\text{O}]{\text{X}_2/\text{Red P}} \text{R}-\underset{\text{X}}{\text{CH}}-\text{COOH}$	(i)	Hell Volhard Zelinsky reaction

89. Option (a) is correct.

The species which contain same number of electrons are called Isoelectronic species.

Species	Number of e^-
Fe^{2+}	$26 - 2 = 24$
Mn^{2+}	$25 - 2 = 23$
O^{2-}	$8 + 2 = 10$
F^-	$9 + 1 = 10$
Na^+	$11 - 1 = 10$
Mg^{2+}	$12 - 2 = 10$
Fe^{3+}	$26 - 3 = 23$

Fe^{2+} and Mn^{2+} are not isoelectronic pair.

90. Option (d) is correct.

Given, $\Lambda_m = 20 \text{ S cm}^2 \text{ mol}^{-1}$

$$\begin{aligned}\Lambda_m^\circ \text{CH}_3\text{OOH} &= \Lambda^\circ \text{CH}_3\text{COO}^- + \Lambda^\circ \text{H}^+ \\ &= 50 + 350 \\ &= 400 \text{ S cm}^2 \text{ mol}^{-1}\end{aligned}$$

From $\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} = \frac{20}{400} = \frac{1}{20}$

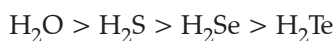
As $K_a = \frac{C\alpha^2}{1-\alpha}$

(Since α is very small $\therefore 1 - \alpha \approx 1$)

$$\begin{aligned}K_a &= C\alpha^2 = 7 \times 10^{-3} \times (1/20)^2 \\ &= 1.75 \times 10^{-5} \text{ mol/L.}\end{aligned}$$

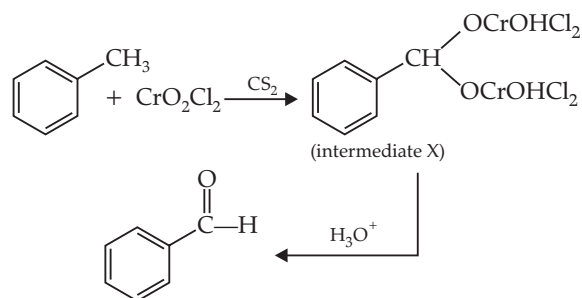
91. Option (c) is correct.

In a modern periodic table, down the group the acidic strength of hydric acid increases because the bond length of $\text{H}-\text{X}$ ($\text{X} = \text{O}, \text{S}, \text{Se}, \text{Te}$) increases due to which bond energy decreases as a result K_a value will increase and $\text{p}K_a$ will decrease. So correct order of $\text{p}K_a$ value will be



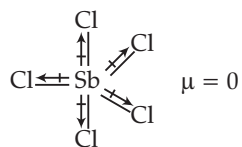
92. Option (b) is correct.

The given reaction is Etard's reaction which is used to prepare aromatic aldehyde.



93. Option (d) is correct.

The molecule which has dipole moment value equal to zero ($\mu = 0$) are considered as Non-polar molecule. In SbCl_5 , net vector summation of bond moments will be zero, so it is a non-polar molecule.

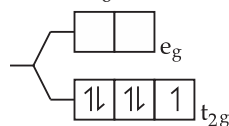
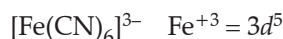


sp^3d , Trigonal bipyramidal

94. Option (a) is correct.

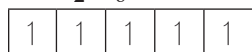
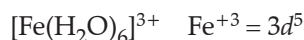
The magnetic moment of the given complexes can be calculated by

$\mu = \sqrt{n(n+2)}$ where n = number of unpaired electrons.



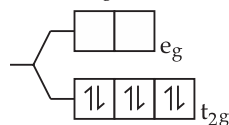
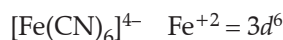
Unpaired electron = 1

$$\mu = \sqrt{n(n+2)} = \sqrt{1(1+2)} = \sqrt{3} = 1.7 \text{ BM}$$

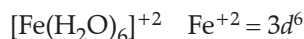


Unpaired electrons = 5

$$\mu = \sqrt{5(5+2)} = \sqrt{5 \times 7} = \sqrt{35} = 5.9 \text{ BM}$$



Unpaired electron = 0, $\mu = 0 \text{ BM}$



Unpaired electron = 4

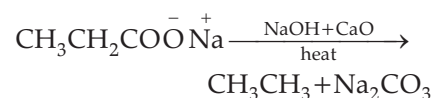
$$\mu = 4.9 \text{ BM}$$

Complex	Number of unpaired e^-	Nature of ligand	$\mu(\text{BM})$
(a) $[\text{Fe}(\text{CN})_6]^{3-}$	1	Strong	1.73

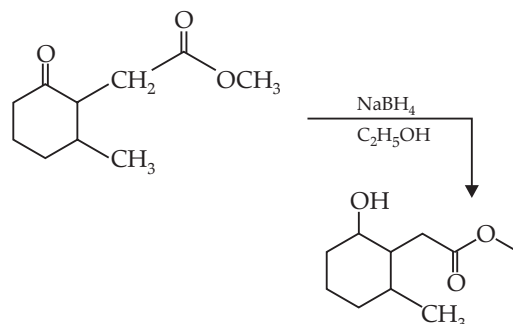
(b) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	5	Weak	5.92
(c) $[\text{Fe}(\text{CN})_6]^{4-}$	0	Strong	0
(d) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	4	Weak	4.90

95. Option (d) is correct.

The given reaction is an example of soda lime decarboxylation reaction where NaOH and CaO are mixed in the ratio of 3 : 1 to convert sodium salt of carboxylic acid into alkane. This reaction is used to prepare both symmetrical as well as asymmetrical alkane.

**96. Option (a) is correct.**

NaBH_4 is a weak reducing agent which can reduce carbonyl group into alcohol but does not reduce esters.

**97. Option (d) is correct.**

From Raoult's law

$$P_{\text{solution}} = P_A^\circ X_A + P_B^\circ X_B$$

Here $P_A^\circ = P_{\text{C}_6\text{H}_6} = 280 \text{ mm Hg}$

$$P_B^\circ = P_{\text{octane}} = 420 \text{ mm Hg}$$

$$n_{\text{C}_6\text{H}_6} = 3, n_{\text{octane}} = 2$$

$$\text{Total mole} = 3 + 2 = 5$$

$$X_{\text{C}_6\text{H}_6} = \frac{3}{5}, X_{\text{octane}} = \frac{2}{5}$$

$$P_{(\text{solution})} = 280 \times \frac{3}{5} + 420 \times \frac{2}{5}$$

$$= 168 + 168 = 336 \text{ mm Hg}$$

98. Option (d) is correct.

For spontaneous process, $\Delta S_{\text{total}} > 0$ and since irreversible process is always spontaneous

$$\therefore \Delta S_{\text{total}} > 0$$

For isothermal process, $\Delta T = 0$. So, $\Delta U = 0$

99. Option (d) is correct.

$$\left[\begin{aligned} n_{O_2} &= \frac{\text{Mass of } O_2}{\text{Molar mass}} = \frac{4}{32} = \frac{1}{8} \text{ mol} \\ n_{H_2} &= \frac{\text{Mass of } H_2}{\text{Molar mass}} = \frac{2}{2} = 1 \text{ mol} \end{aligned} \right]$$

$$n_{\text{total}} = n_{O_2} + n_{H_2} = \frac{1}{8} + 1 = \frac{9}{8} \text{ mol}$$

$$P_t V = n_t RT - \text{ideal gas equation}$$

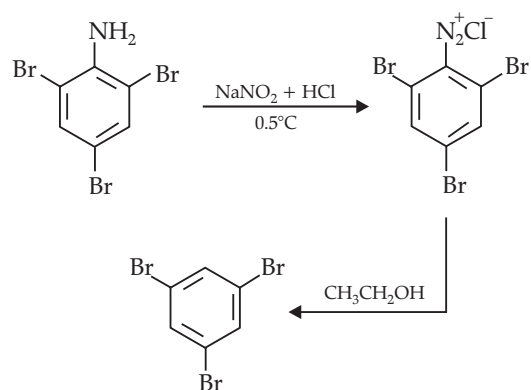
$$T = 273 \text{ K}$$

$$V = 1.0 \text{ L}$$

$$P_t = \frac{\frac{9}{8} \text{ mol} \times 0.082 \text{ L} \cdot \text{atm} \times 273 \text{ K}}{1 \text{ L} \times \text{K} \cdot \text{mol}}$$

$$= 25.18 \text{ atm.}$$

100. Option (c) is correct.



Here, replacement of $-N_2^+Cl^-$ takes place with CH_3CH_2OH . So correct answer is CH_3CH_2OH .

BOTANY

101. Option (d) is correct.

The Punnett square helps to predict the output of a genetic cross easily. Reginald Punnett proposed this method. It is a graphical representation used to calculate the probable genotypes of offspring in genetic analysis. Mendel uses it to understand the monohybrid cross between pure-tall and dwarf breeding plants. It also helps to understand the gamete produced by parents, the formation of the zygote, the genotype of F_1 and F_2 plant.

102. Option (a) is correct.

Some liverworts reproduce asexually by the formation of special structures called gemmae. Gemmae are asexual buds with green and multicellular structures developed in a small receptacle called gemma cups. Gemma cups are located on the thallus. When it detached from the parent plant. It germinates to produce a new individual plant.

103. Option (a) is correct.

Sewall Wright proposed the concept of genetic drift. It is a changes in the frequency of alleles in a population because of random mutations. It occurs by chance in the gametes due to crossing over and results in genetic recombination. When such gametes unite to form zygotes, they differ from generation to generation. The genetic drift has two consequences called the bottleneck effect and the founder's effect. The migration of a small population is involved in the Founder's effect, when a small population is isolated from a large population due to migration and form a colony in a new location with a new gene pool.

104. Option (d) is correct.

The ability of a plant to show different pathways in response to changes in the environment or different phases leads to form different types of structures. This is called for example, heterophylly of cotton, coriander, and larkspur. The shape of the leaves of the juvenile plant differs from the shape of the mature plant.

105. Option (b) is correct.

Meristematic tissue consists of cells that are active and continuously divides. Simple tissue consists of cells similar in structure and function. Vascular tissues are complex permanent tissue and have different types of cells and sclereids are sclerenchymatous cells and they are dead cells. They have thick cells wall and narrow lumen.

106. Option (b) is correct.

Transfer of pollen grains from the anther to the stigma of the same flower is called autogamy. Transfer of pollen grains from the anther to the stigma of another flower of the same species is called geitonogamy. Transfer of pollen grain from the anther of one flower to the stigma of the different plant is called xenogamy. This type of pollination gives a chance to bring a genetically different pollen grain to reach the stigma.

107. Option (b) is correct.

Polymerase chain reaction includes three sequential steps called denaturation, annealing, and extension.

At 94°C , the two complementary strands are separated. This is called denaturation.

The process of binding primer to single strand template DNA is called annealing. The extension of DNA by the addition of nucleotides at -3'OH end of the primer based on template DNA in the presence of Taq DNA polymerase is called extension.

108. **Option (c) is correct.**

Lenticels are helpful for gaseous exchange. Phellogen is also called cork cambium. Phellem, white on the inner side differentiates into secondary cortex or phelloderm. The cell wall of the cork consists of suberin and is impervious to water.

109. **Option (b) is correct.**

As per Gause's theory of the Principle of Competitive Exclusion, when two closely related species compete for the same resources they cannot co-exist, and the weaker one is eliminated. But the recent studies show that they do not rule out the presence of interspecific competition and lead to co-existence. This type of interaction is called resource partitioning. The two species will avoid competition by choosing a different time for feeding or foraging patterns.

110. **Option (c) is correct.**

The isolated DNA fragments cannot be seen in visible light without staining. Hence, the electrophoresed DNA fragments are stained with ethidium bromide. When such DNA strands are exposed to ultraviolet rays they can be seen in bright orange colour. So, DNA strands on staining with ethidium bromide appear as bright orange bands.

111. **Option (d) is correct.**

Algae is useful in several ways. Half of the total carbon dioxide on the earth is fixed by algae by photosynthesis. About 70 marine algae species act as for several marine organisms and play a key role in the food chain. Brown and red algae produce a huge quantity of water-holding material called hydrocolloids. Carrageen is produced by red algae, and algin is produced by brown algae. They are used for commercial purposes.

112. **Option (d) is correct.**

Sometimes stamens of the flower are attached or united with other parts of the flower such as the petal. When stamens are attached to the petal, they are called epipetalous. When the stamens are united into one bunch, they are called monadelphous in two bunches they are called diadelphous. The diadelphous stamens are common characteristic in pea plants.

113. **Option (d) is correct.**

The term Auxins is derived from the Greek language. It means 'to grow', and it was

from human urine. Auxins can be natural and synthetic. Indole-3-acetic acid (IAA) and Indole butyric acid are examples of natural auxins. Naphthalene acetic acid and 2, 4-Dichlorophenoxyacetic (2, 4-D) are synthetic auxins. The auxin 2, 4-D is used as weedicide in agriculture to destroy weeds in the field. It can destroy the weeds of dicotyledonous plants in cereal crops.

114. **Option (c) is correct.**

The organisms with male and female sex organs are called monoecious. The organism with either male or female sex organs is called dioecious organisms. Chara plant is monoecious and has both male and female reproductive organs on the same plant. It is a bisexual plant.

115. **Option (c) is correct.**

The biochemical substances in an organism are called metabolites. Metabolites are of two types, which are primary and secondary metabolites. Carbohydrates, amino acids, lipids, nitrogen bases, nucleosides, and nucleotides are some primary metabolites. They play an important role in normal physiological processes. The materials which are required for growth and development are called secondary metabolites. These are useful to human welfare. Alkaloids, rubber, latex, flavonoids, antibiotics, scents, gums, spices and coloured pigments are examples of secondary metabolites.

116. **Option (d) is correct.**

Francis Crick proposed the Central dogma in molecular biology. It states the flow of genetic information from DNA to RNA to Protein. The transfer of information from DNA to DNA is called replication. The transfer of information from DNA to primary transcript (*mRNA*) is called transcription. It contains coding regions called exons and non-coding regions called introns. The process of synthesizing a protein from *mRNA* is called translation. So, the complete flow chart of central dogma is in the order of a) Replication; b) Transcription; c) Translation; and d) Protein.

117. **Option (a) is correct.**

Garner and Allard discovered the process of photoperiodism for the first time in the tobacco plant. Some plants need to expose to light for a well-defined critical period. Such a phenomenon is called photoperiodism. The exposure of a plant to light for a certain period will induce flowering. Leaves are the sites of the perception of light. Some phytohormones migrate from leaves to shoot apices to induce flowering when exposed to light.

118. Option (a) is correct.

The amount of biomass or organic matter produced per unit area for a certain period during photosynthesis is called primary production. It is measured in terms of weight. The rate of biomass production is called productivity. It is represented in terms of $g^{-2} yr^{-1}$. It is divided into gross primary productivity (GPP) and net primary productivity (NPP). A certain amount of GPP is used by the plant during respiration (R). The gross primary productivity minus respiration is equal to the net primary productivity. In the equation $GPP - R = NPP$, the letter 'R' denotes respiratory loss.

119. Option (d) is correct.

Cristae are inner foldings found in the inner mitochondrial membrane. Thylakoids are flat disk-like membranous sac-like structures in the stroma of plastids. The centromere is the constriction in the chromosome which holds two arms of chromatids together. Cisternae are disc-shaped sacs in the Golgi complex. They contain secretion like enzymes or hormones to be transported to another part of the cell.

120. Option (b) is correct.

Ectocarpus belongs to the class Phaeophyceae. They are very simple branched, filamentous, and represented by kelps. It stores food as complex carbohydrates in the form of laminarin or mannitol.

121. Options (b) is correct.

In Angiosperms, phloem tissue is made up of sieve tube elements, companion cells, phloem parenchyma, and phloem fibres. Mature sieve elements contain a peripheral cytoplasm, large vacuole but do not have a nucleus.

122. Option (d) is correct.

The process of inducing mutation artificially by using certain chemicals by gamma radiation, and selecting and utilizing plants with desirable character as a source of breeding is called mutation breeding. This type induced breeding is carried out by gamma rays in mung bean, which is resistant to yellow mosaic virus and powdery mildew.

123. Option (c) is correct.

All C_4 plants show the presence of a characteristic feature called Kranz Anatomy. It shows the presence of bundle sheath cells that form several layers around the vascular bundles. They are characterized by having a large number of chloroplasts in their leaves. The vascular bundles are surrounded by large parenchymatous tissue,

which is again surrounded by mesophyll cells. Bundle sheaths have large chloroplasts and grana while mesophyll cells have small chloroplasts. Sorghum is a kind of C_4 plant. The first stable product of the C_4 plant is oxaloacetic acid.

124. Option (b) is correct.

The attraction between water molecules due to hydrogen bonds is called cohesion. The attraction between polar surfaces or molecules is called adhesion. Surface tension states that water molecules are attracted more in the liquid phase than the gaseous phase. Loss of water from the margin of the leaf in the form of tiny water drops is called guttation.

125. Option (c) is correct.

To cut DNA with a restriction enzyme, it should be pure and free from other impurities. The plasma membrane RNA, proteins, polysaccharides, and lipids are treated with enzymes like lysozyme, cellulose, chitinase, and proteases. RNA can be removed with ribonuclease. Finally, the purified DNA is precipitated by adding chilled ethanol. Change - to fine threads of DNA.

126. Option (b) is correct.

The interaction between the population of two different species is called interspecific interaction. It can be beneficial, detrimental, or neutral. It can be represented by the sign of '+', '-', and '0' respectively. In amensalism, one species is harmed, whereas the other is unaffected. This type of interaction will prevent the growth of other species by releasing certain chemicals.

127. Option (d) is correct.

Meiosis ensures the production of the haploid phase in the life cycle. It involves two sequential cycles of nuclear division called meiosis I and meiosis II.

In meiosis II, during anaphase II, sister chromatids are held together with centromere. During this phase, the splitting of the centromere of each chromosome takes place. It allows moving chromosomes towards the opposite poles.

128. Option (c) is correct.

A person born with genetic defects can be cured by injecting cells with a corrected gene is called gene therapy. The defective gene is replaced by a normal gene in gene therapy. It also involves the delivery of a normal gene into an embryo or the individual to take over the function and compensate for the non-functional gene. This therapy involves

the amplification of the target gene in an individual's tissue to treat disease.

129. **Option (d) is correct.**

The organisms require a continuous supply of nutrients for growth, reproduction, and to regulate severe body functions. The number of nutrients like carbon, phosphorous, nitrogen, calcium, etc., present in the soil at any given time is called the standing state. It differs in the different ecosystem and seasonal basis.

130. **Option (c) is correct.**

The measure of the amount of food available in an ecosystem is called biomass. The pyramid of biomass represents the relations between the quantity of living matter at the different trophic levels. The pyramid of biomass in the sea is generally inverted due to the higher biomass of fishes than the phytoplankton.

131. **Option (b) is correct.**

An embryo sac is present at the centre the of each ovule. The embryo sac consists of the gametophytic cell. The female gametophyte undergoes three mitotic cell divisions to form eight nucleate stages of the embryo sac. Among these, two cells at the center fused to form polar nuclei. Three cells move towards the top, called antipodals. The remaining three cells are forming egg apparatus with one egg and two synergids at the micropylar end. Thus, a typical angiosperm embryo sac at maturity consists of 7 cells with 8 nuclei.

132. **Option (b) is correct.**

A chromosome consists of two chromatids held together by a centromere. Depending on the position of the centromere, the chromosomes are classified into four types. They are metacentric, sub-metacentric, acrocentric, and telocentric. In the metacentric chromosome, the centromere is located at the centre and forms two equal arms of chromatids.

133. **Option (a) is correct.**

The process of producing similar types of spores is called homosporous plants, as in pteridophytes. The process of producing two kinds of spores like macro and microspores, is called heterosporous plants. In genera like *Selaginella* and *Salvinia* produce two types of spores called microspores and macrospores.

134. **Option (c) is correct.**

Protoplast fusion is applied to obtain a hybrid variety called pomato. Totipotency is used in plant tissue culture. In this method, a cell from the explant can develop into a whole plant. Meristem culture is used to obtain tissues or

plants free from viruses. Micropropagation is used to obtain somaclones.

135. **Option (d) is correct.**

Polymerase Chain Reactions is used to multiply the gene of interest in vitro by using two sets of primers. This technique is used to detect gene mutation, molecular diagnosis, and gene amplification. Except for the purification of isolated proteins.

136. **Option (c) is correct.**

The process of copying genetic information from one strand of the DNA into RNA is called transcription. It is carried out by the transcription unit. The transcription unit has three regions in DNA called a promoter, structural gene, and terminator. RNA polymerase can catalyze the process of elongation in transcription. But the initiation and termination depend on the initiation factor and termination factor. The termination factor can also be called the Rho factor. When RNA polymerase binds to the Rho factor, the process of transcription is terminated in bacteria.

137. **Option (d) is correct.**

A probe is a radiolabelled ssDNA or RNA molecule. It is allowed to find the mutated gene by hybridizing to its complementary DNA in a clone of cells. It can be detected by autoradiography. But mutated genes cannot be found on photographic film, because the probe does not have complementarity with the mutated gene.

138. **Option (c) is correct.**

There are three types of RNA polymerases in the nucleus. They are RNA polymerase I, II, and III. RNA polymerase I transcribe *rRNA*. RNA polymerase II transcribes the precursor of *mRNA*, which *PSII*, while in the stroma lamellae membrane, is also called heterogeneous nuclear RNA. RNA polymerase III transcribes *tRNA*, *5S rRNA*, and small nuclear RNAs.

139. **Option (d) is correct.**

The meristematic tissue is present in the mature regions of roots and shoots and form the woody axis, and appears later than the primary meristem, known as secondary meristem. The cambium cells present in the dicot stem between the primary xylem and phloem are called intrafascicular cambium. The cells of medullary rays joint these cambial rings and become meristematic.

140. **Option (d) is correct.**

The equation $N_t = N_0 e^{rt}$ represents the exponential growth of a population.

N_t = Population density after time t

N_0 = Population density at time zero
 r = Intrinsic rate of natural increase and
 e = The base of natural logarithms

141. Option (c) is correct.

The synthesis of ATP is associated with the chemiosmotic hypothesis. It is linked with the development of proton gradient across the intermembrane space of the mitochondria. It is established by oxidation-reduction reactions. In Electron Chain each NADH + H^+ produces 3 ATP, and one molecule of $FADH_2$ produces 2 ATP molecules. They are produced when proton gradients of passed through ATP synthase.

142. Option (a) is correct.

In grana, lamellae contain both the PSI and PSII, while in the stroma lamellae membrane, PSII and NADP reductase are absent. The cyclic photophosphorylation takes place at light wavelengths beyond 680 nm. It occurs on at PSI, but not on PSII. Both of them are involved in non-cyclic photophosphorylation.

143. Option (c) is correct.

Different spores are produced by a different structure called fruiting bodies. The sexual cycle involves the three following steps. They are plasmogamy karyogamy, and meiosis in a zygote. The fusion of protoplasm between motile and non-motile gametes is called plasmogamy.

144. Option (c) is correct.

DNA fingerprinting can identify the differences in certain specific regions of DNA sequences called repetitive DNA. These small stretches of DNA are repeated several times to form a bulk peak of DNA called satellite DNA. Based on the composition of the base, length of a segment, and the number of repetitive units, the satellite DNA is of several types like micro-satellite and mini-satellite, etc. But DNA fingerprinting involves in finding differences the sequence of repetitive DNA.

145. Option (b) is correct.

A protein is made of polymerization of several amino acid molecules linked by peptide bonds. Unsaturated fatty acids contain one or more double bonds between the two carbon atoms. In a nucleic acid, a phosphate molecule links 3'- carbon of one nucleotide to the 5'- carbon of another sugar molecule of nucleotide to form an ester bond. A glycosidic bond links several monosaccharides to form a large complex polymer called a polysaccharide.

146. Option (b) is correct.

Nitrococcus oxidizes ammonia to nitrite, and it is converted into nitrate by *Nitrobacter*. *Rhizobium* bacteria help in converting atmospheric nitrogen to ammonia. *Thiobacillus* helps in the process of denitrification.

147. Option (a) is correct.

The viability of retaining pollen grains on the stigma depends on prevailing temperature and humidity. In some plants, Rosaceae, Leguminosae, and Solanaceae, stigma maintain the viability of pollen grains for several months. But in rice, pollen grains lose viability in 30 minutes.

148. Option (b) is correct.

The process of inactivating certain gene expressions by inserting recombinant DNA into them is called insertional inactivation. For the selection of recombinant DNA, the inactivation of antibiotics is a complicated process, and it requires simultaneous plating on two plates with different antibiotics. The alternative selective marker is developed by inserting recombinant DNA into the coding sequence of enzyme β -galactosidase.

The pBR³²² is used as a cloning vector. In this vector, the β -galactoside gene is inserted into the ampicillin resistance gene using Pst I, and the recombinant. *E. Coli* will not express the ampicillin-resistant gene. The bacteria are susceptible to ampicillin antibiotics. So, It will not confer ampicillin resistance to the host cell.

149. Option (b) is correct.

The floral formula of different families:

Brassicaceae family $\oplus \overline{\sigma} K_{2+2} C_4 A_{2+4} \underline{G}_{(2)}$

Solanaceae family $\oplus \overline{\sigma} K_{(5)} C_{(5)} A_5 \underline{G}_{(2)}$

Fabaceae family $\overline{\sigma} K_{(5)} C_{1+2+(2)} A_{(9)+1} \underline{G}_1$

Liliaceae family $\oplus \overline{\sigma} P_{(3+3)} A_{3+3} \underline{G}_{(2)}$

150. Option (d) is correct.

During the S phase, replication of DNA occurs. During the G₂ phase, the synthesis of protein and RNA takes place. Quiescent stage, the cell undergoes an active stage, and it remains undivided. During this phase, the cell remains metabolically active. The G₁ phase occurs between mitosis and the initiation of DNA replication. The cell grows to maximum size during this phase.

151. Option (a) is correct.

When a double-stranded RNA is breakdown into small pieces these pieces are called small interfering RNAs (siRNA). The siRNAs are used to silence and bind to complementary mRNA. It is not required for the synthesis of protein.

152. Option (c) is correct.

The intestinal mucosa contains goblet cells. They secrete a juice, and intestinal juice and it is referred to as succus entericus. It contains a variety of enzymes like maltase, dipeptidase, nucleosidase, and lipases.

153. Option (c) is correct.

The single DNA-dependent RNA polymerase can catalyze the transcription of all varieties of RNA in bacteria. It is also called a holoenzyme made up of several polypeptides. It is made up of two subunits of an alpha, a single subunit of beta, beta prime, and omega fragment. Those all together can catalyze the process of initiation, elongation, and termination in the process of transcription.

154. Option (b) is correct.

In a cockroach, the entire foregut is lined with a cuticle. Gastric caecae is present at the junction of the foregut and midgut. Malpighian tubules are located at the junction of the midgut and hindgut.

155. Option (d) is correct.

Metamerism is a common characteristic of the phylum Annelida. Canal system is present in phylum Porifera. Comb plates are found in the body of Ctenophores, useful for locomotion. Cnidoblasts are found in the family members of coelenterates. Hence, they are called cnidarians.

156. Option (c) is correct.

Biofortification is a method of breeding crops, to enhance their nutritional values. This can be done by selective breeding or through genetic engineering and breeding crops with higher vitamins, minerals, proteins, and healthy fats, which improve public health. This method is not used to create a plant resistance against the diseases.

157. Option (a) is correct.

The plant products are of two types called primary metabolites and secondary metabolites. Pigments, alkaloids, terpenoids, toxins, lectins, drugs, and polymeric substances are secondary metabolites. Ricin is a kind of toxin obtained from *Ricinus*. But it is not a drug.

158. Option (b) is correct.

Microbes are used for industrial and commercial production of some organic acids, alcohols, and enzymes. *Aspergillus niger* is used to produce citric acid. *Acetobacter aceti* is used to produce acetic acid. *Clostridium butylicum* is used to produce butyric acid, and *Lactobacillus* is used to produce lactic acid.

159. Option (d) is correct.

Insulin is a hormone produced by the beta-cell

of the Islet of Langerhans. It is synthesized as pro-insulin. It has C-peptide. In the initial stage, it comprises two polypeptides, called A-peptide and B-peptide, along with a stretch of C peptide. Later these two polypeptides are linked by disulphide linkage by removing a stretch of C peptide. So C-peptide is not present in the mature insulin.

160. Option (b) is correct.

When percentage saturation of haemoglobin with oxygen is plotted against the pO_2 , a sigmoid curve is formed. The Sigmoid curve can also be called the oxygen dissociation curve. It is used in studying the effect of factors like pCO_2 , H^+ concentration, etc. In the alveoli, high pO_2 , low pCO_2 , less H^+ , and low temperature will favour the formation of oxyhaemoglobin.

161. Option (d) is correct.

Physalia is known as the Portuguese man of war. *Limulus* is called a living fossil. *Ancylostoma* is a kind of roundworm called hookworm. *Pinctada* is called pearl oyster, belongs to phylum Mollusca.

162. Option (b) is correct.

If a cell in a fruit fly with 8 chromosomes is at the G_1 phase, the number of chromosomes remains the same after the S phase. Because during the S phase, the amount of DNA is double by the process of replication. But the number of chromosomes remains the same.

163. Option (b) is correct.

The skeletal system of birds is very long, hollow with air cavities called pneumatic. Pneumatic cavities are present in birds like, *Neophron*.

164. Option (d) is correct.

The thickness of the ozone layer in a column of air from the ground to the top region of the atmosphere is measured in Dobson units.

165. Option (b) is correct.

The process of coagulation of blood involves, several complex enzymes in a sequence of reactions. Thrombokinase converts prothrombin into thrombin in the plasma. This thrombin converts the inactive form of fibrinogen into fibrins. These fibrins form a network of threads on the surface of ruptured blood vessels and form a clot.

166. Option (d) is correct.

In the case of HIV/AIDS, an Enzyme-linked immunosorbent assay was used for early detection of an infection. This technique is used to detect in the early stages of the disease. It is detected by antibodies produced by the host against the pathogen by the immune system.

167. **Option (c) is correct.**

The membranous organelles are different in terms of their structure and functions. They together form endomembrane system, because their functions are coordinated, it includes the endoplasmic reticulum, Golgi complex, lysosome, and vacuole.

168. **Option (d) is correct.**

According to Chargaff's rule, DNA consists of a 1:1 (base pair rule) ratio of pyrimidine and purine bases. The amount of guanine is equal to the amount of cytosine. Similarly, the amount of thymine is equal to adenine.

Number of Adenine = number of thymine

If the number of (A) is 30%, then the number of thymine would be 30%

$$\begin{aligned}\text{Therefore} &= 100 - (A + T) \\ &= 100 - (30 + 30) \\ &= 100 - 60 = 40\end{aligned}$$

The percentage of (C) and (G) = 40

Then (C) and (G) would be 20%.

T=30%, A=30%, G=20%, and C=20%

169. **Option (d) is correct.**

The restriction enzyme endonucleases are called molecular scissors, which cut the DNA at a specific sequence. The restriction endonucleases can find specific nucleotide recognition sequences. The sequence of the first half of one strand will be like a mirror image with the second half of the complementary strand. This is called a palindromic nucleotide sequence.

170. **Option (d) is correct.**

The intercalated discs are located only in cardiac muscles. They are absent in smooth muscles. They are involuntary muscles present in the uterus, alimentary canal, gall bladder, and wall of the blood vessels.

171. **Option (c) is correct.**

Vaults, diaphragms, and cervical caps are temporary contraceptive methods adapted for females. They block the entry of sperm.

IUDs enhance the phagocytosis of sperms in the uterus. They act as a spermicide.

Vasectomy is a surgical method of contraception adapted for males. Vas deferens are cut and tied up with a small incision on the scrotum.

Tubectomy is a surgical method of contraception adapted for females. A small portion of the fallopian tube is cut and tied up through an incision in the abdomen or through the vagina.

172. **Option (d) is correct.**

Polymerase chain reaction includes three steps called denaturation, annealing, and extension.

At 94°C, the two complementary strands are separated. This is called denaturation. If this temperature is not maintained in the beginning, it will affect the process of denaturation.

173. **Option (d) is correct.**

Sickle cell anaemia is an example of autosomal recessive disorder. It is caused due to frameshift mutation, which leads to the replacement of valine in the place of glutamic acid. It is transmitted from parent to offspring when both the partners are heterozygous carriers. It is expressed only in the homozygous ($Hb^S Hb^S$) individuals that are about only 25%.

	Male	×	Female
Parents:	$Hb^A Hb^S$		$Hb^A Hb^S$
Gametes:	Hb^A Hb^S		Hb^A Hb^S

♀ \ ♂	Hb^A	Hb^S
Hb^A	$Hb^A Hb^A$ Normal	$Hb^A Hb^S$ Carrier
Hb^S	$Hb^A Hb^S$ Carrier	$Hb^S Hb^S$ Sickle cell

Normal: 25%
Carriers: 50%
Sickle cell: 25%

174. **Option (a) is correct.**

Echinoderms are triploblastic and coelomates with true coelom. Roundworms have an organ-system level of body organization. The water vascular system is a characteristic of Echinoderms. The above three statements are correct.

175. **Option (a) is correct.**

Zona pellucida is a glycoprotein polymer capsule surrounding the secondary oocyte. It is an antral follicle, and the acellular layer contains receptors (ZP3) for the binding of sperm. It contains enzymes that catalyze the sperm to penetrate the oocyte during fertilization and remains even after ovulation.

176. **Option (c) is correct.**

The common hepatic duct and cystic duct unite together to form bile duct. The bile duct and main pancreatic duct reach the major duodenal papilla. These two ducts join each other and expand to form

hepatopancreatic ampulla. The sphincter of Oddi is a muscular structure located at the junction hepatopancreatic duct and duodenum.

177. **Option (d) is correct.**

Myasthenia gravis is an autoimmune disorder. It mainly affects the neuromuscular junction results in fatigue, weakening, and paralysis of skeletal muscle.

178. **Option (a) is correct.**

A person with 'AB' blood groups is called "Universal recipients". Because they do not have anti-A and anti-B antibodies in their plasma, their bodies can accept blood from all blood groups.

179. **Option (b) is correct.**

In animal cells, replication of DNA occurs during the S phase of the cell cycle. At the same time, centrioles also duplicate in the cytoplasm and move towards the opposite poles.

180. **Option (a) is correct.**

The Juxtaglomerular cells in the kidneys produce a peptide hormone known as erythropoietin. This hormone stimulates the process of erythropoiesis.

181. **Option (a) is correct.**

The housefly belongs to the phylum – Arthropoda, the class – Insecta, the order – Diptera, the family – Muscidae, and the Genus – *Musca*. The scientific name of a housefly is *Musca domestica*.

182. **Option (d) is correct.**

Some diseases are transmitted through sexual intercourse. Such diseases are called sexually transmitted diseases or venereal diseases. It can spread by the transfusion of blood from an infected person to a healthy person and from an infected mother to a fetus.

183. **Option (b) is correct.**

The pressure exerted by an individual gas in a mixture of gases is called the partial pressure of that gas. It is represented with the letter 'p'. The partial pressure of oxygen and carbon dioxide at the alveoli are $pO_2 = 104$ and $pCO_2 = 40$, respectively

184. **Option (c) is correct.**

The process of preventing fertilization from preventing pregnancy in women is called contraception. It is the most widely accepted method to delay pregnancy or space for children. LNG-20 is a kind of contraceptive. It is used as an intrauterine device. It releases a hormone called levonorgestrel 20 mg. It acts as phagocytosis of sperms so that a woman can avoid pregnancy.

185. **Option (d) is correct.**

Diakinesis is the last stage of meiotic prophase I. During this phase, the terminalisation of chiasmata takes place. Chromosomes are fully condensed, and the meiotic spindle is assembled to separate homologous chromosomes.

186. **Option (a) is correct.**

The process of delivering a baby is called parturition. It is induced by a complex neuroendocrine mechanism. An increase in estrogen and progesterone ratio, synthesis of prostaglandins, and release of oxytocin will induce the contraction of the uterus. This can lead to parturition. But prolactin does not play any role in parturition. It promotes the growth of the mammary gland and the formation of milk.

187. **Option (b) is correct.**

In earthworms, prostomium serves as a covering for the mouth. It looks like a wedge and is used to open cracks in the soil into which it can crawl. It has receptors and can also act as a sensory structure.

188. **Option (b) is correct.**

Adenosine deaminase is an enzyme which is very crucial for the immune system. Defects in this gene of adenosine deaminase result in a disease called severe combined immunodeficiency disease. This occurs due to dysfunction of the immune system.

189. **Option (b) is correct.**

Adaptive radiation is the process of evolution of different species in a geographical area and radiating to other areas—for example, Darwin's finches.

The species with analogous organs exhibit convergent evolution. The organs are anatomically not similar, but have same functions. E.g., wing of birds and bats.

The species with homologous organs exhibit divergent evolution. The organs are anatomically similar but perform different functions in different organisms—bone of man and whale.

Evolution by anthropogenic action occurs due to human interference, such as selecting antibiotic-resistant microbes, herbicides, and pesticide-resistant varieties.

190. **Option (d) is correct.**

Mammals from cold climates possess shorter ears and limbs to reduce heat loss. This is called Allen's Rule.

Physiological adaptation can be observed in Kangaroo rats.

Behavioural adaptation can be seen in the desert lizard.

Biochemical adaptation can be observed in marine fishes living at the deep bottom of seas and oceans.

191. **Option (c) is correct.**

Histone octamers consist of units of 8 histone molecules with a pair of each protein H2A, H2B, H3, and H4. H1 histone present outside of the octamer connects the DNA with the octamer. Histones consist of basic amino acid residues such as lysine and arginine with a side chain. They have a positive charge in the side chain. But they are not acidic.

192. **Option (b) is correct.**

At high altitudes, mountaineers experience altitude sickness due to a lack of sufficient oxygen. At such altitude, atmospheric pressure is very less. Hence, the person feels nausea, fatigue, and heart palpitation.

Hence, both the Assertion (A) and Reason (R) are correct.

193. **Option (b) is correct.**

Multiple Ovulation Embryo Transfer Technology is used for herd improvement. The cow is given FSH – a hormone to induce follicular and superovulation but not LH. Each cow produces about 6-8 eggs per cycle. 8-32 celled embryos are transferred to the surrogate mother. The cow is fertilized by artificial insemination.

194. **Option (c) is correct.**

Filariasis is caused by *Wuchereria bancrofti*. Amoebic dysentery is caused by *Entamoeba histolytica*. Pneumonia is caused by a bacteria called *Streptococcus Pneumoniae*. Ringworm is caused by fungi that belongs to genera *Microsporum*, *Epidermophyton*, and *Trichophyton*.

195. **Option (d) is correct.**

Lecithin is a kind of phospholipid. It has phosphorus and a phosphorylated organic compound. Ex. Soyabean oil. It is present in neural tissues with a more complex structure. Glycerol is an example of simple lipid. It can

also be called trihydroxy propane. Several lipids have both glycerol and fatty acids.

196. **Option (c) is correct.**

The ovary produces the hormone relaxin in the later phase of pregnancy. After ovulation, the ruptured Graafian follicles are converted into corpus luteum. The corpus luteum produces a hormone called relaxin.

197. **Option (c) is correct.**

In the epithelium and other tissues, three types of junctions are present. They are tight junctions, adhering junctions, and gap junctions. Tight junctions prevent the leakage of substances across a tissue. Gap junctions help in communication with each other by connecting the cytoplasm of adjoining cells. It helps in the rapid transfer of ions, small molecules, and other big molecules.

198. **Option (a) is correct.**

The scapula is a triangular flat bone located between the second and seventh ribs. Cranium is formed by the fusion of fibrous joints by dense connective tissue. Sternum is a flat bone on the ventral midline of thorax. The vertebral column is formed by the joining of cartilaginous joints between the adjacent vertebrae.

199. **Option (b) is correct.**

During muscular contraction, the following events occur in the following sequence order. There are disappearing 'H' zone, decrease in the width of the 'I' band, myosin hydrolyzes ATP to release ADP and inorganic phosphate, and Z-lines are pulled inward attached to actin filaments.

200. **Option (a) is correct.**

The code which has at least three letters is called the triplet code. There are about 64 codons to code for 20 amino acids. Among them, AUG has an initiator dual function. It codes for Methionine and acts as an initiator codon. The codon 'AAA' and 'AAG' code for amino acid lysine. So, statement I is incorrect, but statement II is correct.

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